



TORCH-

TORCH-X

USER OPERATION MANUAL

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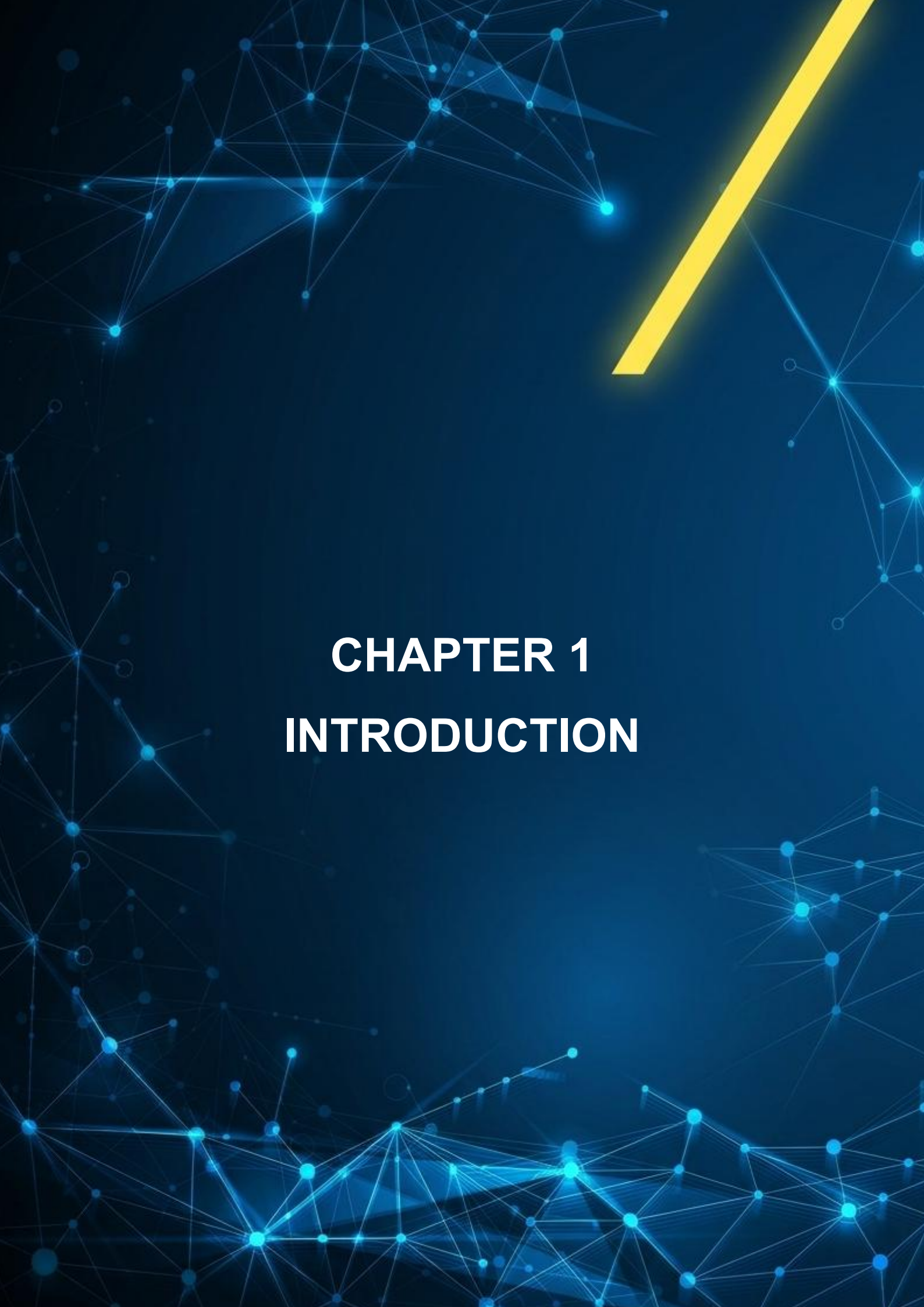
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CHAPTER 1

INTRODUCTION

1.1 PURPOSE OF THIS MANUAL

The purpose of this manual is to provide DITA gun users with comprehensive guidance for the operation and use of the Torch-X™ Artillery system as implemented on the 155 mm DITA platform.

1.2 ARRANGEMENT OF THIS MANUAL

This section introduces the Torch-X™ Artillery system as implemented on the 155 mm DITA platform and establishes the context for the entire manual. It defines the purpose and structure of the document while providing essential background information required for its effective use. This manual includes the following chapters:

Chapter 1 - Introduction

This chapter provides general information on the scope and arrangement of the manual, list of applicable documents, general safety instructions, definition of warnings, cautions and notes, and list of abbreviations.

Chapter 2 - GENERAL DESCRIPTION

This chapter provides a comprehensive overview of the Torch-X™ Artillery system as implemented on the 155 mm DITA platform. It outlines the system architecture, operational purpose, and key capabilities that enable advanced command and control of artillery fire missions within a networked battlefield environment.

Chapter 3 - HUMAN-MACHINE INTERFACE

This chapter defines the Human-Machine Interface (HMI) of the Torch-X™ Artillery system as implemented on the 155 mm DITA platform. It provides a detailed description of the graphical user interface, system screens, and user interaction mechanisms that enable efficient operation of the system under operational conditions.

Chapter 4 - USER GUIDE

This chapter provides comprehensive operational guidance for the effective use of the Torch-X™ Artillery system on the 155 mm DITA platform. It describes the step-by-step procedures required to perform system operations, from initial setup through execution and management of fire missions.

1.3 APPLICABLE DOCUMENTATION

N/A

1.4 WARNINGS, CAUTIONS AND NOTES

Warnings and cautions precede the text and follow the paragraph heading to which they apply. Notes may precede or follow applicable text, depending upon the material to be highlighted. Warnings, cautions or notes are concise statements used to emphasize important or critical data. The definitions of warning, caution and note are as follows:

WARNING

A warning highlights an essential operating or maintenance procedure, practice, condition, statement, etc., which, if not strictly observed, could result in injury to, or death of, personnel or long term health hazards.

CAUTION

A caution highlights an essential operating or maintenance procedure, practice, condition, statement etc., which, if not strictly observed, could result in damage to, or destruction of, equipment or loss of mission effectiveness.

NOTE

A note highlights or clarifies an essential system description or operating or maintenance procedure, condition or statement.

1.5 LIST OF ABBREVIATIONS AND TERMS

1.5.1 Terms

Term	Meaning	Description
Live Mission	-	An operational fire mission executed using live ammunition. The system fully controls and supports the firing process, including ballistic calculations, loading, aiming, and firing functions.
Practice Mission	-	A training mission used for simulation and crew proficiency. Ballistic solutions are calculated, but weapon firing circuits remain inhibited to prevent live fire.
Standard Mode	-	A fire mission execution mode in which the operator manually controls and authorizes each stage of the firing sequence, including aiming, loading, and firing.
Semi-Automatic Mode	-	A mission execution mode in which the system automates preparation and loading processes, while the operator retains firing authorization.
Automatic Mode	-	A mission execution mode in which the system automatically performs aiming, loading, and firing operations with minimal operator intervention.
CFF	Call for Fire	A structured request for artillery fire support transmitted through the command-and-control network.
Quick Task	-	A rapidly generated operational task that can be assigned directly from the tactical map, such as ISR, Engagement, or Evacuation.
LOS	Line of Sight	A terrain analysis function used to determine whether a direct visual path exists between two geographic locations.
Dominated Area	-	A terrain analysis product identifying areas visible from a selected observation point based on terrain elevation data.
MV	Muzzle Velocity	The speed of a projectile as it exits the weapon barrel. Used by the ballistic engine to improve firing solution accuracy.
Optimized Fire Solution	-	The primary ballistic solution selected by the system based on mission requirements, ammunition availability, safety constraints, and environmental conditions.
Grid Correction	-	A fire adjustment applied to an active mission in order to correct the point of impact based on observer feedback or battlefield conditions.

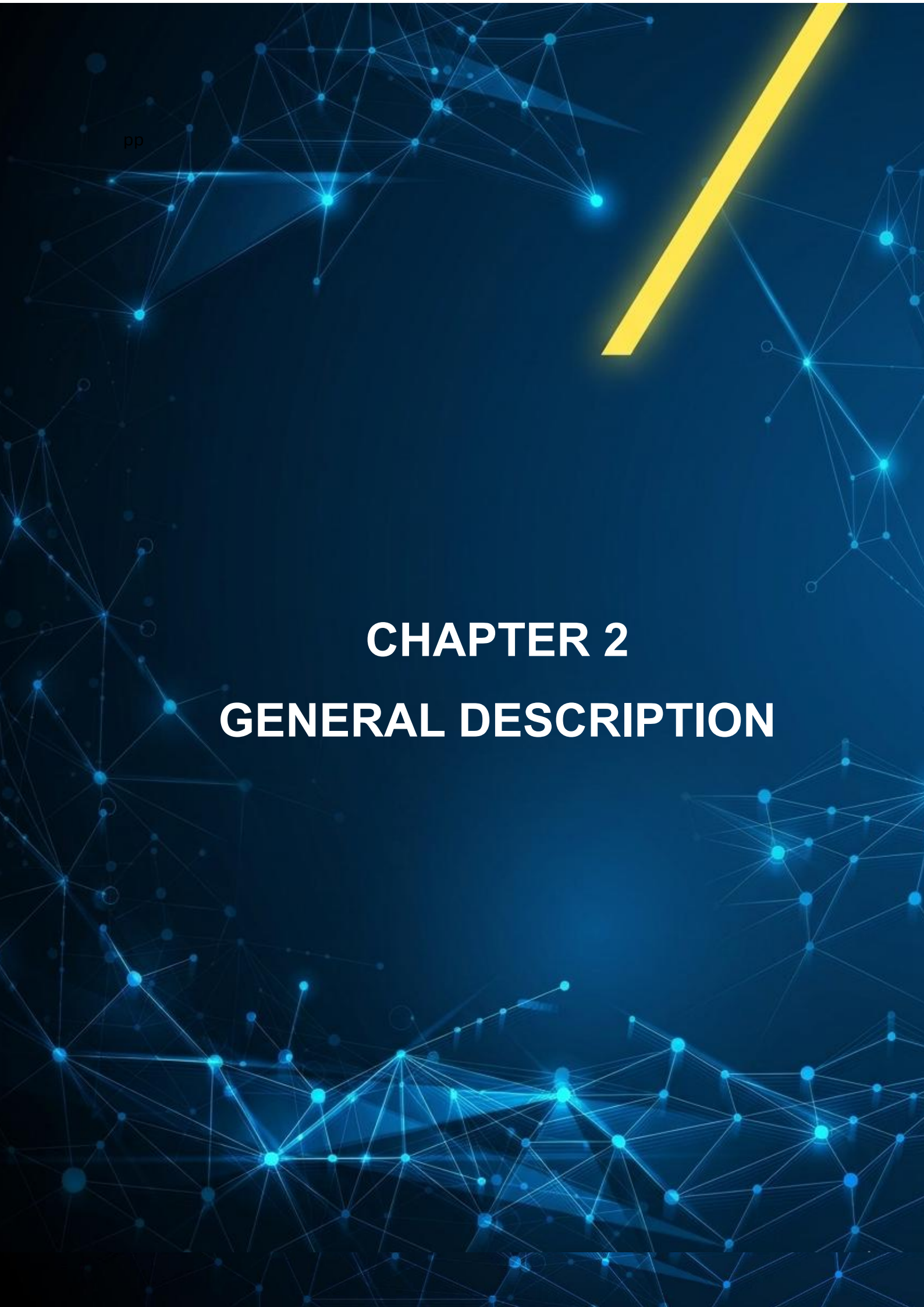
Term	Meaning	Description
METCM	Meteorological Computer Message	A set of data, describing meteorological conditions in various heights around an area known as the Center of Validity (CoV). These meteorological conditions serve as input parameters for various ballistic calculations for the purpose of increasing the accuracy of artillery fire, and allowing friendly radar systems to better detect and identify friendly artillery shells while in mid-flight for the purpose of performing artillery adjustments. METCM objects may be created via a meteorology device or entered manually into the system. These objects are distributed between all adjacent artillery unit.
METEO11	Meteorological Message Type 11	A meteorological message containing atmospheric data such as wind, temperature, and air density at different altitudes for ballistic computation purposes.
Fire Mission	-	A tactical mission created to engage a target using indirect fire. It includes target data, ammunition selection, ballistic calculations, and mission execution parameters.
Fire Plan	-	A component of a fire mission that defines ammunition type, fuze settings, number of rounds, firing rate, and engagement method.
MRSI	Multiple Rounds Simultaneous Impact	A firing technique in which multiple rounds are launched on different trajectories so that they impact the target simultaneously.
MQE	Minimum Quadrant Elevation	A safety-related firing parameter that defines the minimum permissible elevation angle for a given firing sector or mission.
TOT	Time on Target	A coordinated fire support technique in which rounds are planned to arrive at the target at a specific designated time.
AOI	Area of Interest	A designated operational area used for monitoring, planning, analysis, and situational awareness on the tactical map.
Common Fire Mission	-	A standard fire mission created through the tactical map or mission management workflow for engaging a designated target.

1.5.2 Abbreviations

Abbreviation	Meaning
AOI	Area of Interest
AMSL	Above Mean Sea Level
BCP	Battery Command Post
CAS	Close Air Support
C4I	Command, Control, Communications, Computers and Intelligence
C4ISR	Command, Control, Communications, Computers, Intelligence, Surveillance and Reconnaissance
CFF	Call for Fire
CEP	Circular Error Probable
CoV	Center of Validity
DCB	Data Collection Box
DEM	Digital Elevation Model
DFS	Direct Fire Control System
DTG	Date-Time Group
DTM	Digital Terrain Model
EAPE	Estimated Altitude Position Error
EHPCEP	Estimated Horizontal Position Circular Error Probable
EVAC	Evacuation
ETC	Enhanced Tactical Computer
FCS	Fire Control System
FDC	Fire Direction Center
FFE	Fire for Effect
FO	Forward Observer
GIS	Geographic Information System
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HMI	Human-Machine Interface
IBIT	Initiated Built-In Test
INS	Inertial Navigation System
ISR	Intelligence, Surveillance and Reconnaissance
LACU	Loading Automat Control Unit
LOS	Line of Sight
LUT	Last Update Time
MED	Medical Support

MEDEVAC	Medical Evacuation
METCM	Meteorological Computer Message
METEO11	Meteorological Message Type 11
MGRS	Military Grid Reference System
MQE	Minimum Quadrant Elevation
MRSI	Multiple Rounds Simultaneous Impact
MV	Muzzle Velocity
MVR	Muzzle Velocity Radar
MVW	Muzzle Velocity for Work
NATO	North Atlantic Treaty Organization
OPORD	Operation Order
RAS	Robotic and Autonomous Systems
SAM	Surface-to-Air Missile
SPG	Self-Propelled Gun
TOT	Time on Target
UAV	Unmanned Aerial Vehicle
UTC	Coordinated Universal Time
UTM	Universal Transverse Mercator
WGS	Weapon Guidance System

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CHAPTER 2

GENERAL DESCRIPTION

2.1 SCOPE

This chapter provides a comprehensive overview of the Torch-X™ Artillery system as implemented on the 155 mm DITA platform. It outlines the system architecture, operational purpose, and key capabilities that enable advanced command and control of artillery fire missions within a networked battlefield environment.

2.2 GENERAL OVERVIEW

Elbit Systems' **Torch-X™** is an advanced, modular Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance (C4ISR) and battle management software framework.

It connects combat units, sensors, and weapons across air, land, and sea, delivering real-time situational awareness and AI-driven decision support.

Torch-X operates across several specialized defense domains to shorten the "sensor-to-shooter" loop:

- **Mounted:** Integrates sensors, weapons, and communications into combat vehicles for networked command and control.
- **Dismounted:** Connects individual soldiers on the battlefield, providing Blue Force Tracking, navigation, augmented reality, and 3D visualization.
- **Artillery & Fires:** Provides digital fire control and management for indirect fire and Close Air Support (CAS) to optimize precision and coordination.
- **Maritime & Borders:** Fuses data from various physical and electronic sensors to provide a unified operational picture for coastal and naval protection.
- **Robotic and Autonomous Systems (RAS):** Allows commanders to control and integrate unmanned platforms.

The **Torch-X Artillery application** enables the gun crew to create fire missions and fire plans, communicate with allied forces via an integrated messaging system (chat), manage and monitor onboard ammunition inventory, and input meteorological data.

The application provides **real-time situational awareness** by displaying precise positioning, azimuth, and elevation data, while computing advanced ballistic firing solutions and monitoring critical weapon system statuses.

The gun crew can manage tactical maps and execute vital operational commands—all delivered through a modern, ergonomic, and intuitive user interface designed with high-visibility for **easy and fast operation under combat conditions**.

Specifically for the **DITA gun platform**, Torch-X is designed to autonomously operate the Loader and Control Unit (LACU) system to execute fire missions with minimal user interaction.



Figure 2-1: Torch-X™ Artillery Application

2.3 OPERATIONAL NEED

The **Torch-X™** Artillery primary operational need is to enable the DITA gun crew to engage artillery targets quickly with high accuracy.

In the modern networked warfare, there is a critical need for battlefield digitization and frameworks that enhance combat effectiveness while reducing the cognitive load on personnel at all echelons.

By minimizing manual interaction with both the gun and C4I systems, Torch-X provides optimal execution of fire missions while maintaining constant control over safety and system status.



Figure 2-2: 155 mm DITA Self Propelled Gun (SPG) C4I Suite

2.4 MAIN FEATURES

- Provides four distinct autonomy levels for managing missions: Manual, Standard, Semi-Automatic, and Automatic
- Key technical features include sophisticated ballistic algorithms, rapid navigation and deployment procedures, and NATO interoperability standards.
- Rapid sensor-to-shooter loops and superior situational awareness.
- One-command firing procedures, such as "Load" for Semi-Automatic or "Start" for Automatic modes.
- Automatic safety cessation, where firing procedures stop automatically if technical errors or safety issues are detected.
- Enables gun crew to manage fire missions through four levels of autonomy, which are designed to minimize user interaction and optimize mission execution.

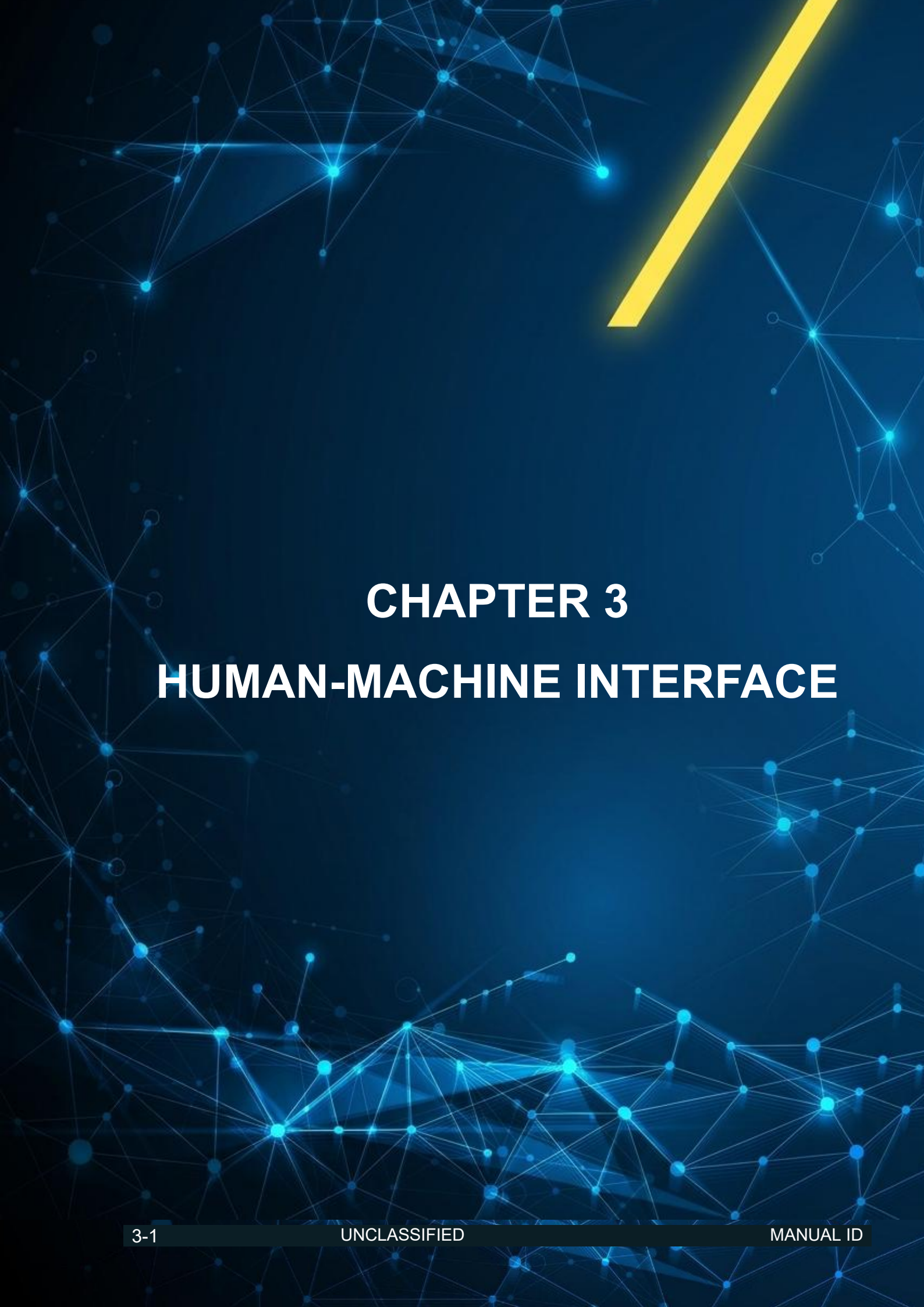
2.5 INTERFACES

Torch-X is built on the E-CIX™ open architecture framework, for seamless integration with a wide array of hardware and software.

It interfaces directly with platform control systems, computing terminals, and radio communications. The system features integration with various sensors, including INS, MVR, WGS, and DCB.

Furthermore, it connects multiple echelons and roles, from Forward Observers (FO) to Battery Command Posts (BCP) and battalion Firing Direction Centers (FDC).

For the DITA platform, the core interface involves the autonomous operation of the LACU system to manage firing timing and rates.



CHAPTER 3

HUMAN-MACHINE INTERFACE

3.1 SCOPE

This chapter defines the Human-Machine Interface (HMI) of the Torch-X™ Artillery system as implemented on the 155 mm DITA platform. It provides a detailed description of the graphical user interface, system screens, and user interaction mechanisms that enable efficient operation of the system under operational conditions.

3.2 SCREENS TREE

(See Figure 3-1)

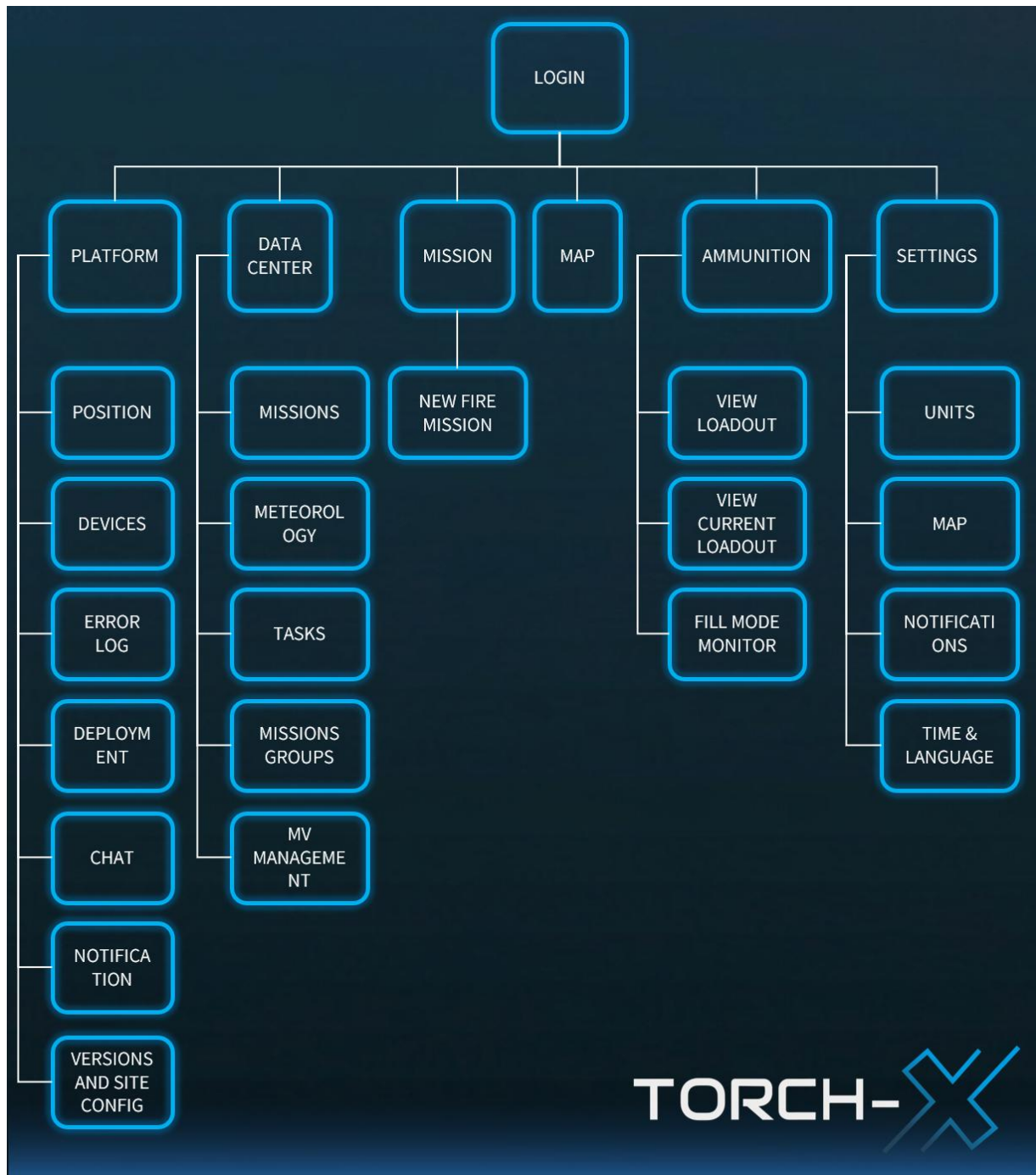


Figure 3-1: Screens Tree

3.2.1 Screen Layout

(See Figure 3-2 and Table 3-1)

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Figure 3-2: Screen Layout

Table 3-1: Screen Layout

No.	Item	Description
1	Application Header	Displays critical system parameters and alerts while navigating between different application screens. [refer to section 3.2.1.1].
2	Screen/Window Name	Displays the current screen or menu name.
3	Main Work Area	Displays active operational screens. Serves as the central interface for the application's ongoing operations, automatically resizing its layout to accommodate side panels (like the Side Map Panel) when expanded.
4	Split-Screen Button (<)	Resizes and splits the screen layout to display the tactical map alongside the active workspace.
5	Main Screens Navigation Bar	Enables the user to switch instantly between application screens with a single tap. [refer to section 3.2.1.3]

3.2.1.1 Application Header

(See Figure 3-3 Table 3-2)



Figure 3-3: Application Header

Table 3-2: Application Header

No.	Item	Description
1	TORCH-X Logo	When tapped, opens the Software Versions And Site Config window [refer to section 3.2.1.1.1]
2	Connectivity Indicator	Displays the current connectivity status between the TORCH-X application and external network entities, including local systems, Battery, friendly units.
3	Beacon Indicator	Provides a real-time status overview of the Ballistic Engine (Beacon). The Beacon is responsible for executing the artillery firing algorithms and ballistic software package, generating accurate firing solutions and supporting all ballistic computations required for mission planning and execution on the DITA artillery system. States:  - the Beacon is offline, disconnected, or has a critical software/hardware failure.  - the Beacon is fully connected, online, and functional.
4	Navigation Indicator	Provides the operator with a real-time overview of the platform's self-positioning and spatial orientation status, specifying the active data source.  - No valid self-positioning data (either automated or manual).  - platform's self-positioning coordinates are not derived from automated sensors (GPS/INS) but were manually input by the operator.  - platform's self-positioning data is currently derived from the Global Positioning System (GPS). Tapping the Navigation Indicator opens info window:

No.	Item	Description
		Navigation Status Manual Position: No Selected Is Connected: No Is Valid Position: Yes Is Stable Mode: No Is Position Fixed Manually: No • Manual Position: Indicates whether the current coordinates were manually entered by the crew rather than retrieved from automated navigation systems. • Is Valid Position: A dynamic status confirming whether the current positioning data is reliable and valid for firing operations. • Is Position Fixed Manually: Indicates whether a manual override or calibration of the platform's position was performed by the operator (e.g., during GPS/INS degradation). • Selected Is Connected: Displays the connectivity status (Yes/No) of the selected navigation/positioning sensor. • Is Stable Mode: Indicates whether the navigation system has completed its alignment process and reached a stable state required for accurate movement or firing.
5	Platform Indicator	Displays the geographical coordinates of the platform using the Universal Transverse Mercator (UTM) coordinate system. •  - displayed when the operator presses EXIT POSITION button. Indicates that the vehicle has left its firing position or is in transit, declaring the stored position data invalid. •  - displayed when the operator presses SET POSITION button. Indicates that the platform is deployed in a defined firing position, the coordinates are locked into the system, and the howitzer is ready for ballistic calculations. Tapping the Platform indicator window opens an info window: • Current Position: Displays the platform's live self-positioning coordinates (derived from the active navigation source). • Save as Position: Displays the position coordinates saved in the system from the last time SET POSITION was executed.
6	Mission Type	Defines the tactical objective of the fire mission: Practice or Live.
7	Mission Activation Status	Indicates the current phase and operational state (Waiting, TBD) of the selected fire mission
8	Mission Autonomous Mode	Displays the automation state of the DITA howitzer's systems selected by the user: Standard mode, Semi-automatic mode, or Automatic mode. .

No.	Item	Description
9	Chat icon	Opens the Chat Window. This window provides an instant-messaging interface for communication between the platform crew and external command elements or other units. [refer to section 3.2.1.1.2].
10	Notifications icon	Opens the Notifications window that categorizes all system notifications into four tabs: ALL , Threats , Caution , and Info . [refer to section 3.2.1.2.3]
11	System Clock	Displays synchronized system time in a 24-hour format (HH:MM:SS).

3.2.1.1.1 System Version and Site Config Window

(See Figure 3-4 and Table 3-3)

9002-087v01

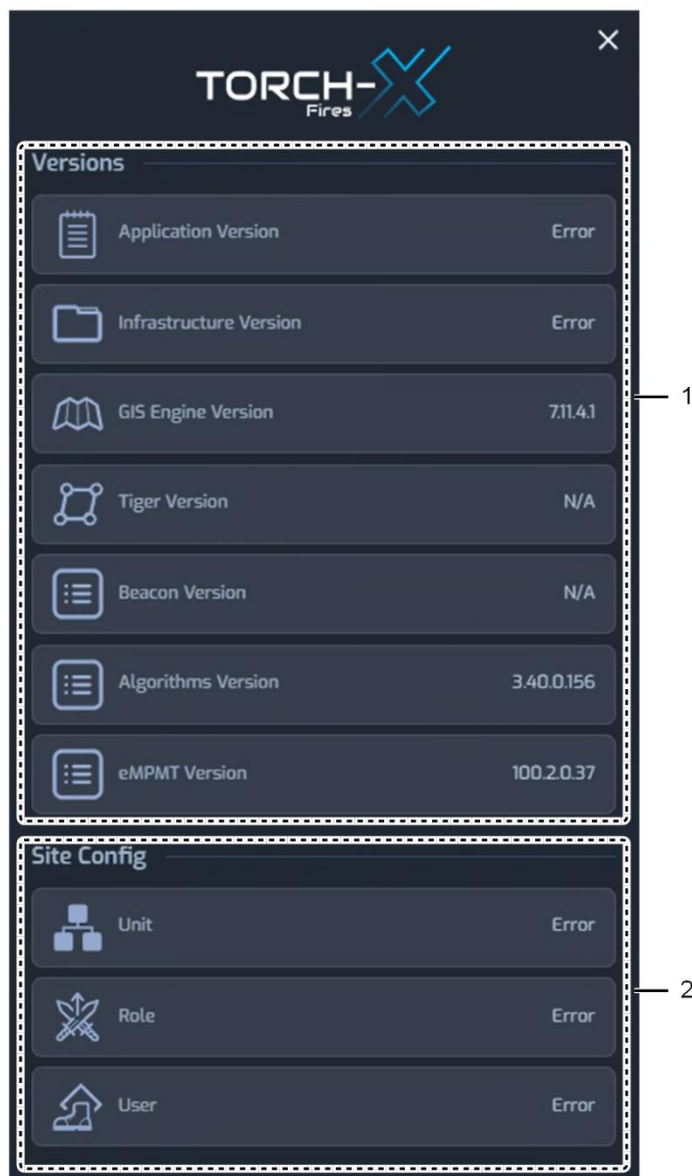


Figure 3-4: System Version and Site Config Window

Table 3-3: System Version and Site Config Window

No.	Item	Description
1	Versions Section	Provides a centralized view of the current software/firmware versions running in various system components: • Application Version - The version of the TORCH-X application. • Infrastructure Version - The version of the core system Infrastructure. • GIS Engine Version - The version of the Geographic Information System (GIS) engine used for the digital mapping, terrain analysis, and spatial data • Tiger Version - The core software infrastructure that serves as the secure runtime environment and management layer for the TORCH-X). • Beacon Version - The version of the beacon module. • Algorithms Version - The version of the core fire-control calculation engine, ballistic modules, and trajectory computation software. • eMPMT Version - The version of the eMPMT engine.
2	Site Config	A core infrastructure component that dictates the application's operational profile based on the specific asset type it is deployed on within the artillery network. This configuration dynamically adapts the HMI, data flows, and command capabilities. • Unit- displays the tactical designation or ID of the specific DITA artillery battery • Role - displays the operational profile assigned to the application (e.g., Commander, Gunner) • User – displays the current user logged into the system.

3.2.1.1.2 Chat Window

(See Figure 3-5 and Table 3-4)

9002-048v01

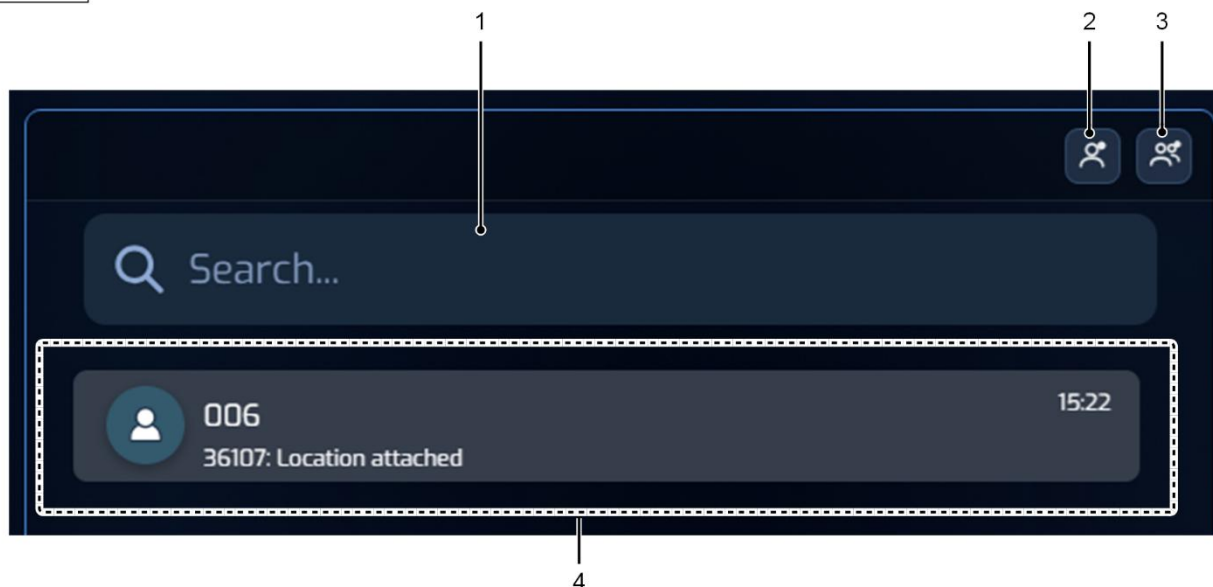


Figure 3-5: Chat Window

Table 3-4: Chat Window

No.	Item	Description
1	Search Field	Enables the user to type and quickly filter specific user within the list. Tapping the user record, opens a conversation window [refer to section 3.2.1.2].
2		When clicked, opens a list of available network users to initiate a conversation.
3		When clicked, Opens a list of available network users to create a group chat.
4	Network Users	Displays a list of available network users to initiate a conversation. Tapping the user record, opens a conversation window [refer to section 3.2.1.2.2]. User type icon is displayed next to each user record: Group () or Individual User ()

3.2.1.1.3 User Chat Window

(See Figure 3-6 Table 3-5)



Figure 3-6: User Chat Window

Table 3-5: User Chat Window

No.	Item	Description
1	 (Pick Entity)	Enables the user to quickly choose an entity directly from the tactical map interface.
2	Search Field	Enables the user to type and quickly filter specific user within the list.
3	Network Users	Displays a list of available network users to initiate a conversation. Tapping the user record, opens a conversation window [refer to section 3.2.1.2]. User type icon is displayed next to each user record: Group () or Individual User ()

3.2.1.2 User Conversation Window

(See Figure 3-7 and Table 3-6)

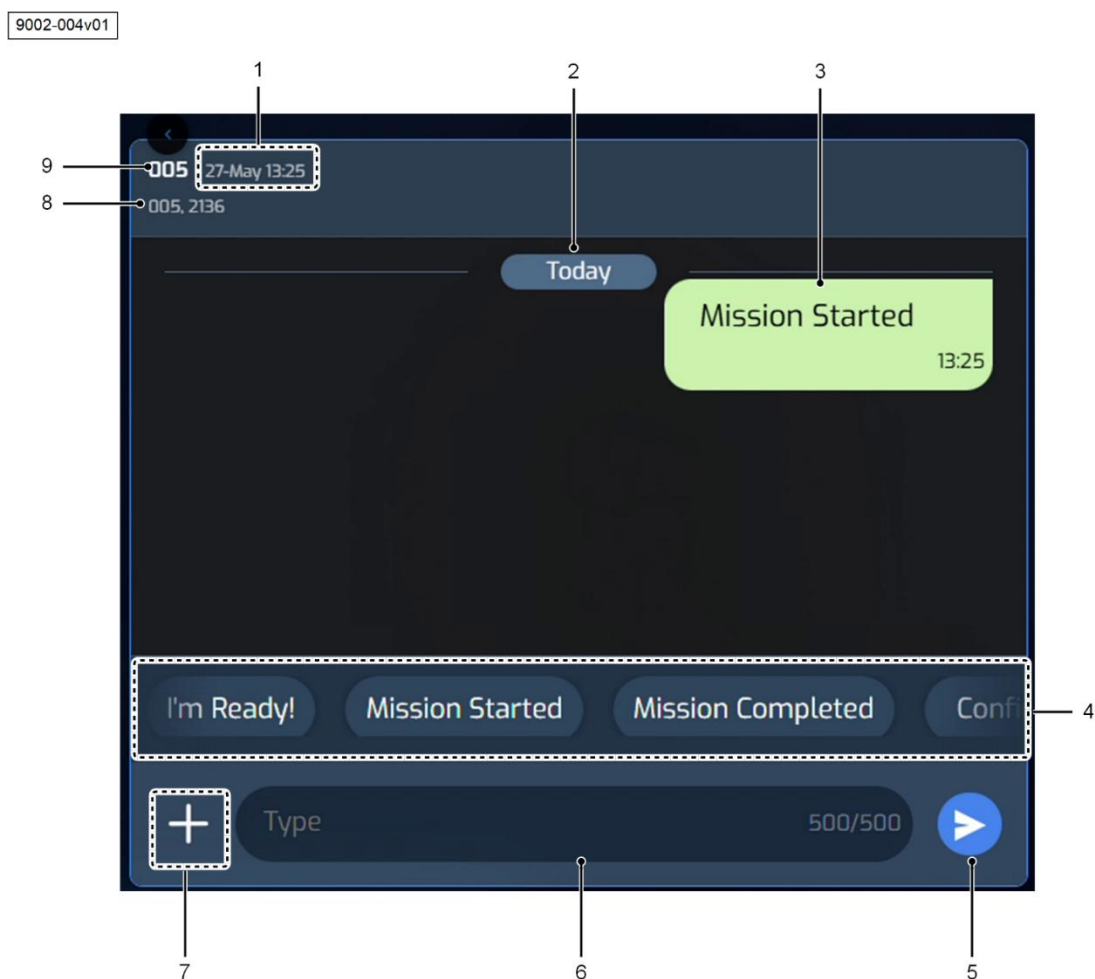
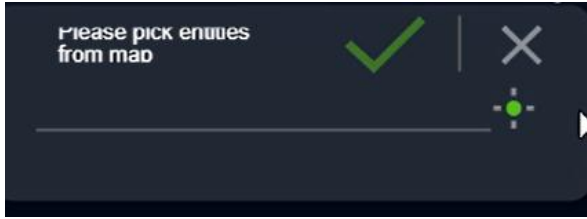


Figure 3-7: User Conversation Window

Table 3-6: User Conversation Window

No.	Item	Description
1	Message Timestamp	Shows the specific creation/transmission time next to each individual message.
2	Date Separator	Visual markers dividing the chat history by day (e.g., "Today", "Yesterday", or specific dates).
3	Chat Bubbles	Displays the content of sent and received text messages along with the time of transmission (HH:MM).
4	Quick Presets	A menu containing pre-configured, standard operational message templates (Presets) for rapid response and minimal typing under combat conditions.
5	SEND Button	Sends the typed message or attached tactical data over the active radio/network link.
6	Text Input Field	Enables the user to type manual responses or compose new messages (up to 500 characters).
7	"+" Button	Opens an action menu containing the following options: • PASTE: Inserts copied text data from the system clipboard into the input field. • ADD MEDIA: enables the attachment of external media/files. • ADD LOCATION: enables the user to select specific coordinates from the map. When clicked, a pop-up message appears: "Please Select a Location from the Map". • ADD ENTITY: enables the user to select specific entity from the map. When clicked, a pop-up message appears: "Please Pick entities from Map". 
8	Recipient Details	Displays the identity, tactical callsign, or user profile of the message recipient.
9		

3.2.1.2.1 Group Chat Window

(See Figure 3-8 and Table 3-7)

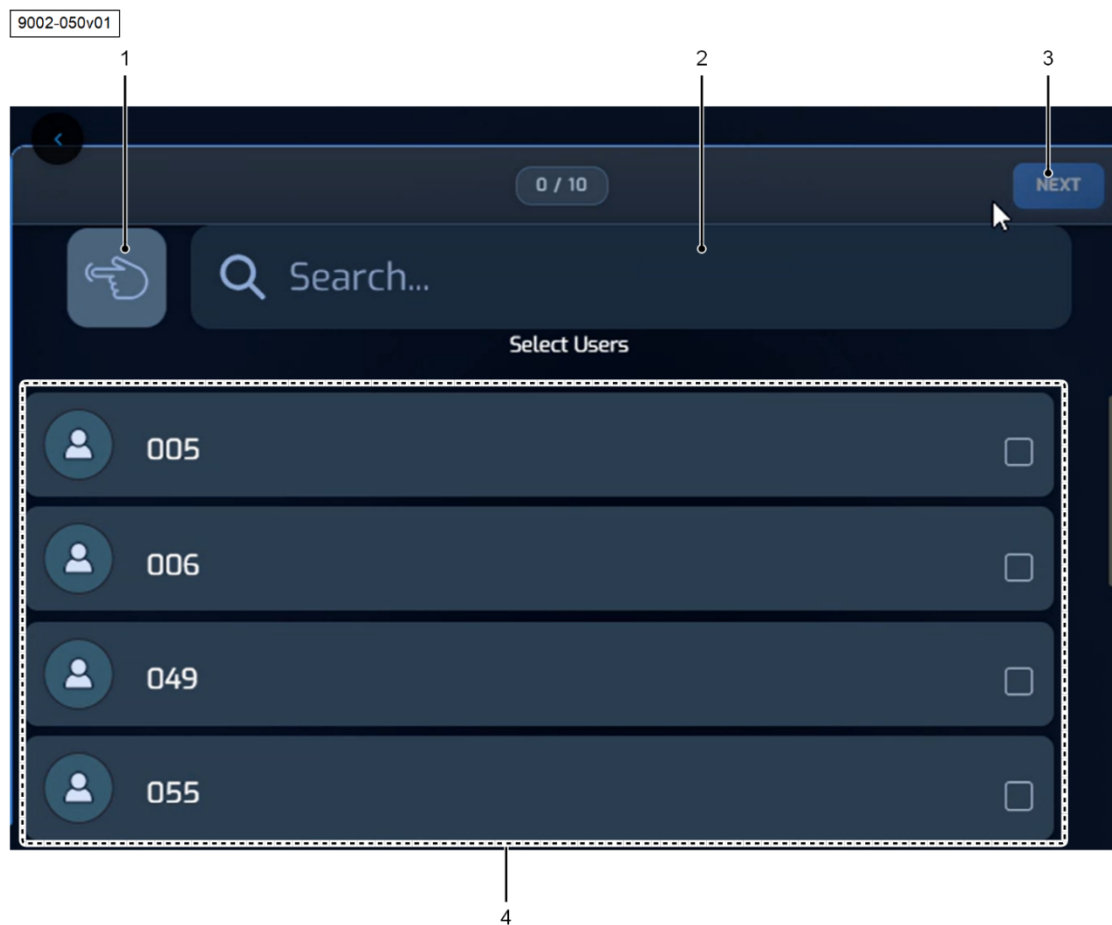


Figure 3-8: Group Chat Window

Table 3-7: Group Chat Window

No.	Item	Description
1	 (Pick Entity)	Enables the user to quickly choose an entity directly from the tactical map interface.
2	Search Field	Enables the user to type and quickly filter specific user within the list.
3	NEXT Button	Opens a new group window for entering the group details such as group Subject. In addition, the window displays the selected users.
4	Network Users	Displays a list of available network users with checkboxes to select and add multiple contacts to the group chat. The user can select up to 10 participants to add to the group.

3.2.1.2.2 Group Conversation Window

(See Figure 3-9 and Table 3-8)

9002-051v01

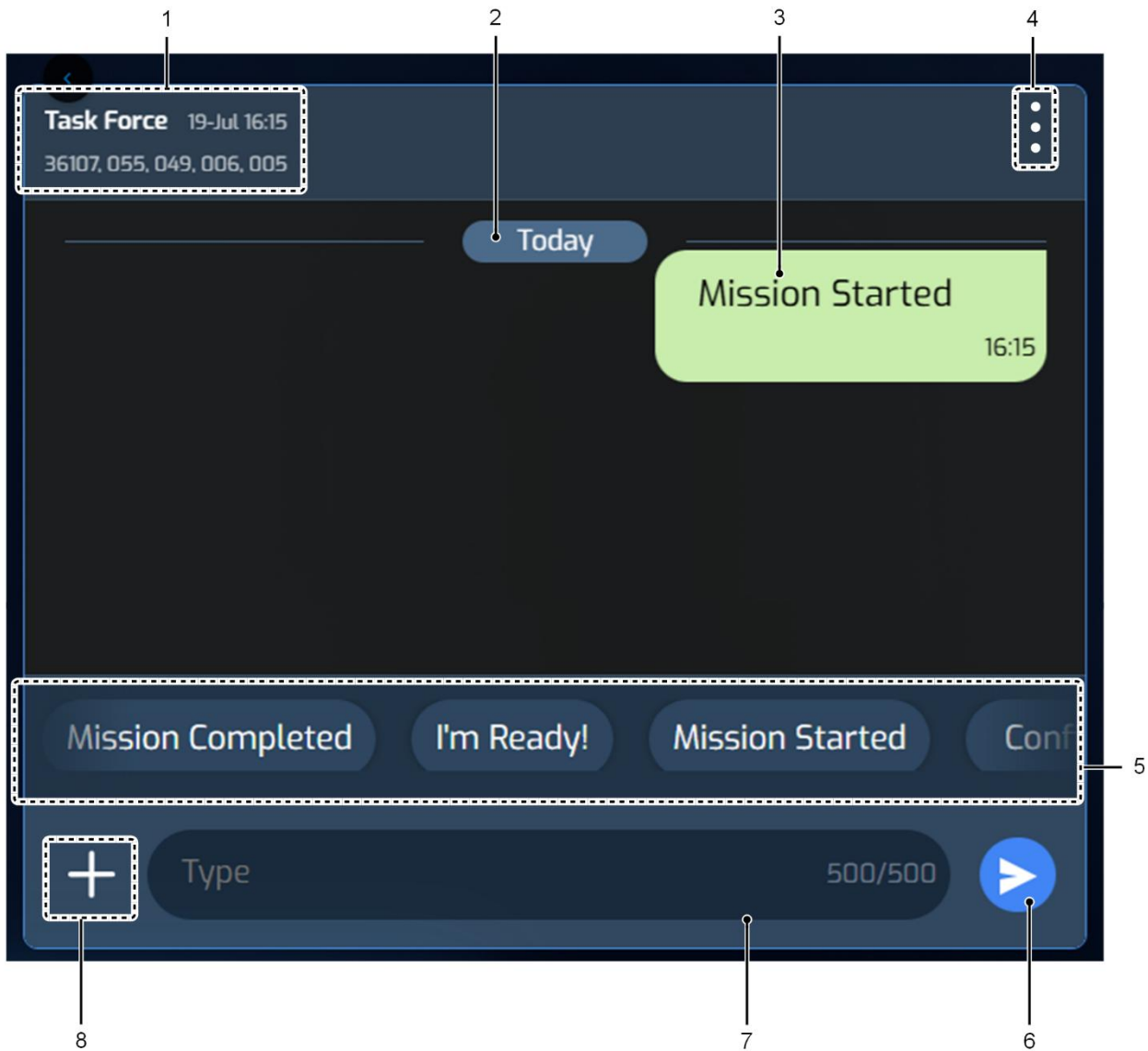
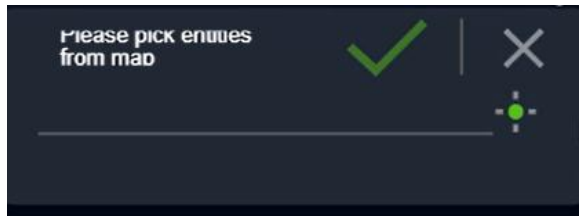


Figure 3-9: Group Conversation Window

Table 3-8: Group Conversation Window

No.	Item	Description
1	Message Details	Displays the identity, tactical callsign, or user profile of the message sender and the the specific message timestamp.
2	Date Separator	Visual markers dividing the chat history by day (e.g., "Today", "Yesterday", or specific dates).
3	Chat Bubbles	Displays the content of sent and received text messages along with the time of transmission (HH:MM).
4	Action Context Menu	Opens a menu with the following options: • Properties Opens the "Add Participant" window to view current participants, remove them from the group, or modify the group's subject/name. • Participant Opens the chat window to select and add new participants to the group, or remove existing ones. • Exit Group Disconnects the platform/unit from the active group.
5	Quick Presets	A menu containing pre-configured, standard operational message templates (Presets) for rapid response and minimal typing under combat conditions.
6	SEND Button	Sends the typed message or attached tactical data over the active radio/network link.
7	Text Input Field	Enables the user to type manual responses or compose new messages (up to 500 characters).
8	"+" Button	Opens an action menu containing the following options: • PASTE: Inserts copied text data from the system clipboard into the input field. • ADD MEDIA: enables the attachment of external media/files. • ADD LOCATION: enables the user to select specific coordinates from the map. When clicked, a pop-up message appears: "Please Select a Location from the Map". • ADD ENTITY: enables the user to select specific entity from the map. When clicked, a pop-up message appears: "Please Pick entities from Map".



3.2.1.2.3 Notifications Window

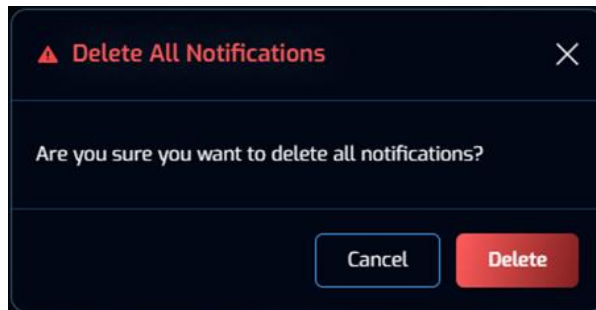
(See Figure 3-10 and Table 3-9)

9002-018v01



Figure 3-10: Notifications Window

Table 3-9: Notifications Window

No.	Item	Description
1	Search Button	Opens the search field used to filter and locate specific alerts or messages within the notification log by typing keywords.
2	Group By Button	Opens a drop-down menu: • Notification Type: Groups notifications by their severity or operational category (e.g., Threats, Caution, Info). • Read Status: Groups notifications by their review state.
3	Action Button	Opens a drop-down menu: • Acknowledge All: Marks all active notifications as read/viewed. • Delete All: Clears all notifications from the current view or log. 
4	Notifications Tabs	Categorizes all system notifications into four tabs: • All Tab - Displays a unified, chronological log of all system notifications • Threats Tab - Displays critical tactical alerts that may required user intervention (e.g., incoming counter-battery fire, or immediate target updates).

No.	Item	Description
		• Caution Tab - Displays operational warnings and system status alerts (e.g., LACU Error, , or Partly DTM check). • Info Tab - Displays general status updates and non-critical system logs (e.g., Platform Connected, WGS is connected).
5	All/ Threats/ Caution/ Info Message	Displays a list of messages (All, Threats, Caution, or Info) based on the currently selected tab.

3.2.1.2.4 Notifications Window – Message

(See Figure 3-11 and Table 3-10)

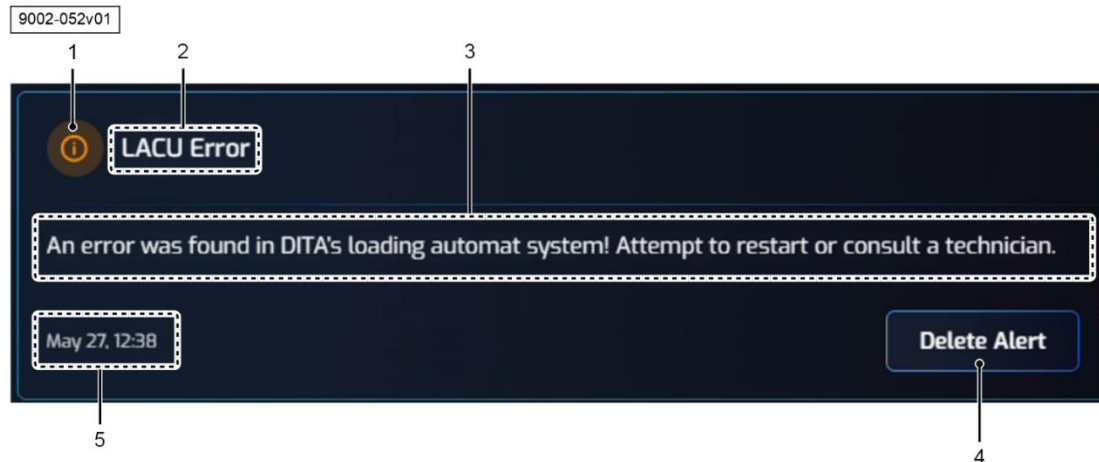


Figure 3-11: Notifications Window – Message

Table 3-10: Notifications Window – Message

No.	Item	Description
1	Info Icon	Indicates an error, warning, or alert condition that may require user attention or corrective action.
2	Alert Name	Indicates the event/Alert Name.
3	Alert Description	A detailed text description providing operational context and specific data about the alert.
4	Delete Alert Button	Removes the alert from the active view.
5	Time Stamp	Indicates the time (HH:MM) and date of the alert.

3.2.1.2.5 Notifications Pop-up Message

(See Figure 3-12 and Table 3-11)

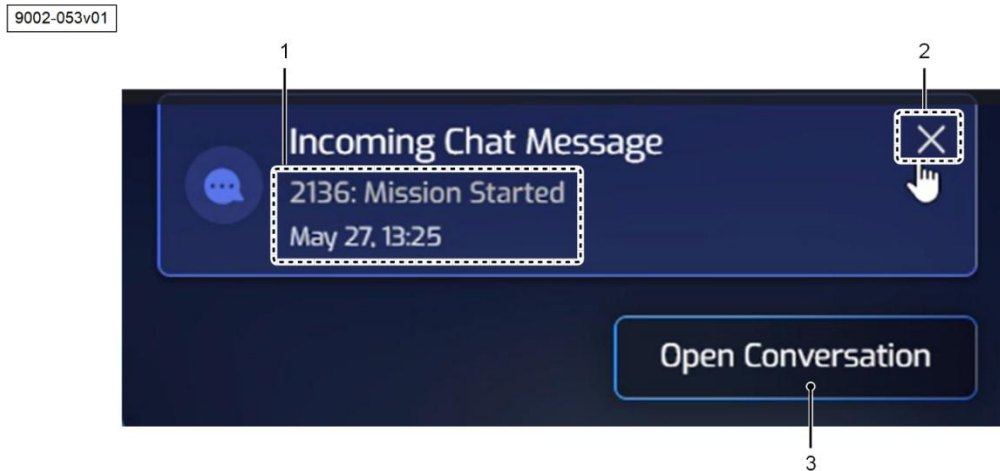


Figure 3-12: Notifications Pop-up Message

Table 3-11: Notifications Pop-up Message

No.	Item	Description
1	Sender Identifier	Displays the sender ID (e.g., unit name, or battalion number), the message (e.g., "Mission Started") and timestamp.
2	X Button	Hides the pop-up message from the current view.
3	Open Conversation Button	Opens the active conversation thread with the specific sender.

3.2.1.3 Main Screens Navigation Bar

(See Figure 3-13 and Table 3-12)

9002-005v01

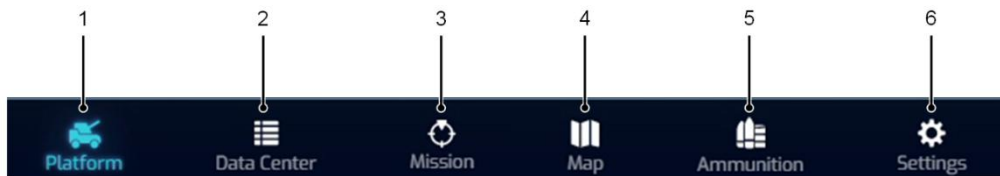


Figure 3-13: Main Screens Navigation Bar

Table 3-12: Main Screens Navigation Bar

No.	Item	Description
1	Platform	When Clicked, displays the Platform Screen. This screen serves as the primary screen after user login, providing a centralized, real-time overview of the system physical status, devices status, spatial orientation and location, and system health. For detailed description, refer to section 3.2.2.
2	Data Center	When Clicked, displays the Data Center Screen. This screen displays system records and stored data, organized into the following categories: Fire Missions, Mission Groups, Meteorological Telegrams, Tasks, Ammunition Muzzle Velocity. For detailed description, refer to section 3.2.4.
3	Mission	When Clicked, displays the Mission Screen. [refer to section 3.2.5]
4	Map	When Clicked, displays the Map Screen. [refer to section 3.2.3]. This screen displays the operational map, enabling the user to perform geospatial actions and utilize integrated map tools.
5	Ammunition	When Clicked, displays the Ammunition Management Screen. [refer to section 3.2.7]
6	Settings	When Clicked, displays the Settings Screen. [refer to section 3.2.5]

3.2.2 Platform Screen

(See Figure 3-14 and Table 3-13)

9002-002v01



Figure 3-14: Platform Screen

Table 3-13: Platform Screen

No.	Item	Description
1	Platform Management Bar	Monitors system operational status, coordinates, and altitude while enabling the user to toggle between system Position modes. [refer to section 3.2.2.1]
2	Roll Section	Displays numerical and graphical indication of the vehicle's Roll angle relative to the horizontal plane. Current Value: Represents the real-time, actual physical Roll angle of the Vehicle as measured by the sensors (INS).
3	Fire Elevation (mil-East) and Pitch Section	Displays the vertical angle (elevation) of the gun barrel, measured by the INS in Angular Mills according to the Eastern European standard. Current Value: Represents the real-time, actual physical elevation angle of the barrel as measured by the turret sensors (INS). Mission Value: Represents (Yellow Color) the target/computed elevation angle required by the ballistic computer to execute the current loaded fire mission. In addition, it Provides a dynamic graphical representation of the DITA HOWTIZER and displays the real-time elevation angle relative to the horizontal plane. - The crew uses this field to verify that the physical barrel elevation (Current) aligns perfectly with the calculated firing solution (Mission) before initiating a fire sequence. Current [Pitch]: Displays the real-time Pitch angle of the vehicle chassis relative to the geographic horizon.

No.	Item	Description
4	Fire Azimuth (mil-East) Section	Displays the Azimuth angle of the turret, measured in Angular Mils according to the Eastern European standard. • Current Value: Represents the real-time, actual physical Azimuth angle of the turret as measured by the turret sensors (INS, Shaft Encoders). • Mission Value: Represents (Yellow Color) the target/computed Azimuth angle required by the ballistic computer to execute the current loaded fire mission. In addition, this section provides a real-time, visual representation of the turret's traverse angle (azimuth) relative to the vehicle's chassis and operational sectors [refer to section 3.2.2.2].
5	Platform Management Menus	Provides quick access to critical parameter updates during platform deployment and prior to fire mission execution: • Position Menu - enables the user to monitor, select, and manage the spatial coordinates and elevation data of the platform. [refer to section 3.2.2.3] • Devices Menu - provides connectivity and link status of seven peripheral devices integrated with the Fire Control System (FCS) for continuous monitoring. In addition, it enables the user to enter, configure or update relevant data required for the device's operation. • Error Log Menu - aggregates real-time diagnostic data and fault codes from all connected devices. • Deployment Menu - includes dedicated functions required to manage the platform's deployment status such as firing position setting/Exiting, defining the firing sector orientation, and updating critical ballistic parameters essential for accurate fire control. [refer to section 3.2.2.6]

3.2.2.1 Platform Screen Management Bar

(See Figure 3-15 and Table 3-14)



Figure 3-15: Platform Screen Management Bar

Table 3-14: Platform Screen Management Bar

No.	Item	Description
1	Screen/Window Name	Displays the current screen or menu name.
2	Coordinates and Altitude Display	Provides a continuous, real-time display of the system current global position and altitude (in meters), synchronized via the onboard GPS/GNSS and Inertial Navigation System (INS) or manual entry.
3	Travel Lock Status indicator	Enables the user to monitor the travel lock state (open/closed).
4	System Operational Status	- Displays visual confirmation for the crew, indicating whether the platform is fully prepared to accept and execute fire missions (READY) or if the automated firing sequence is currently interlocked (NOT READY). - This operational readiness is transmitted to the Battery Command Post (BCP) to provides the Battery Commander with a comprehensive Operational Picture for efficient fire mission allocation.
5	Exit Position / SET Position	Toggles the system between two modes: SET Position: sets and locks the howitzer's current geographic coordinates and azimuth—derived from the integrated Inertial Navigation System (INS) and GPS—establishing them as the platform's official firing position reference. The sampled position is rendered on the tactical map display for visual cross-verification by the crew. Exit Position: commands the DITA system to immediately exit its current firing posture, clear its locked and stored position data, and prepare for movement. For example, for rapid displacement (shoot-and-scoot maneuver).

3.2.2.1.1 SET POSITION Concept

GPS and INS provide the platform's current position and orientation; however, these sources are **not always fully consistent**. Positioning errors, sensor inaccuracies, signal degradation, and navigation drift may cause small differences between the reported coordinates and the platform's actual location.

While these deviations may be acceptable for navigation purposes, they can significantly affect ballistic calculations. Artillery fire-control computations require a **verified and reliable firing position** to ensure accurate firing solutions.

For this reason, the operator must first verify the platform location and confirm that the displayed coordinates represent the platform's **true position**. Once the operator is satisfied with the location data, the SET POSITION command is executed.

When **SET POSITION** is pressed, the system copies the coordinates and altitude currently displayed in the **Position** menu (depending the Location Source: INS, GPS, or manual) and stores them as the official firing position and the unit position (BCP). The selected data is transferred to the **Stored Location** field in the **Deployment** menu.

From that point onward, the Stored Location becomes the authoritative reference position used by the ballistic engine for all fire-control calculations, regardless of subsequent changes or minor fluctuations in the GPS or INS reported position. This ensures that all firing solutions are calculated from a stable, verified, and known platform location.

3.2.2.2 Azimuth Widget

(See Figure 3-16 and Table 3-15)

9002-009v01



Figure 3-16: Azimuth Widget

Table 3-15: Azimuth Widget

No.	Item	Description
1	White Arc	Marks the current angle in azimuth measured by the system INS.
2	Turret Schema	Displays outline of the DITA howitzer. The chassis is fixed forward (pointing to the 0-mil mark), while the turret and barrel dynamically rotate to reflect the actual azimuth angle.
3	Operational & Restricted Sectors	Shaded Red Zone (No-Fire / Restricted Sector): Visually defines the safety limits or dead zones where firing is blocked due to physical obstructions or chassis protection. Clear Zone (Firing Sector): Highlights the authorized and safe traverse envelope available for weapon engagement.
4	Mil-Spec Compass Ring	A standard artillery Mil scale (0 to 6400 mils) with Key directional markers indicate directions: 0/6400 (Front/North), 1500 (E) (East), 3000 (S) (South), and 4500 (W) (West).
5	Chassis Heading	Marks the vehicle's true centerline (0 mils) (Blue color).
6	Mission Heading	Marks the mission calculated angle (Yellow color) in azimuth.

3.2.2.3 Position Menu

(See Figure 3-17 and Table 3-16)

9002-011v01

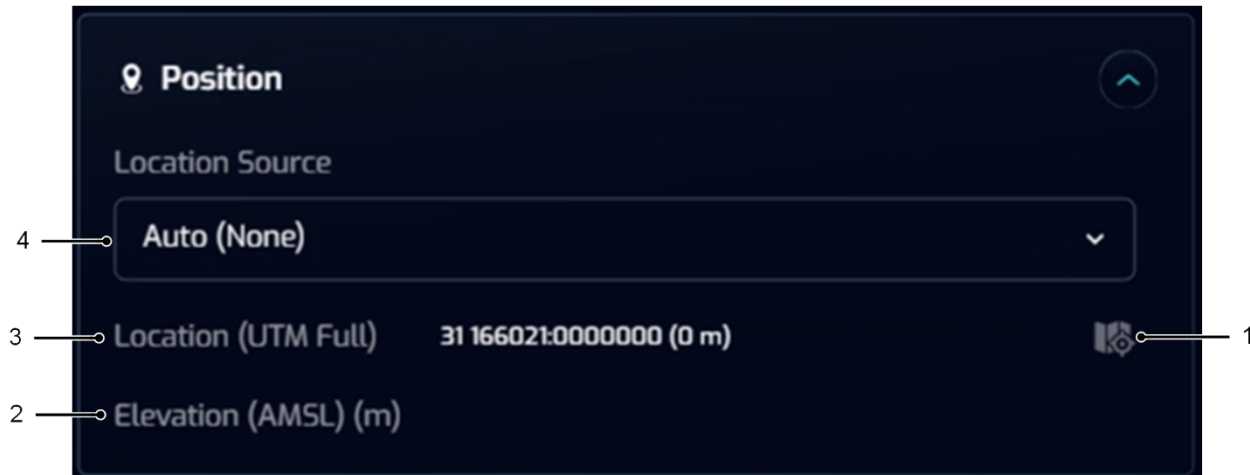


Figure 3-17: Position Menu

Table 3-16: Position Menu

No.	Item	Description
1	Pinpointing from Map ()	Quick-action button that enables the user to tap directly on the tactical map to automatically set coordinates. When clicked, the screen layout changes to display the tactical map alongside the active workspace.
2	Location (UTM Full)	Displays the precise geographical coordinates of the platform using the Universal Transverse Mercator (UTM) coordinate system.
3	Elevation (AMSL) (m)	Displays the platform's altitude/elevation in meters Above Mean Sea Level (AMSL).
4	Location Source	Defines the active positioning input: • Auto (None) - system-selected optimal source (<u>default</u>) – INS. • GPS: live satellite data from the onboard navigation system (GPS). • Manual/MAP – user-defined coordinates: manual entry or directly from the map.

3.2.2.4 Devices Menu

(See Figure 3-18 and Table 3-17)

9002-012v01

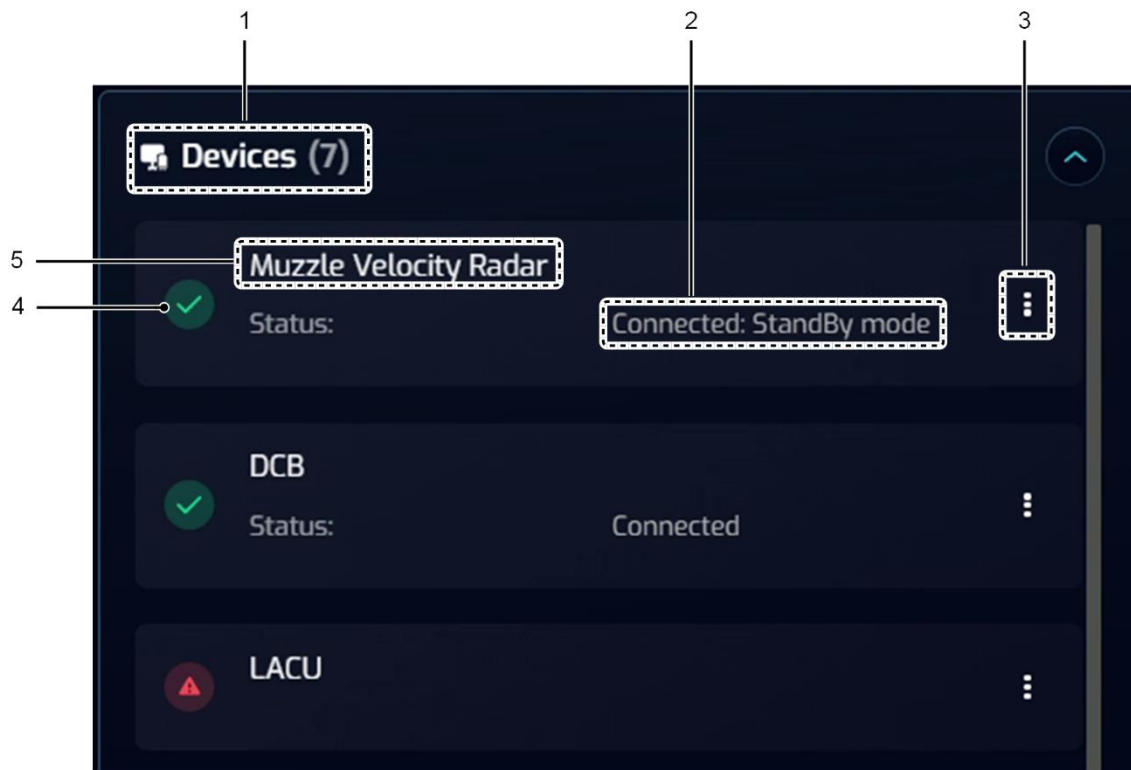


Figure 3-18: Devices Menu

Table 3-17: Devices Menu

No.	Item	Description
1	Window Name	Displays the device name. The dropdown menu contains seven devices: • Muzzle Velocity Radar (MVR) – Measures the actual speed of the projectile as it leaves the barrel. This real-time data is instantly fed into the fire control system to adjust ballistic calculations and maximize firing accuracy for subsequent projectiles. • Data Collection Box (DCB) - functions as the central network switch and communication gateway for the DITA platform. The main onboard hardware devices (LACU, MVR, INS, WGS, filling boxes, etc.) route through the DCB, making it the <u>critical link to the application</u>. • Loading Automat Control Unit (LACU) - responsible for the automated ammunition handling and ballistic execution sequence within the DITA turret (Conveyor Management, Loading Supervision). • Filling Boxes - An auxiliary control device that enables the ammunition crew to manually control the conveyors and move through the ammunition magazine during the loading process. The device allows the crew to record the ammunition loaded into each magazine slot, including shell and propellant configurations. In addition, the commander may define a desired ammunition load plan, and the Filling Boxes guide the

No.	Item	Description
		crew during loading to ensure the correct ammunition is placed in the designated magazine slots according to the approved loadout plan. • Weapon guidance system (WGS) - manages and operates the DITA automated turret, providing the crew with monitoring data and direct control over turret movement (interfaces the INS). It translates ballistic data into precise mechanical adjustments, ensuring proper azimuth and elevation alignment for accurate firing operations. • INS - calculates real-time positioning, navigation, and aiming data for the DITA platform. It operates in two primary modes: (1) <u>Navigation Mode</u>: Provides platform positioning coordinates, roll, and pitch/elevation angles. (2) <u>Laying Mode</u>: Provides orientation data specific to the gun barrel for accurate target engagement. • Direct Fire Control System (DFS) – manages the <u>direct-fire</u> engagement cycles. Upon activation, it isolates the LACU from the main C4I system and verifies real-time shell and charge data for ballistic calculations. After firing, the DFS automatically reports the ammunition consumption and system status back to the main C4I application.
2	Connection Status	Provides a color-coded, visual representation of the operational connection state to the selected device: • Disconnected - The device is disconnected. There is no communication or data flow. • Connected - The device is connected and fully operational. • Standby mode – the device is powered and interfaced with the C4I, but some functions are halt. • Fault - Indicates a device-level issue. The physical or network connection exists, but the device is reporting an internal fault.
3	Properties Form	Opens a dedicated properties/configuration window for the chosen device. This enables the user to view and modify technical and operational parameters.
4	Health Icon	Provides a color-coded, visual representation of the operational connection state to the selected device: •  - The device is disconnected. There is no communication or data flow. •  - The device is connected and fully operational. •  - Indicates a device-level issue. The physical or network connection exists, but the device is reporting an internal fault/state.
5	Device Name	Displays the device name or identifier.

3.2.2.4.1 Devices Menu – MVR Form

(See Figure 3-19 and Table 3-18)

9002-070v01

The screenshot shows the 'MVR' form interface. It has a dark blue background with white text and input fields. At the top left is the 'MVR' logo. The form is divided into two main columns. The left column contains: 'SOFTWARE VERSION' (callout 8), 'MVR Errors' (callout 7), and 'Current Velocity' (callout 6). The right column contains: 'MVR Status' (callout 1) showing 'Connected: Measure mode', 'MVR Errors Comments' (callout 2), and 'Expected MV' (callout 3) showing '550'. At the bottom right are 'Save' (callout 5) and 'Cancel' (callout 4) buttons.

Figure 3-19: Devices Menu – MVR Form

Table 3-18: Devices Menu – MVR Form

No.	Item	Description
1	MVR Status	Displays the communication status of the MVR (Connected / Disconnected).
2	MVR ERRORS COMMENTS	Textual breakdown of the active error code for troubleshooting by the crew or technical echelon.
3	EXPECTED MV	Displays the expected muzzle velocity calculated based on the selected shell type and charge zone, serving as the baseline for pre-fire ballistic computations.
4	Cancel	Discards all changes made and closes the form.
5	Save	Saves the current data.
6	CURRENT VELOCITY	Displays the actual muzzle velocity (in m/s) measured by the MVR during the last fired shell.
7	MVR ERRORS	Displays active hardware or software error codes reported by the MVR.
8	SOFTWARE VERSION	Displays the current firmware/software version installed on the MVR unit.
9	Connect to mvr	Commands the system to establish or re-establish communication with the Muzzle Velocity Radar (MVR). This function is typically used when the MVR is disconnected or communication with the device has been lost.

3.2.2.4.2 Devices Menu – DFS Form

(See Figure 3-20 and Table 3-19)

9002-092v01

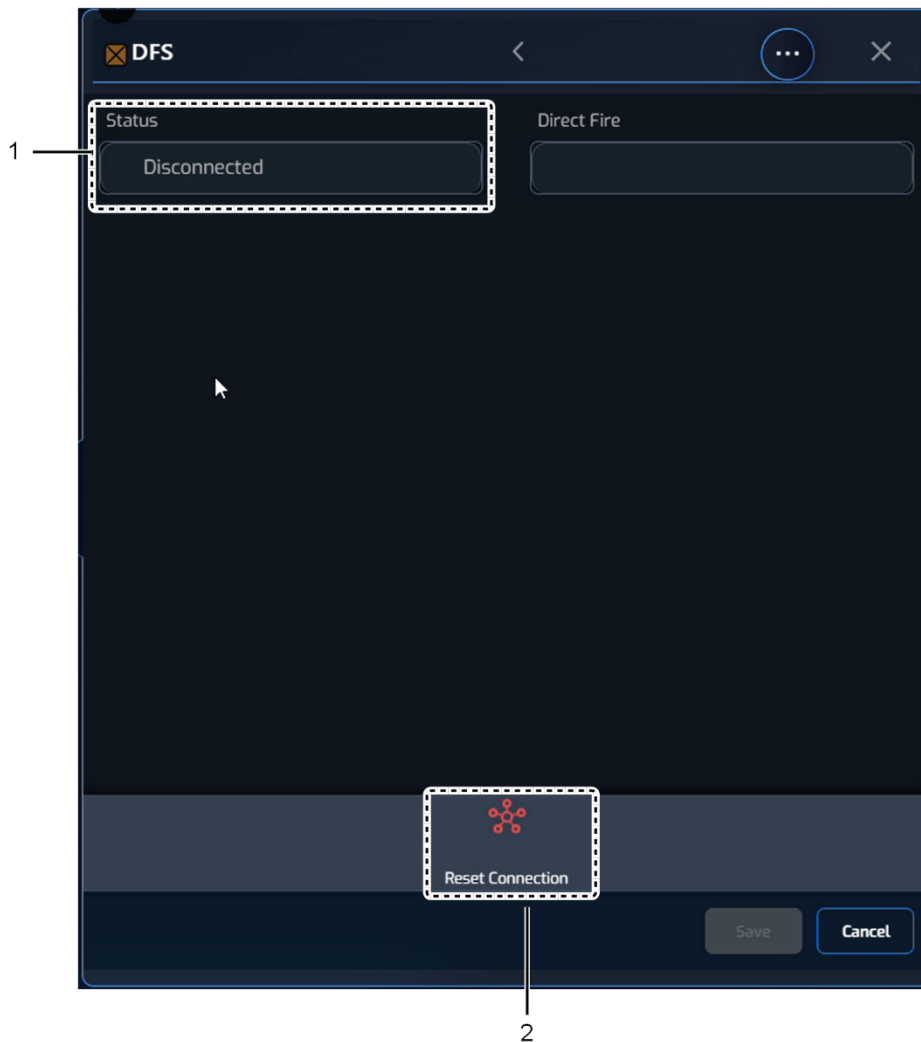


Figure 3-20: Devices Menu – DFS Form

Table 3-19: Devices Menu – DFS Form

No.	Item	Description
1	Status	Displays the current operational and connectivity status of the DFS
2	Reset Connection	Attempts to re-establish communication with the DFS by resetting the active connection and initiating a new connectivity session.

3.2.2.4.3 Devices Menu – INS Form

(See Figure 3-21 and Table 3-20)

9002-095v01

The screenshot displays the 'Main' section of the INS Form. It features a dark-themed interface with several input fields and status indicators. Numbered callouts point to specific elements:

- 1**: Points to the 'Working Mode' input field.
- 2**: Points to the 'Align Mode' input field.
- 3**: Points to the 'Navigation Mode' input field.
- 4**: Points to the 'Orientation Data Validation' input field.
- 5**: Points to the 'Connection Status' field, which displays 'Not Connected'.
- 6**: Points to the 'Align Time' input field.
- 7**: Points to the 'Travel Lock Status' input field.

Below the input fields, there are expandable sections for 'Position' and 'Comments'. At the bottom, there is a row of three circular icons labeled 'IBIT', 'Override Alert', and 'Reset Distance', followed by 'Save' and 'Cancel' buttons.

Figure 3-21: Devices Menu – INS Form - Main Section

Table 3-20: Devices Menu – INS Form - Main Section

No.	Item	Description
1	Working Mode	Displays the current operational mode of the INS subsystem.
2	Align Mode	Displays the active INS alignment mode used to establish platform position and orientation.
3	Navigation Mode	Displays the active navigation mode used by the INS for position and orientation data generation.
4	Orientation Data Validation	Displays the validity status of the platform orientation data (Azimuth, Roll, and Pitch) provided by the INS.
5	Connection Status	Displays the communication status of the INS (Connected / Disconnected).
6	Align Time	Displays the elapsed or required time for completion of the INS alignment process.
7	Travel Lock Status	Displays the current state of the gun travel lock mechanism.

9002-096v01

The screenshot displays the 'Position Section' of the INS Form. It features a dark blue background with white text and input fields. The form is organized into two columns. The left column contains fields for 'Datum Type', 'GPS Satellites', 'Altitude (m)', 'EAPE (m)', 'Barrel Pitch (mil-East)', 'Estimated Azimuth Error (mil-East)', and 'Estimated Roll Error (mil-East)'. The right column contains fields for 'GPS Status', 'EHPCEP (m)', 'Barrel Azimuth (mil-East)', 'Barrel Roll (mil-East)', 'Estimated Pitch Error (mil-East)', and 'Traveled Distance (m)'. A 'Location (UTM Full)' section is highlighted with a dashed white border and contains four input fields and a document icon. A 'Comments' section is at the bottom. Numbered callouts (1-14) point to specific elements: 1 points to the 'Datum Type' field; 2 points to the 'GPS Satellites' field; 3 points to the 'Location (UTM Full)' section; 4 points to the 'Altitude (m)' field; 5 points to the 'EAPE (m)' field; 6 points to the 'Barrel Pitch (mil-East)' field; 7 points to the 'Estimated Azimuth Error (mil-East)' field; 8 points to the 'Estimated Roll Error (mil-East)' field; 9 points to the 'Traveled Distance (m)' field; 10 points to the 'Estimated Pitch Error (mil-East)' field; 11 points to the 'Barrel Roll (mil-East)' field; 12 points to the 'Barrel Azimuth (mil-East)' field; 13 points to the 'EHPCEP (m)' field; and 14 points to the 'GPS Status' field.

Figure 3-22: Devices Menu – INS Form – Position Section

Table 3-21: Devices Menu – INS Form – Position Section

No.	Item	Description
1	Datum Type	Displays the active geodetic datum used by the navigation system for coordinate calculations and positioning data.
2	GPS Satellites	Displays the number of GPS satellites currently tracked and used for position calculation.
3	Location (UTM Full)	Displays the platform's current position in full Universal Transverse Mercator (UTM) coordinates.
4	Altitude (m)	Displays the platform altitude above Mean Sea Level (AMSL) in meters.
5	EAPE (m)	Displays the Estimated Altitude Position Error (EAPE), indicating the expected vertical positioning accuracy in meters.
6	Barrel Pitch	Displays the current elevation angle of the gun barrel as measured by the INS, relative to the horizontal plane.
7	Estimated Azimuth Error	Displays the estimated accuracy deviation of the barrel azimuth measurement, indicating the expected error in the reported azimuth value.
8	Estimated Roll Error	Displays the estimated accuracy deviation of the platform or barrel roll measurement, indicating the expected error in the reported roll value.
9	Traveled Distance	Displays the cumulative distance traveled by the platform since the last reset or reference point.
10	Estimated Pitch Error	Displays the estimated accuracy deviation of the barrel pitch measurement, indicating the expected error in the reported elevation value.
11	Barrel Roll (mil-East)	Displays the current barrel roll angle measured by the INS in Eastern Mils, indicating the lateral tilt of the weapon relative to its longitudinal axis.
12	Barrel Azimuth (mil-East)	Displays the current gun barrel azimuth measured by the INS in Eastern Mils.
13	EHPCEP (m)	Displays the Estimated Horizontal Position Circular Error Probable (CEP), indicating the expected horizontal positioning accuracy in meters.
14	GPS Status	Displays the operational status of the GPS receiver and the availability of satellite-based positioning data.

9002-097v01

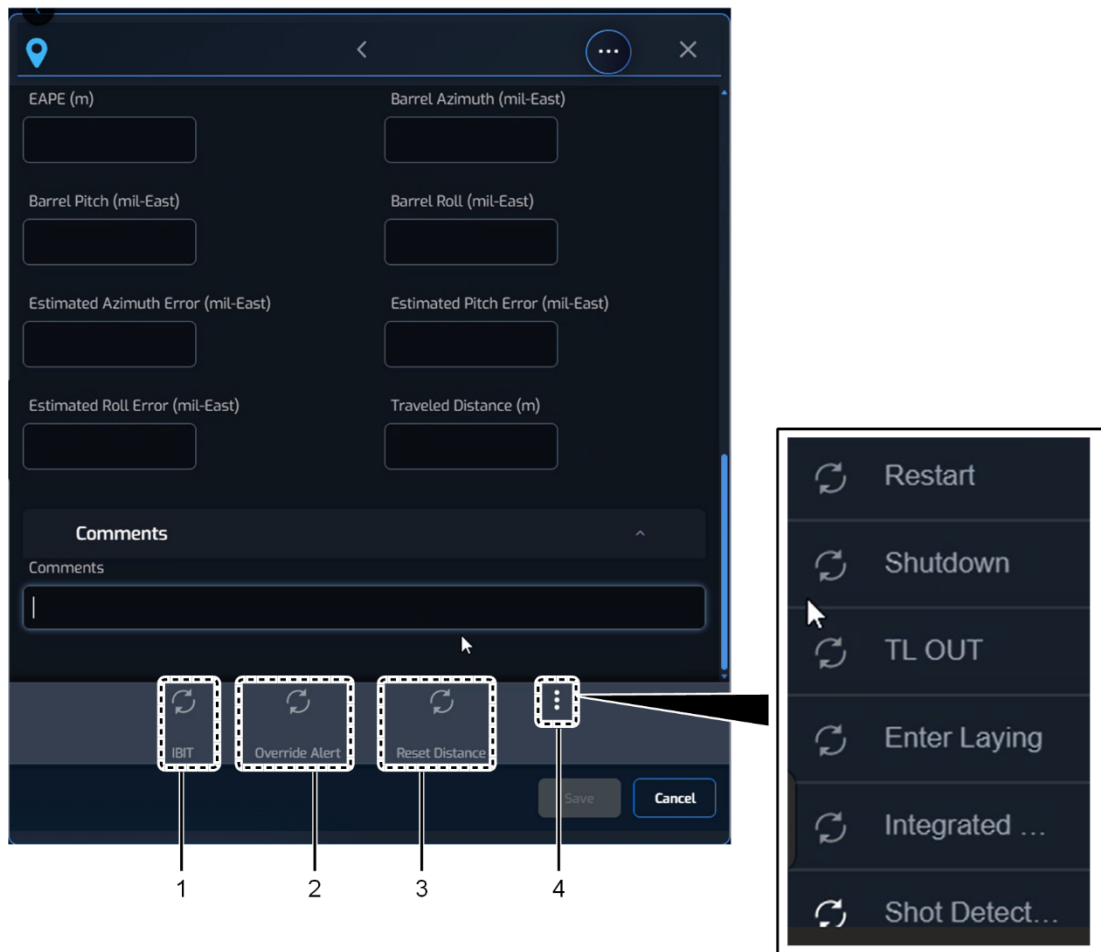


Figure 3-23: Devices Menu – INS Form – Control Buttons Section

Table 3-22: Devices Menu – INS Form – Control Buttons Section

No.	Item	Description
1	IBIT	Initiates the Initiated Built-In Test (IBIT) procedure to verify the operational status and integrity of the INS subsystem. The test performs diagnostic checks and reports detected faults or system readiness.
2	Override Alert	Acknowledges and overrides the selected INS alert or warning, allowing the operator to continue system operation subject to applicable operational procedures and permissions.
3	Reset Distance	Resets the accumulated traveled distance counter and starts a new distance measurement cycle from zero.
4	Actions	Opens a context menu containing additional INS-related commands, maintenance functions, and diagnostic operations available to the user: • Restart - Restarts the INS subsystem and reinitializes its navigation and orientation functions. This action is typically used to recover from faults or communication issues. • Shutdown - powers down the INS subsystem and terminates all active navigation, positioning, and orientation processes. • TL OUT - Commands the system to release or disengage the Travel Lock mechanism, allowing the weapon system to transition from transport configuration to operational status. • Enter Laying - Switches the INS to Laying Mode, enabling the provision of precise barrel orientation, azimuth, and elevation data required for weapon alignment and target engagement. • Integrated Mode - Commands the INS to operate in Integrated Mode, utilizing all available navigation sources and sensors to provide optimized positioning and orientation data. • Shot Detection - Enables the Shot Detection function, allowing the INS to detect and record weapon firing events for navigation compensation, system monitoring, and ballistic processing.

3.2.2.4.4 Devices Menu – WGS Form

(See Figure 3-24 and Table 3-23)

9002-093v01

Figure 3-24: Devices Menu – WGS Form

Table 3-23: Devices Menu – WGS Form

No.	Item	Description
1	Status	Displays the current operational and connectivity status of the WGS
2	Park Position	Commands the weapon system to move the turret and barrel to the predefined transport/parking position.
3	Lay-on target	Commands WGS to automatically align the turret and barrel to the calculated azimuth and elevation required for the selected target.
4	Stop	Immediately halts the current weapon guidance or movement operation and suspends all active automatic positioning commands.
5	Release Barrel Lock	Releases the barrel locking mechanism, enabling the weapon system to move the barrel for aiming, laying, or operational procedures.

3.2.2.4.5 Devices Menu – LACU Form

(See Figure 3-25 and Table 3-24)

9002-094v01

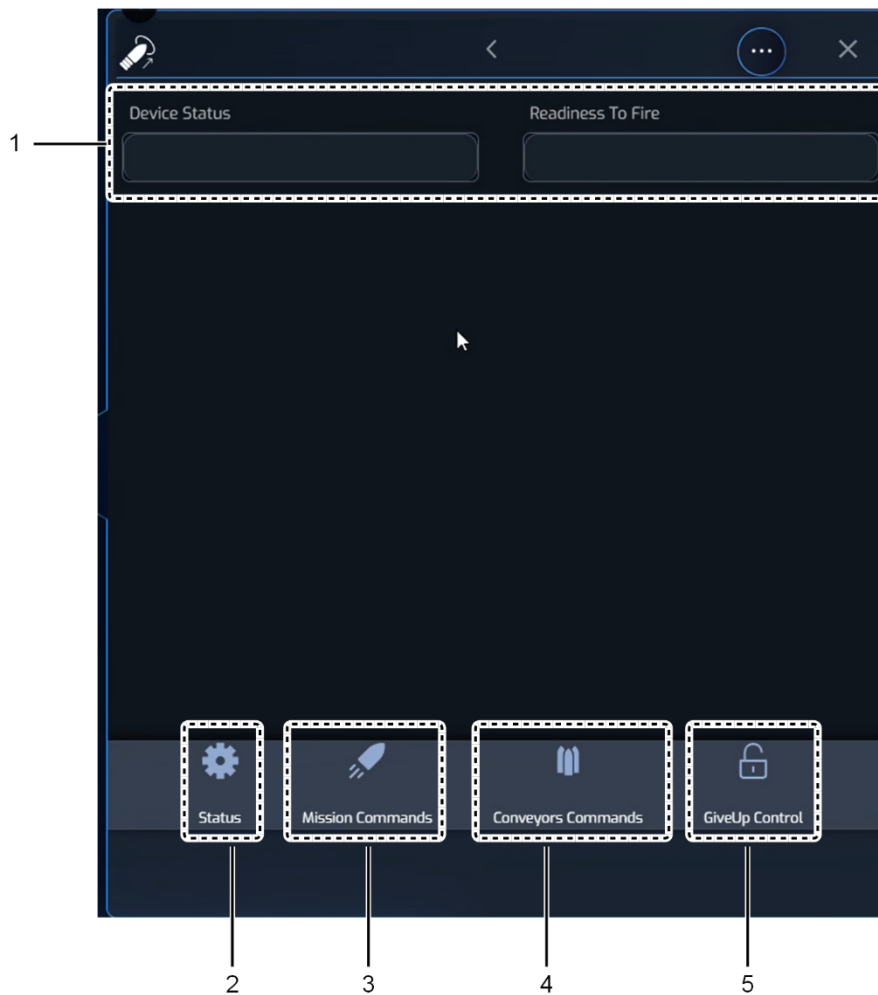


Figure 3-25: Devices Menu – LACU Form

Table 3-24: Devices Menu – LACU Form

No.	Item	Description
1	Status	Displays the current operational and connectivity status of the LACU.
2	Status	Opens a dedicated panel containing the current operational and connectivity status of the LACU, indicating whether the subsystem is connected, available, and ready to perform automated loading and mission execution functions.
3	Mission Commands	Opens a dedicated panel containing mission-related commands and controls used by the LACU during fire mission execution.
4	Conveyor Commands	Opens a dedicated panel containing commands for controlling the ammunition conveyor system, including ammunition movement and magazine handling operations.

No.	Item	Description
5	GiveUp Control	Releases LACU control authority and returns control of the subsystem to another authorized system, mode, or operator.

3.2.2.5 Error Log Menu

This menu aggregates real-time diagnostic data and fault codes from all connected devices. The events are classified by color code:

- **Green (OK):** Nominal operation; system/device functioning normally.
- **Blue (General Info):** Standard status updates and non-critical notifications.
- **Orange (Important):** Warning state; non-critical fault requiring attention without halting operations.
- **Red (Critical):** Major failure; requires immediate operator intervention.

3.2.2.6 Deployment Menu

(See Figure 3-26 and Table 3-25)

9002-014v01

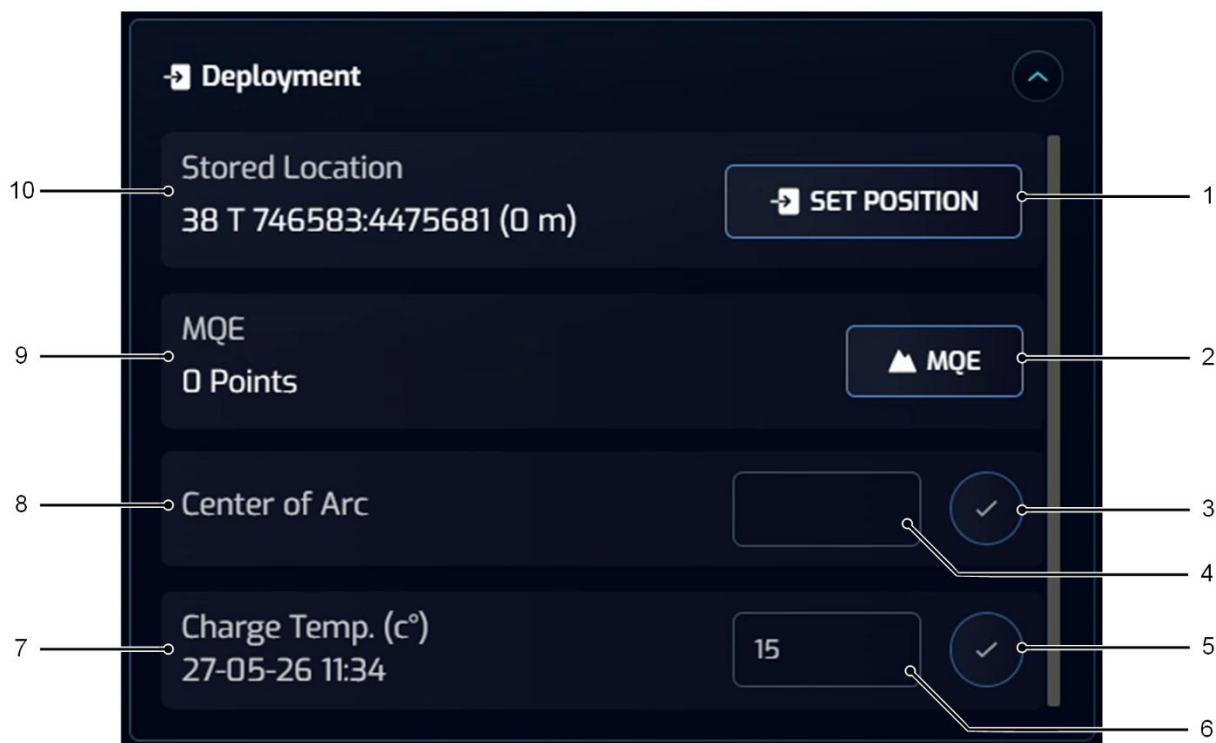


Figure 3-26: Deployment Menu

Table 3-25: Deployment Menu

No.	Item	Description
1	SET POSITION/EXIT POSITION.	Toggles the system between two modes: • Set Position: sets and locks the howitzer's current geographic coordinates and azimuth—derived from the integrated Inertial Navigation System (INS) and GPS—establishing them as the platform's official firing position reference. The sampled position is rendered on the tactical map display for visual cross-verification by the crew. • Exit Position: commands the DITA system to immediately exit its current firing posture, clear its locked and stored position data, and prepare for movement. For example, for rapid displacement (shoot-and-scoot maneuver).
2	MQE Button	Opens the MQE sub-window for detailed management. [refer to section 3.2.2.6.1].
3	✓ (Confirm)	Confirms the typed value in Center of Arc field.
4	COA Input Field	Used for manually entering the Center of Arc value (in mils/degrees).
5	✓ (Confirm)	Confirms the typed value in Charge Temp. (c°) field.

No.	Item	Description
6	Charge Temp. Input Field	Used for manually entering a new ammunition temperature value for accurate ballistic calculation.
7	Charge Temp. (c°) Field	Displays the current charge temperature in degrees Celsius (°C) together with A timestamp (Date and Time) showing the last update. NOTE: By default , the temprature data remains valid for 2 hours . NOTE: The standard propellant temperature is 21.1°C. All propellant temperature corrections are calculated relative to this reference temperature.
8	Center of Arc Field	Enables the user to input and view the primary firing sector orientation (azimuth of the central lay line for the battery/gun), recived from the BCP.
9	MQE Field	Displays the Minimum Quadrant Elevation (MQE) value currently stored in the system.
10	Stored Location Field	Displays the system saved coordinates in a standard Military Grid Reference System (MGRS) format, followed by the Altitude data (above sea level) in meters. NOTE: If the INS-reported position deviates by <u>more than 50</u> meters from the coordinates stored in the STORED LOCATION field, the STORED LOCATION value is automatically cleared.

3.2.2.6.1 MQE Window

(See Figure 3-27 and Table 3-26)

9002-015v01

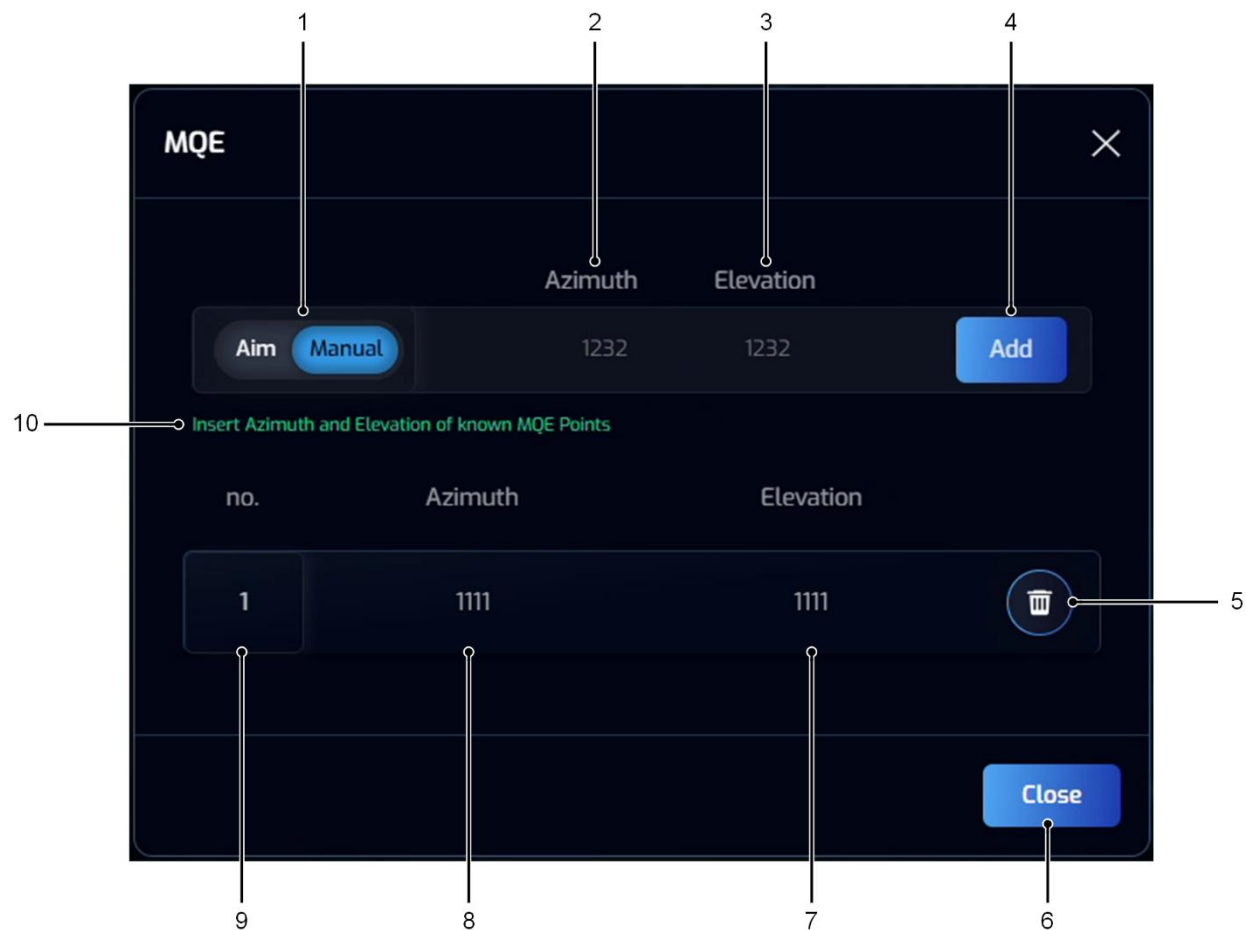


Figure 3-27: MQE Window

Table 3-26: MQE Window

No.	Item	Description
1	Aim/Manual	Enables the user to toggle between automated alignment targeting (AIM) from INS and direct user input (MANUAL)
2	Azimuth Input Field	- Used for manually entering the required Azimuth angle 0-6400 (in mils). - Active only in MANUAL mode.
3	Elevation Input Field	- Used for manually entering the required Elevation angle 0-6400 (in mils). - Active only in MANUAL mode.
4	Add Button	When clicked, validates and sets the manually entered Azimuth and Elevation values into the active list.
5	Trash Icon	When clicked, removes the record from the MQE window.

No.	Item	Description
6	Close Button	Closes the MQE sub-window and returns the user to the Deployment menu.
7	Elevation Value	Displays the manually added Elevation and Azimuth values.
8	Azimuth Value	
9	No.	An incremental index showing the record number.
10	Note Field	Displays instructions for the user.

3.2.3 Map Screen

(See Figure 3-28 and Table 3-27)

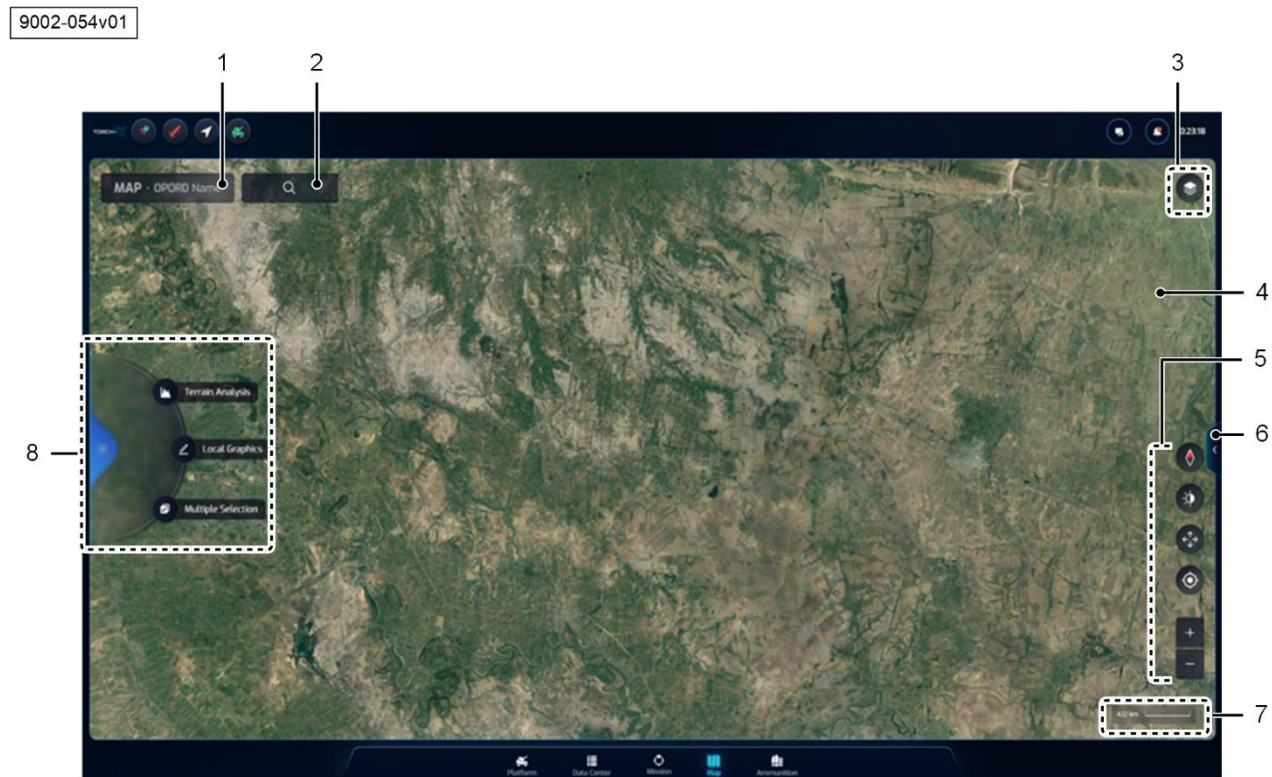
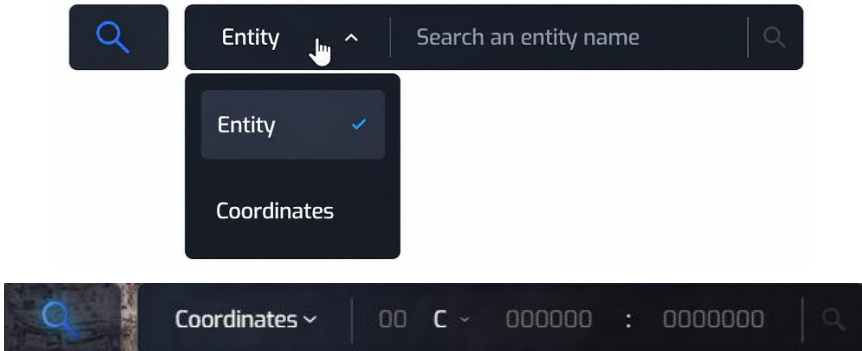


Figure 3-28: Map Screen

Table 3-27: Map Screen

No.	Item	Description
1	MAP OPORD Name Label	Displays the identifier or name of the currently active Operation Order (OPORD). This ensures the crew is viewing the correct tactical overlays, control measures, and graphics associated with the specific mission.
2	Search Button	Opens a search field and displays the virtual keyboard, enabling the user to quickly search for a tactical entity, location, or geographic coordinates within the system. 

No.	Item	Description
3	Map Layers Button	Opens the Map Layers window [refer to section 3.2.3.1]. This enables the user to toggle various visual overlays on or off.
4	Tactical Map	Serves as the primary visual interface where all positional data, targets, and map objects are displayed. It provides the user with real-time situational awareness including operational terrain, data layers, and tactical overlays. NOTE: right- tapping the map opens the Map Summary window [refer to section 3.2.3.4].
5	Map Functions Bar	Enables the operator to control and operate various map functions [refer to section 3.2.3.2].
6	Sliding Drawer Button	Toggles the interface layout between a Full-Screen Map View and a Split-Screen / Dashboard View for multitasking.
7	Map Scale Indicator	Provides a dynamic visual reference for measuring distances.
8	Map Tools Menu	Provides quick access to advanced utilities. It contains the following functional menus: • Terrain Analysis: Enables the user to perform measurements and analysis of local terrain. • Local Graphics: Enables the user to draw, edit, and overlay tactical graphics (roads, fields, houses), markers, or annotations directly onto the map screen. [refer to section 3.2.3.3]. • Multiple Selection: Provides tools to select, group, and manage multiple map elements or targets simultaneously.

3.2.3.1 Map Layers Window

(See Figure 3-29 and Table 3-28)

9002-090v01

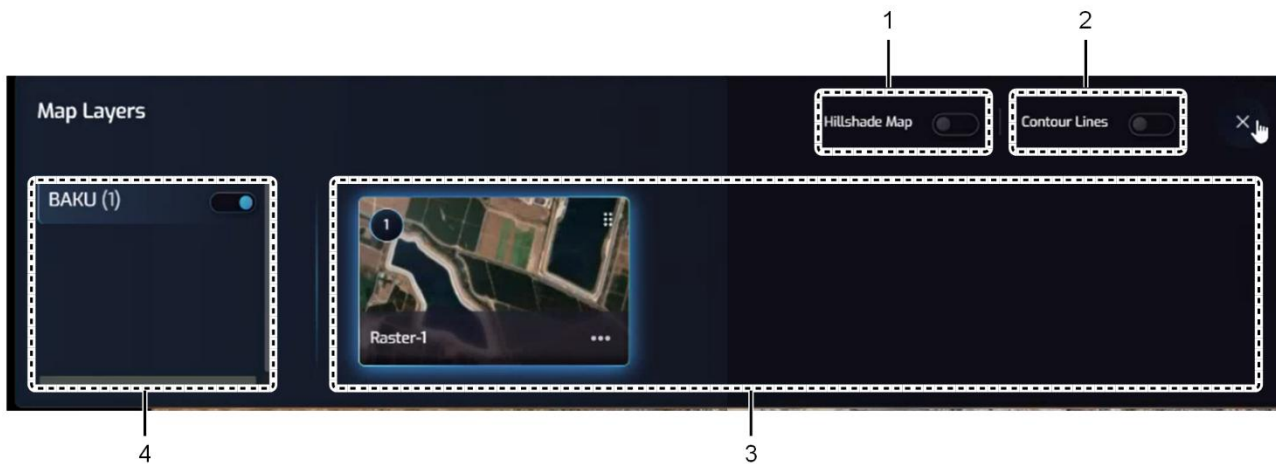


Figure 3-29: Map Layers Window

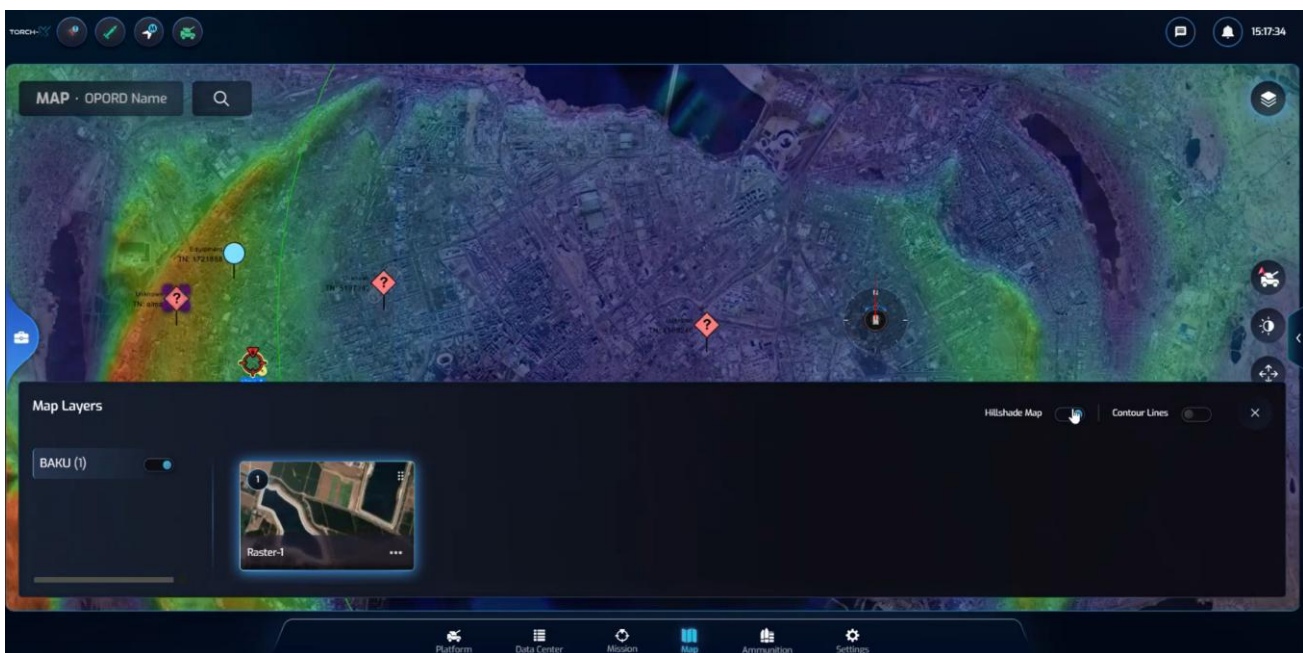


Figure 3-30: Map Layers Window - Hillshade Map

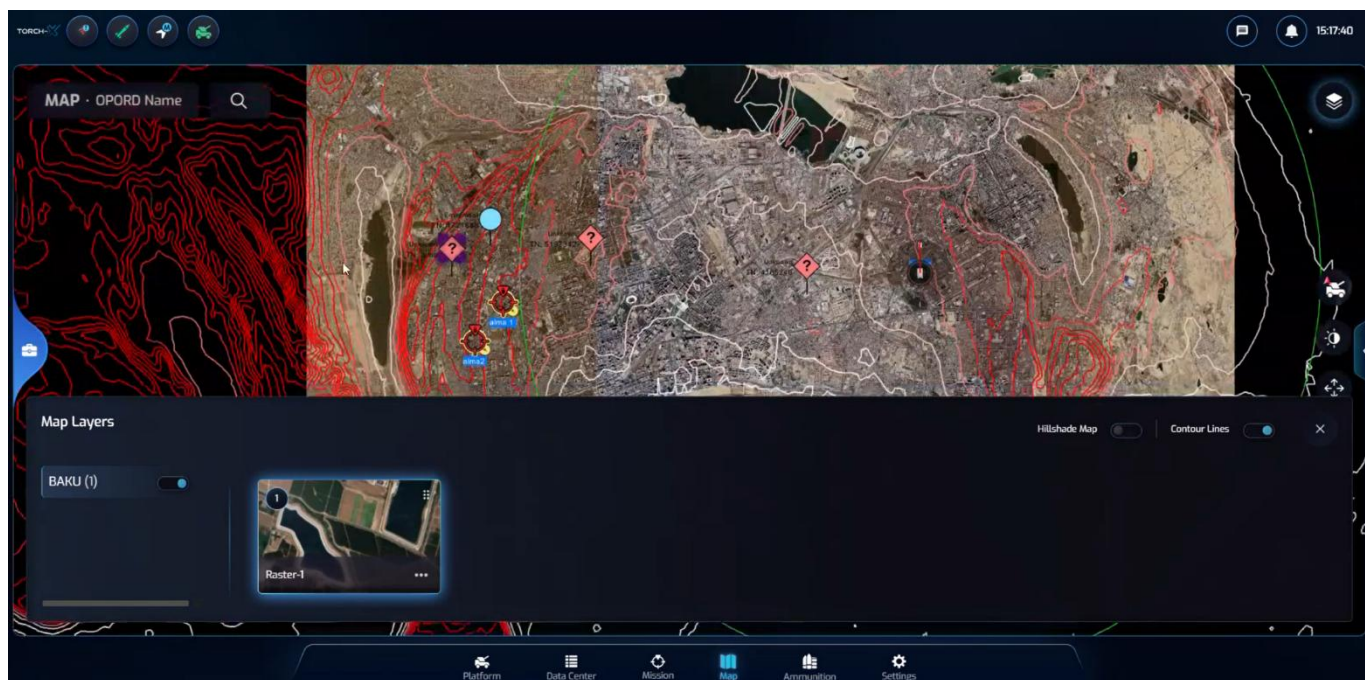


Figure 3-31: Map Layers Window - Contour Lines

Table 3-28: Map Layers Window

No.	Item	Description
1	Hillshade Map Button	Toggles the visual overlay of hillshading (shaded relief) on the digital map display (See Figure 3-30). It processes digital elevation model (DEM) data to simulate shadows and highlights based on terrain topography.
2	Contour Lines	Toggles the overlay of topographic contour lines (isolines connecting points of equal elevation) on the digital map display (See Figure 3-31) to enable precise elevation and slope analysis.
3	Map/Layer Preview	Provides a preview of the selected map or layer. The preview includes the following annotations: • Layer Name – Displays the name of the selected layer (e.g., RASTER-1). • Zoom To Fit (...) – Automatically adjusts the map scale and position to display the entire selected map or layer extent within the current map view. •  – Displays or hides the selected layer on the tactical map.
4	Loaded Maps / Layers	Displays all maps and map layers currently loaded into the system. Each layer includes a dedicated Toggle control that enables the user to show or hide the selected layer on the tactical map display.

3.2.3.2 Map Functions Bar

(See Figure 3-32 and Table 3-29)

9002-055v01

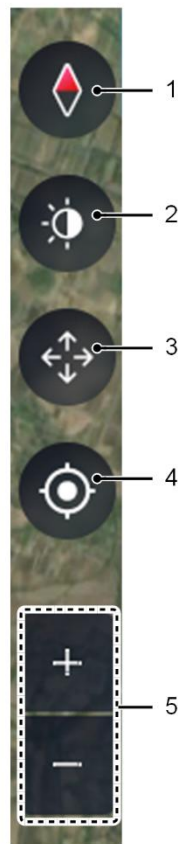


Figure 3-32: Map Functions Bar

Table 3-29: Map Functions Bar

No.	Item	Description
1	North Arrow / Compass	Indicates map orientation. Tapping the icon opens a menu to select the map's reference orientation:  - The map remains fixed with North always facing the top of the screen.  - The map rotates in real-time so that the turret/barrel azimuth always faces upward, facilitating intuitive target engagement.  - The map rotates based on the DITA platform's driving direction or chassis front.

No.	Item	Description
2	Map Contrast / Brightness Toggle	Adjusts the map display contrast/day-night mode to optimize visibility under changing ambient light conditions. 
3	Pan / Tilt Control	Enables the user to navigate the map by panning the view horizontally and vertically, and to adjust the map display tilt angle. 
4	Center on Platform (My Location):	Centers the map view onto the platform's current GPS/INS position coordinates.
5	Zoom Controls (+ / -)	 -Zooms in for a high-resolution, detailed view.  - Zooms out.

3.2.3.3 Map Tools Menu - Local Graphics

(See Figure 3-33 and Table 3-30)

9002-098v01



Figure 3-33: Map Tools Menu - Local Graphics

Table 3-30: Map Tools Menu - Local Graphics

No.	Item	Description
1	Geometric Shapes	Opens a context menu containing tactical graphic overlays and geometric shapes (e.g., arrows, polygons, circles, and squares) for manual map annotation.
2	Properties Window	Opens the properties window for the selected map graphic, enabling the user to modify its visual attributes such as line thickness, line color, line type, fill type, and other appearance settings. This function is available only after a shape or graphic object has been created or selected on the map.
3	Shape center (UTM Full)	Displays the UTM coordinates representing the exact geographic center of the selected tactical shape or overlay.
4	Delete	Permanently deletes the selected tactical shape, or drawing overlay, from the map.
5	Confirm	Validates and saves the current input, configuration changes, or map modifications.
6	Exit	Discards any unsaved changes or inputs and closes the Local Graphics menu.

3.2.3.4 Map Summary Toolbar

(See Figure 3-34 and Table 3-31)

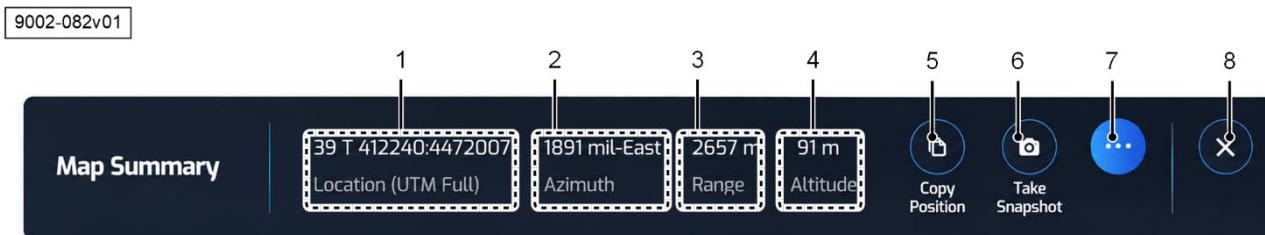
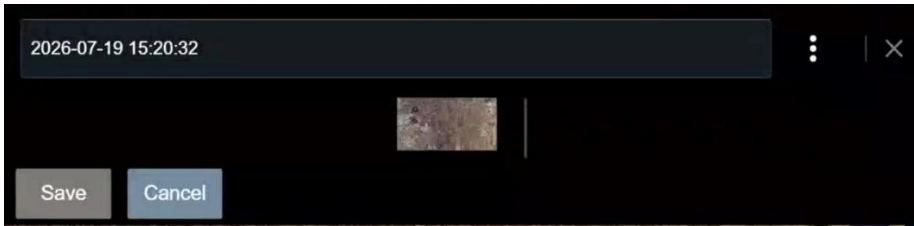


Figure 3-34: Map Summary Toolbar

Table 3-31: Map Summary Toolbar

No.	Item	Description
1	Location	Displays the geographic coordinates (e.g., Grid Reference/UTM) of the position clicked by the mouse cursor on the tactical map.
2	Azimuth	Displays the horizontal angle (heading or bearing) from the platform to the position clicked by the mouse cursor on the tactical map, measured in mils or degrees.
3	Range	Displays the line-of-sight distance from the platform to the position clicked by the mouse cursor on the tactical map or specified coordinates.
4	Altitude	Displays the elevation above sea level or ground level of the position clicked by the mouse cursor on the tactical map.
5	Copy Position Button	Copies the active geographic coordinates to the system clipboard for rapid transfer or data entry.
6	Take Snapshot Button	Captures and saves a current visual or data record of the map interface.  Tapping the  button opens a context menu containing editing tools for the captured screen image: • Undo – Reverts the last modification made to the image. • Redo – Reapplies the most recently undone modification. • Crop – Enables the user to trim the image and retain only the selected area. • Paint – Enables freehand drawing and annotation directly on the image. after tapping this option, a dedicated slider is displayed to enable the user to modify the line color and adjust the stroke thickness. • Write – Enables the user to add text annotations to the image. after tapping this option, a dedicated slider is displayed to enable the user to modify the text color and size.

No.	Item	Description
7	Action Menu Button	Opens a context menu with the following options: • Track () - enables the user to create and place tactical entities on the map, such as hostile forces, friendly units, logistics assets, and other operational objects. • Quick Task () - enables the operator to instantly generate and distribute operational tasks with a single action: Evacuation, Engagement, ISR, etc. • CFF () - opens Call For Fire (CFF) request form that enables the Forward Observer (FO) or Battery Command Post (BCP) to generate and transmit structured fire requests tailored to the following operational types: FFE (Fire for Effect), Smoke, Illumination, Registration and to selected the assault weapon: DITA, Nora, etc. • Common Fire Mission () - opens the New Fire Mission wizard for creating a new fire mission.
8	X Button	Discards any unsaved changes or inputs and closes the Map Summary menu.

3.2.3.5 Map Annotations and Tactical Symbols

(See Figure 3-35 and Table 3-32)

9002-091v01

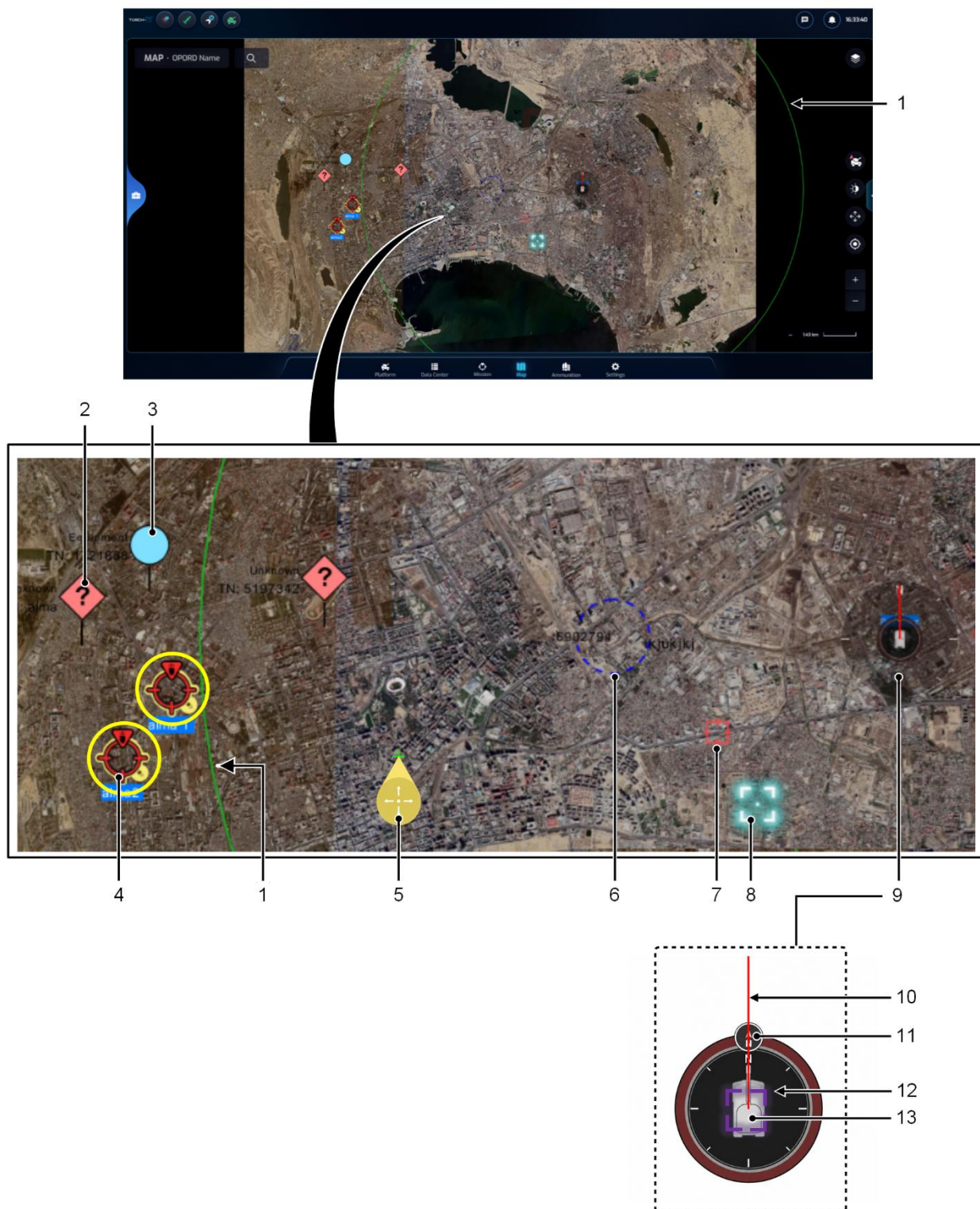
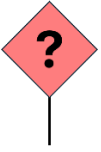


Figure 3-35: Map Annotations and Tactical Symbols

Table 3-32: Map Annotations and Tactical Symbols

No.	Item	Description
1	AOI (Area of Interest) Radius	The green circle surrounding the platform entity symbol indicates the Area of Interest (AOI). It defines an approximate 10 km operational radius around the platform for situational awareness and tactical monitoring.
2	Track Entity 	Represents a tactical entity on the digital map, such as a friendly force, hostile unit, vehicle, target, or other operational asset.
3	TBD	TBD
4	Target Symbol 	- Represents a designated hostile target. - The yellow circle surrounding the symbol indicates the target <u>safety</u> radius. - The mission name is displayed below the symbol identifies the fire mission associated with the target.
5	Reference Area 	Indicates an Area of Interest (AOI) on the tactical map.
6	Shape Object	Represents a graphical shape or annotation created by the user on the tactical map, such as a circle, rectangle, line, arrow, polygon, or freehand graphic. These objects are used to support mission planning, area marking.
7	Quick Task Symbol 	Represents an operational task assigned on the tactical map, such as Evacuation, Medical Support (MEDEVAC), Intelligence, Surveillance and Reconnaissance (ISR), or Engagement.
8	Selection Marker	Indicates the exact map location selected by the user while creating or editing a target, entity, or point of interest. The marker serves as a visual confirmation of the currently selected position.
9	Platform Self-Entity	Graphical representation of the self platform (DITA self-propelled howitzer) on the tactical map. The symbol displays the platform silhouette, current gun barrel orientation, and a north reference indicator.
10	Red Azimuth Line	Indicates the current gun barrel azimuth direction relative to the platform and map orientation.
11	North Indicator (N)	Indicates the geographic North direction on the map and serves as a reference for orientation and azimuth measurements.

No.	Item	Description
12	Platform Center Marker (purple cross)	The indicates the platform's center point.
13	Platform Direction Indicator	Displays the platform silhouette and indicates the current vehicle heading and direction of movement on the tactical map

3.2.4 Data Center Screen

This screen displays system records and stored data, organized into the following categories: Fire Missions, Mission Groups, Meteorological Telegrams, Tasks, Ammunition Muzzle Velocity.

3.2.4.1 Data Center – List Management Window – Meteorology

(See Figure 3-36 and Table 3-33)

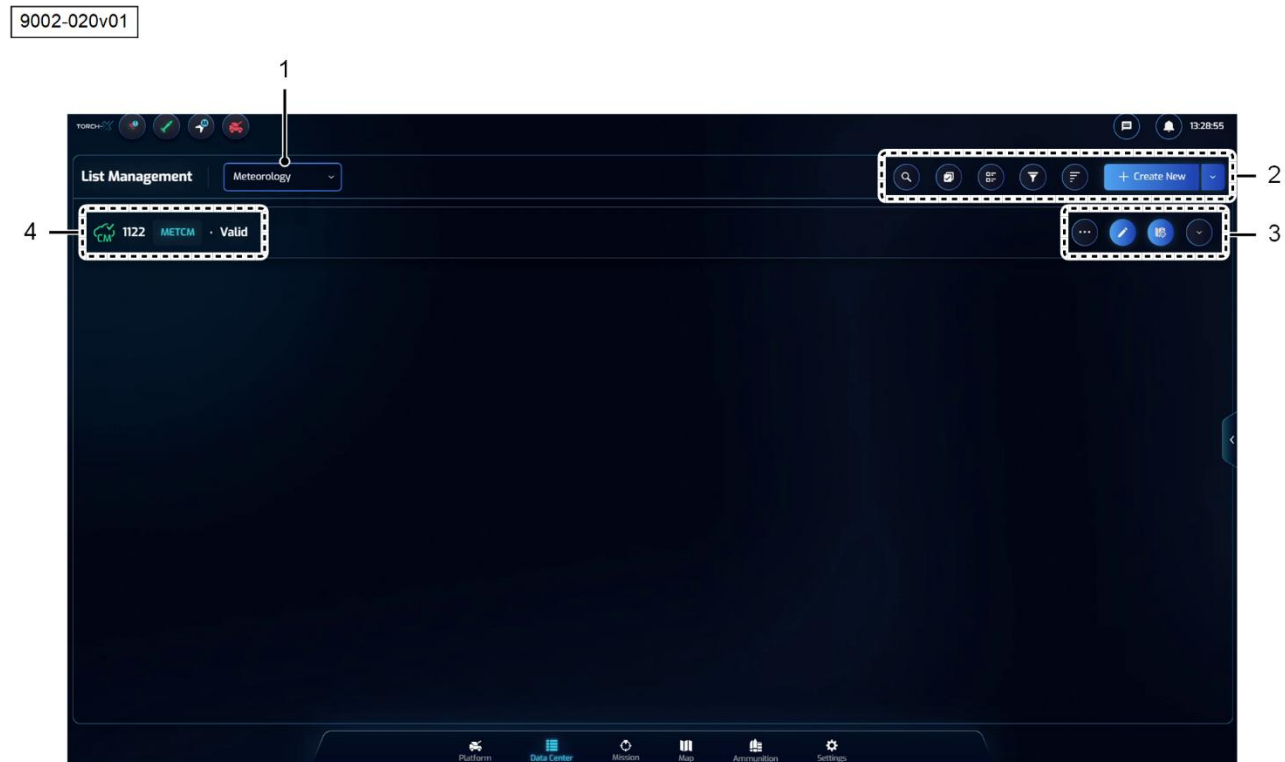


Figure 3-36: Data Center – List Management Window – Meteorology

Table 3-33: Data Center – List Management Window – Meteorology

No.	Item	Description
1	List Management Drop-down Menu	- Enables the user to select the displayed list in the main table view: Missions – displays the list of missions. Meteorology – displays the list of Meteorological telegrams. Tasks - displays the list of tasks. Mission Groups - displays the list of mission groups. MV Management - displays the list of ammunitions.
2	Meteorology Action Toolbar	Enables the user to perform operations on the active list: Search () - Opens a dynamic text filtering field to quickly locate a specific record by name, ID, or parameters Select All / Multi-Select () - Enables multi-selection mode to perform batch actions on multiple records simultaneously. Group By () - Enables the user to organize and aggregate the displayed records within the main table view according to: Type, creator, expired or cancelled. Filter () - enables the user to filter the list based on specific criteria (e.g., date, status, or source). Sort () - Controls the sorting order of the list (e.g., update date, A-Z, or creation time). Create New () – Opens a window for manual creation of new METCM meteorological telegram [refer to section 3.2.4.3.3]. Tapping the (▼) arrow enables creation of METEO11 meteorological telegram [refer to section 3.2.4.3.2]
3	Meteorology Telegram Toolbar	Enables the user to perform the following actions on the Meteorology Telegram: Properties - Opens the properties window of the meteorological message. Meteo Form Lines - Opens a detailed tabular view showing atmospheric data organized by altitude layers. Go to Location - switches to the tactical map view and centers the display on the coordinates of the meteorological station. Extend Validity - enables the operator to manually extend the operational validity of the current meteorological message by a defined duration of 1, 2, 4, or 8 hours. Set as Not applicable - Flags the meteorological message as invalid for ballistic use without deleting it from the system.

No.	Item	Description
		• Delete – deletes the selected meteorological message from the Data Center list.
4	Meteorology Telegram Details	Provides a quick view information for each record [refer to section 3.2.4.3.1].

3.2.4.2 Meteorological Message Sources

A meteorological message may be generated using one of the following methods:

- 1) **Standard Meteorological Message** - Use a pre-defined meteorological message containing a complete atmospheric profile.
- 2) **Meteorological Station Input (if available)** - A meteorological station may generate a meteorological message in a standardized TXT file format containing the complete atmospheric profile, including ground-layer data, atmospheric layers, altitude, pressure, wind, temperature, location, validity period, timestamp, and other relevant meteorological parameters. The generated file can be imported into the TORCH-X and used as the meteorological source for ballistic calculations..
- 3) **Local Atmospheric Measurements (Manual Creation)** - Create a meteorological message using locally measured weather data at the firing position. The first atmospheric layer is generated from measurements obtained by available instruments, such as:
 - Barometer (air pressure)
 - Psychrometer / Hygrometer (humidity)
 - Thermometer (temperature)

The collected data is used as the baseline atmospheric layer for ballistic calculations.

3.2.4.3 Meteorological Message Validity

A meteorological message is subject to both **temporal** validity and **spatial** validity:

- Temporal Validity – A meteorological message remains valid for 2 hours from its effective time. After the validity period expires, the message is no longer used for ballistic calculations.
- Spatial Validity (see Figure 3-37) – The meteorological message is valid only within its defined Area of Validity (AoV). For the message to be applied in ballistic calculations, **both the firing platform (1) and the target location (2) must be located within the validity radius of the meteorological message (3)**. If either the platform or the target is outside this area, the message is considered not applicable and will not be used by the ballistic engine.

9002-0105v01

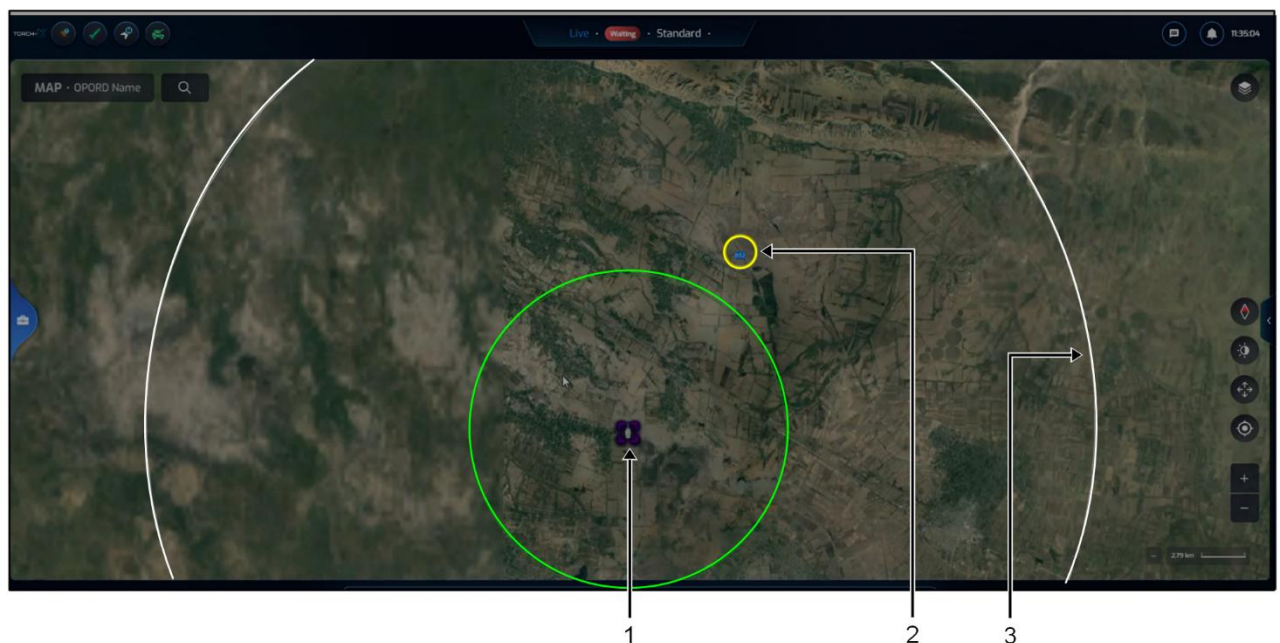


Figure 3-37:– Meteorology - Spatial Validity

3.2.4.3.1 Meteorology Telegram Details

(See Figure 3-38 and Table 3-34)

9002-089v01

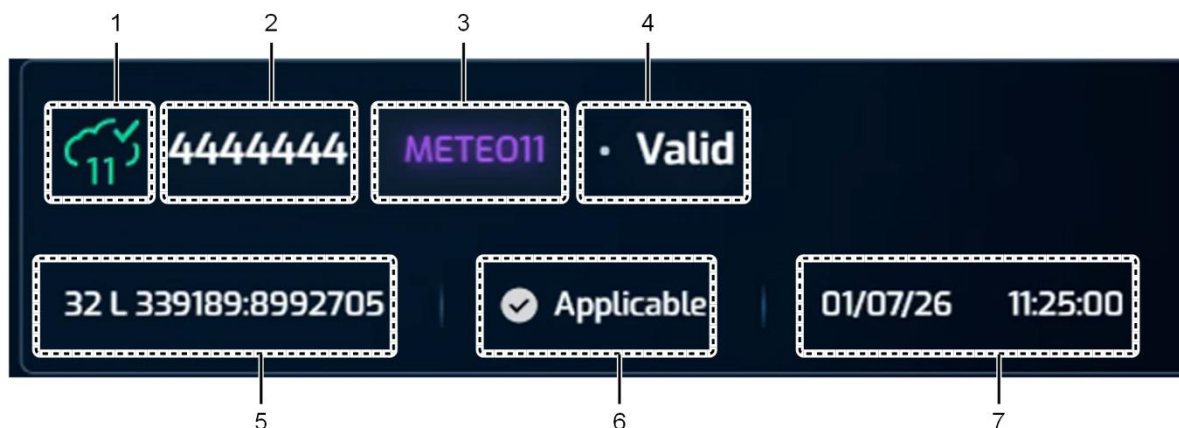


Figure 3-38: Data Center – List Management Window – Meteorology Telegram Details

Table 3-34: Data Center – List Management Window – Meteorology Telegram Details

No.	Item	Description
1	Meteorological Telegram Status	A visual icon representing the message validity status:  The data meets all validation parameters and available for ballistic computations.  The data is incomplete or has exceeded its expiration time.  The data is in its initial editing or configuration stage.
2	Name	The unique identifier or title assigned to the specific meteorological message.
3	Meteorological Telegram Type	  () - a text label stating the data format standard (e.g., METCM or METEO11).
4	Operational State	The operational state of the message: <u>Draft</u>: The data is in its initial editing or configuration stage. <u>Valid</u>: The data meets all validation parameters and available for ballistic computations. <u>Invalid</u>: The data is incomplete or has exceeded its expiration time.
5	Position Data	Displays the geographical position of the meteorological station.
6	Applicability Status	Not Applicable - The meteorological telegram is excluded from fire mission computations and cannot be used as the active meteorological data source. Applicable - The meteorological telegram is valid and may be used for fire mission computations.

No.	Item	Description
7	Date and Time	Displays the date and time indicating when the meteorological message becomes effective.

3.2.4.3.2 Create New METEO11 Form

(See Figure 3-39, Figure 3-40 and Table 3-35, Table 3-36)

9002-046v01

Create New METEO11 · Step 1/2: Properties

9 — Choose a method

Manual Import

8 — *Telegram Name

7 — *Location

00 C 000000 : 0000000

7 — *Station Altitude

- 0.0 +

6 — *Start Time Expiration Time

05/27/26 13:31 05/27/26 15:31

5 — *Ground Temperature Deviation *Ground Pressure Deviation

- 0 + - 0 +

← Previous Cancel Save Next →

4 3 2 1

Figure 3-39: Create New METEO11 Form (1 of 2)

Table 3-35: Create New METEO11 Form (1 of 2)

No.	Item	Description
1	Next	Displays the next properties window within the METEO11 screen.
2	Save	Validates and stores the telegram in the system, instantly syncing it with the Fire Control System (FCS) to update ballistic trajectory calculations.
3	Cancel	Closes the METEO11 Properties Screen without saving the changes.
4	Previous	Displays the previous properties window within the METEO11 screen.
5	Ground Temperature Deviation	Enables the user to set the actual ground temperature deviation from the standard atmosphere.
6	Ground Pressure Deviation	Enables the user to set the actual ground barometric pressure deviation from the standard atmosphere.
7	Start Time	Enables the user to set the date and time indicating when the meteorological message becomes effective.
8	Expiration Time	Enables the user to set the date and time defining when the message expires.
9	Station Altitude	Enables the user to set the altitude of the meteorological station above sea level (in meters) (for Ground Pressure calculation).
10	Location	Used to set the coordinates of the meteorological station. Supports two entry methods: - Manual coordinate input. - Direct "Map Click" ( selection from the C4I tactical map.
11	Telegram Name	A text field for entering name for the meteorological message.
12	Choose Method	Used to define the meteorological data source: IMPORT: loads meteorological data from a received file. For imported files, a new field appears: Imported file: METCM3 1 (1).txt MANUAL: Unlocks fields for manual user entry.

9002-099v01

Create New METEO11 Step 2/2: Input Required Lines

Line	Altitude (m)	Air Density (kg/m³)	Temperature (°C)	Wind Direction (°)	Wind Speed (m/s)
0	02				
1	04				
2	08				
3	12				
4	16				
5	20				
6	24				
7	30				

← Previous Cancel Save Done

Figure 3-40: Create New METEO11 Form (2 of 2)

Table 3-36: Create New METEO11 Form (2 of 2)

No.	Item	Description
1	Line	The index number of the atmospheric layer/zone.
2	Altitude (m)	The altitude of the radiosonde at the sampled atmospheric layer, measured in meters above the meteorological station.
3	Air Density (kg/m ³)	The absolute air density at the specified altitude.
4	Temperature (°C)	The air temperature at the specified layer.
5	Wind Direction (°)	The direction from which the wind is blowing at that altitude, measured in degrees (0°–360°).
6	Wind Speed (m/s)	The velocity of the crosswind/headwind, measured in meters per second, used to calculate drift corrections.
7	Done	Finalizes the meteorological data entry process. Tapping this button performs a system validation check on all mandatory fields and atmospheric layers.
8	Save	Validates and stores the telegram in the system, instantly syncing it with the Fire Control System (FCS) to update ballistic trajectory calculations.
9	Cancel	Closes the METEO11 Properties Screen without saving the changes.
10	Previous	Displays the previous properties window within the METEO11 screen.

3.2.4.3.3 Create New METCM Form

(See Figure 3-41, Figure 3-42 and Table 3-37, Table 3-38)

9002-0100v01

Create New METCM • Step 1/2: Properties

1 Choose a method

Manual Import

2 *Telegram Name

3 *Location

00 C ~ 000000 : 000000

4 *Station Altitude

— 0.0 +

*Station Air Pressure

— 0 +

5 *Start Time

05/27/26 12:04

Expiration Time

05/27/26 14:04

6

7

8 ← Previous

9 Cancel

10 Save

11 Next →

Figure 3-41: Create New METCM Form (1 of 2)

Table 3-37: Create New METCM Form (1 of 2)

No.	Item	Description
1	Choose a Method	Used to define the meteorological data source: • IMPORT: loads meteorological data from a received file loaded into the tactical computer. • MANUAL: Unlocks fields for manual user entry.
2	Telegram Name	A text field for entering name for the meteorological message.
3	Location	Used to set the coordinates of the meteorological station. Supports two entry methods: • Manual coordinate input. • Direct "Map Click" () selection from the C4I tactical map.
4	Station Altitude	Enables the user to set the altitude of the meteorological station above sea level (for Ground Pressure calculation).
5	Start Time	Enables the user to set the date and time indicating when the meteorological message becomes effective.
6	Expiration Time	Enables the user to set the date and time defining when the message expires.
7	Station Pressure Air	Enables the user to manually enter the absolute barometric pressure at the meteorological station's actual altitude.
8	Previous	Displays the previous window.
9	Cancel	Closes the METCM window without saving the changes and returns to Data Center screen.
10	Save	Saves the current METCM data.
11	Next	Displays the next properties window within the METCM screen.

9002-0101v01

Create New METCM · Step 2/2: Input Required Lines

Line	Wind Direction (10mil)	Wind Speed (Knot)	Temperature (0.1°K)	Air Pressure (Millibar)
0				
1				
2				
3				
4				
5				

Figure 3-42: Create New METCM Form (2 of 2)

Table 3-38: Create New METCM Form (2 of 2)

No.	Item	Description
1	Line	The index number of the atmospheric layer/zone.
2	Wind Direction	The direction from which the wind is blowing at that altitude, measured in degrees (0°–360°).
3	Wind Speed	The velocity of the crosswind/headwind, measured in meters per second, used to calculate drift corrections.
4	Temperature	The air temperature at at the specified layer.
5	Air Pressure	The air pressure at the specified altitude.
6	Done	Finalizes the meteorological data entry process. Tapping this button performs a system validation check on all mandatory fields and atmospheric layers.
7	Save	Validates and stores the telegram in the system, instantly syncing it with the Fire Control System (FCS) to update ballistic trajectory calculations.
8	Cancel	Closes the METCM Properties Screen without saving the changes.
9	Previous	Displays the previous properties window within the METCM screen.

3.2.4.4 Data Center – List Management Window – Missions

(See Figure 3-43 and Table 3-39)

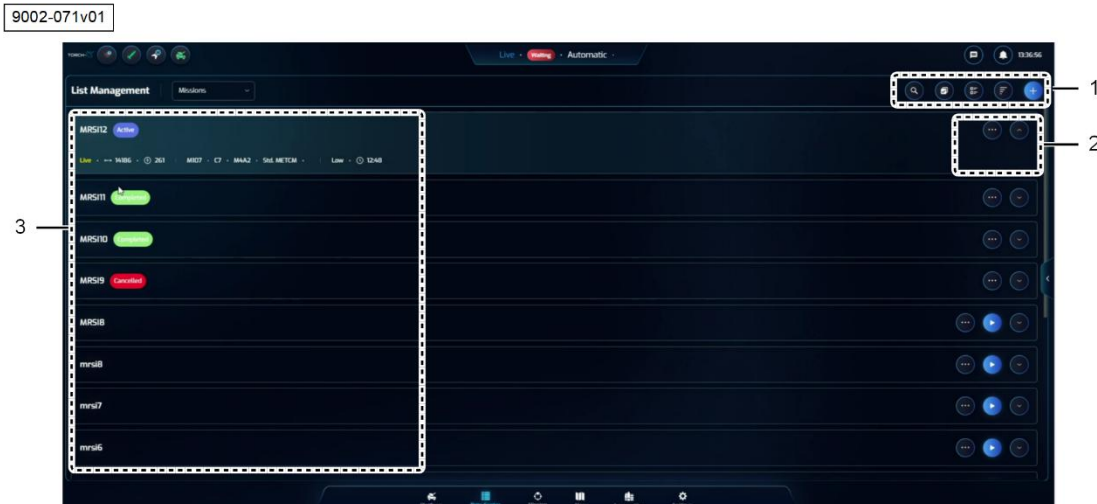


Figure 3-43: Data Center – List Management Window – Missions

Table 3-39: Data Center – List Management Window – Missions

No.	Item	Description
1	List Management – Missions Action Toolbar	Enables the user to perform operations on the active list: Search (🔍) - Opens a dynamic text filtering field to quickly locate a specific record by name, ID, or parameters Select All / Multi-Select (📋) - Enables multi-selection mode to perform batch actions on multiple records simultaneously. Group By (📊) - Enables the user to organize and aggregate the displayed records within the main table view according to: Mission Status, Time on Target, Mission Type. Filter (🔽) - enables the user to filter the list based on specific criteria (e.g., date, status, or source). Sort (☰) - Controls the sorting order of the list (e.g., update date, A-Z, or creation time). Create New (+) – Opens a window for manual creation of New Fire Mission. [refer to 3.2.5.1.4]
2	Mission Toolbar	Action Menu (⋮) – opens a context menu that enables the user to Delete the selected Mission or Go to Location (target location) 📄 - Initiate fire mission computation.

No.	Item	Description
		 - activates the computed fire mission for live execution and switches the interface to the active MISSION screen (see 3.2.5.1.5).  - expands a details panel directly beneath the row, displaying the critical tactical and ballistic parameters of the selected mission (see Figure 3-44).
3	Missions List	Displays a list of the missions created by the user. Each mission recorded in the list displays an operational status: Active: The fire mission is currently active. Completed: The designated number of rounds has been successfully fired, and the mission is archived. Cancelled: The mission was aborted prior to or during execution—either manually by the operator or via a override command from the Battery Command Post (BCP). Calculated: The firing solution (including quadrant elevation, azimuth, fuse setting, and optimal charge zone) has been successfully computed by the Fire Control Computer and is validated. Selecting a fire mission entry expands a details panel directly beneath the row, displaying the critical tactical and ballistic parameters of the selected mission (See Figure 3-44 and Table 3-40).

9002-083v01

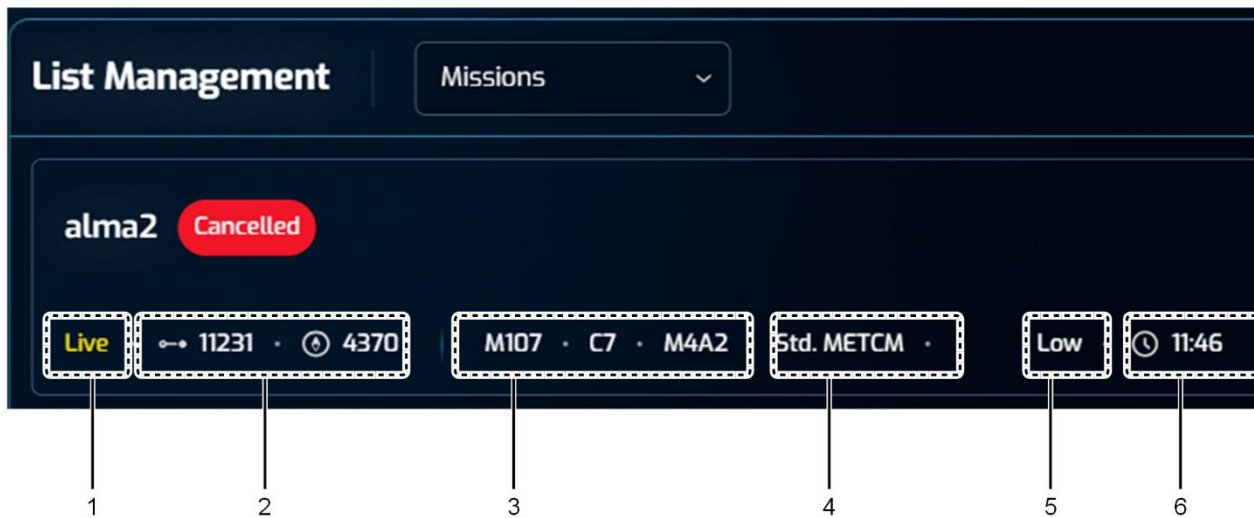


Figure 3-44: Data Center – List Management Window – Mission Details Panel

Table 3-40: Data Center – List Management Window – Mission Details Panel

No.	Item	Description
1	Mission Purpose	Defines the operational context of the fire mission: <u>Live</u> Standard combat mode. The Fire Control Computer transmits firing data directly to the unmanned turret, authorizing automated shells loading, automated laying, and weapon discharge upon trigger commands. <u>Practice</u>: Training and simulation mode. The software fully processes the fire mission, calculates ballistic trajectories, and simulates target engagement on the screen, but mechanically isolates the automatic loader and firing circuits to prevent accidental live fire.
2	Target Data	Displays the range and azimuth to the target.
3	Ammunition Data	Specifies the allocated loadout for the mission: Shell type, Fuse type, Propellant type, Charge type.
4	Meteorological Message	Displays the meteorological message type (METCM, METEO11).
5	LOW	Displays the mission trajectory type: LOW or HIGH.
6	Timestamp	The exact time the mission was created.

3.2.4.4.1 Data Center – List Management Window – Create New Missions Group Form

(See Figure 3-45 and Table 3-41)

This window enables the gun commander to aggregate multiple individual fire missions into a single operational or administrative group. Fire missions are typically grouped based on shared ballistic or tactical parameters, such as: Ammunition & Charge Optimization, Mission Type, focusing on a single target area for coordinated mass fire, etc.

9002-084v01

The screenshot shows a 'Create New Missions group' window. It features a 'Group Name' text field (1) containing 'TAM', an 'Autonomy Level' dropdown menu (2) set to 'Automatic', and a 'Missions' table (3) with columns 'Name', 'Proj Category', 'Proj Amount', and 'Order'. A '+ Add Mission' button is located in the table area. A search modal (4) is open, showing a search bar and 'Cancel' and 'Add' buttons. At the bottom of the window are 'Cancel' (6) and 'Create' (5) buttons.

Figure 3-45: Data Center – List Management Window – Create New Missions Group Form

Table 3-41: Data Center – List Management Window – Create New Missions Group Form

No.	Item	Description
1	Group Name Field	Alphanumeric text input field for entering the unique identifier or callsign of the group.
2	Autonomy Level Dropdown Menu	Defines the mission autonomy level: • Standard Mode: Manual operation, requiring operator control and validation for each stage of the firing sequence (Aim, Load, Shoot). • Semi-Automatic Mode: Assisted operation, where the system automates calculations and physical preparations, requiring operator confirmation to shoot (tapping the “PRESS TO SHOOT” button (). • Automatic Mode: Fully autonomous operation, minimizing operator intervention by executing the complete firing cycle automatically upon mission approval. • Manual - A <u>non-autonomous</u> mode. In this mode, the user loads ammunition manually. All other platform operations can be executed via the TORCH-X C4I system according to the user command. • The user can execute firing procedures <u>directly</u> through the TORCH-X C4I application. In standard and semi-automatic firing modes, user initiate and manage the mission entirely via the TORCH-X C4I application (interfacing the DITA LACU), except the physical firing action itself. • Upon creating or receiving a mission, the firing mode is automatically set to the default configuration (except for adjustment plans, which are always set to standard mode). • If a fault is detected (e.g., a LACU error, safety event, or a "Check Fire" command), firing procedures are <u>automatically aborted</u>, and the user receives an immediate alert to determine how to proceed. • The user can stop the firing procedures at any moment using a single command.
3	Add Mission Button	Opens a search window used to locate existing fire missions within the system and add the selected missions to the current mission group.
4	Search Missions Window	Dynamic search and filtering window used to quickly locate existing fire missions from the system's database.
5	Create Button	Saves the newly mission group to the database and displays it in the List Management Screen – Mission Groups.
6	Cancel Button	Closes the window and aborts the process without saving data.

3.2.4.5 Data Center – List Management Window – Tasks

(See Figure 3-46 and Table 3-42)

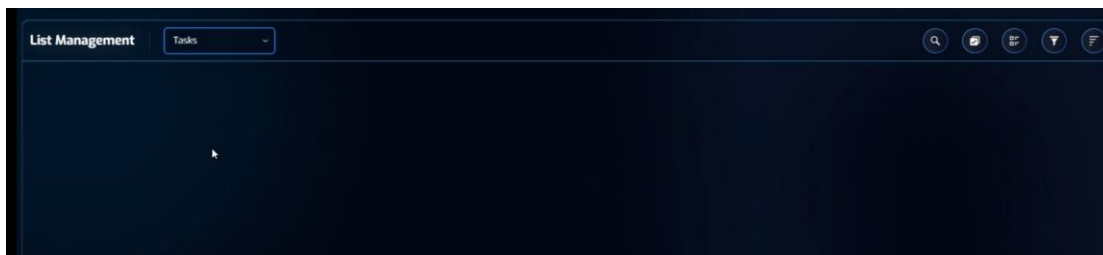


Figure 3-46: Data Center – List Management Window – Tasks

Table 3-42: Data Center – List Management Window – Tasks

No.	Item	Description
1	List Management – Tasks Action Toolbar	Enables the user to perform operations on the active list: • Search () - Opens a dynamic text filtering field to quickly locate a specific record by name, ID, or parameters • Select All / Multi-Select () - Enables multi-selection mode to perform batch actions on multiple records simultaneously. • Group By () - Enables the user to organize and aggregate the displayed records within the main table view according to: Mission Status, Time on Target, Mission Type. • Filter () - enables the user to filter the list based on specific criteria (e.g., date, status, or source). • Sort () - Controls the sorting order of the list (e.g., update date, A-Z, or creation time).
2	Tasks List	Displays a list of the missions created by the user.

3.2.4.6 Data Center – List Management Window – Mission Groups

(See Figure 3-47 and Table 3-43)

9002-0102v01

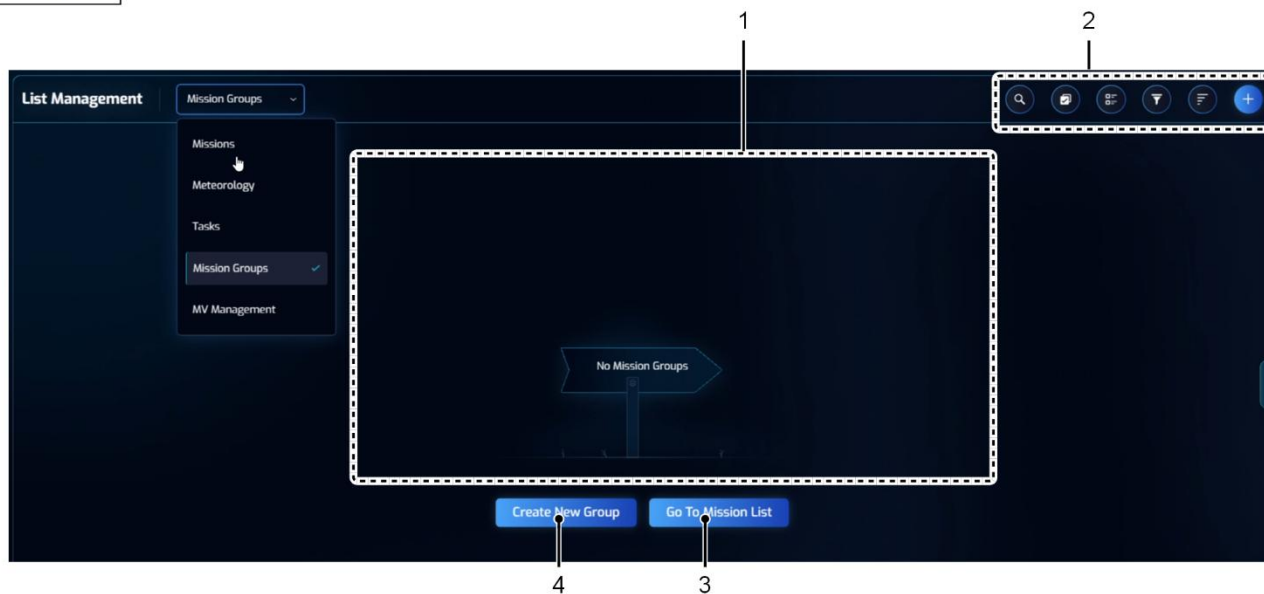


Figure 3-47: Data Center – List Management Window – Mission Groups

Table 3-43: Data Center – List Management Window – Mission Groups

No.	Item	Description
1	Mission Groups List	Displays a list of the Mission Groups created by the user.
2	List Management – Mission Groups Action Toolbar	• Search () - Opens a dynamic text filtering field to quickly locate a specific record by name, ID, or parameters • Select All / Multi-Select () - Enables multi-selection mode to perform batch actions on multiple records simultaneously. • Group By () - Enables the user to organize and aggregate the displayed records within the main table view according to: Mission Status, Time on Target, Mission Type. • Filter () - enables the user to filter the list based on specific criteria (e.g., date, status, or source). • Sort () - Controls the sorting order of the list (e.g., update date, A-Z, or creation time). • Create New () – Opens a window for manual creation of new Mission Group. [refer to section 3.2.4.4.1]
3	Go To Mission List	Navigates to Mission List Screen [refer to section 3.2.4.4].

No.	Item			Description
4	Create Button	New	Group	Opens the New mission groups window [refer to section 3.2.4.4.1].

3.2.4.7 Data Center – List Management Window – MV Management

(See Figure 3-48 and Table 3-44)

9002-085v01



Figure 3-48: Data Center – List Management Window – MV Management

Table 3-44: Data Center – List Management Window – MV Management

No.	Item	Description
1	Filter By Section	Enables the user to filter the displayed ammunition table rows based on specific categories or combination: shell, propellant, charge.
2	Ammunitions List	Displays a list of authorized, pre-calculated ammunition configurations and combinations. These combinations are directly mapped from digital Firing Tables compiled during technical testing. The FCS references these tables for ballistic calculations. For more detailed information on Ammunition Parameters fields, refer to section 3.2.4.7.1.

3.2.4.7.1 Data Center – MV Management - Ammunition Parameters

(See Figure 3-49 and Table 3-45)

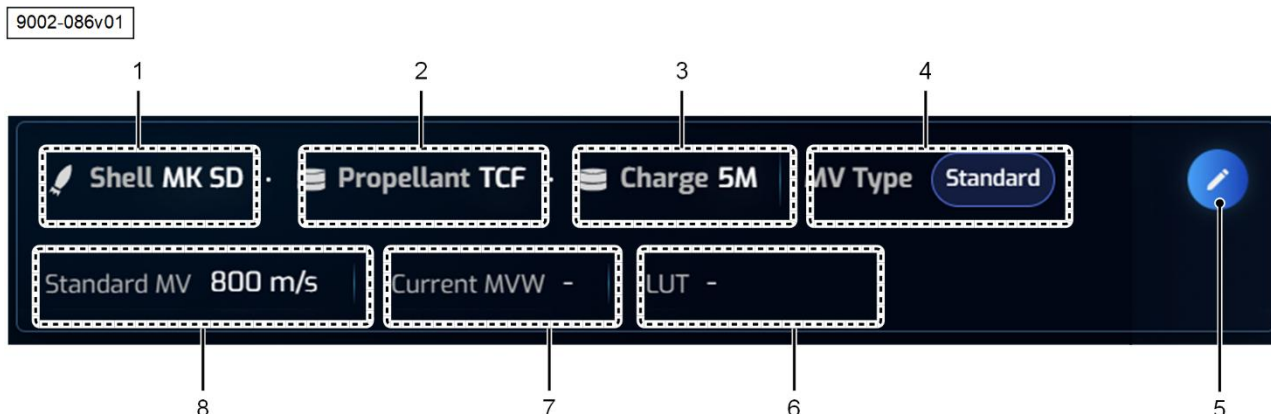


Figure 3-49: Data Center – MV Management - Ammunition Parameters

Table 3-45: Data Center – MV Management - Ammunition Parameters

No.	Item	Description
1	Shell Type	Displays the shell type: M107, MK SD or MK DV
2	Propellant Type	Displays the propellant type: TCF or M4A2.
3	Charge Type	Displays the charge type: C1-C7 or M1-M2.
4	MV Type	Displays the Muzzle Velocity type: Standard - Indicates that the muzzle velocity value is derived from the standard firing tables. Manual - Upon manual data entry by the user. Calibrated - Indicates that the muzzle velocity value was automatically calibrated by the system using MVR measurements obtained during firing.
5		Opens the Update MV Data window, enabling the operator to manually modify the ammunition's Muzzle Velocity (new MVW parameter). 
6	Last Update Time (LUT)	Displays the date and time of the most recent update to the muzzle velocity value.

No.	Item	Description
7	Current Muzzle Velocity for Work (MVW)	Displays the baseline muzzle velocity from the standard firing tables. This value is used as the reference for the Expected MV calculation during mission execution. A hyphen ("-") denotes that no radar-measured variance is currently active or captured.
8	Standard MV	Displays the baseline Standard Muzzle Velocity (MV) in meters per second (m/s). NOTE: Serves as the starting reference point for the ballistic computing before applying the Current MVW and LUT corrections.



3.2.5 Mission Screen

(See Figure 3-50 and Table 3-46)

9002-0104v01

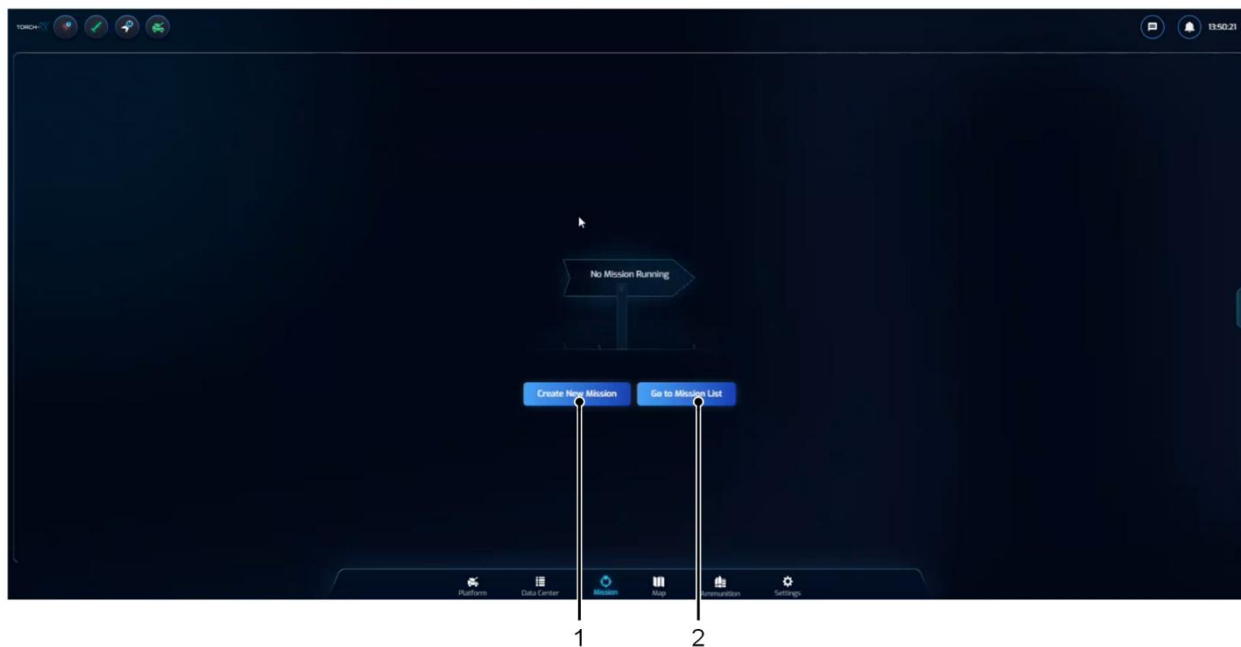


Figure 3-50: Mission Screen

Table 3-46: Mission Screen

No.	Item	Description
1	Create new mission	Opens the New Fire Mission wizard for creating a new fire mission
2	Go to Mission List	Navigates to the Mission List screen (in Data Center Screen) and displays all available fire missions.

3.2.5.1.1 New Fire Mission Screen - Target Selection

(See Figure 3-51 and Table 3-47)

9002-077v01

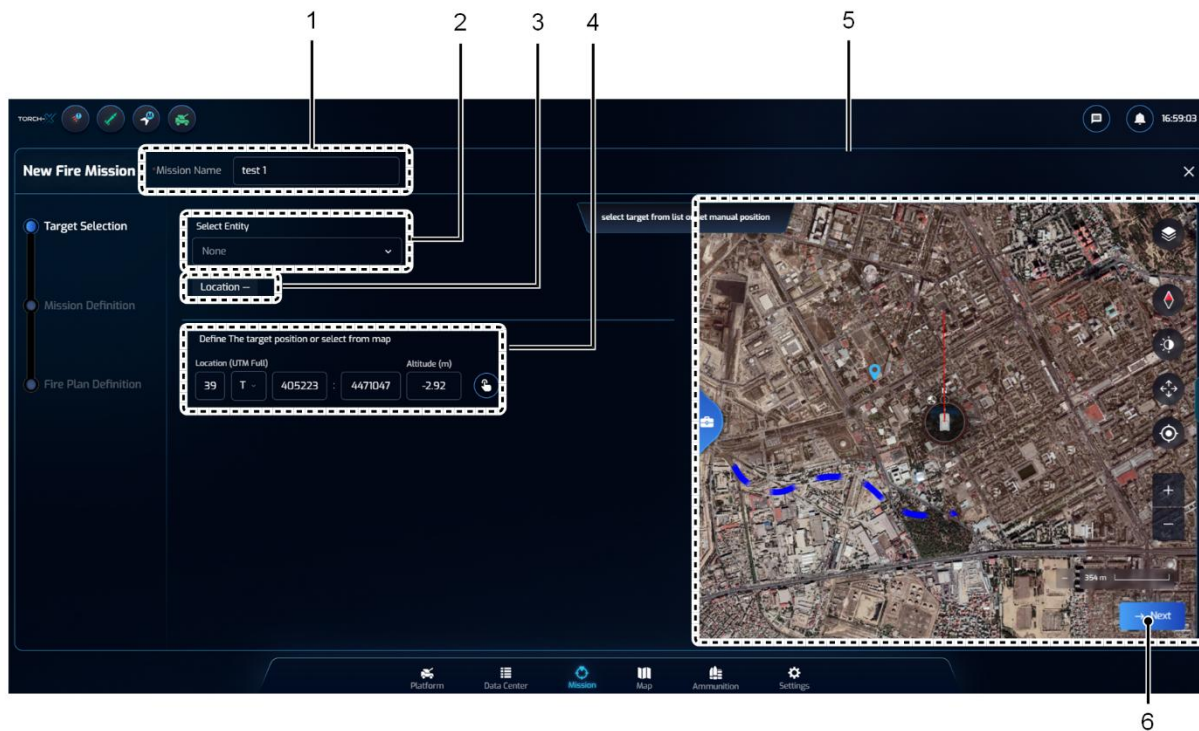


Figure 3-51: New Fire Mission Screen - Target Selection

Table 3-47: New Fire Mission Screen - Target Selection

No.	Item	Description
1	Mission Name Field	Alphanumeric text input field for entering the unique identifier or callsign of the fire mission.
2	Select Entity Drop-down Menu	Enables the user to select an entity from the available entities list to be associated with the fire mission.
3	Location	Displays the platform's current self-location coordinates, representing the user's own position as derived from the active positioning source (GPS, INS, or manually entered coordinates).
4	Target Position Fields	Enables the user to define the target position via two methods: Manual Entry: Direct input of full UTM coordinates and altitude in meters (M) above sea level. Via the Map: Direct capture of coordinates and altitude by selecting the target on the Tactical Map.
5	Tactical Map	Enables quick target location acquisition by capturing coordinates and altitude directly.
6	Next Button	Displays the next window (Mission Definition) within the New Fire Mission wizard.

3.2.5.1.2 New Fire Mission Screen - Mission Definition

(See Figure 3-52 and Table 3-48)

9002-075v01

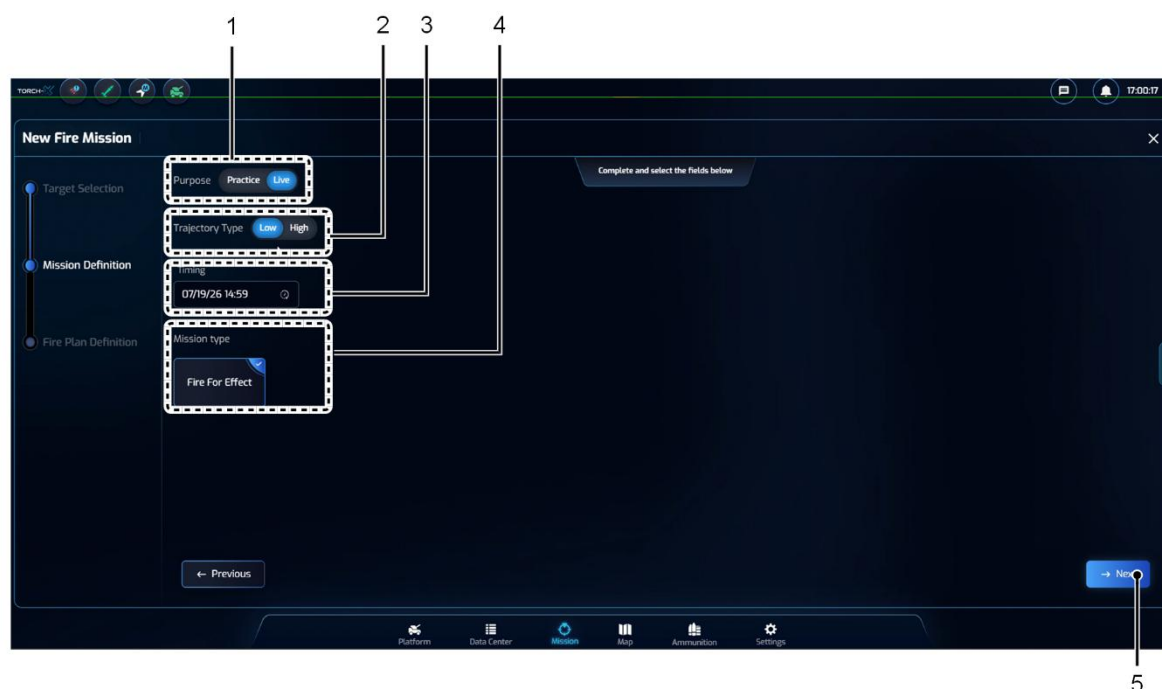


Figure 3-52: New Fire Mission Screen - Mission Definition

Table 3-48: New Fire Mission Screen - Mission Definition

No.	Item	Description
1	Purpose	Defines the tactical objective of the fire mission: Practice or Live. Live: Standard combat mode. The Fire Control Computer transmits firing data directly to the unmanned turret, authorizing automated shells loading, automated laying, and weapon discharge upon trigger commands. Practice: Training and simulation mode. The software fully processes the fire mission, calculates ballistic trajectories, and simulates target engagement on the screen, but mechanically isolates the automatic loader and firing circuits to prevent accidental live fire.
2	Trajectory Type	Defines the ballistic flight path for the DITA howitzer (e.g., Low Angle, High Angle) to optimize terrain clearance.
3	Timing	Opens an integrated Date & Time Picker overlay for selecting the mission execution time and date: - Calendar grid view for selecting the day, month, and year. - Vertical scrolling interface for selection of hours and minutes.
4	Mission Type	Classifies the mission type as Fire For Effect .
5	Next Button	Displays the next window (Fire Plan Definition) within the New Fire Mission wizard.

3.2.5.1.3 New Fire Mission Screen – Fire Plan Definition

(See Figure 3-53 and Table 3-49)

9002-076v01



Figure 3-53: New Fire Mission Screen – Fire Plan Definition

Table 3-49: New Fire Mission Screen – Fire Plan Definition

No.	Item	Description
1	Fire Plan Properties	Displays a tabular overview of all fire plans assigned to the mission together with key parameters of each fire plan in structured columns. A single fire mission can contain multiple fire plans. Each Fire plan contains an Action Menu (⋮) button that opens a context menu providing advanced options: • Edit - Opens the Fire Plan window to modify the selected fire plan parameters. • Delete- Deletes the selected fire plan record from the fire mission's fire plan list.
2	Add fire plan Button	Opens the New Fire Plan Window [refer to 3.2.5.1.4].
3	Save Button	Saves the current fire mission data.
4	Play Button	Activates the fire mission. Mission status changes from planning phase (Data Center Screen) execution phase (Mission Screen). NOTE: Only one fire mission can be Activated at any given time.

3.2.5.1.4 New Fire Plan Window

(See Figure 3-54 and Table 3-50)

9002-027v01

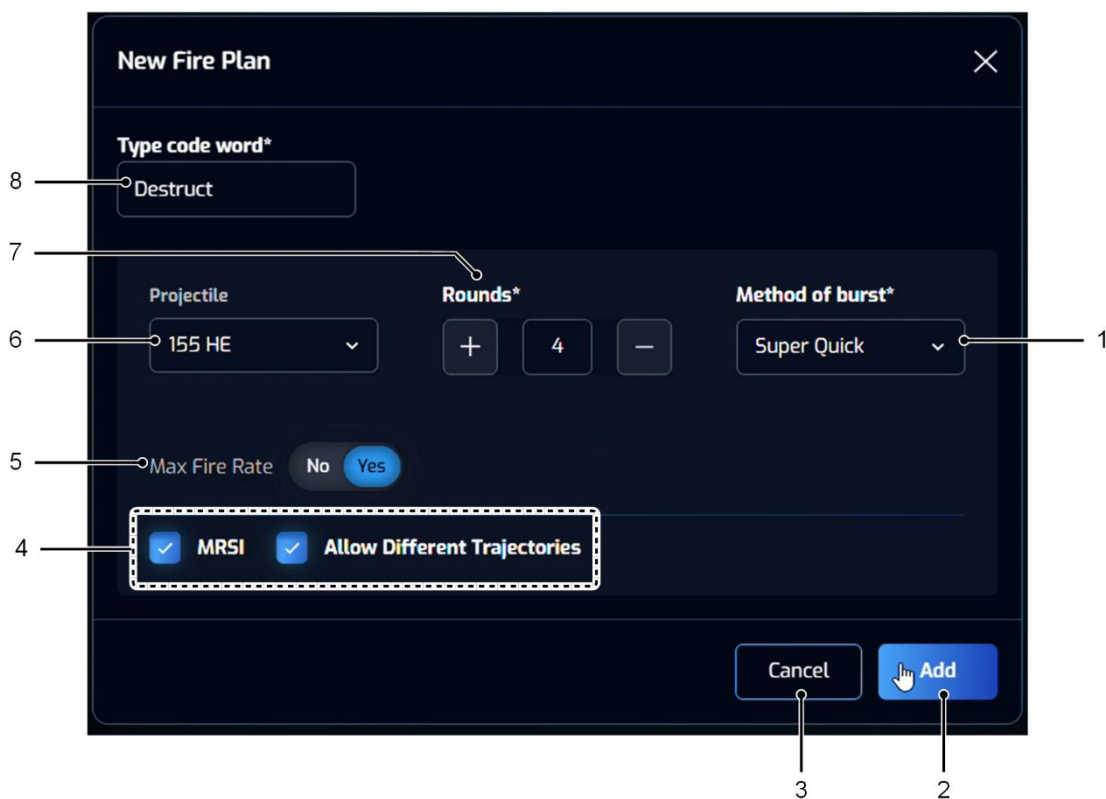
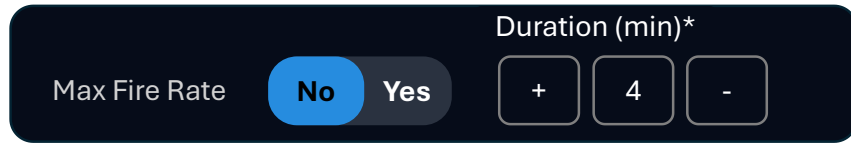


Figure 3-54: New Fire Plan Window

Table 3-50: New Fire Plan Window

No.	Item	Description
1	Method of burst Menu	Sets the Burst firing mode: (1) Delay (2) Proximity (3) Super Quick (4) Quick (5) Air ,etc.
2	Add	Adds the fire plan to the current fire mission.
3	Cancel	Closes the fire plan window without saving the changes and returns to new fire mission window.
4	MRSI Checkbox	Activates the Multiple Rounds Simultaneous Impact (MRSI) function, automatically calculating varied ballistic trajectories to ensure all specified rounds hit the target at the exact same second. Note: Checking the MRSI checkbox opens an additional “ Allow Different Trajectories ”. When enabled, the system calculates and utilize a combination of different ballistic trajectories (e.g., mixing low-angle and high-angle fire) within the same fire plan.
5	Max Fire Rate Toggle Button	Enables/Disables the maximum fire rate option (supported by the DITA automatic loading system):

No.	Item	Description
		• Yes – Fire Rate is set to Maximum. • No – opens a counter used to define the duration (in minutes) between rounds. NOTE: The system handles fire rate logic via the conditional MAX FIRE RATE toggle. When set to YES, the system ignores timing constraints and fires the designated rounds at peak mechanical autoloader speed. When toggled to NO, the interface exposes a DURATION field; the operator inputs the total time allowed to expend the allocated rounds, triggering a timing algorithm within the Fire Control Computer that automatically calculates and schedules the precise intervals between each loading and firing cycle.  When a mission is executed under MAX FIRE RATE = YES, the Fire Control Computer bypasses timed delays and fires continuously based on subsystem readiness loops: It monitors the WGS and the LACU concurrently; the instant the WGS reports AIMED (barrel locked on target) and the LACU reports READY TO FIRE (shell successfully chambered and breech closed), the system automatically triggers the firing mechanism for the next round.
6	Projectile Menu	A dropdown menu used to select the required ammunition type (155 HE).
7	Rounds Incremental Counter	Used to specify the number of rounds to be fired.
8	Type Code Word Field	Used for typing a unique tactical identifier or code name to the specific fire mission. Note: fields with astrix (*) - mandatory input field.

3.2.5.1.5 New Fire Mission Screen – Play

(See Figure 3-55 and Table 3-51)

9002-058v01

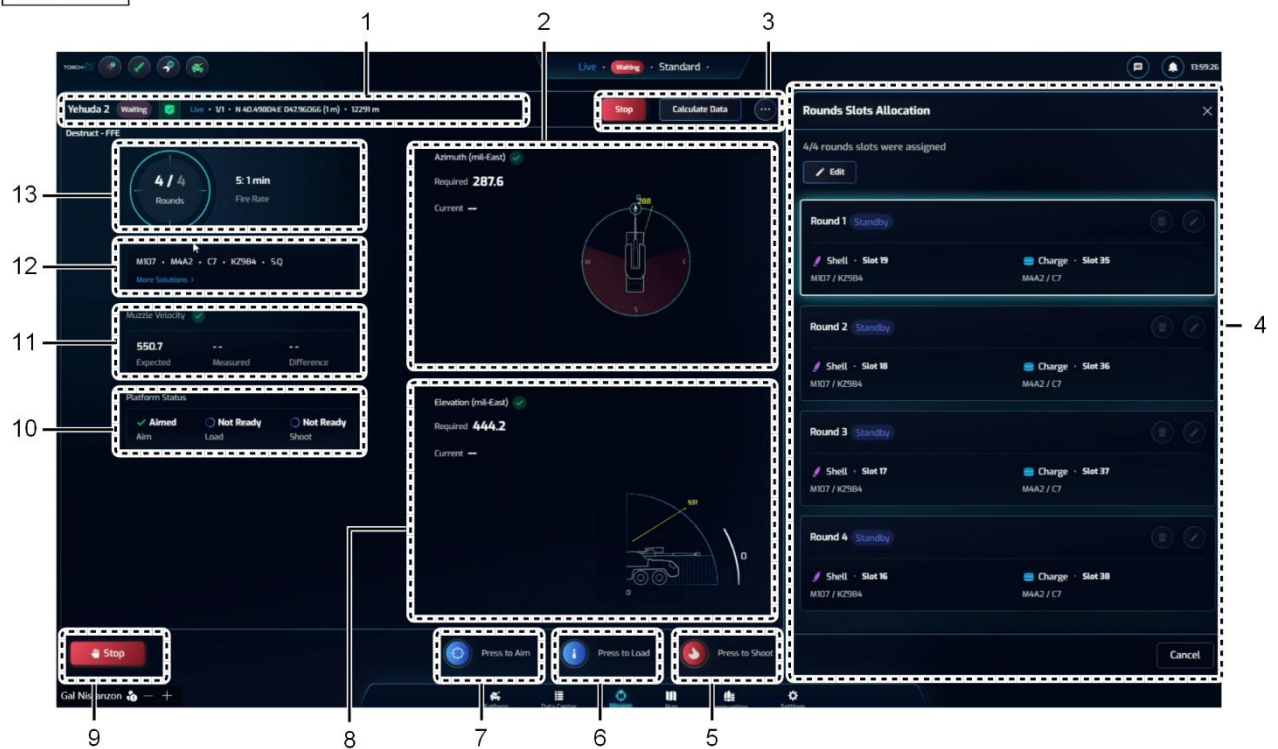
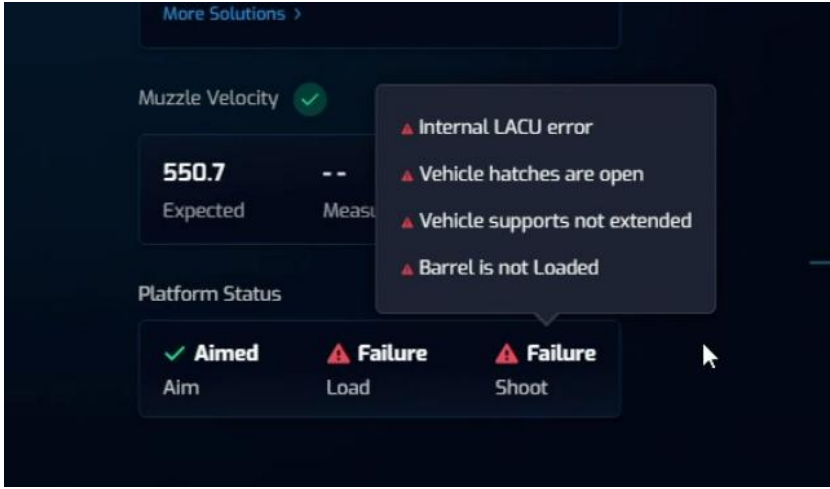


Figure 3-55: New Fire Mission Screen – Play

Table 3-51: New Fire Mission Screen – Play

No.	Item	Description
1	Mission Information	Provides information regarding the active mission: • Mission Name - The name of the active fire mission. • Status: [WAITING] - Indicates the mission is queued or awaiting execution. • Mission Safety Button () - Visual indicator of system and weapon safety checks status. •  - Safety checks failed. Firing is blocked. •  - Firing safety checks are fully validated. •  (warning) - Minor issue detected (e.g., uncalculated fire correction exists). • Tapping this button opens the Mission Information and Status panel [refer to section 3.2.5.1.6]. • Mission Purpose [Live/Practice] -: Defines the operational context of the fire mission: • Target Coordinates - The geographic location (UTM/MGRS) of the target.
2	Azimuth Targeting Infographic	Graphical overlay displaying Required (calculated) vs. Current (measured) Azimuth. •  - serves as the positive system lock indicator, confirming that the DITA turret and barrel are precisely AIMED onto the computed target coordinates. •  - the DITA turret and barrel are NOT AIMED onto the computed target coordinates.
3	Mission Controls	Enables the user to perform actions on the active mission: • Calculate Data Button - Computes the ballistic and firing solutions for the mission. This function is disabled if a shell is currently loaded in the chamber. • Action (...) Button - Opens secondary commands for the active mission [refer to section 3.2.5.1.7].
4	Rounds Slots Allocation Window	Displays and manages all ammunition rounds assigned to the Active fire mission, tracking their designated slots in the automatic loading system [refer to section 3.2.5.1.10].
5	Press to Shoot	Initiates the firing sequence of the loaded round. To prevent unintentional firing, the operator must continuously press and hold the button until the circular progress indicator is fully completed.
6	Press to Load	Initiates the automatic round loading sequence.
7	Press to Aim	Automatically aims/lays the gun barrel onto the calculated azimuth and elevation.

No.	Item	Description
8	Elevation Targeting Infographic	Graphical overlay displaying Required (calculated) vs. Current (measured) Elevation.  - serves as the positive system lock indicator, confirming that the DITA turret and barrel are precisely AIMED onto the computed target coordinates.  - the DITA turret and barrel are NOT AIMED onto the computed target coordinates.
9	Stop Button	Immediately halts the active mission firing sequence and suspends all automated loading and firing mechanisms (transmitting STOP command to WGS and LACU subsystems).
10	Platform Status	 - Aim Field: Ready - A dynamic status indicator (yellow color) confirming that the WGS has successfully completed all physical and computational adjustments and is ready. Aimed –upon pressing the "PRESS TO AIM" button () , this indication confirms that the system has successfully completed the automatic azimuth and elevation adjustments, precisely aligning the barrel with the target coordinates. Failure - indicating a technical or safety fault within a subsystem that prevents the aiming process. Not Ready - A dynamic status indicator confirming that the WGS sub-system is not ready. - Load Field: Ready - A dynamic status indicator (yellow color) confirming that the LACU is ready. Loaded - upon pressing the "PRESS TO LOAD" button () , this indication confirms that the system has successfully completed the

No.	Item	Description
		chambering of the ammunition (shell and propellant charge), locked the breech, and is fully prepared for firing. • Not Ready - A dynamic status indicator confirming that the LACU sub-system is not ready. • Failure – indicating a technical or safety fault within a subsystem that prevents the loading process (e.g. Internal LACU error, Vehicle hatches are open, Vehicle supports not extended). • Shoot Field: • Not Ready - Indicates that one or more firing prerequisites have not been met, preventing the system from executing a firing command. • Ready – Indicates that all firing prerequisites have been complied, including weapon readiness, ammunition loading completion, safety validation, and subsystem availability, allowing the system to execute a firing command. • Firing - upon pressing the "PRESS TO SHOOT" button (), this indication confirms the firing sequence has been performed. • Failure – indicating a technical or safety fault within a subsystem that prevents the firing, such as hatches open, or LACU Failure, etc. 
11	Muzzle Velocity (MV) Tracking	Displays information regarding the MV: • Expected - The calculated muzzle velocity based on charge, temperature, and wear. • Measured - The actual velocity detected by the MVR unit upon firing. • Difference – the delta between expected and measured velocity.
12	Firing Solution Management	Displays the current Firing Solution ballistic data, ammunition profile, and fire plan. In addition, includes the option to view more firing solutions (e.g., for different charge zones) (refer to section 3.2.5.1.8).
13	Ammunition & Firing Rate Dashboard	Provides information regarding the ammunition and fire rate for the active fire mission: • Ammunition Counter (X / Y) - Displays the number of rounds fired out of the total allocated (e.g., 2 / 4).

No.	Item	Description
		Rate of Fire - Displays the fire rate (e.g., 1:1, 5:1). 5:1 specifies a cycle of 5 rounds per minute (one round every 12 seconds).

3.2.5.1.6 Mission Information and Status Panel

(See Figure 3-56, Figure 3-57, Table 3-52 and Table 3-53)

Selecting the Map Data & Correction tab automatically centers the map view on the designated mission target.

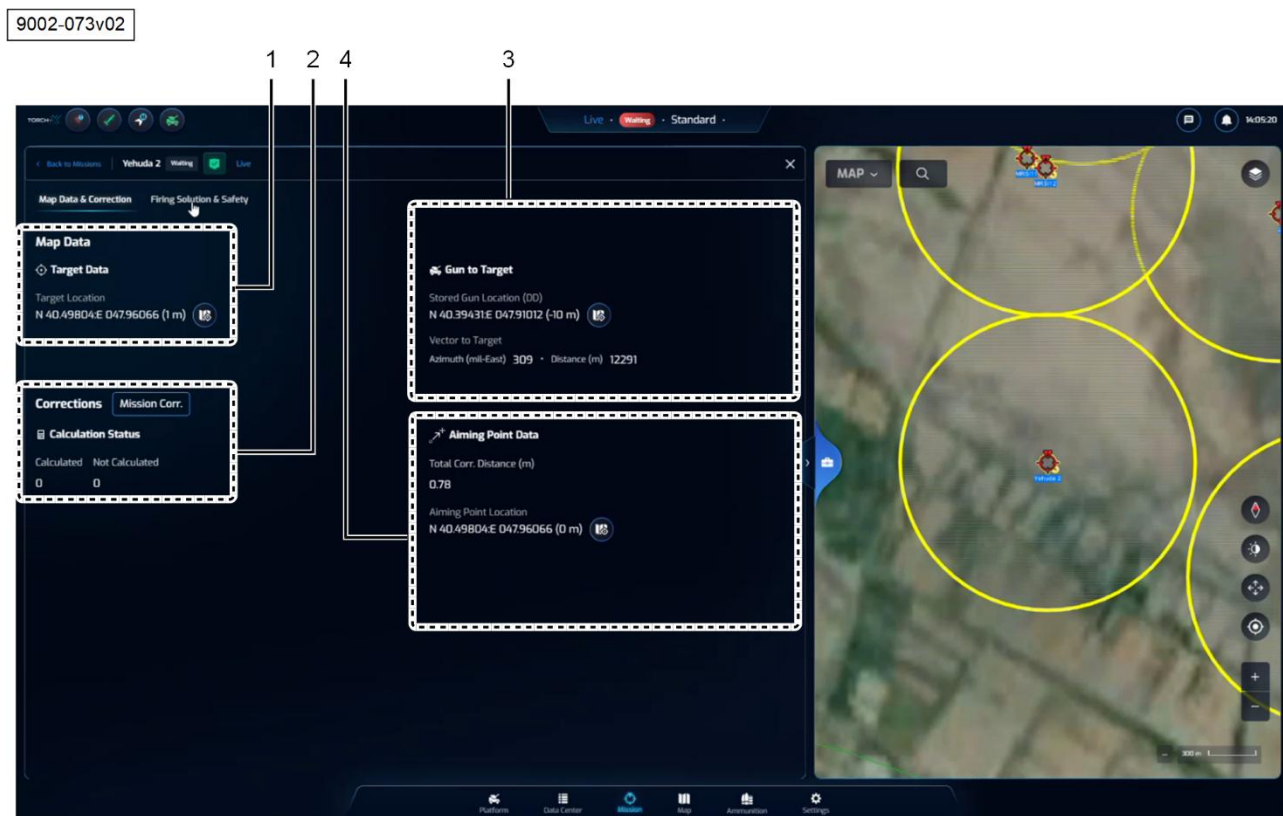


Figure 3-56: Mission Information and Status Panel - Map Data & Correction Tab

Table 3-52: Mission Information and Status Panel - Map Data & Correction Tab

No.	Item	Description
1	Map Section	Displays the precise geographical location data and coordinates of the designated target.
2	Corrections Section	Displays fire correction data, tracking both calculated ballistic adjustments and manual non-calculated corrections.
3	Gun to Target	Displays the gun stored location and the calculated vector to the target data, including azimuth and distance.
4	Aiming Point Data	Displays aiming point parameters, including the total baseline distance and the specific coordinates of the aiming point.

9002-074v01

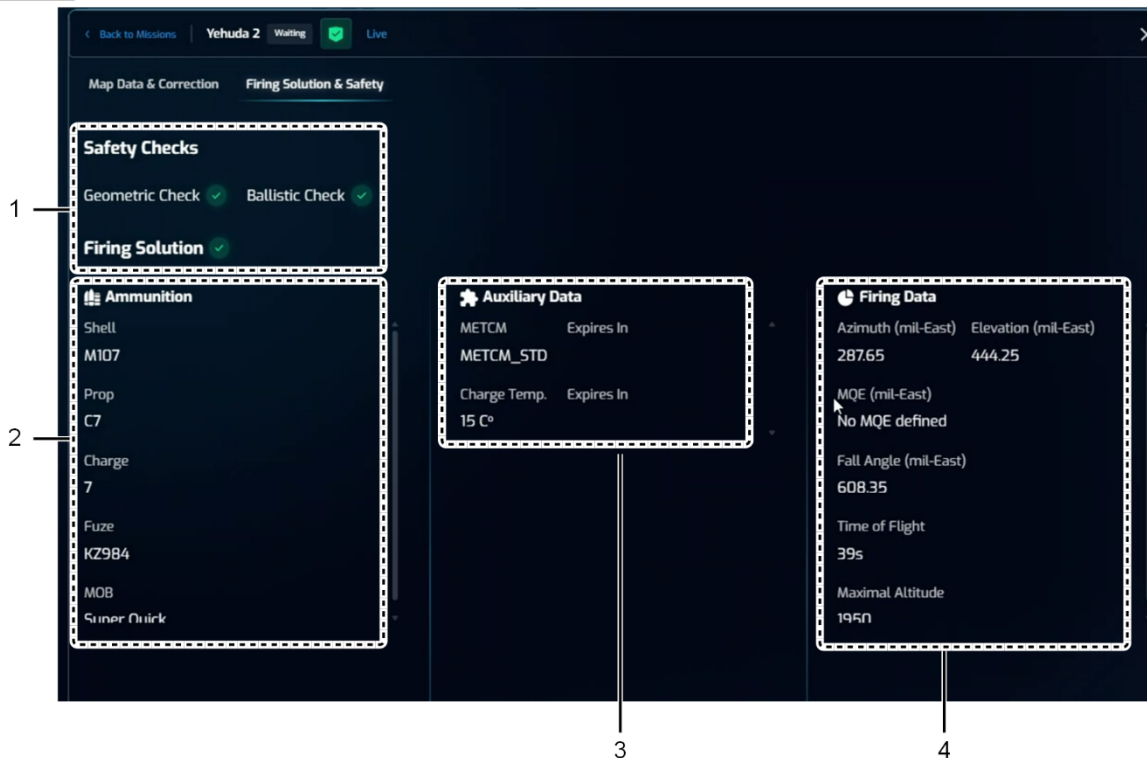


Figure 3-57: Mission Information and Status Panel - Firing Solution & Safety Tab

Table 3-53: Mission Information and Status Panel - Firing Solution & Safety Tab

No.	Item	Description
1	Safety Checks Section	Monitors and displays the status of safety verification mechanisms. It validates ballistic solutions against safety zones, ensures geometric constraints prevent physical interference or firing into restricted sectors, and confirms that all fire control parameters remain within safe operational thresholds. Geometric Check - Performs a geometric evaluation to verify that the calculated firing solution falls strictly within the authorized boundaries of the Hazards zone. Ballistic Check - Validates the firing parameters against aerodynamic and environmental data.
2	Ammunition Section	Displays configuration data for the ammunition combination allocated to the active fire mission. This includes the selected shell type, propellant charge type, fuze settings and the designated burst firing mode (such as Super Quick).
3	Auxiliary Data Section	Displays environmental data for the active fire mission, including the meteorological message, charge temperature, and a validity field indicating data expiration.
4	Firing Data Section	Displays calculated firing and platform navigation data required for target engagement such as platform azimuth and elevation values, the platform's current altitude, and the MQE..

3.2.5.1.7 Mission Controls - Action Button

(See Figure 3-58 and Table 3-54)

9002-072v01

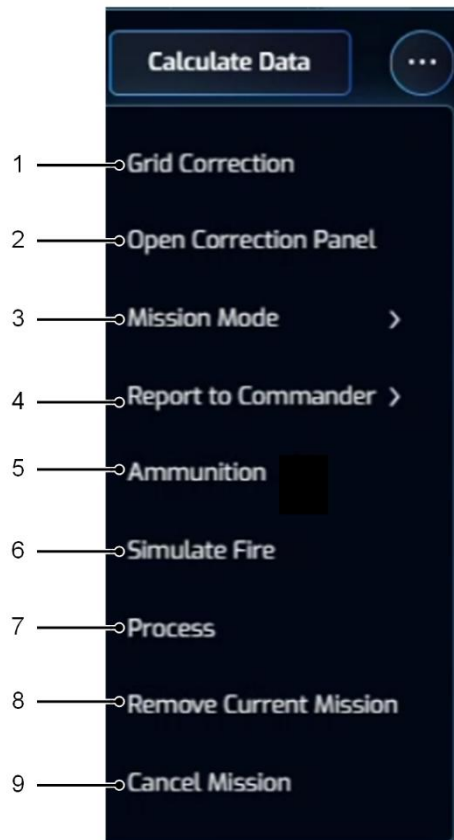


Figure 3-58: New Fire Mission Screen – Mission Controls – Action Button

Table 3-54: New Fire Mission Screen – Mission Controls – Action Button

No.	Item	Description
1	Grid Correction	Opens the Grid Correction Request Window that enables the user to enter a fire correction [refer to section 3.2.5.1.9].
2	Open Correction Panel	Opens the Corrections window to view all fire corrections [refer to section 3.2.5.1.13].
3	Mission Mode	Enables the operator to select the level of system autonomy for fire mission execution. Available modes include: • Standard Mode: Manual operation, requiring operator control and validation for each stage of the firing sequence (Aim, Load, Shoot). • Semi-Automatic Mode: Assisted operation, where the system automates calculations and physical preparations, requiring operator confirmation to shoot (tapping the “PRESS TO SHOOT” button (). • Automatic Mode: Fully autonomous operation, minimizing operator intervention by executing the complete firing cycle automatically upon mission approval. In this mode, the following <u>Attention window</u> appears:

No.	Item	Description
		 • Fire Single Round: Commands the system to fire exactly one 155mm shell. • Fire Entire Plan: Commands the system to execute the entire pre-loaded fire plan sequentially without further manual intervention. • Fire: Commands the system to initiate the selected firing type (Fire Single Round or Fire Entire Plan). • Cancel: Closes the window without saving any changes.
4	Report to Commander	Opens a context menu to send tactical status reports to the commander via the data link: Ready, Shot, Cease Fire, Misfire, and Rounds Complete.
5	Ammunition	Opens the Rounds Slots Allocation Window [refer to section 3.2.5.1.10].
6	Simulate Fire	Manually reports a fired round to update the ammo count and mission progress (if the automatic LACU sensor or MVR fails to detect the shot).
7	Process	Recomputes the fire mission data following updates to the firing solution or ammunition allocation.
8	Remove Current Mission	Manually terminates the currently active fire mission. Upon explicit operator confirmation via a pop-up prompt, the system updates the mission status to Complete and reports that all allocated rounds for this specific mission have been successfully fired.
9	Cancel Mission (Dynamic Button)	Cancels the active mission. The button Changes from Cancel Mission to End Mission after the final round is reported ("Rounds Complete" message in the Report to Commander option).

3.2.5.1.8 Fire Solution Window

(See Figure 3-59 and Table 3-55)

9002-079v01

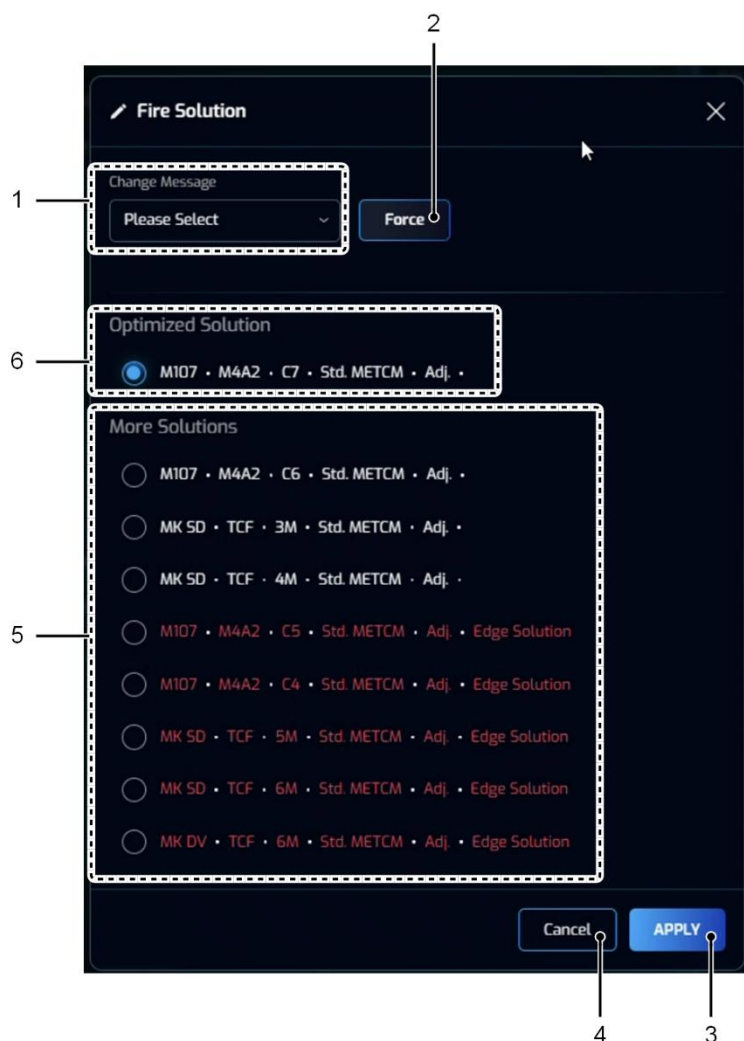


Figure 3-59: New Fire Mission Screen – Play - Fire Solution Window

Table 3-55: New Fire Mission Screen – Play - Fire Solution Window

No.	Item	Description
1	Change Message	Enables the user to select an alternative meteorological message to be used as the environmental data source for ballistic calculations associated with the active fire mission.
2	Force	Sets the selected meteorological message as the active environmental data source for the current fire mission. Upon tapping, the system recalculates the firing solution using the selected meteorological data and updates all affected ballistic parameters.
3	Apply	Implements any modified settings.
4	Cancel	Discards all pending changes made and closes the window.
5	More Solution	Displays alternative fire solutions for user selection. • White Font Options: Indicate solutions that are valid/Available for execution. • Red Font Options: Indicate firing solutions that cannot be calculated accurately or do not meet operational, ballistic, or safety constraints.
6	Optimized Solution	Displays the prime, system-calculated fire solution, optimized based on current operational parameters (e.g., target type, platform position, available ammunition inventory, firing modes, and meteorological data).

3.2.5.1.9 Grid Correction Request Window

(See Figure 3-60 and Table 3-56)

9002-078v01

Figure 3-60: Grid Correction Request Window

Table 3-56: Grid Correction Request Window

No.	Item	Description
1	Observation Target Azimuth	Displays the azimuth from the observer to the target used as the reference for the correction.
2	Increase / Decrease	Defines whether the correction increases or decreases the firing parameter.
3	Reference	Displays the user, observer, or system entity that entered or transmitted the fire correction request.

No.	Item	Description
4	Participate ID	Displays the identifier of the participating unit or observer submitting the correction.
5	Program ID	Displays the identifier of the fire correction program or mission associated with the request.
6	Up Down Value	Specifies the correction value to move the impact point closer to or farther from the observer-target line.
7	Right Left Value	Specifies the lateral correction value to shift the impact point right or left of the target.
8	Increase Decrease Value	Specifies the numerical adjustment value applied to the selected correction parameter.
9	Main or Illum	Defines whether the correction applies to the Main fire mission or to an Illumination mission.
10	Calculator Button	Calculates the fire correction values and computes ballistic adjustments based on the data entered in the Grid Correction Request form. The calculated correction can then be applied to the active fire mission to update the firing solution.
11	Save Button	Validates and saves the entered fire correction data to the active mission.
12	Cancel Button	Closes the Fire Correction Request window without saving any changes

3.2.5.1.10 New Fire Mission Screen – Play – Rounds Slots Allocation

(See Figure 3-61 and Table 3-57)

9002-059v01

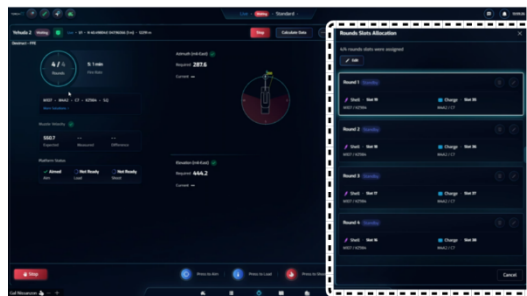
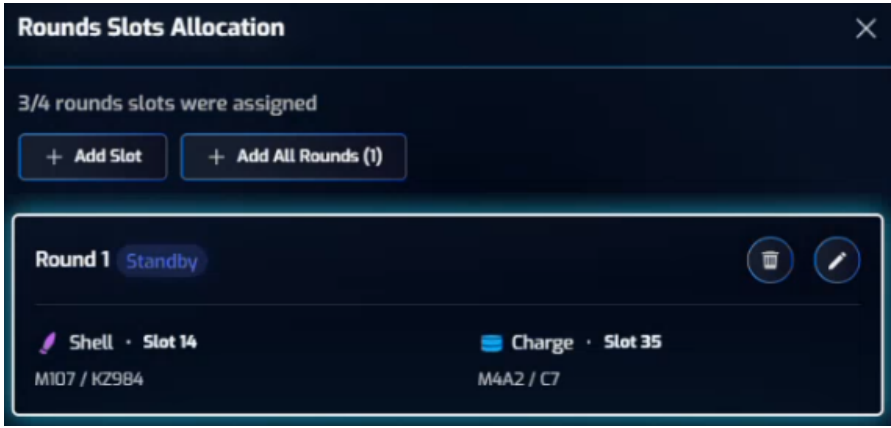


Figure 3-61: New Fire Mission Screen – Play – Rounds Slots Allocation

Table 3-57: New Fire Mission Screen – Play – Rounds Slots Allocation

No.	Item	Description
1	Edit Button	Tapping this button toggles the window layout to edit mode. It enables individual round modification (EDIT / REMOVE) and activates the Add Slot button for adding new rounds.  • Add Slot Button - Opens a window for allocating an additional ammunition slot or round to the fire mission. •  Button - Opens the Edit Round window for editing the rounds/charges. [refer section 3.2.5.1.11]. •  Button - deletes the selected round allocation data from the rounds allocation window. • Add All Rounds Button - Automatically adds all rounds defined in the active fire mission fire plan to the Rounds Slots Allocation window.
2	Round Properties	Displays technical and ballistic data for the selected round in the mission list. [refer to section 3.2.5.1.12].
3	Cancel	Closes the window without saving any changes.

3.2.5.1.11 Edit Round Window

(See Figure 3-62 and Table 3-58)

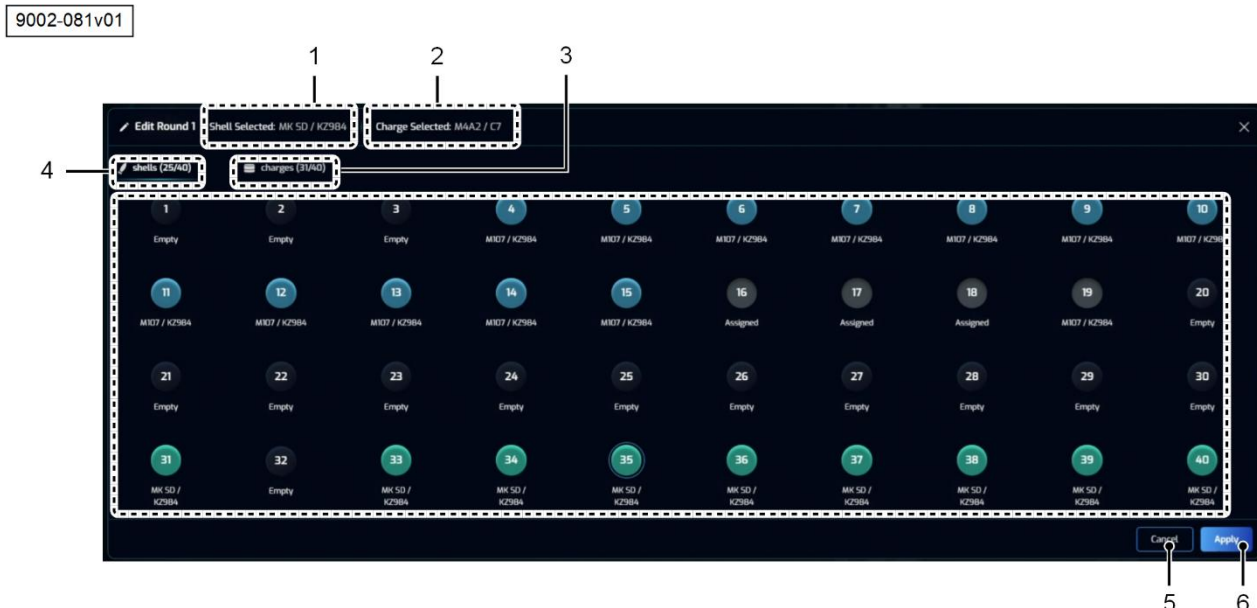


Figure 3-62: New Fire Mission Screen – Play – Rounds Slots Allocation - Edit Round Window

Table 3-58: New Fire Mission Screen – Play – Rounds Slots Allocation - Edit Round Window

No.	Item	Description
1	Shell/Fuze Selected	Displays the currently selected shell type along with its configured fuze.
2	propellant/Charge Selected	Displays the currently selected propellant type and its configured charge.
3	Charge Tab	Displays a slot-based layout of the propellant magazine, showing the specific propellant and charge configuration loaded in each slot.
4	Shell Tab	Displays a slot-based layout of the ammunition magazine, showing the specific shell type and fuze configuration loaded in each slot.
5	Cancel	Closes the window without saving any changes.
6	Apply	Saves all changes made and updates the round allocations.

3.2.5.1.12 New Fire Mission Screen – Play – Rounds Slots Allocation – Round Properties

(See Figure 3-63 and Table 3-59)

9002-062v01



Figure 3-63: New Fire Mission Screen – Play – Rounds Slots Allocation – Round Properties

Table 3-59: New Fire Mission Screen – Play – Rounds Slots Allocation – Round Properties

No.	Item	Description
1	Delete Round Button	Deletes the selected round record from the assigned fire mission list
2	Edit Round Properties Button	Enables editing the round properties.
3	Charge Info	Displays ballistic information regarding the charge: charge type and Zone.
4	Shell Info	Displays ballistic information regarding the shell: shell type, fuze type, firing mode
5	Round Name	displays the name of the round
6	Status Label	Indicates the current operational state of the round (e.g., Standby, Shot, Loaded).

3.2.5.1.13 New Fire Mission Screen – Play – Fire Corrections

(See Figure 3-64 and Table 3-60)

9002-060v01

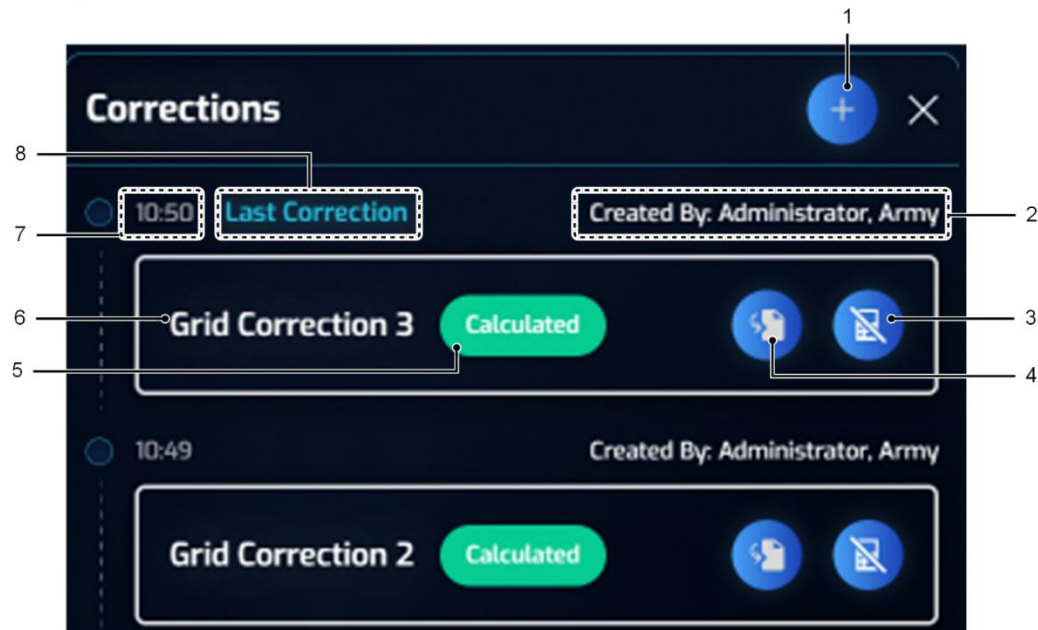


Figure 3-64: New Fire Mission Screen – Play – Fire Corrections

Table 3-60: New Fire Mission Screen – Play – Fire Corrections

No.	Item	Description
1	+ Button	Opens a window for applying a new fire correction
2	Created By Field	Indicates the user that entered the correction.
3	Calculate Fire Correction Button	Calculates the Fire Correction values and computes ballistic adjustments based on the entered correction data. Tapping the calculator icon again removes the fire correction from the ballistic calculations. NOTE: Entered corrections will not be applied to the firing data until they are calculated.
4	View Correction Details Button	Opens the Grid Correction Request Window for editing the fire correction details.
5	Calculation Status	Displays whether the correction has been calculated and integrated into the ballistic solution (e.g., Grid Correction Request).
6	Fire Correction Name	Displays the fire correction unique name.
7	Fire Correction Time Stamp	Displays the exact time the correction was applied.
8	Fire Correction Order	Indicates the most recent correction that was executed.

3.2.6 Settings Screen

(See Figure 3-65 and Table 3-61)

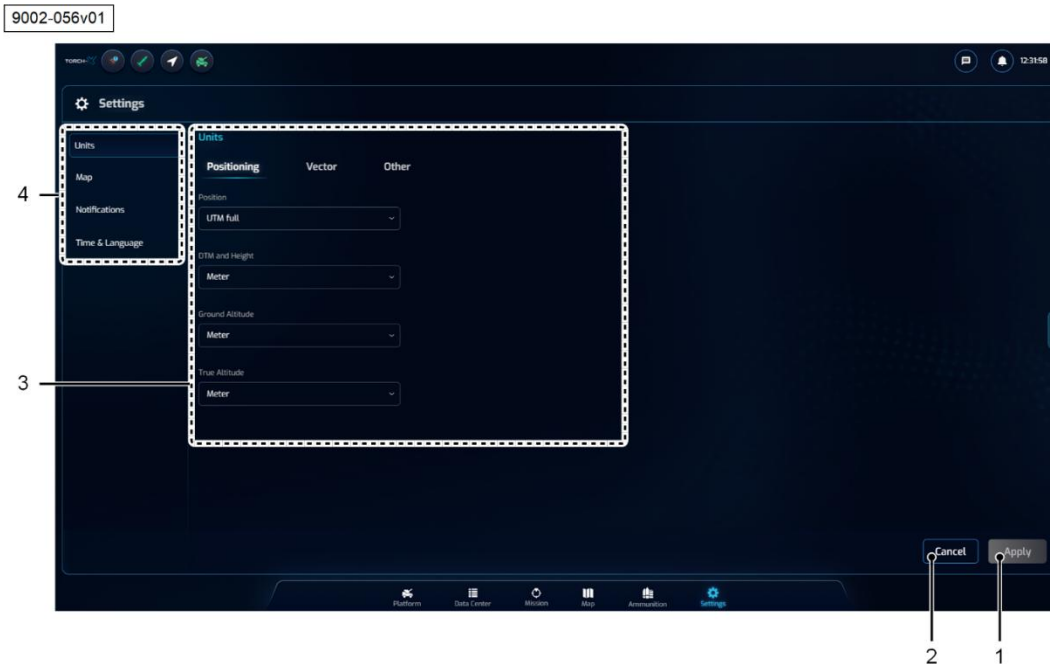


Figure 3-65: Settings Screen

Table 3-61: Settings Screen

No.	Item	Description
1	Apply	Implements any modified settings without closing the configuration window
2	Cancel	Discards all pending changes made and closes the window.
3	Sub-menu Panel	Enables the user to configure operational, technical, and communication parameters. The display updates dynamically based on the user's menu selection.
4	Navigation Panel	Provides access to system settings panels. Tapping on the menus opens a side sub-menu containing dedicated tabs to manage operational, technical, and communication parameters.

3.2.6.1.1 Settings Screen – Units Menu – Positioning Tab

(See Figure 3-66 and Table 3-62)

9002-057v01

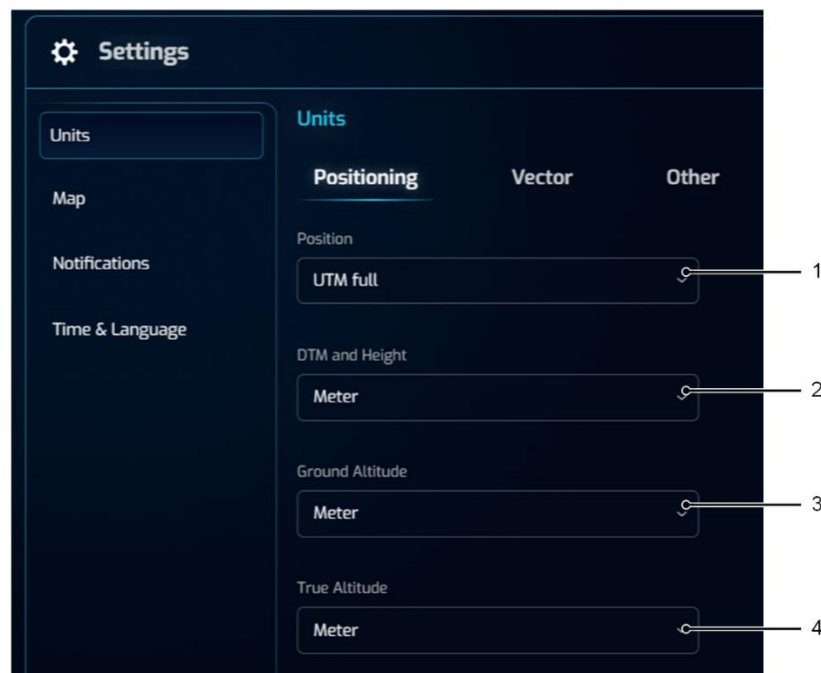


Figure 3-66: Settings Screen – Units Menu – Positioning Tab

Table 3-62: Settings Screen – Units Menu – Positioning Tab

No.	Item	Description
1	Position	Sets the measurement unit for Position values (e.g., MGRS, UTM accurate, or S-42).
2	DTM and Height	Sets the measurement unit for DTM and Hight values (e.g., Meters, Centimeter, or Feet).
3	Ground Altitude	Sets the measurement unit for Ground Altitude values (e.g., Meters, Yard, or Feet).
4	True Altitude	Sets the measurement unit for True Altitude values (e.g., Meters, Centimeter, or Feet).

3.2.6.1.2 Settings Screen – Units Menu – Vector Tab

(See Figure 3-67 and Table 3-63)

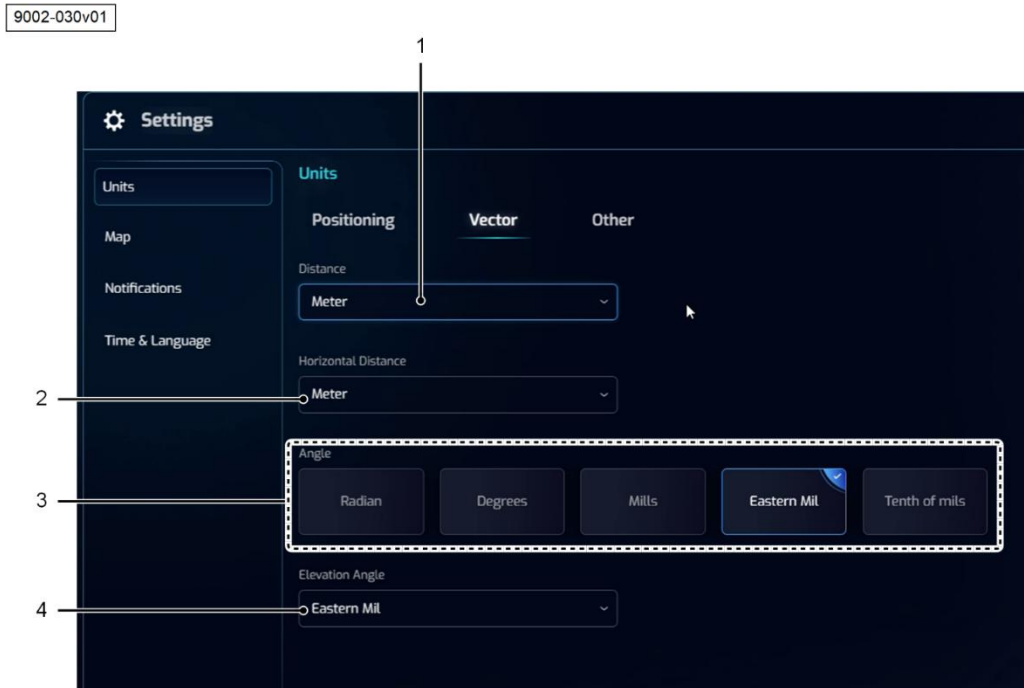


Figure 3-67: Settings Screen – Units Menu – Vector Tab

Table 3-63: Settings Screen – Units Menu – Vector Tab

No.	Item	Description
1	Distance	Sets the measurement unit for Distance values (e.g., Meters, Kilometers, or Nautical Miles).
2	Horizontal Distance	Sets the primary measurement unit for Horizontal Distance values (e.g. Feet, Miles, or Centimeter).
3	Angle	Sets the measurement unit according to user selection: (1) Radian (2) Degrees (3) Mills (4) Eastern Mil (5) Tenth of mils.
4	Elevation Angle	Sets the measurement unit for Elevation Angle values (e.g., Mills, Degrees, or Eastern Mil)

3.2.6.1.3 Settings Screen – Units Menu – Other Tab

(See Figure 3-68 and Table 3-64)

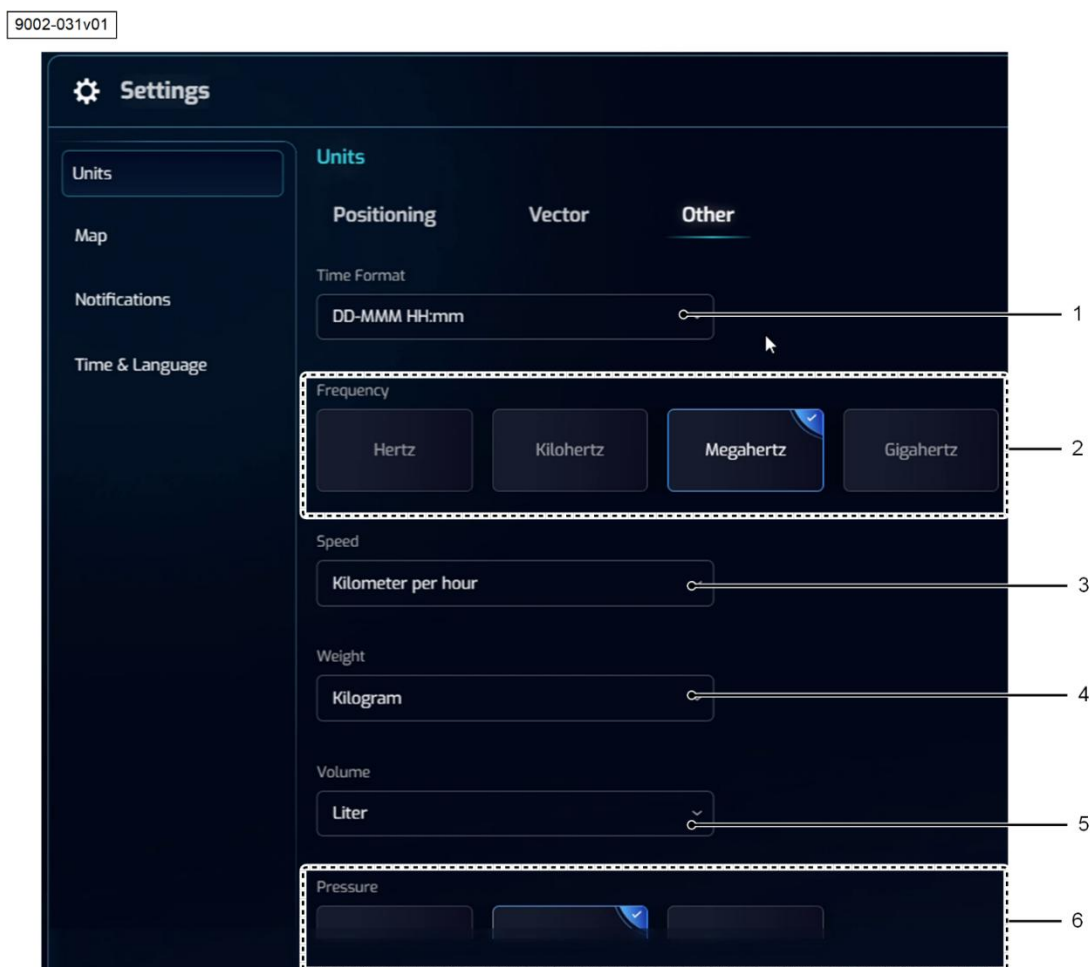


Figure 3-68: Settings Screen – Units Menu – Other Tab

Table 3-64: Settings Screen – Units Menu – Other Tab

No.	Item	Description
1	Time Format	Configures the system-wide time display layout (e.g., DD-MMM HH:mm format, DTG, UTC).
2	Frequency	Sets the measurement unit for the Frequency according to user selection: (1) Hertz (2) KiloHertz (3) Megahertz (4) gigahertz (5).
3	Speed	Sets the measurement unit for the Speed values (e.g. Kilometers per Hour [km/h], Meters per Second [m/s], or Naval Knots).
4	Weight	Sets the measurement unit for the Weight values (e.g. Kilogram, pound, or Tonne).
5	Volume	Sets the measurement unit for the Volume values (e.g. Liter, Pint, or Gallon).
6	Pressure	Sets the measurement unit for the Frequency according to user selection: (1) Pascal (2) Milibar (3) Inches Hg (4).
7	Temparture	Sets the measurement unit for the Frequency according to user selection: (1) Celsius (2) Kelvin (3) Fahrenheit (4) 10Kelvin.

3.2.6.1.4 Settings Screen – Map Menu – Map & Grid Tab

(See Figure 3-69 and Table 3-65)

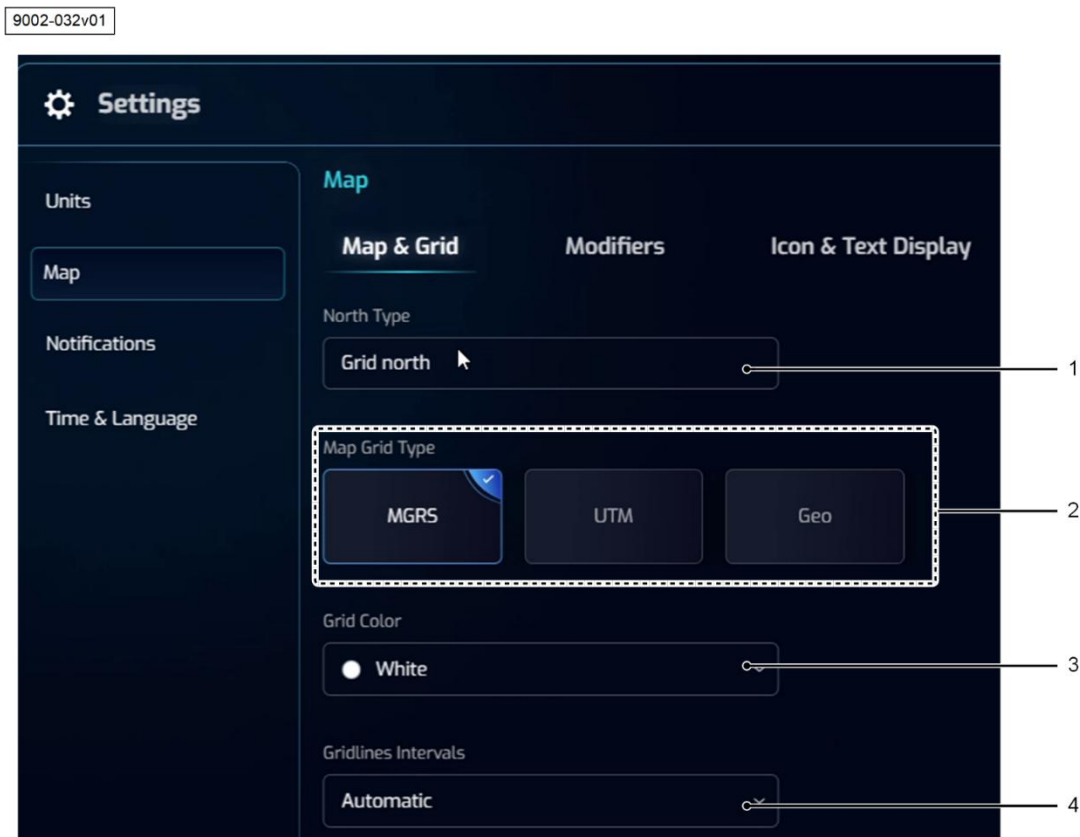


Figure 3-69: Settings Screen – Map Menu – Map & Grid Tab

Table 3-65: Settings Screen – Map Menu – Map & Grid Tab

No.	Item	Description
1	North Type	Aligns the system's directional reference (north) with the vertical grid lines of the active map (e.g., UTM or MGRS) [see Map Grid Type (2)].
2	Map Grid Type	Sets the map grid type according to user selection: (1) MGRS, (2) UTM, (3) Geo.
3	Grid Color	Sets the map grid color according to user selection from various colors.
4	Gridlines Intervals	Configures the spacing of tactical gridlines overlaid on the map display, enabling the user to adjust the visual coordinate resolution based on the current operational scale (e.g., intervals of 100m, 1km, or 10km).

3.2.6.1.5 Settings Screen – Map Menu – Modifiers Tab

(See Figure 3-70 and Table 3-66)

9002-033v01

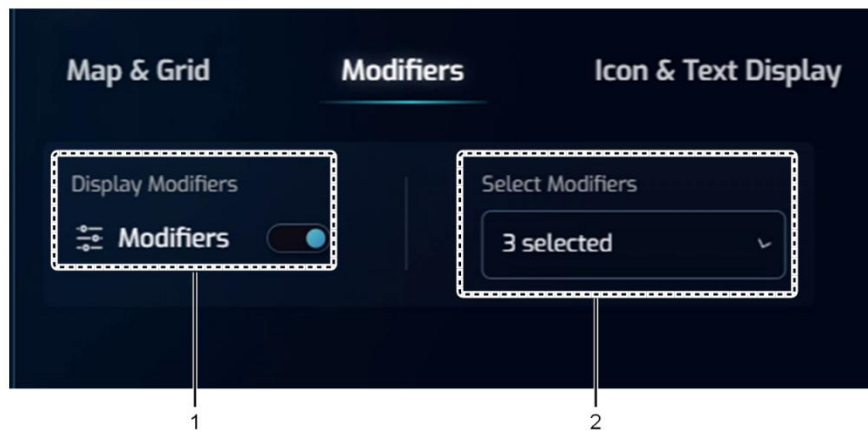


Figure 3-70: Settings Screen – Map Menu – Modifiers Tab

Table 3-66: Settings Screen – Map Menu – Modifiers Tab

No.	Item	Description
1	Display Modifiers Toggle Button	Enables or disables the visibility of all selected text modifiers across the interface.
2	Select Modifiers Menu	Displays a summary of active filters (e.g., "3 selected") and expands into a multi-select panel containing the following: • Search Bar: enables the user to type and quickly filter specific options within the list. • Select All / Clear All: Quick-action buttons to instantly check or uncheck all available options in the list. • Options Checkboxes: Individual category toggles.

3.2.6.1.6 Settings Screen – Map Menu – Icon & Text Display Tab

(See Figure 3-71 and Table 3-67)

9002-034v01

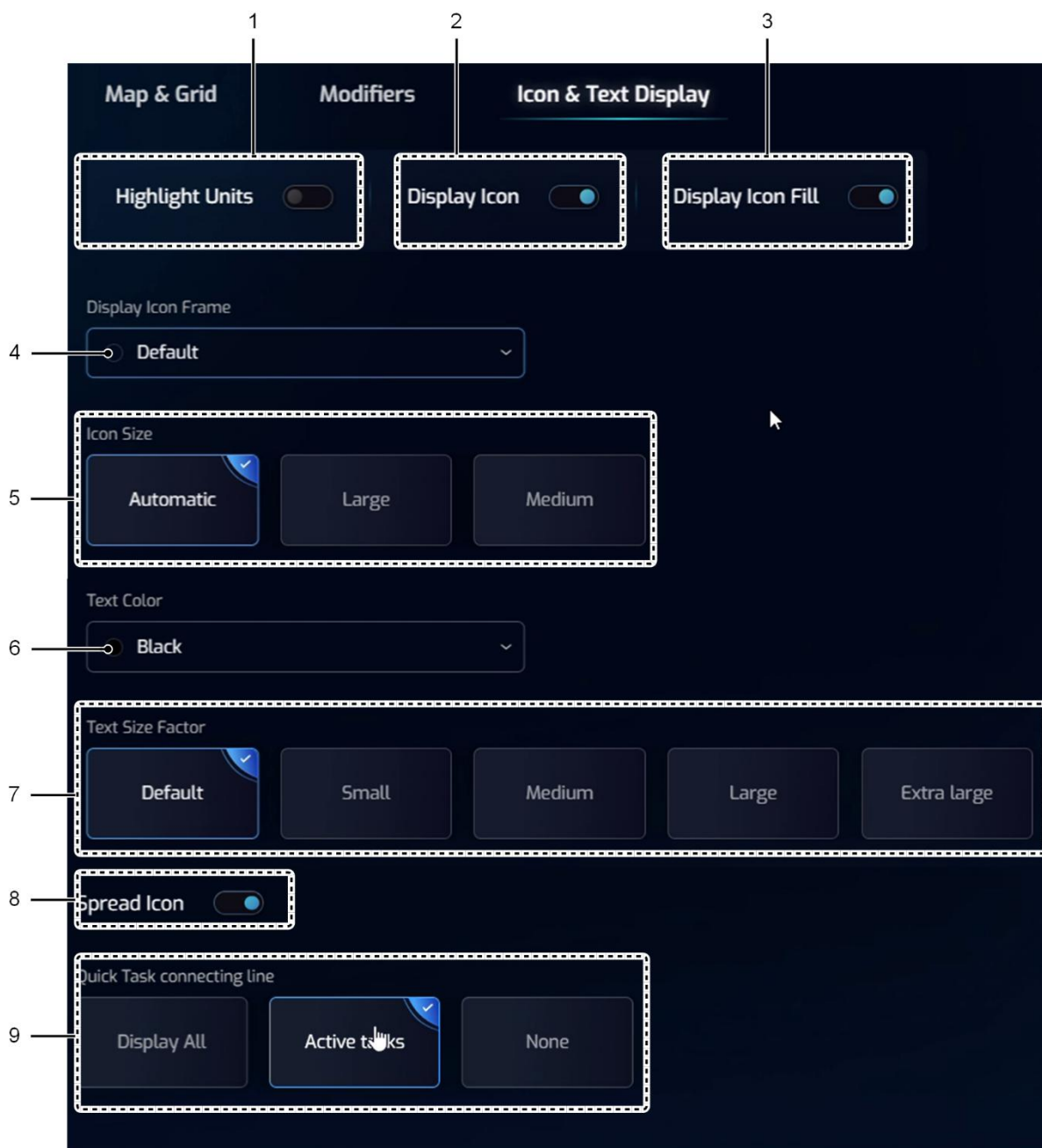


Figure 3-71: Settings Screen – Map Menu – Icon & Text Display Tab

Table 3-67: Settings Screen – Map Menu – Icon & Text Display Tab

No.	Item	Description
1	Highlight Units Toggle Button	Enables or disables unit highlighting in the map.
2	Display Icon Toggle Button	Enables or disables map icon visibility.
3	Display Icon Fill Toggle Button	Enables or disables map icon color fill.
4	Display Icon Frame Menu	Enables the user to configure the Icon Frame in the map: (1) Default (2) Black (3) White (4) None.
5	Icon Size	Enables the user to configure the icon size in the map from the following options: (1) Automatic (2) Large (3) Medium
6	Text Color	Enables the user to configure the text color in the map from various colors.
7	Text Size Factor	Adjusts the global font scale in the map interface to adjust readability based on operational conditions: (1) Small (2) Medium (3) Large (4) Extra Large
8	Spread Icon	Spreads out overlapping map icons to ensure clarity and distinct readability in high-density areas.
9	Quick Task Connecting Line	Enables the user to configure the task connecting line in the map: (1) Display All (2) Active tasks (3) None.

3.2.6.1.7 Settings Screen – Notifications Menu –Visual Tab

(See Figure 3-72 and Table 3-68)

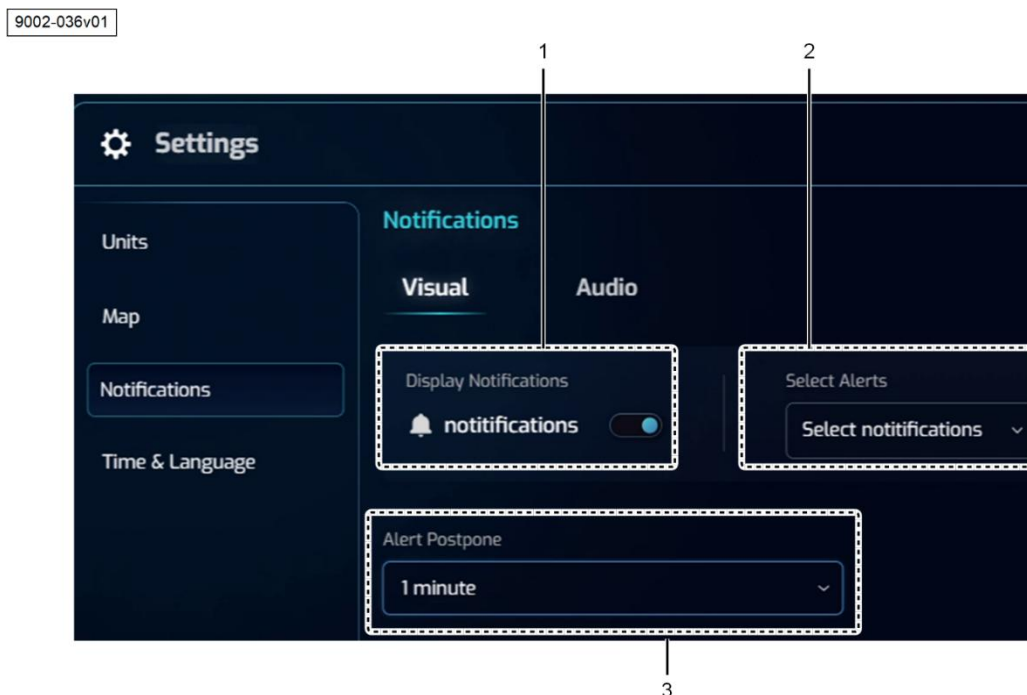


Figure 3-72: Settings Screen – Notifications Menu –Visual Tab

Table 3-68: Settings Screen – Notifications Menu –Visual Tab

No.	Item	Description
1	Display Notifications Toggle Button	Toggles the visibility of real-time pop-up notifications on the main interface.
2	Select Alerts	Displays a summary of active filters (e.g., "3 selected") and expands into a multi-select panel containing the following: Search Bar: enables the user to type and quickly filter specific options within the list. Select All / Clear All: Quick-action buttons to instantly check or uncheck all available options in the list. Options Checkboxes: Individual category toggles.
3	Alert Postpone	Enables the user to postpones the active system alert for a specific duration (e.g. 1 minute, 10 minutes, or 15 minutes).

3.2.6.1.8 Settings Screen – Notifications Menu –Audio

(See Figure 3-73 and Table 3-69)

9002-037v01

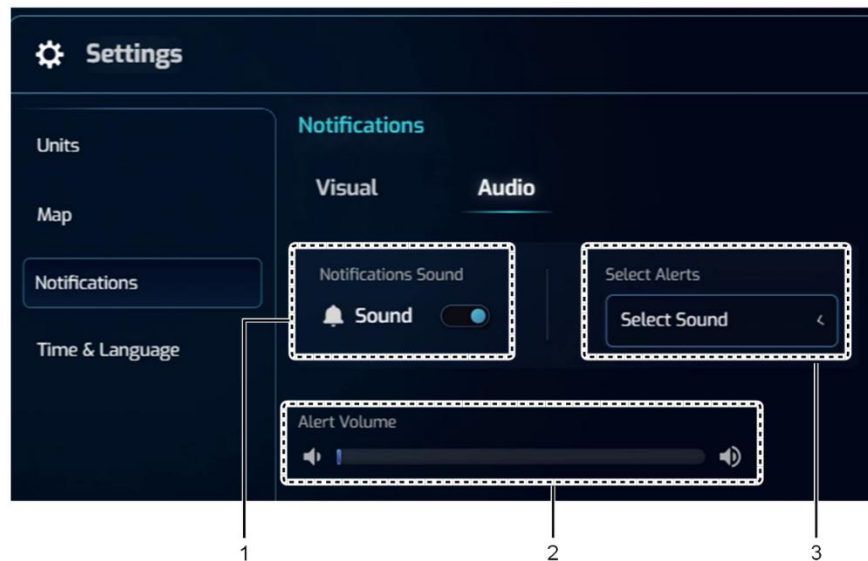


Figure 3-73: Settings Screen – Notifications Menu –Audio

Table 3-69: Settings Screen – Notifications Menu –Audio

No.	Item	Description
1	Notifications Sound Toggle Button	Toggles the audible alerts for incoming system notifications.
3	Alert Volume	Enables the user to adjust the master volume level for tactical alerts.
2	Select Alerts	Displays a summary of active filters (e.g., "3 selected") and expands into a multi-select panel containing the following: • Search Bar: enables the user to type and quickly filter specific options within the list. • Select All / Clear All: Quick-action buttons to instantly check or uncheck all available options in the list. • Options Checkboxes: Individual category toggles.

3.2.6.1.9 Settings Screen – Time & Language Menu –Clocks

(See Figure 3-74 and Table 3-70)

9002-060v01

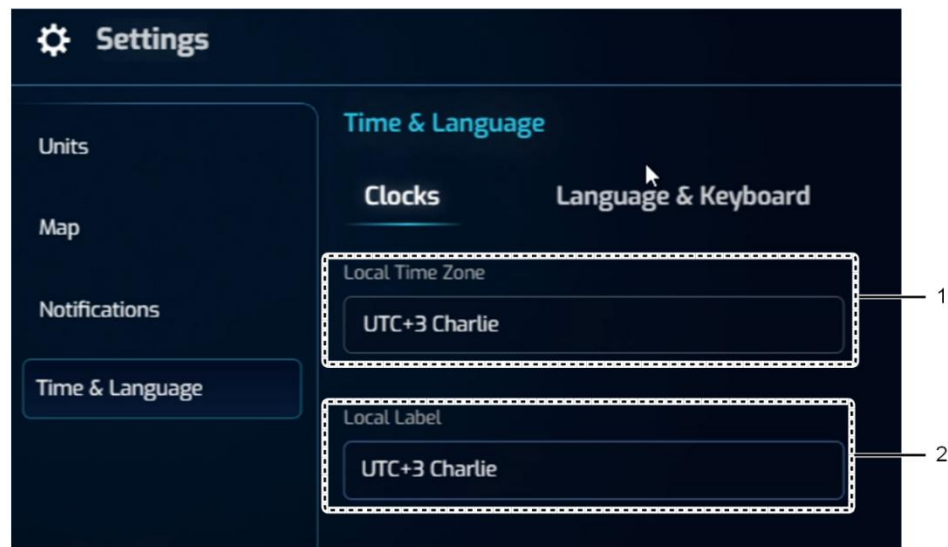


Figure 3-74: Settings Screen – Time & Language Menu –Clocks

Table 3-70: Settings Screen – Time & Language Menu –Clocks

No.	Item	Description
1	Local Time Zone	Sets the operational time zone for the system using standard UTC and Military Time Zone Alpha suffixes (e.g., Charlie for UTC+3).
2	Local Label	Defines the custom text label applied to timestamps across the application interface.

3.2.6.1.10 Settings Screen – Time & Language Menu –Language & Keyboard

(See Figure 3-75 and Table 3-71)

9002-038v01

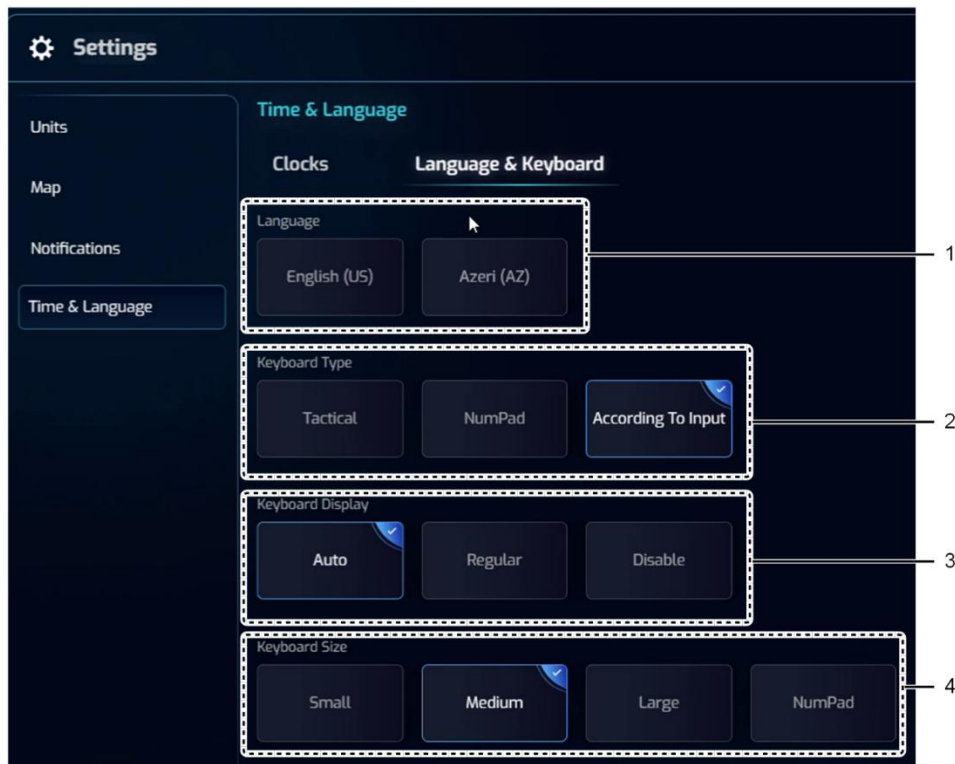


Figure 3-75: Settings Screen – Time & Language Menu –Language & Keyboard

Table 3-71: Settings Screen – Time & Language Menu –Language & Keyboard

No.	Item	Description
1	Language	Enables the user to switch the system interface language between English and Azeri (AZ) . <u>Note</u> : Changing the language triggers an automatic application restart to apply the chosen language.
2	Keyboard Type	Defines the layout profile based on data entry needs: Tactical: Customized layout optimized for rapid artillery data entry. Numeric (Numpad): Dedicated number pad for entering coordinates, charge levels, and mission data. According to Input: Context-aware layout that dynamically switches based on the selected text field.
3	Keyboard Display	Controls the virtual keyboard's activation: Auto: Automatically displays the keyboard when an input field is selected. Manual (On): Keeps the keyboard permanently accessible or visible. Off: Disables the virtual keyboard.
4	Keyboard Size	Scales the keyboard size to maintain usability: (1) Small (2) Medium (3) Large (4) NumPad

3.2.7 Ammunition Management Screen

(See Figure 3-76 and Table 3-72)

This screen provides a real-time visual summary of the DITA howitzer's onboard combat load.

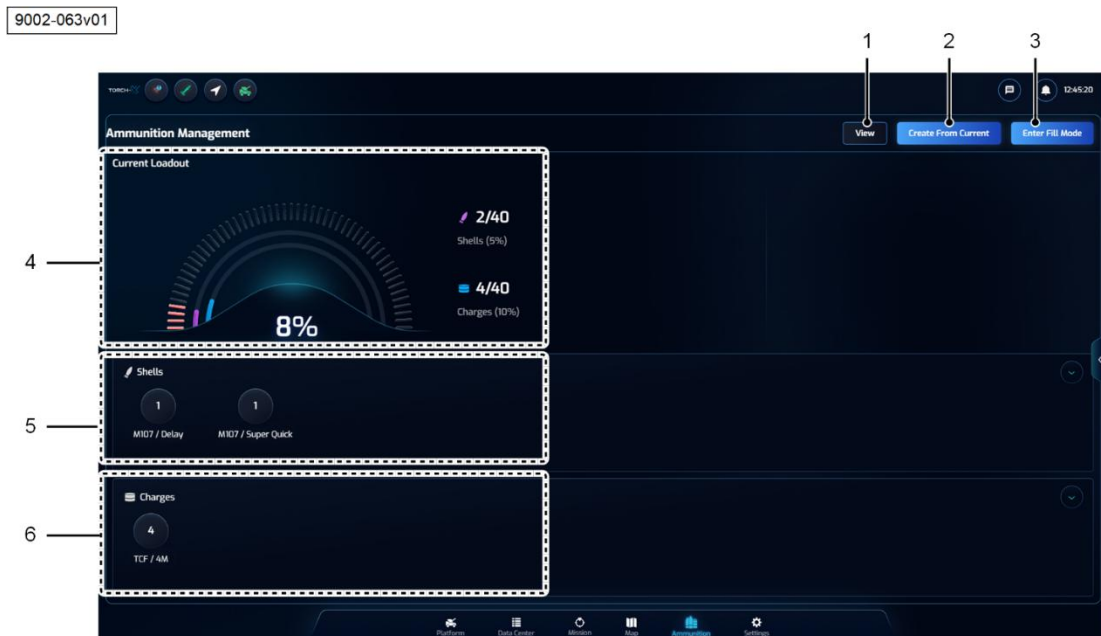


Figure 3-76: Ammunition Management Screen

Table 3-72: Ammunition Management Screen

No.	Item	Description
1	View Button	Opens the View Loadout screen. This screen enables the user to view the active combat loadout and select specific ammunition slots to view inventory details [refer to section 3.2.7.1.1].
2	Create From Current Button	Opens the View Current Loadout screen [refer to section 3.2.7.1.2]. This screen enables the user to <u>manually</u> define and update the ammunition currently loaded in the magazine for each slot. This function is primarily used when automatic ammunition reporting from the Filling Boxes subsystem is unavailable (e.g., due to a communication failure), allowing the crew to manually synchronize the system inventory with the actual magazine loadout.
3	Enter Fill Mode Button	Opens the Fill Mode Monitor screen [refer to section 3.2.7.1.3].
4	Current Loadout Section	Displays the exact remaining quantities of Shells and Charges. It includes a dynamic graphical Infographic dashboard showing remaining inventory as a percentage (%) of maximum capacity.
5	Shells Section	Displays the Shells inventory, categorized by type and Method of burst settings (e.g., M107/Delay, or M107/ Super Quick):
6	Charges Section	Displays the propellant inventory, categorized by type and charge (e.g., TCF/4M):

3.2.7.1.1 View Loadout Screen

(See Figure 3-77 and Table 3-73)

9002-064v01

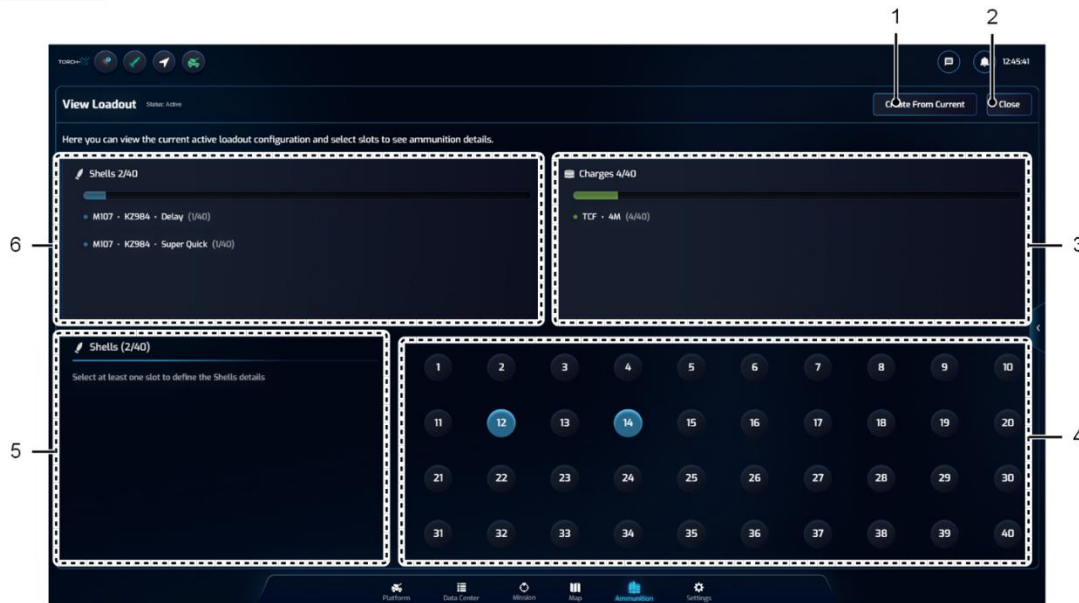
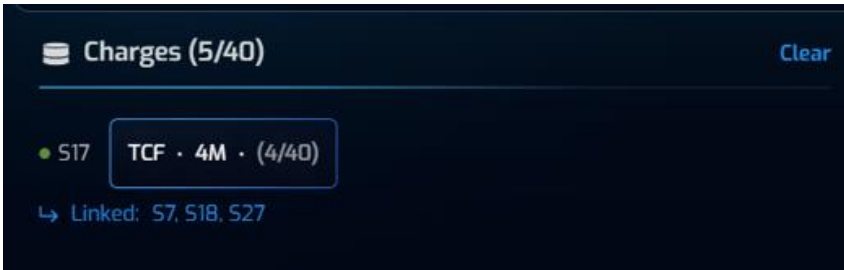


Figure 3-77: View Loadout Screen

Table 3-73: View Loadout Screen

No.	Item	Description
1	Create From Current Button	Opens the View Current Loadout screen [refer to section 3.2.7.1.2].
2	Close Button	Closes the View Loadout screen and returns to Ammunition Management screen.
3	Charges 0/40 Section	Provides a categorized breakdown of the charges inventory, displaying individual combination of Propellant · Charge alongside their exact current quantities.
4	Charges/Shells Magazine Slots	A visual representation of the DITA automatic magazine layout, consisting of 40 individual, numbered cell indicators. Each numbered cell corresponds to a specific physical storage slot (1 to 40) within the automated shell magazine or charge compartment. The magazine layout uses Color-Coding: <u>Purple Slots</u>: Indicate magazine cells loaded with M4A2 Type Propellant Charges. <u>Green Slots</u>: Indicate magazine cells loaded with TCF Type Propellant Charges.

No.	Item	Description
5	Dynamic Details Window	After tapping a slot in the charges magazine slots layout, enables the user to view specific parameters (slot ID, propellant type, charge type, quantity) and details for either shells or charges, depending on the active inventory tab (Shells/charges).  The Clear button clears all assigned charges/shells parameters and data from the selected magazine slots, resetting them to an Empty status. Linked? When the Shells tab is active, the user can view specific parameters (slot ID, shell weight, fuze type, quantity)  Linked?
6	Shells Section 0/40	Provides a categorized breakdown of the shell inventory, displaying individual combination of Shell · Fuze · Method of burst alongside their exact current quantities.

3.2.7.1.2 View Current Loadout Screen

(See Figure 3-78, Figure 3-79 and Table 3-75, Table 3-74)

9002-065v01

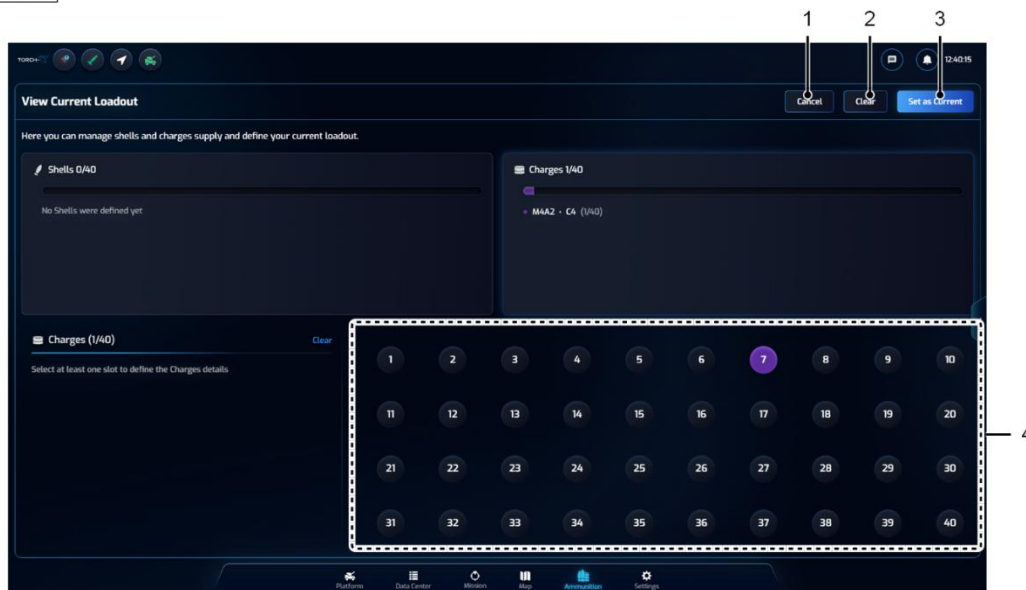
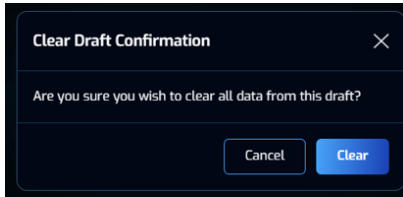


Figure 3-78: View Current Loadout Screen

Table 3-74: View Current Loadout Screen

No.	Item	Description
1	Cancel	Closes the View Current Loadout screen without saving any changes.
2	Clear	Resets all input fields and selection criteria within the window to their default/blank states. 
3	Set as Current	Designates the selected ammunition configuration as the active loadout for the next scheduled fire mission.
4	Charges Magazine Slots	A visual representation of the DITA automatic magazine layout, consisting of 40 individual, numbered cell indicators. Each numbered cell corresponds to a specific physical storage slot (1 to 40) within the automated shell magazine or charge compartment. The magazine layout uses Color-Coding: <u>Purple Slots</u> : Indicate magazine cells loaded with M4A2 Type Propellant Charges. <u>Green Slots</u> : Indicate magazine cells loaded with TCF Type Propellant Charges.

No.	Item	Description
		When the user selects a specific slot, the system opens the Edit Slot Window, [refer to Figure 3-79] enabling the user to: • Modify the specific type of charge/propellant assigned to the selected slot. • Redefine or update the charge/propellant configuration as needed.

9002-066v01

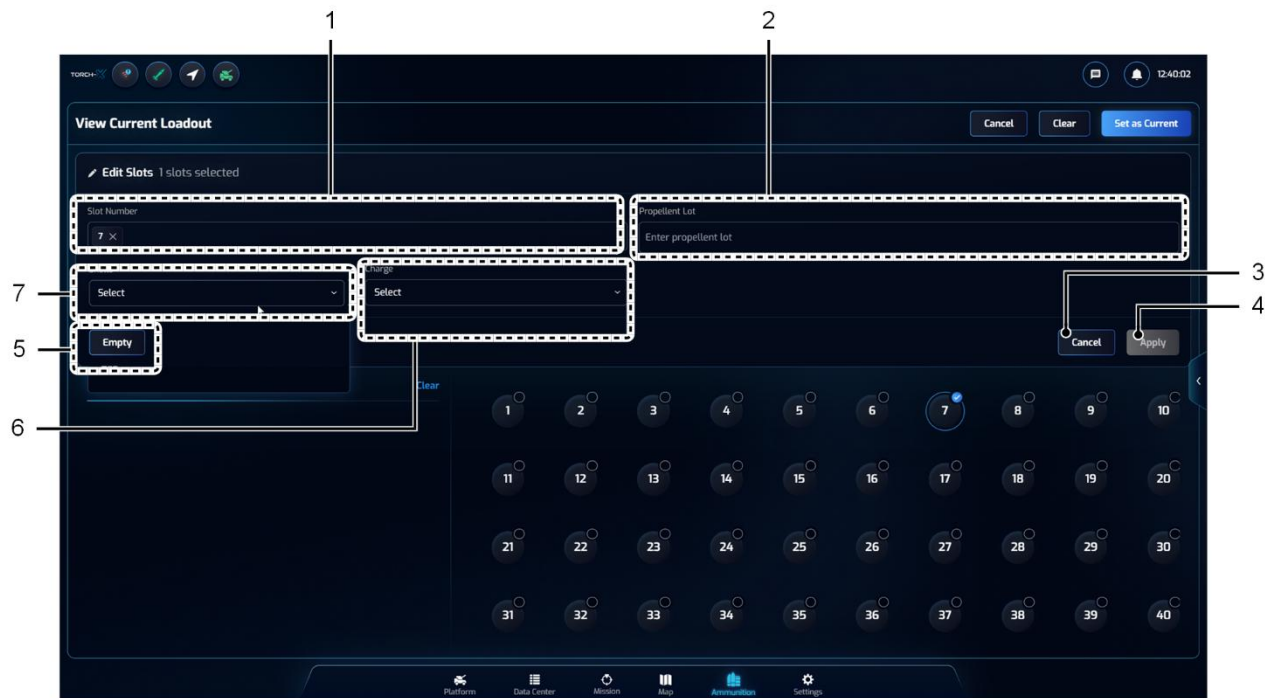


Figure 3-79: View Current Loadout Screen - Charges View - Edit Slots Window

Table 3-75: View Current Loadout Screen - Charges View - Edit Slots Window

No.	Item	Description
1	Slot Number Field	Displays the specific magazine slot numbers selected by the user for parameter configuration.
2	Propellant Lot Field	Alphanumeric text input field for entering the manufacturer batch/lot number of the charges (critical for precise muzzle velocity variance calculations).
3	Cancel	Closes the Edit Slots Window without saving any changes.
4	Apply	Saves and applies the configured ammunition parameters to the selected magazine slots.
5	Empty Button	Removes all ammunition data from the container. Clear Confirmation × Are you sure you want to clear all charges slots? This action will remove all ammunition data from the container. Cancel Clear
6	Charges Dropdown Menu	Used to select the Charges type: M1-M7 or C1-C7.

No.	Item	Description
7	Propellant Type Dropdown Menu	Used to select the Propellant type: M4A2 or TCF.

9002-067v01

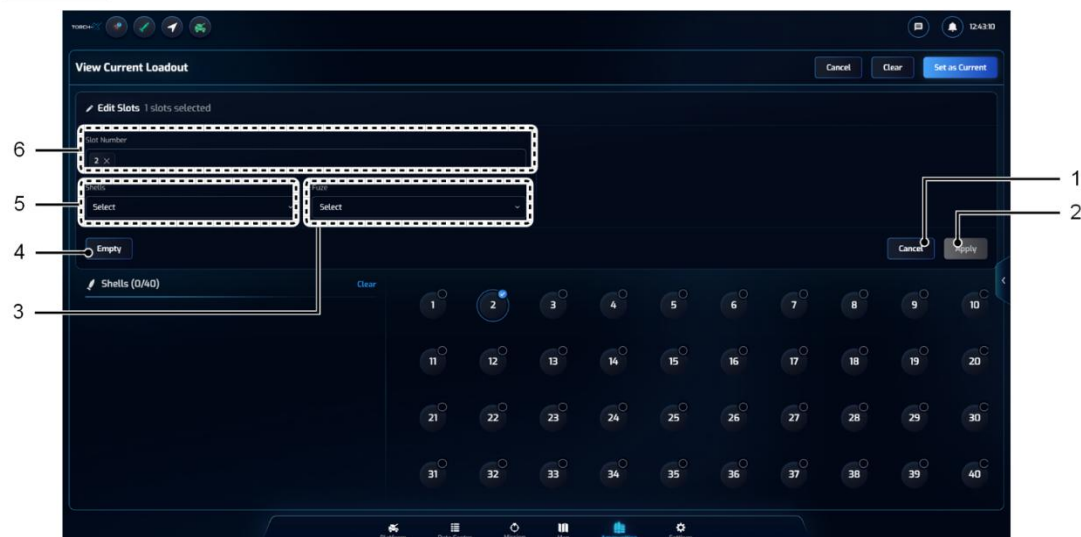


Figure 3-80: View Current Loadout Screen - Shells View - Edit Slots Window

Table 3-76: View Current Loadout Screen - Shells View - Edit Slots Window

No.	Item	Description
1	Cancel	Closes the Edit Slots Window without saving any changes.
2	Apply	Saves and applies the configured ammunition parameters to the selected magazine slots.
3	Fuze Dropdown Menu	Used to select the Fuze type: KZ984.
4	Empty Button	Removes all ammunition data from the container.
5	Shells Dropdown Menu	Used to select the Shell type: M107, MK SD or MK DV.
6	Slot Number Field	Displays the specific magazine slot numbers selected by the user for parameter configuration.

3.2.7.1.3 Fill Mode Monitor Screen

(See Figure 3-81 and Table 3-77)

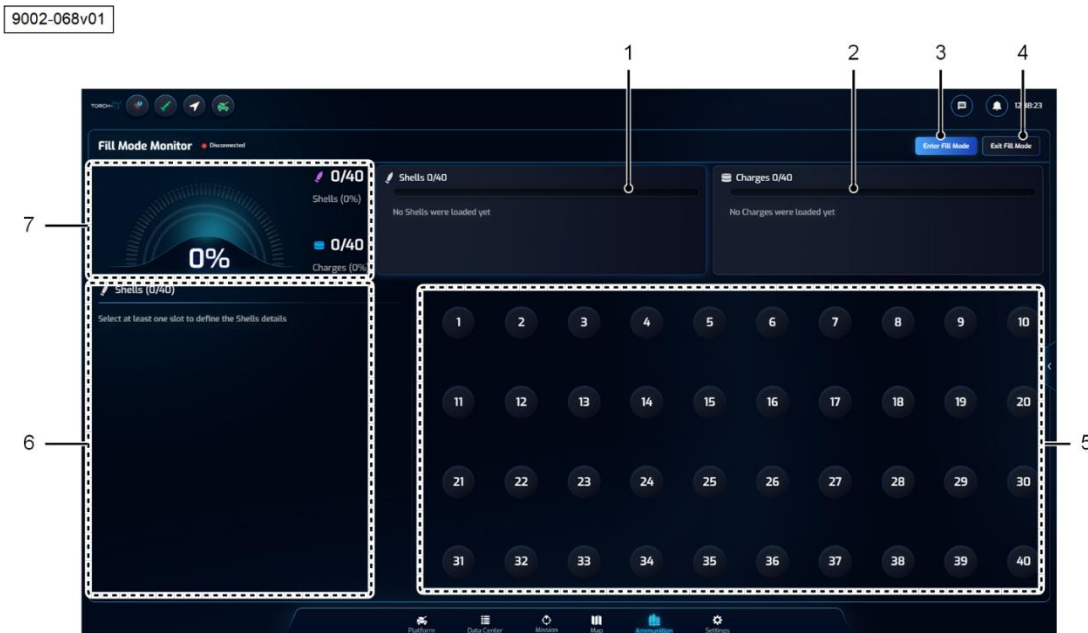


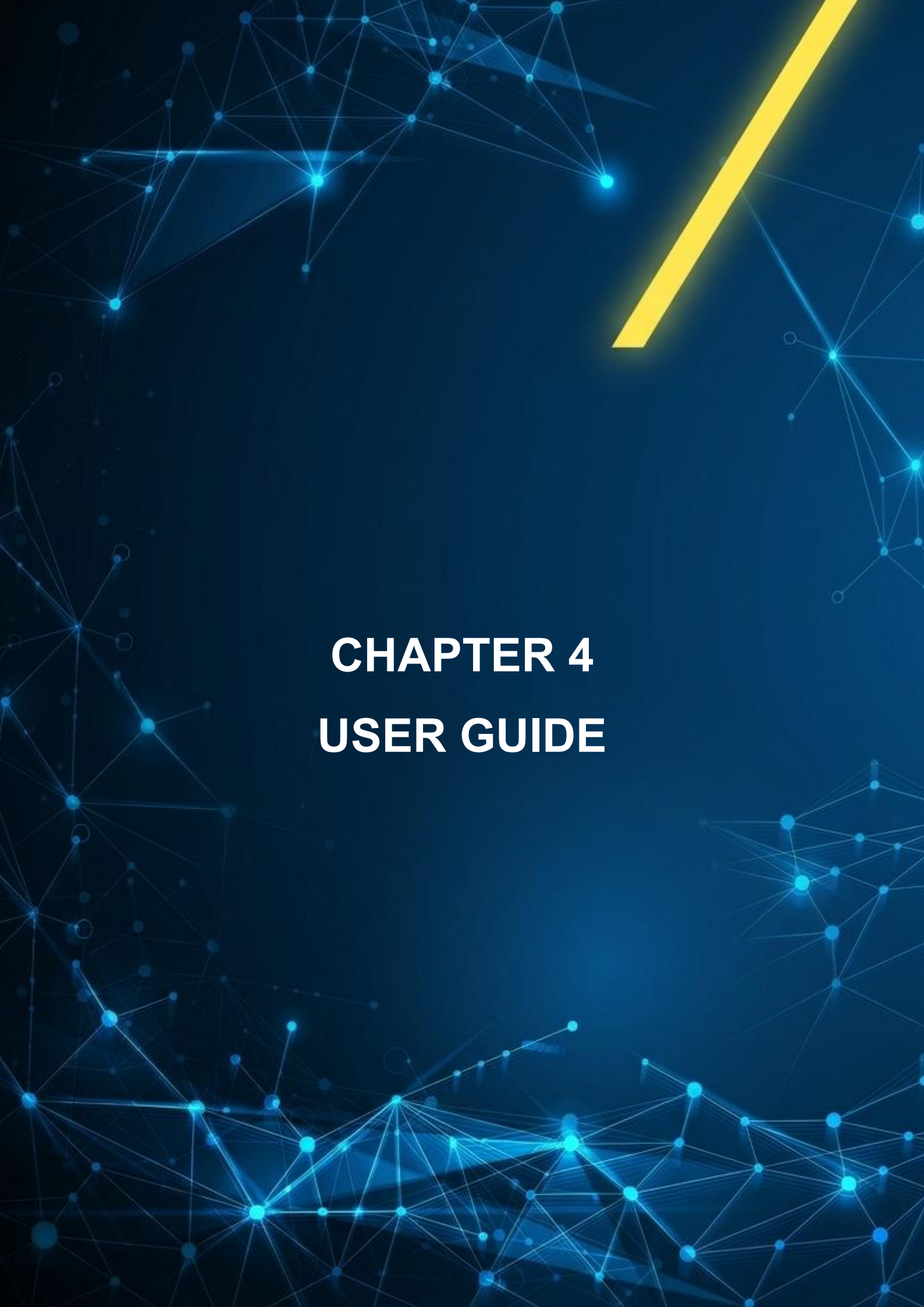
Figure 3-81: Fill Mode Monitor Screen

Table 3-77: Fill Mode Monitor Screen

No.	Item	Description
1	Shells Section	Displays the Shells inventory, categorized by type and Method of burst settings (e.g., M107/Delay, or M107/ Super Quick):
2	Charges Section	Displays the propellant inventory, categorized by type and charge (e.g., TCF/4M):
3	Approve loudout	all Sends the current magazine loading status to the Commander for verification and approval. Upon submission, the Commander Approval screen is displayed on the commander's station, presenting the magazine layout for review and approval.
4	Exit Fill Mode	Returns the user to the Ammunition Management screen.
5	Charges Magazine Slots	A visual representation of the DITA automatic magazine layout, consisting of 40 individual, numbered cell indicators. Each numbered cell corresponds to a specific physical storage slot (1 to 40) within the automated shell magazine or charge compartment. The magazine layout uses Color-Coding: <u>Purple Slots</u> : Indicate magazine cells loaded with M4A2 Type Propellant Charges. <u>Green Slots</u> : Indicate magazine cells loaded with TCF Type Propellant Charges.
6	Shells/Charges Details	Enables the user to define the shells/charges details (depending the selected slot).

No.	Item	Description
7	Current Loadout Section	Displays the exact remaining quantities of Shells and Charges. It includes a dynamic graphical Infographic dashboard showing remaining inventory as a percentage (%) of maximum capacity.

עמוד זה הושאר ריק במתכוון



CHAPTER 4

USER GUIDE

4.1 SCOPE

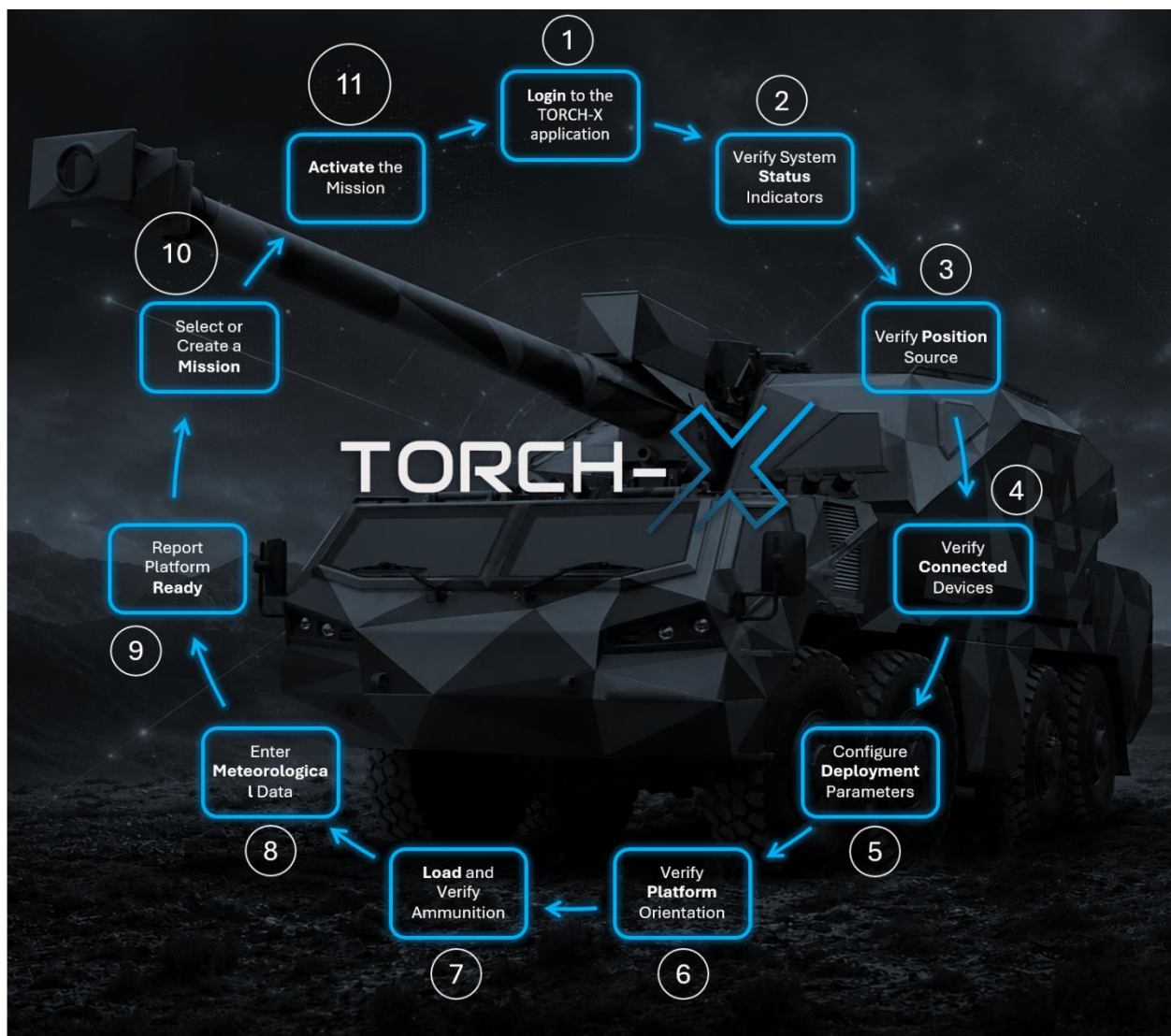
This chapter provides comprehensive operational guidance for the effective use of the Torch-X™ Artillery system on the 155 mm DITA platform. It describes the step-by-step procedures required to perform system operations, from initial setup through execution and management of fire missions.

4.2 USER GUIDE

4.2.1 Preconditions for Fire Mission Execution

- 1) **Beacon** status is Connected and Online.
- 2) All required **devices** are Connected and Operational.
- 3) A valid Self-Position is available (INS/GPS or Manual Position).
- 4) **SET POSITION** has been executed successfully.
- 5) Platform operational status is **READY**.
- 6) A Fire Mission has been created.
- 7) A valid **ballistic solution** has been calculated.
- 8) The mission has been **activated** (Play Mission).

Perform the following procedure after system startup and before receiving or executing a fire mission.



4.2.2 User Login

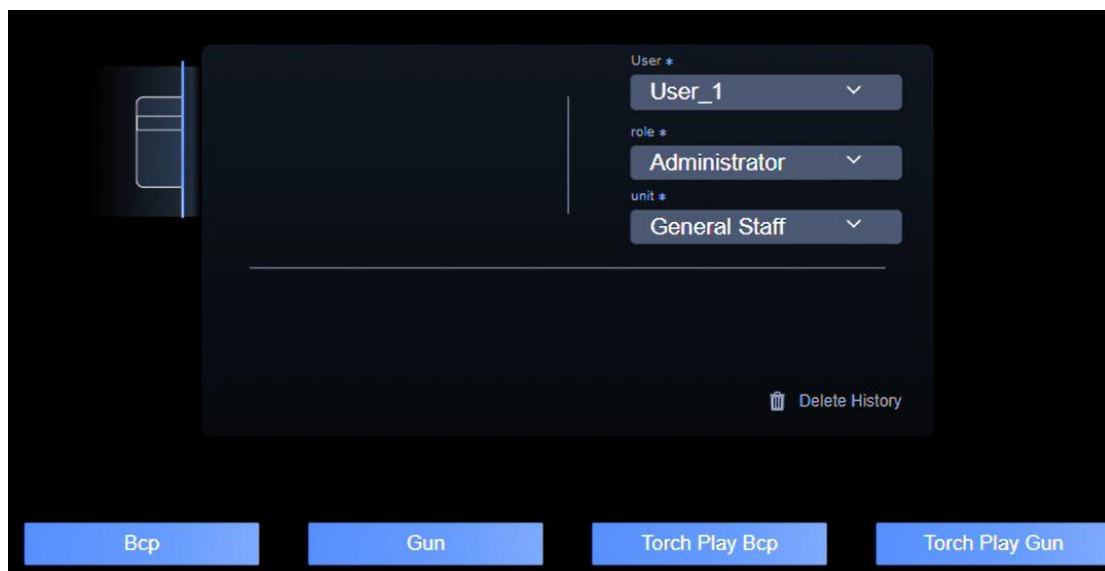
By default, the TORCH-X application is configured to **start automatically** during system startup and perform an automatic user login. Under normal operating conditions, no user action is required and the application opens directly to the operational interface.

However, if the application does not start automatically due to a system restart, configuration issue, or any other operational reason, the user can manually launch the application and perform the login procedure described in this section.

- a. On the **ETC** main screen, tap the **TORCH-X** icon.

Result:

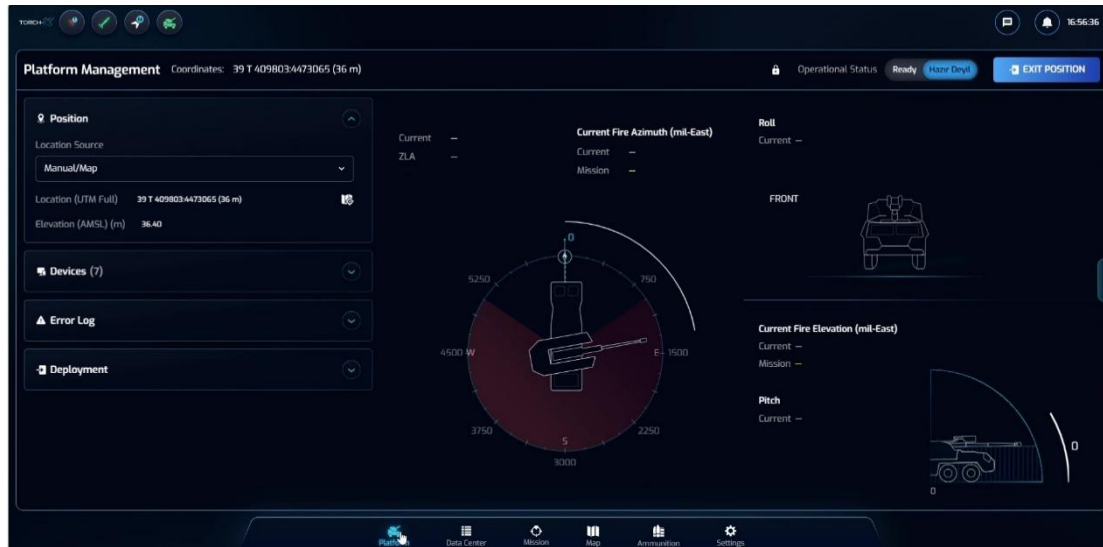
The **LOGIN** screen appears.



- b. Tap the **Torch Play Gun** button.

Result:

After the system loads, the **PLATFORM** screen (the main application screen) appears.



4.2.3 Device Status Check

- On the **Platform** screen, verify that the following indicators are green:

- Connectivity Indicator ()
- Beacon Indicator ()
- Navigation Indicator ()

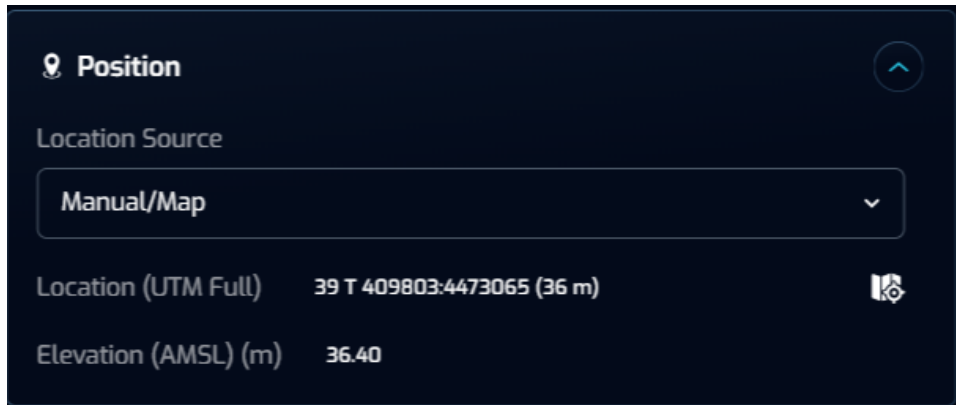
NOTE

The Platform Indicator () will remain inactive (red) until the firing position is set.

- On the **Platform** screen, tap the **Position** menu.

Result:

Position Menu opens:



The screenshot shows a dark-themed interface for the 'Position' section. At the top, there is a location pin icon and the title 'Position'. Below this is a 'Location Source' dropdown menu currently set to 'Manual/Map'. Underneath, two data fields are displayed: 'Location (UTM Full)' with the value '39 T 409803:4473065 (36 m)' and 'Elevation (AMSL) (m)' with the value '36.40'. A small icon of a map with a location pin is visible to the right of the location data.

- c. In the **Location Source** field, select **AUTO**.

Result:

The location source is configured to automatic update mode.

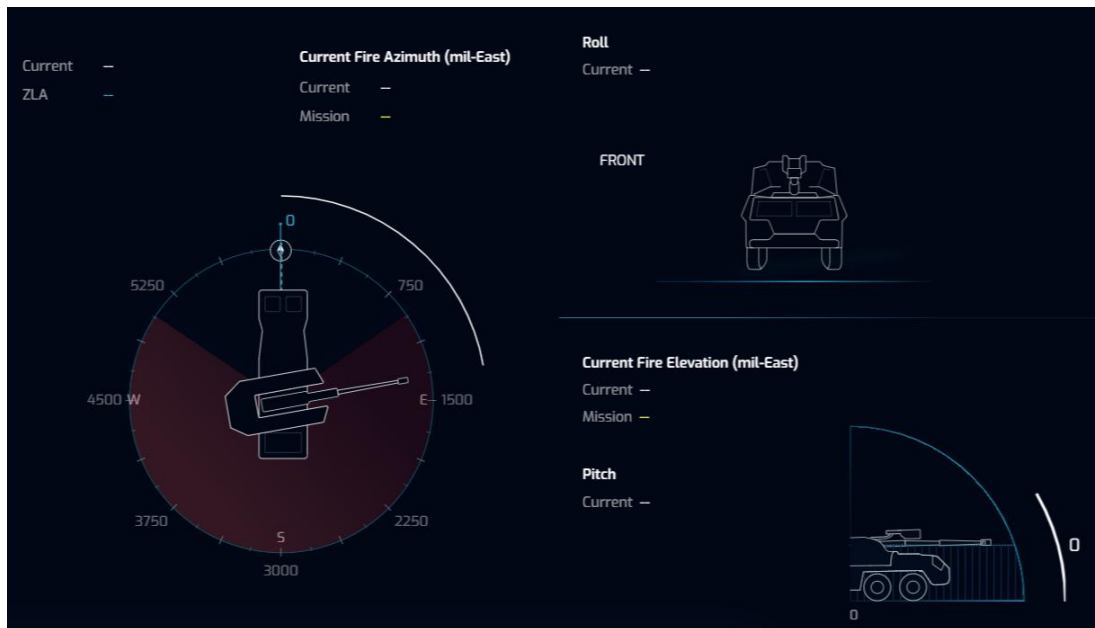
NOTE

- If GPS or INS positioning data is **unavailable** or invalid, the platform location can be entered manually by selecting a position directly on the tactical map.
- After selecting the position, verify that the platform location is updated and a corresponding entity/location is created on the map.

- d. In the **Platform** screen header, verify that the location and altitude data are displayed.

Platform Management Coordinates: 39 T 409803:4473065 (36 m)

- e. Verify that the **azimuth**, roll, **pitch** and **elevation** widgets are displayed correctly and valid and within the operational limits specified for the DITA platform.



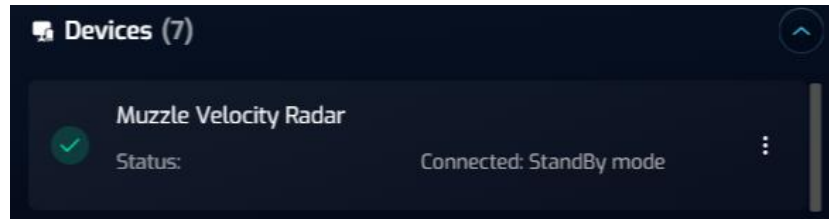
- f. On the **Platform** screen, tap the **Devices** menu.

Result:

Devices Menu Opens:



- g. Verify that all **seven (7)** sub-systems listed display a **Connected** status accompanied by a green connectivity icon ().



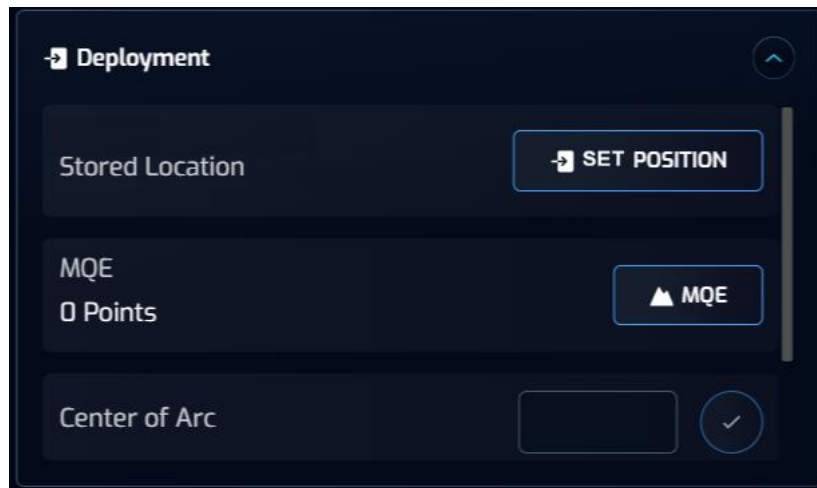
- h. In the **Platform** screen Header, verify that the Travel Lock icon () indicates an "Open" (Unlocked) status.

4.2.4 Deployment Tab Operation

- a. In the **Platform** screen header, tap the **Deployment** menu.

Result:

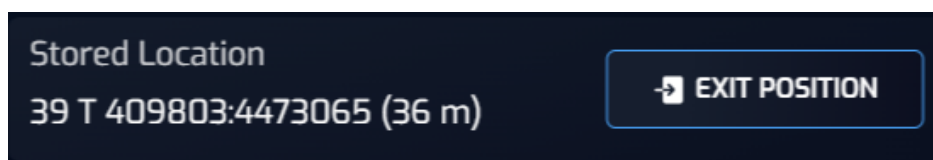
The **Deployment** menu opens.



- b. Tap the **SET POSITION** button.

Result:

- 1) The system acquires the platform's self-position data (Manual, INS, or GPS) to be used during ballistic calculations and firing solutions.
- 2) The **Stored Location** field displays the synchronized coordinates and altitude of the platform.



- 3) The **Platform** Status Indicator () turns green to confirm that the platform position has been successfully set and initialized in the system.

CAUTION

- The commander must verify the reported platform location after executing SET POSITION. Verification should be performed by comparing the displayed coordinates with known terrain features, landmarks, map data, and the **actual** location of the platform.

The commander shall not rely solely on data provided by the GPS, INS, or other navigation sensors.

- Refer to Section 3.2.2.1.1 for additional details.

- c. In the **Deployment** menu, tap the **MQE** button.

Result:

The MQE window opens.

no.	Azimuth	Elevation
No MQE points recorded		

- d. To perform a manual update instead of the default **Aim** mode (data received from the INS), tap the **Manual** option.

Result:

The **Azimuth** and **Elevation** data entry fields become editable.

- e. In the **Azimuth** and **Elevation** fields, enter the desired values (0-6400), and then tap **Add**.

Result:

The manual values are added to the MQE Points table.

- f. Tap **Close** to complete the operation and close the window.

Result:

The MQE window closes, and settings are saved.

- g. Verify that the **Center of Arc** field displays a valid value (received from the **BCP**). If not, enter the value, and then tap the Checkmark

icon ().

- h. In the **Charge Temp. (°C)** field, enter the ammunition temperature in degrees Celsius, and then tap the Checkmark icon ().

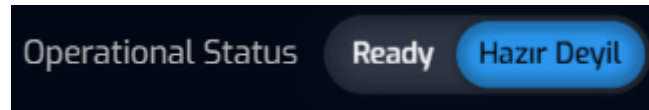
Result:

The temperature value is saved, and the corresponding update date and time appear next to the field.

NOTE

All values **must reflect** the actual firing orientation and operational conditions, and not arbitrary or placeholder values.

- i. When all parameters are verified and complete, in the **Platform** screen Header, tap the Ready button ( ).



Result:

The system's Operational Status changes to **Ready**, indicating that the DITA is fully prepared to receive fire missions.

4.2.5 Filling Ammunition

The ammunition inventory can be set using one of two methods, depending on the availability of communication with the Filling Boxes subsystem.

1) Automatic Mode (Filling Boxes **Connected**)

- a) Verify that the Filling Boxes status indicator is green and communication is established.
- b) Enter Fill Mode.
- c) As ammunition is physically loaded into the magazine, the system automatically detects and displays the shells and propellant charges in the magazine slots.
- d) Verify that the reported ammunition configuration matches the actual ammunition loaded in the platform.
- e) Submit the loadout for commander review and approval, if required.

2) Manual Mode (**No** Filling Boxes **Communication**)

- a) Open Ammunition screen → View Current Loadout.
- b) Select each magazine slot manually.
- c) Define the ammunition parameters for each slot:
 - (1) Shell type
 - (2) Fuze type
 - (3) Propellant type
 - (4) Charge type
- d) Repeat the process until all loaded ammunition is configured in the system.
- e) Tap **Set as Current**.

4.2.5.2 Filling Ammunition - Automatic Mode

Before mission execution, ammunition must be loaded into the platform.

When the Filling Boxes subsystem is **connected** and operational, ammunition loaded into the magazine is automatically detected and displayed in the **Fill Mode Monitor** screen (See Figure 3-81).

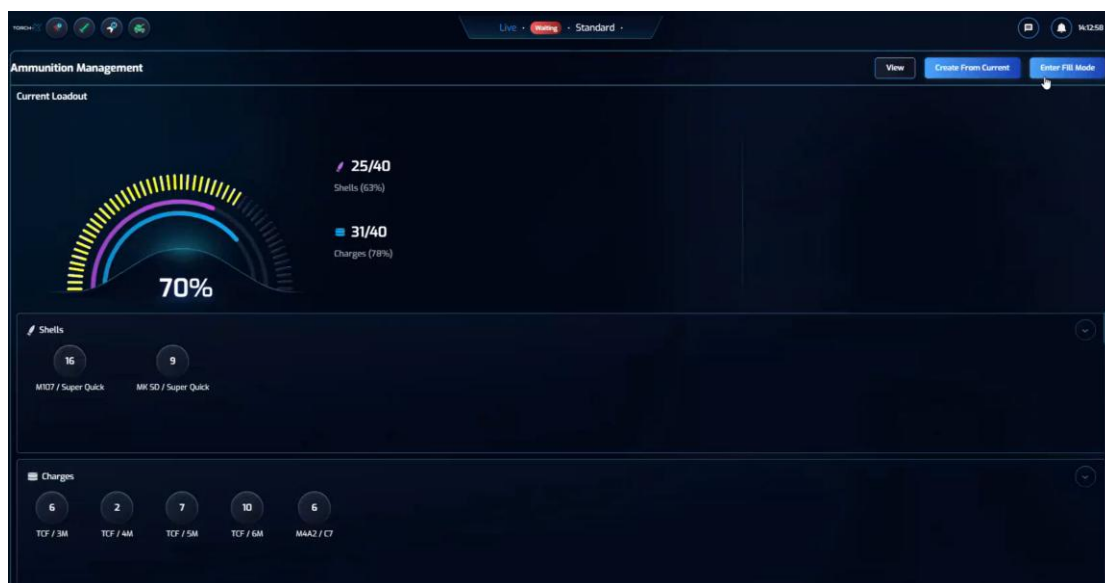
CAUTION

The commander **must verify and approve** the reported ammunition configuration before activating and executing the mission.

- a. In the Main **Screens Navigation Bar**, tap **AMMUNITION**.

Result:

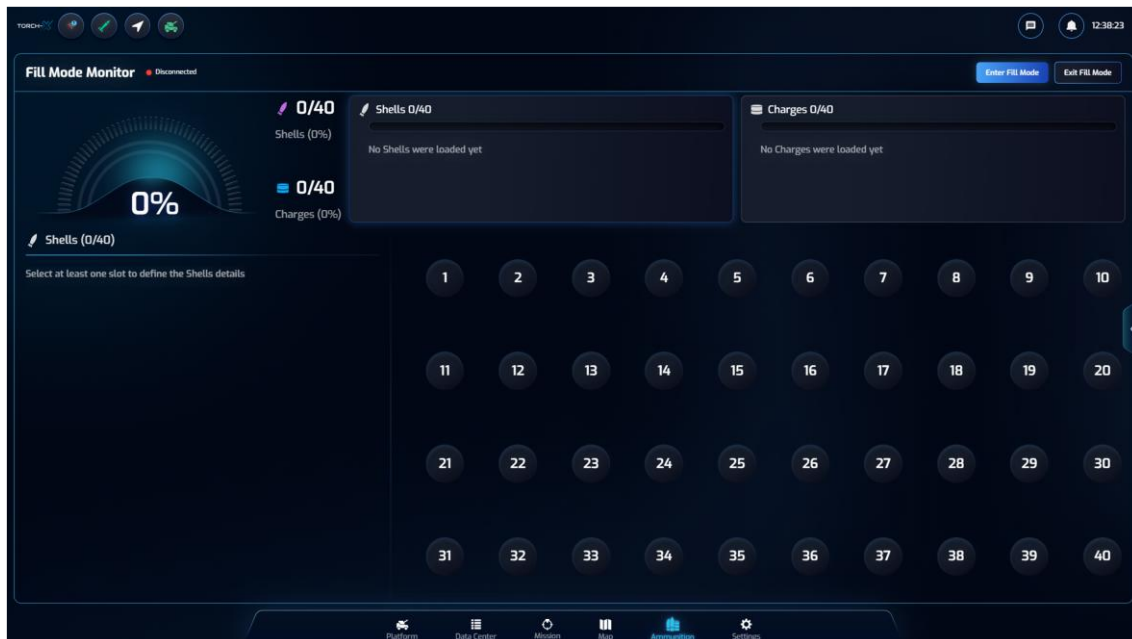
The **Ammunition Management** screen is displayed. The screen displays the current onboard ammunition status, including available shells and propellant charges.



- b. Tap **Enter Fill Mode**.

Result:

- 1) The **Fill Mode Monitor** screen is displayed.
- 2) The **Shells** and **Charges** inventory view is displayed on the Fill Mode Monitor screen.



- c. Verify that the filling boxes device **status indicator** is green (**Fill Mode Monitor** ● Connected). Meaning, the communication with the Filling Boxes system is established and operational.
- d. As ammunition is **physically** loaded into the magazine, the system automatically detects and displays the shells and propellant charges in the magazine slots.
- e. Verify that the reported ammunition configuration matches the actual ammunition loaded in the platform.

CAUTION

Verify that the shell type and fuze configuration entered in the system **match** the ammunition physically loaded into the platform.

4.2.5.3 Filling Ammunition - Manual Mode

If communication with the Filling Boxes subsystem is unavailable, manually configure the ammunition inventory via **Ammunition** → **View Current Loadout**, define the ammunition parameters for each occupied magazine slot, and then tap **Set as Current** to activate the configured loadout.

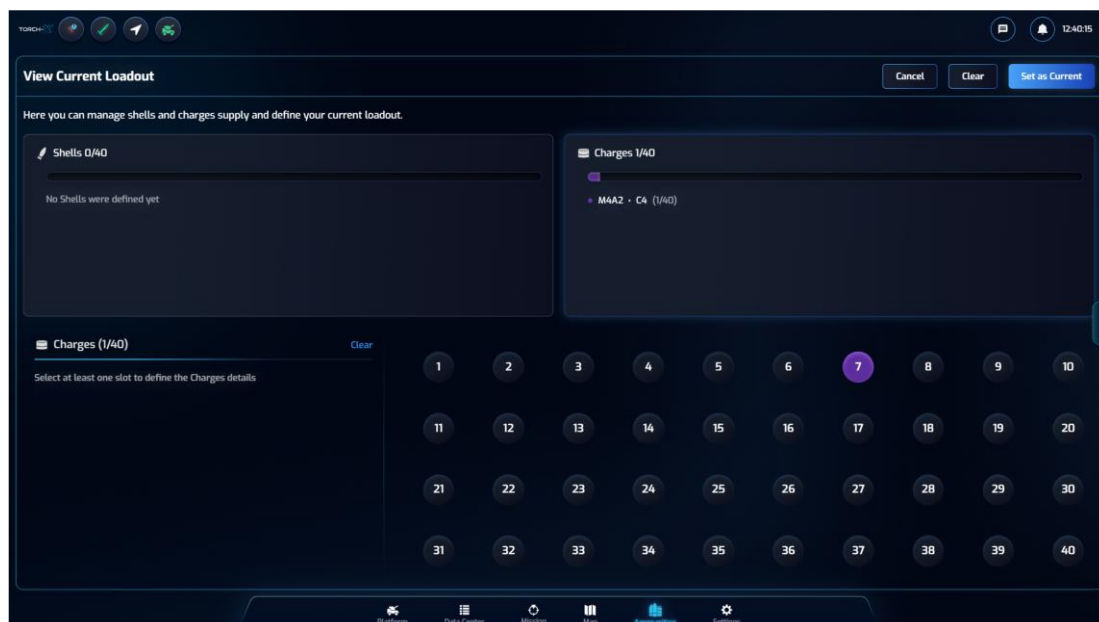
NOTE

- The procedure below describes the configuration of **Shells**. The process for configuring Charges (Propellant Charges) is identical.
- When configuring charges, select the Charges view and enter the relevant charge parameters using the same workflow, controls, and validation process used for shells.

- a. In **Fill Mode Monitor** Screen, Tap **Enter Fill Mode**.

Result:

- 1) The **View Current Loadout** Screen is displayed.
- 2) The system enters fill mode, enabling ammunition loading and monitoring operations.



- b. Select whether to configure **Shells** or **Charges**.

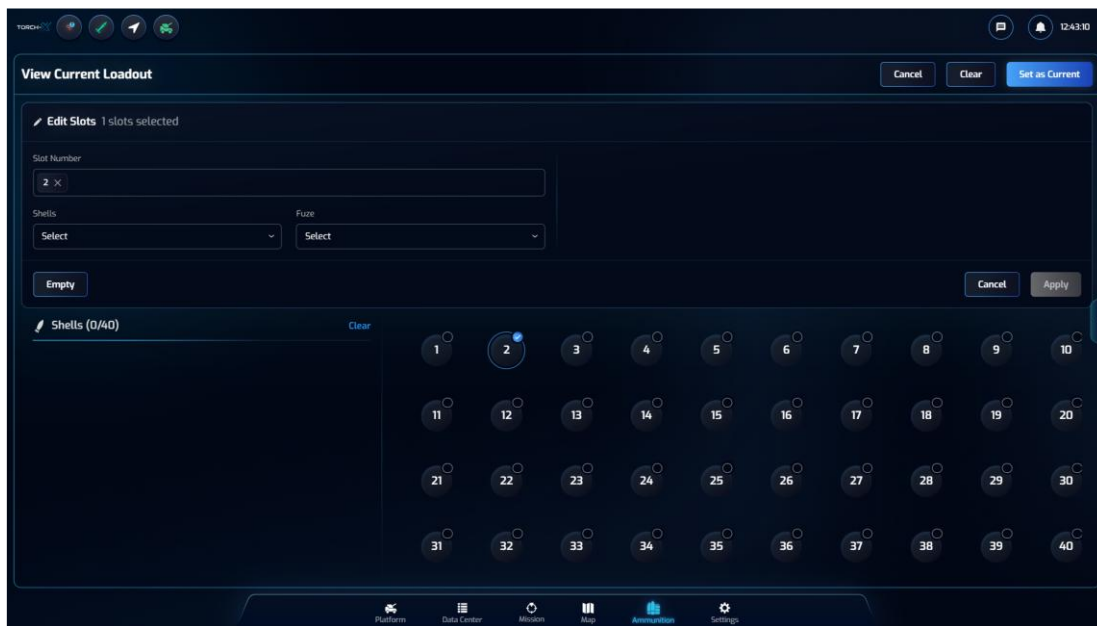
Result:

The magazine layout and slot configuration window update according to the selected inventory type:

- a) **Shells View** – enables configuration of Shell Type and Fuze Type.
- b) **Charges View** – enables configuration of Propellant Type, Charge Type, and Propellant Lot.
- c. Select the one or more magazine slot (1-40) corresponding to the slot that the shell was manually loaded into.

Result:

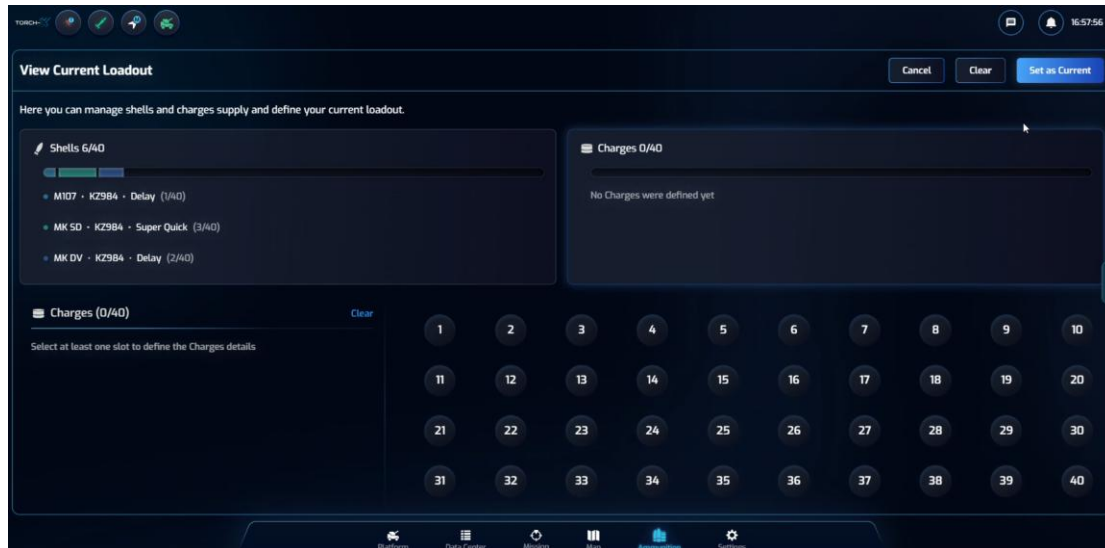
The **Slot Number** field displays the selected magazine slot(s).



- d. In the **Shells** field, select the shell type loaded into the selected slot.
- e. In the **Fuze** field, select the fuze type installed on the selected shell.
- f. Tap **Apply**.

Result:

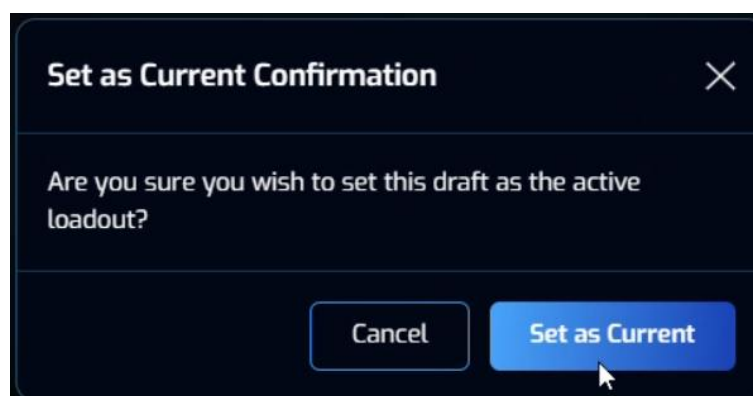
- 1) The selected shell and fuze configuration is assigned to the selected magazine slot(s).
- 2) The selected slot(s) are displayed with the newly assigned ammunition configuration.
- 3) The slot containing the configured shell is **color-coded** to indicate the selected shell type.
- 4) The ammunition inventory is updated.



- g. Repeat this procedure for each shell loaded into the magazine until all required shell slots have been configured.
- h. When done, tap **Set as Current**.

Result:

Pop-up message appears:



- i. Click **Set as Current**.

Result:

The configured ammunition loadout is saved as the active platform inventory and becomes available for ammunition allocation and fire mission execution.

- j. Repeat this procedure for each **Charge** loaded into the magazine until all required charge slots have been configured.

4.2.6 Map Operations

The map screen provides access to a range of map tools that support the user in carrying out various operational tasks, such as navigation, tactical planning, and situational awareness.

4.2.6.1 Track Entity

4.2.6.1.1 Adding New Track Entity

- a. Double- tap the desired location on the tactical map.

Result:

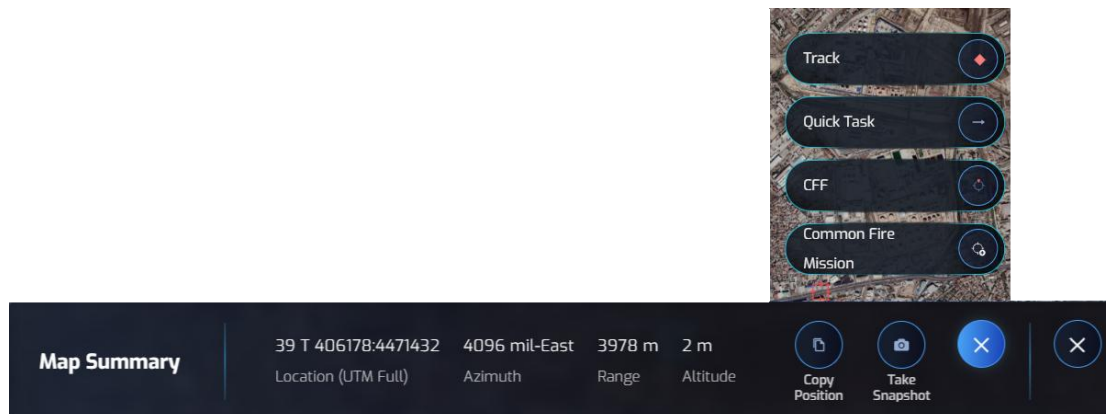
The **Map Summary** window opens.



- b. Tap the  icon.

Result:

Context-menu opens:



- c. Select **Track**.
- d. Tap the location on the map where you want to create the track entity.

Result:

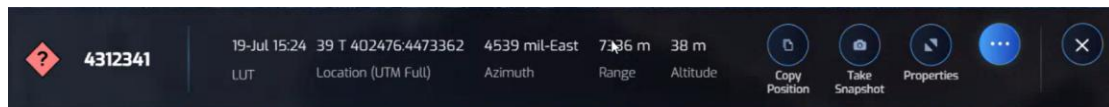
A Track entity icon () is displayed at the selected location on the map.



- e. Tap the **Track** entity displayed on the tactical map.

Result:

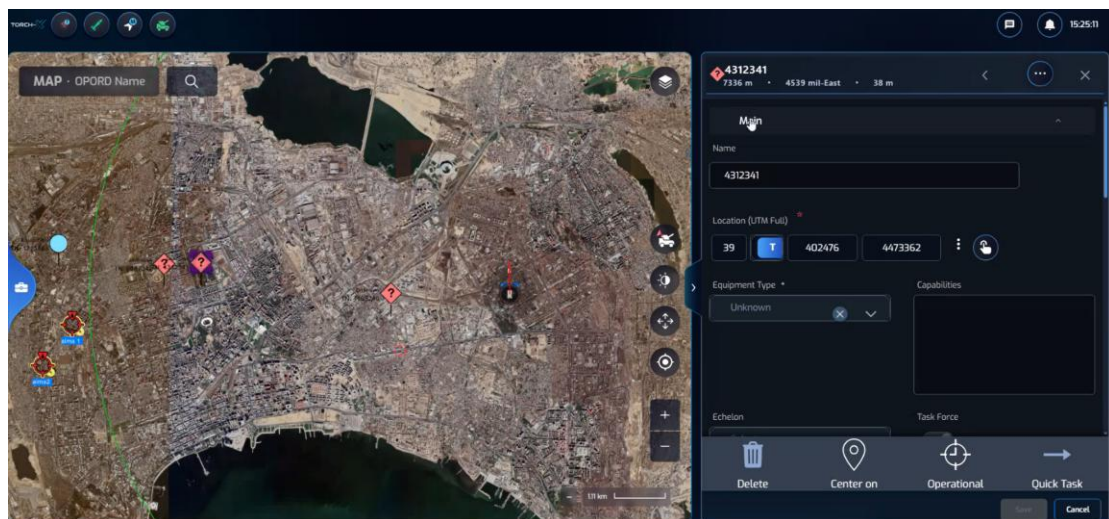
The entity details window opens and displays the **entity ID** together with operational data, including azimuth, location, last update date and time, range, and altitude.



- f. Tap **Properties** (⚙️).

Result:

The Track entity properties window opens.



NOTE

Mandatory fields are identified by a visual indication adjacent to the field (red text).

- g. In the **Name** field, enter the entity name.
- h. If required, in the **Location** field, enter a new location or select a location directly from the tactical map.
- i. In the **Capabilities** field, enter the entity capabilities, remarks, or operational notes.
- j. In the **Equipment Type** field, select the equipment type from the drop-down menu (for example, SAM SA-14).
- k. In the **Echelon** field, select the entity echelon from the available options.
- l. Use the **Task Force** toggle to define whether the entity belongs to a task force.
- m. In the **Standard Identity** field, select the entity classification (for example, Friendly, Hostile, Unknown, or Neutral).
- n. If applicable, in the **Priority** field, use the + and – buttons to set the entity priority level for Investigation/Action.
- o. In the **Identification** field, select the entity category (for example: Vehicle, Person, Animal, Ground, Air, UAV, or other applicable category).
- p. In the **Classification** field, select the entity type (for example: Animal, Vehicle, Group, Armed Person, or other applicable classification).
- q. Use the **Exercise** toggle to define whether the entity is associated with an exercise or a live operational environment.
- r. Continue entering data in the remaining fields as required by the operational scenario and mission requirements.
- s. Tap **Save**.

Result:

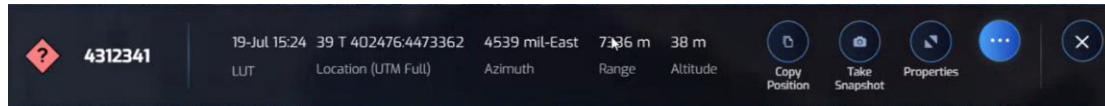
- 1) The entity data is applied to the system,
- 2) The entity is updated on the tactical map with all configured properties and classifications.
- 3) Any visual changes associated with the selected configuration are displayed automatically: Friendly entity – blue color, Hostile entity – red color).

4.2.6.1.2 Actions on Track Entity

- a. Tap the **Track** entity displayed on the tactical map.

Result:

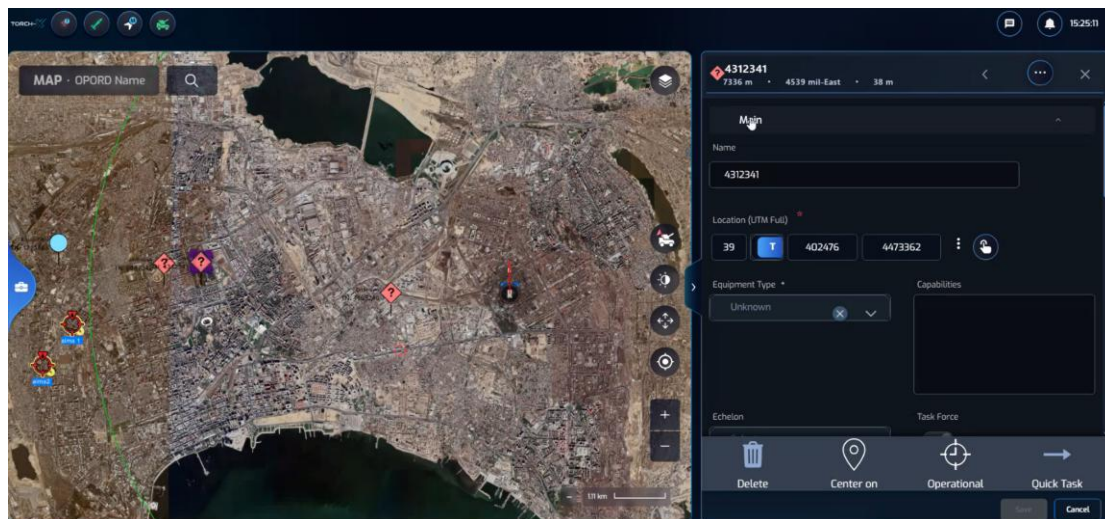
The Entity Details window opens and displays the entity's operational data.



- b. Tap **Properties** (ⓘ).

Result:

The Extended Entity Properties window opens.



- c. To delete the entity, tap **Delete** (🗑️).

Result:

The selected entity is removed from the system and no longer appears on the tactical map.

- d. To center the map display on the entity's location, tap **Center On** (📍).

Result:

The tactical map automatically centers on the entity's current location.

- e. To define the entity's **operational status**, tap **Operational** (⌚).

Result:

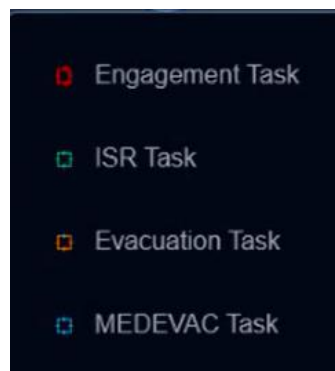
Operational Condition context menu opens:



- f. Select the operational condition of the entity (for example: Destroyed, Planned, Damaged, Operational, or other available condition).
- g. To assign a quick task to the entity, tap **Quick Task** (→).

Result:

The Quick Task context menu opens.



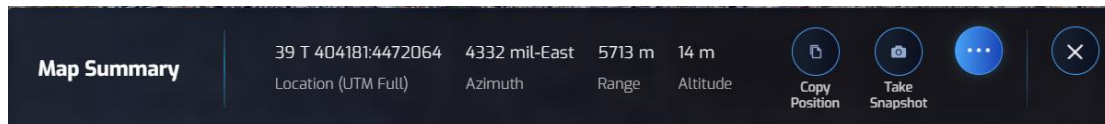
- h. Select the quick task type.
- i. When done, tap **Save**.

4.2.6.2 Creating a Quick Task

- a. Double- tap the desired location on the tactical map.

Result:

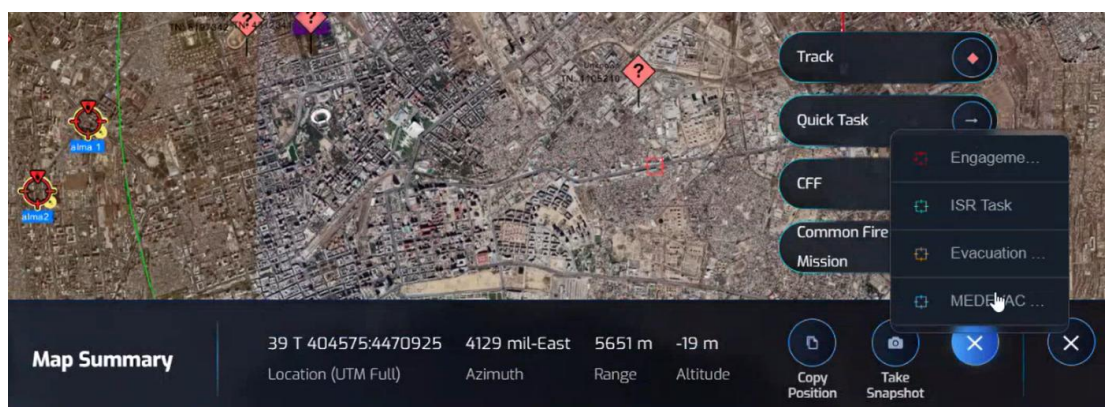
The **Map Summary** window opens.



- b. Tap the  button and select **Quick Task**.

Result:

The Quick Task context menu opens.



- c. Select the required **Quick Task** type (ISR, Evacuation, etc.).
- d. Tap the desired location on the tactical map to create the task entity.

Result:

A Quick Task entity () is created at the selected location, and the task symbol is displayed on the tactical map.

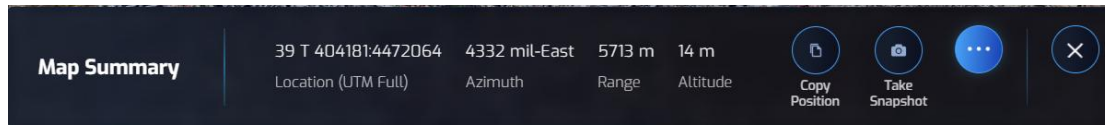


4.2.6.3 Creating a Call for Fire (CFF) Request

- a. Double- tap the desired location on the tactical map.

Result:

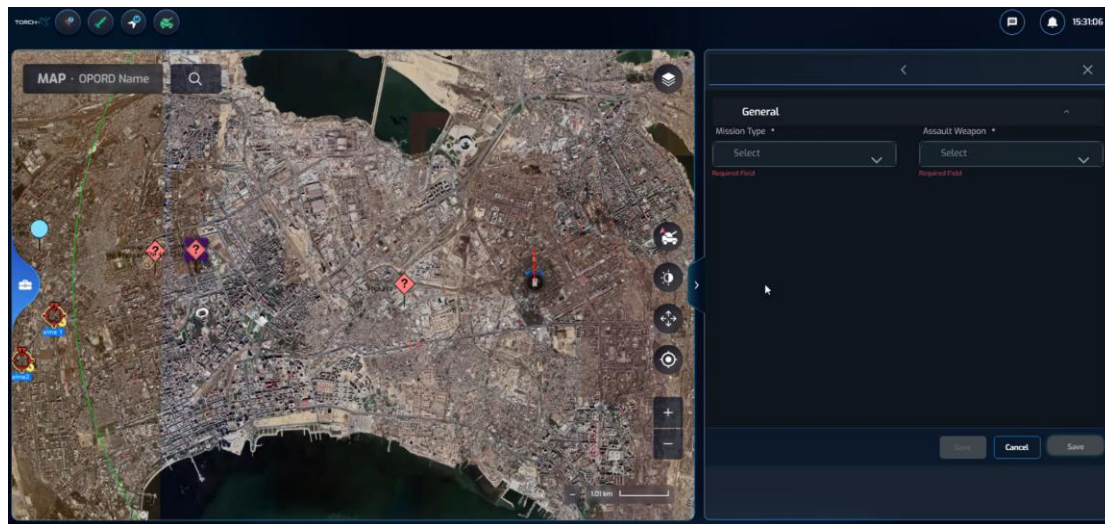
The **Map Summary** window opens.



- b. Tap the  button and select **CFF**.

Result:

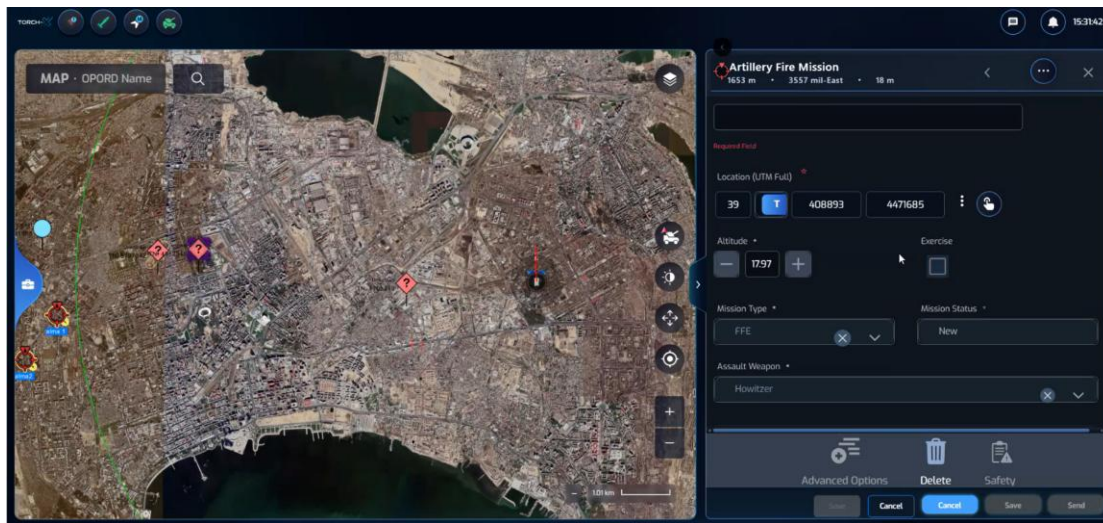
The **CFF** window opens on the right side of the screen.



- c. In the **Mission Type** field, select the required mission type (for example, FFE).
- d. In the **Assault Weapon** field, select the requested weapon system (for example, DITA, NORA, or Mortar).
- e. Tap **Save**.

Result:

- 1) The selected mission type and weapon system are applied to the CFF request.
- 2) The Artillery Fire Mission window opens.



- f. In the **Name** field, enter a name for the fire mission.
- g. In the **Location** field, define the target location by performing one of the following actions:
 - 1) Enter the target coordinates manually.
 - 2) Tap the  icon, and then select the target location directly from the tactical map.
- h. Continue entering the required operational and mission data in the remaining fields.
- i. Tap **Send**.

Result:

- 1) A CFF entity () is created at the selected location.
- 2) The CFF request is transmitted through the C4I network to the designated command-and-control system for further processing and tasking.

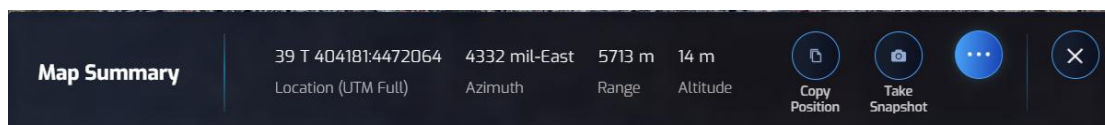
4.2.6.4 Creating a Common Fire Mission

A Common Fire Mission is a tactical fire mission entity typically generated from a Call for Fire (CFF) request. It defines the **target location** and the engagement parameters, including the **type of ammunition** and fire plan to be employed against the target. The mission is usually created based on information reported by a Forward Observer (FO) or another authorized fire support element.

- a. Double- tap the desired location on the tactical map.

Result:

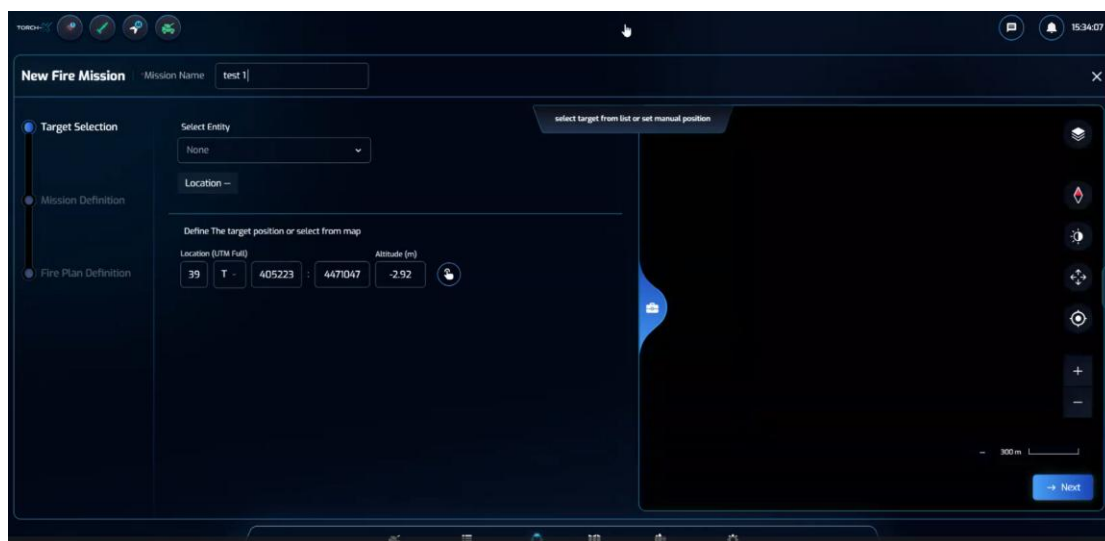
The **Map Summary** window opens.



- b. Tap the  button and select **Common Fire Mission**.

Result:

New Fire Mission Wizard opens:



- c. Complete the New Fire Mission wizard according to the procedure described in Creating a New Fire Mission (see Section 4.2.11).

NOTE

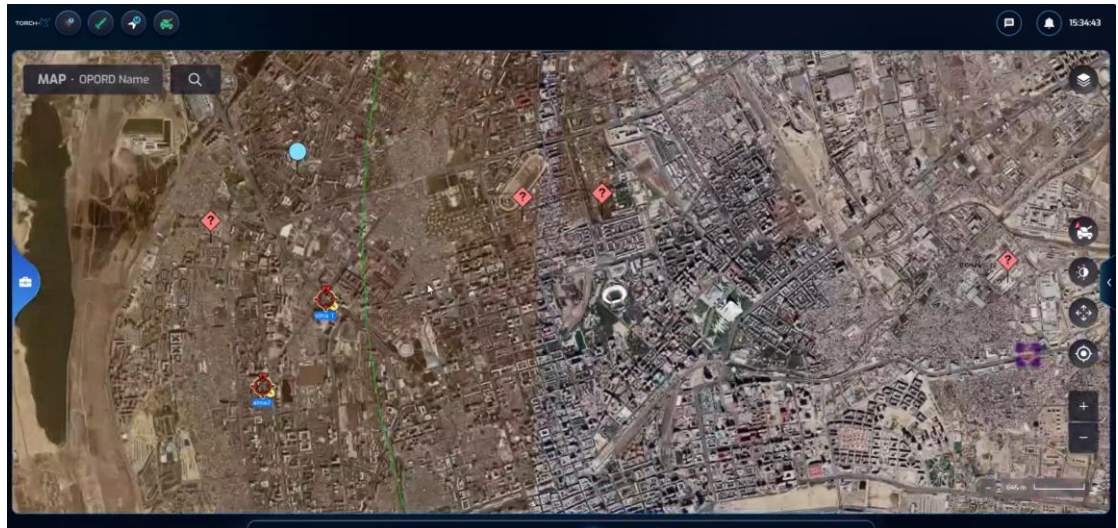
The **New Fire Mission** wizard uses the same workflow, fields, and validation rules as the New Fire Mission wizard described in Section 4.2.11.

4.2.6.5 Opening the Terrain Analysis Tools

- a. In the **Main Screens Navigation** toolbar, tap **Map**.

Result:

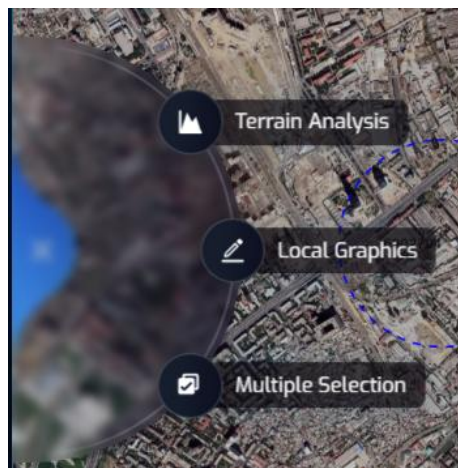
Map Screen appears:



- b. In the **Map** screen, tap the **Toolbox** icon ().

Result:

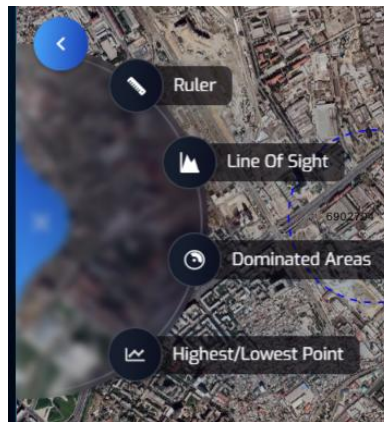
The Map Tools menu opens.



- c. Select **Terrain Analysis**.

Result:

The available terrain analysis tools are displayed.

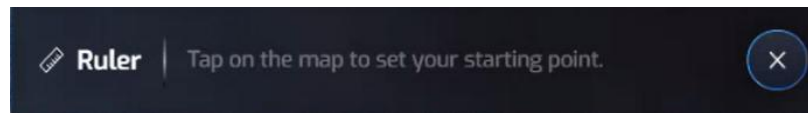


4.2.6.6 Measuring Distance on the Map (Ruler)

- a. Open the Terrain Analysis Tools (see section 4.2.6.5)
- b. Tap **Ruler**.

Result:

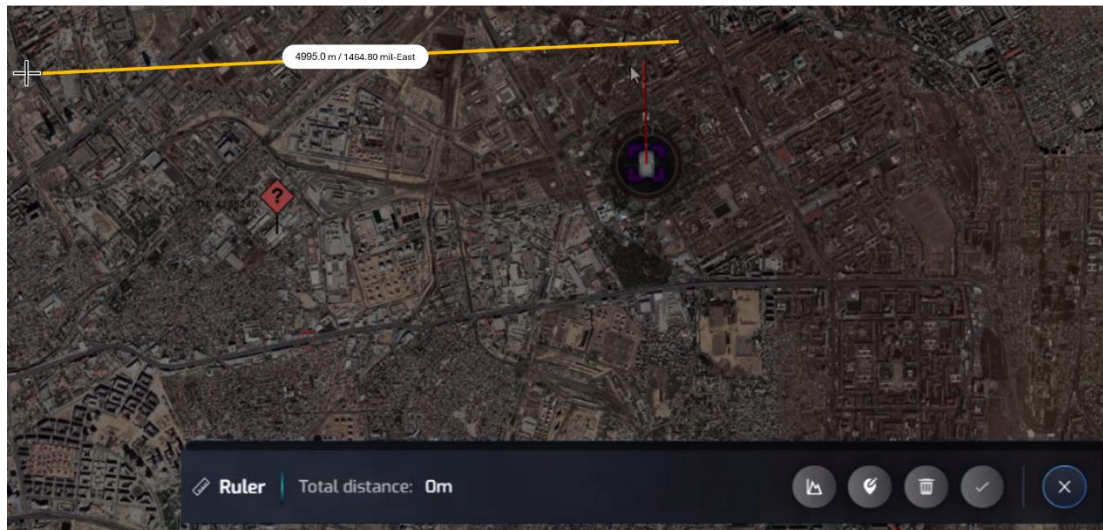
The Ruler tool is activated.



- c. Tap on the map to set the starting point.
- d. Move the cursor to the next point.
- e. Tap the next point on the map. **To continue the measurement**, tap additional locations on the map as required.

Result:

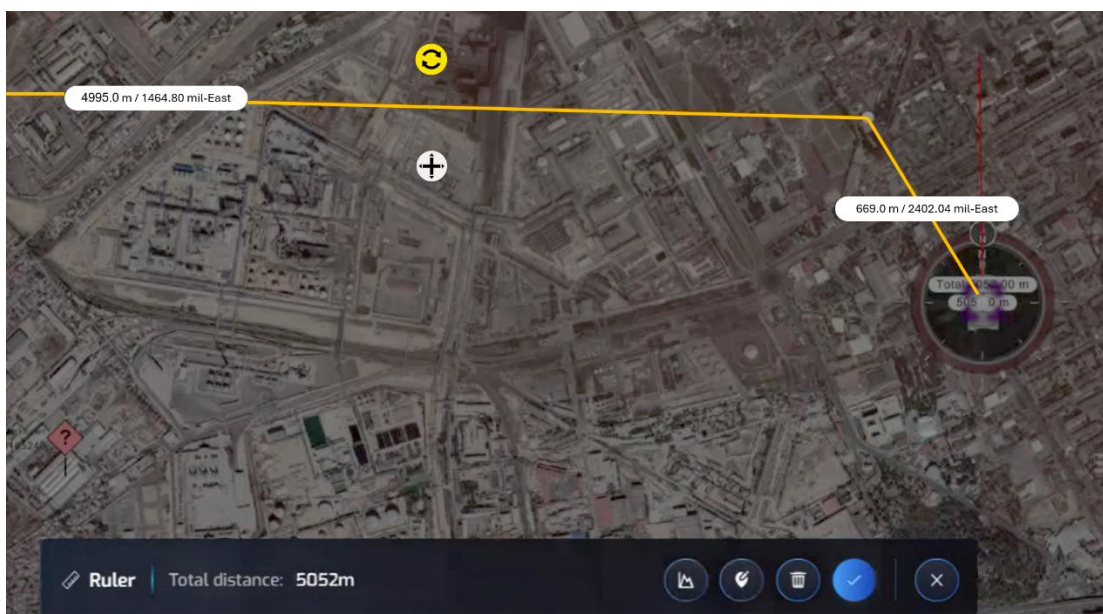
A measurement line is displayed between the selected locations, together with the calculated distance information.



- f. Tap the end point on the map.

Result:

- 1) The **Total Distance** is displayed in the **Ruler** window.
- 2) The measurement line displayed on the map.
- 3) Measurement editing tools, such as Tilt (🔄) and Move (📏), appears next to the line.
- 4) Additional measurement options and controls are displayed in the Ruler window.



- g. To confirm the measurement and complete the operation, tap ().

Result:

The measurement is saved and displayed on the tactical map.

NOTE

Once the measurement is completed, the user can modify the measurement route, reposition measurement points, or perform additional analysis using the tools available in the Ruler window.

4.2.6.6.2 Viewing a Ruler Cross Section

The Cross Section function enables the user to analyze the terrain profile along a defined measurement route during the planning phase. The displayed terrain profile assists in evaluating slopes, elevation changes, and terrain obstacles along the selected route.

Reviewing the cross-section can help identify terrain conditions that may increase the risk of vehicle instability, mobility restrictions, or vehicle rollover during movement.

- a. Tap a measurement line on the map.

Result:

the **Ruler** window opens:



- b. In the **Ruler** window, Tap **Cross Section** ().

Result:

- 1) The system generates and displays a terrain cross-section window along the selected measurement line.
- 2) While reviewing the terrain profile in the Cross Section window, the **current cursor position** on the cross-section graph is synchronized with the measurement line displayed on the tactical map.



- c. Review the terrain profile along the selected route. Identify steep slopes, elevation changes, and terrain features that may affect platform movement.

4.2.6.6.3 Editing Ruler Location

- a. Tap a measurement line on the map.

Result:

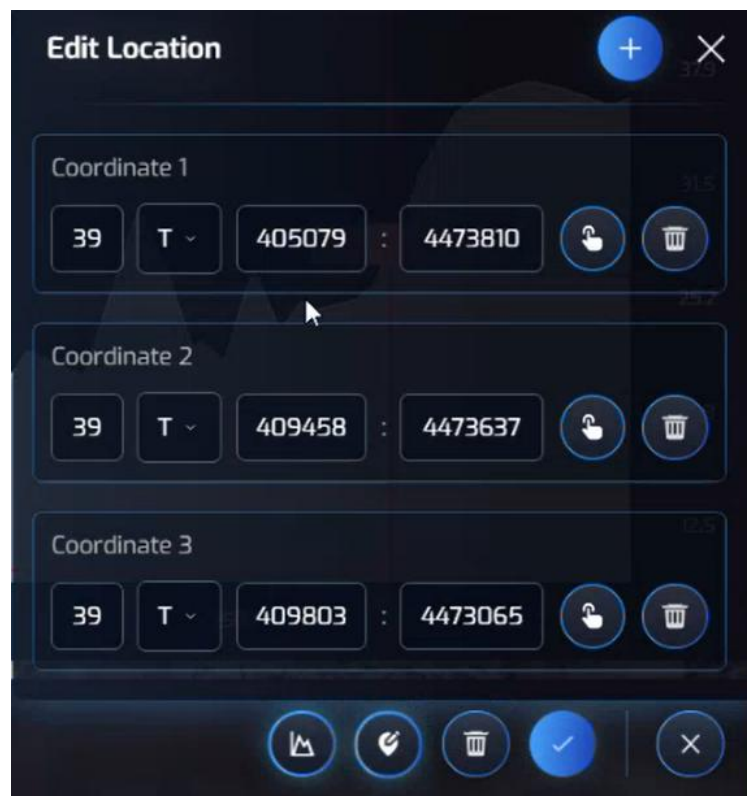
the Ruler window opens:



- b. In the **Ruler** window, Tap **Edit Location** ().

Result:

Edit location window opens:



- c. To **manually** modify a measurement point, enter the required coordinates in the relevant coordinate fields.
- d. To **delete** a measurement point, tap the Delete icon () next to the relevant coordinate.
- e. To select a measurement point **directly from the map**, tap the Pick from Map icon (), and then select the desired location on the tactical map.
- f. To **add a new measurement point**, tap  and enter the coordinates in the new measurement point row.
- g. Tap .

Result:

All modifications are saved, and the updated measurement route is displayed on the tactical map.

4.2.6.6.4 Deleting a Ruler Measurement

- a. Tap a measurement line on the map.

Result:

the Ruler window opens:



- b. In the **Ruler** window, Tap **Delete** ().

Result:

The selected Ruler measurement is removed from the tactical map.

4.2.6.7 Measuring Distance Between Two Objects or Specific Location

This function enables the user to accurately measure the distance between two tactical entities or between a tactical entity and a specific geographic location.

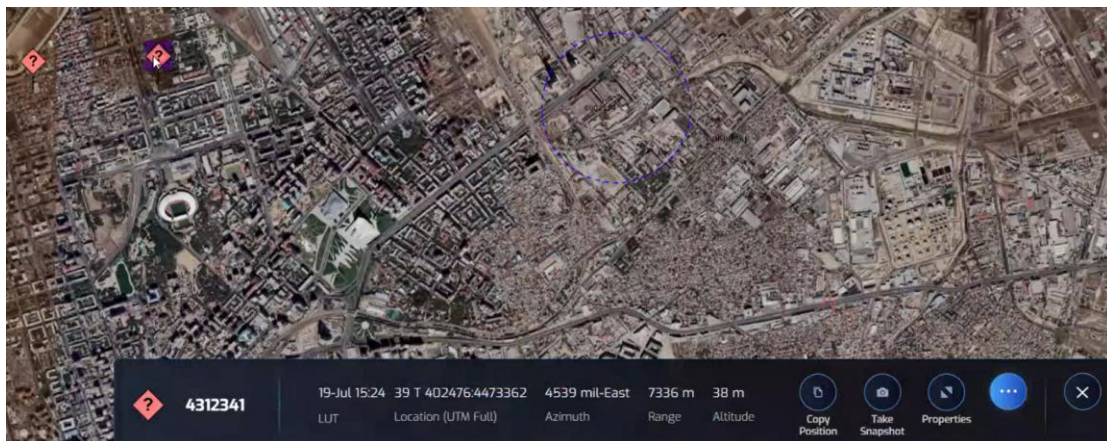
This function is particularly useful when precise manual selection of an entity's position on the tactical map is difficult due to map scale, screen resolution, icon density, or overlapping tactical symbols.

Instead of manually selecting an exact point on the map, the user can directly select a tracked entity as the measurement start point, end point, or both.

- a. Tap the object from which you want to measure the distance.

Result:

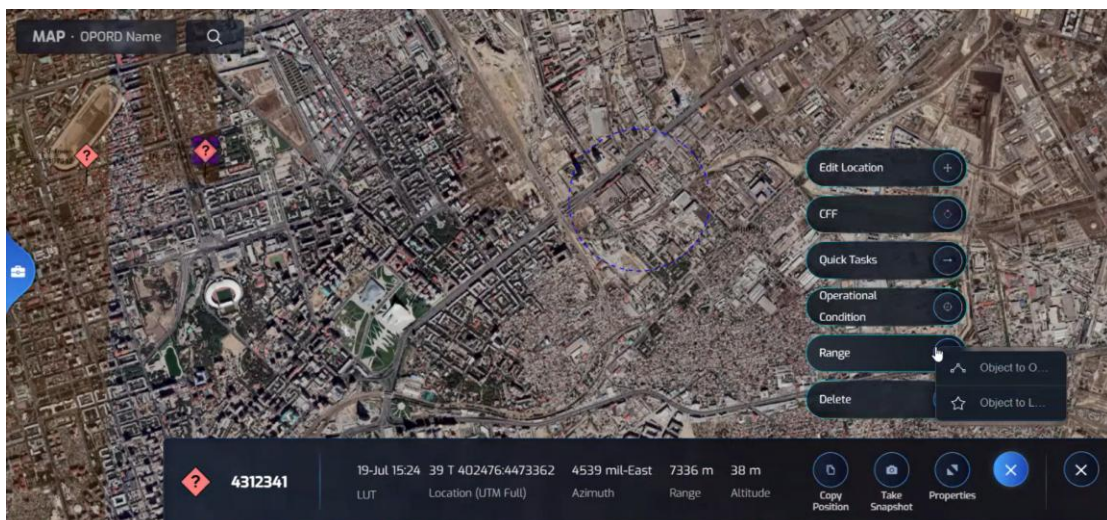
The object Details window opens.



- b. Tap , and then select **Range**.

Result:

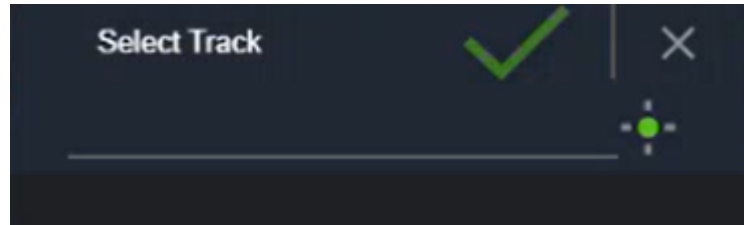
The Context menu opens.



- c. Select **Object-to-Object**.

Result:

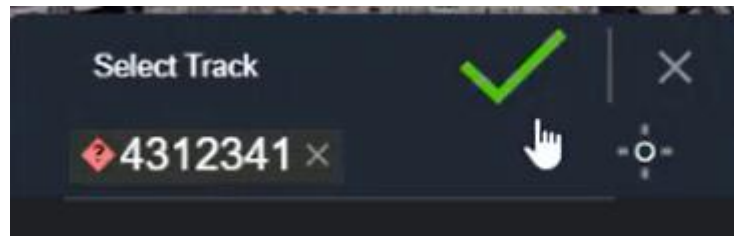
The **Select Track** window opens.



- d. Tap the object to which you want to measure the distance.

Result:

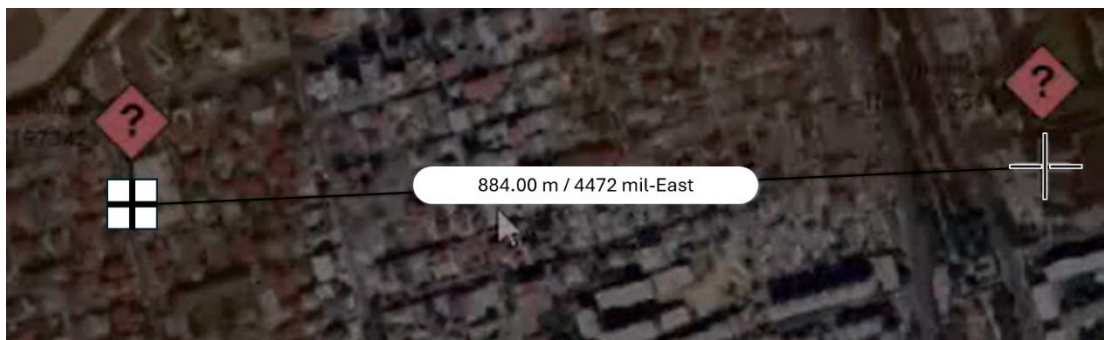
The selected object is displayed in the Select Track window.



- e. Tap ✓.

Result:

A measurement line is drawn between the objects, and the calculated distance is displayed on the line.

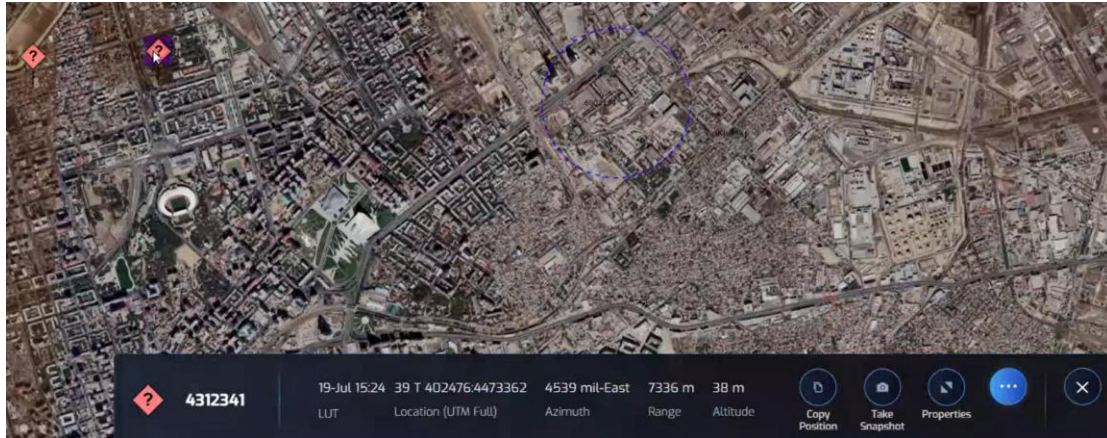


4.2.6.8 Measuring Distance from Object to Specific Location

- a. Tap the object from which you want to measure the distance.

Result:

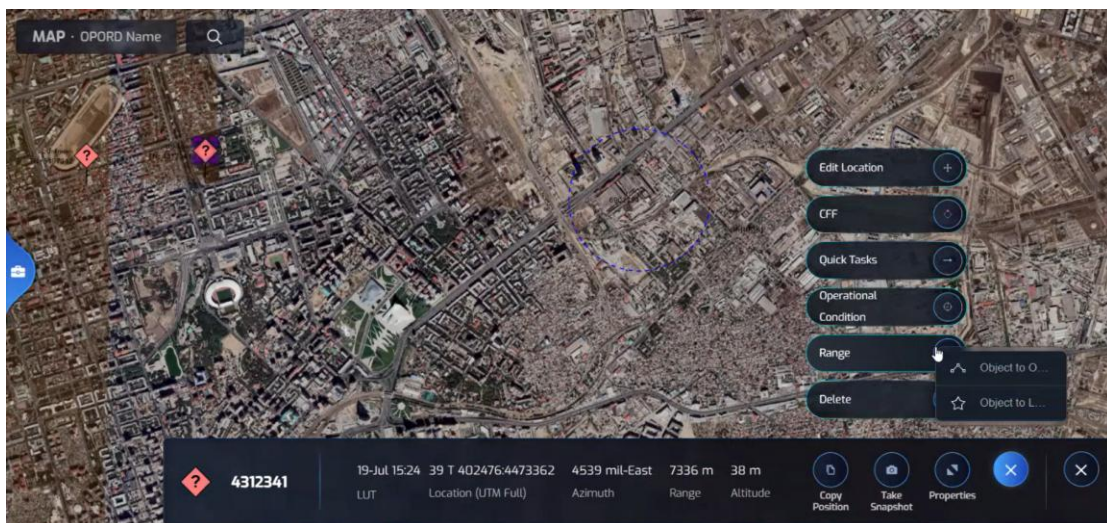
The object Details window opens.



- b. Tap , and then select **Range**.

Result:

The Context menu opens.



- c. Select **Object-to-Location**.

Result:

The **Location** window opens.



- d. Select the destination location by performing one of the following actions:

- Enter the coordinates manually.

- Tap Pick from Map (), and then select the desired location on the tactical map.

Result:

The selected location is displayed in the location field.

- e. Tap ✓.

Result:

A measurement line is drawn between the selected entity and the selected location (indicated with cross ) , and the distance is displayed on the line.



4.2.6.9 Viewing Line of Sight

The Line of Sight function enables the user to evaluate the visibility between two locations on the map and determine whether the line of sight is clear or obstructed by terrain.

- a. Open the Terrain Analysis Tools (see section 4.2.6.5)
- b. Tap **Line of Sight**.

Result:

The Line of Sight tool is activated.



- c. Tap the **starting** location on the tactical map.

Result:

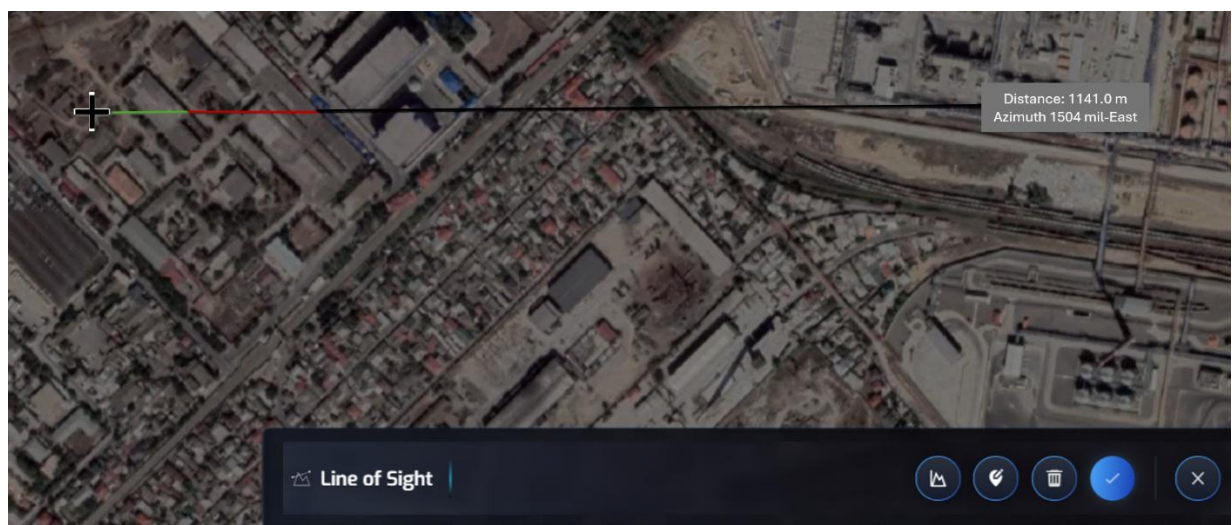
The starting point is defined.

- d. Move the cursor toward the desired end location.

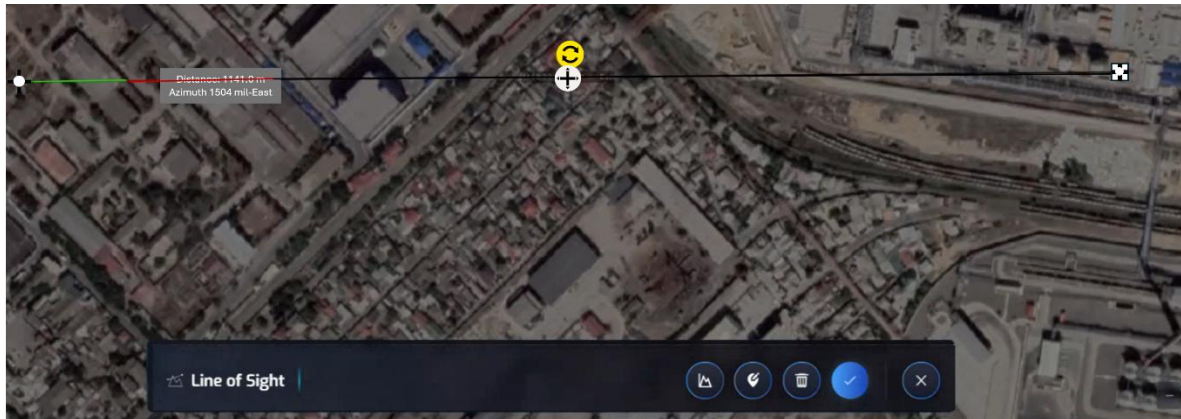
Result:

- 1) A line is displayed on the map between the start point and the cursor position.
- 2) The line is color-coded to indicate line-of-sight status along the route.
 - **Green** – Clear line of sight.
 - **Red** – Obstructed line of sight.
 - **Black** – TBD.

- e. Floating information boxes display the **distance** and **azimuth** values.



- f. Tap the end location on the map.
- g. Result:
 - 1) The line-of-sight analysis is completed and displayed on the map.
 - 2) The system indicates whether the line of sight between the two locations is clear or obstructed.
 - 3) Editing tools, such as Tilt (🔄) and Move (📏), appears next to the line.



NOTE

- The displayed distance and azimuth values are updated dynamically while the user moves the cursor prior to selecting the end point.
- Line-of-sight analysis is based on available terrain elevation data and is intended to support mission planning, observation planning, sensor placement, and route selection.
- The procedures for viewing a **Cross Section**, **editing measurement points**, and **deleting** a Line of Sight (LOS) are identical to those described for the Ruler tool.

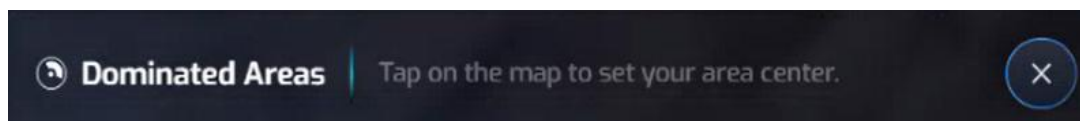
4.2.6.10 Viewing Dominated Areas

The Dominated Areas function enables the user to identify terrain areas that are visible and observable from a selected location. This analysis assists in planning observation posts, sensor placement, firing positions, and tactical maneuver routes.

- a. Open the Terrain Analysis Tools (see section 4.2.6.5)
- b. Tap **Dominated Areas**.

Result:

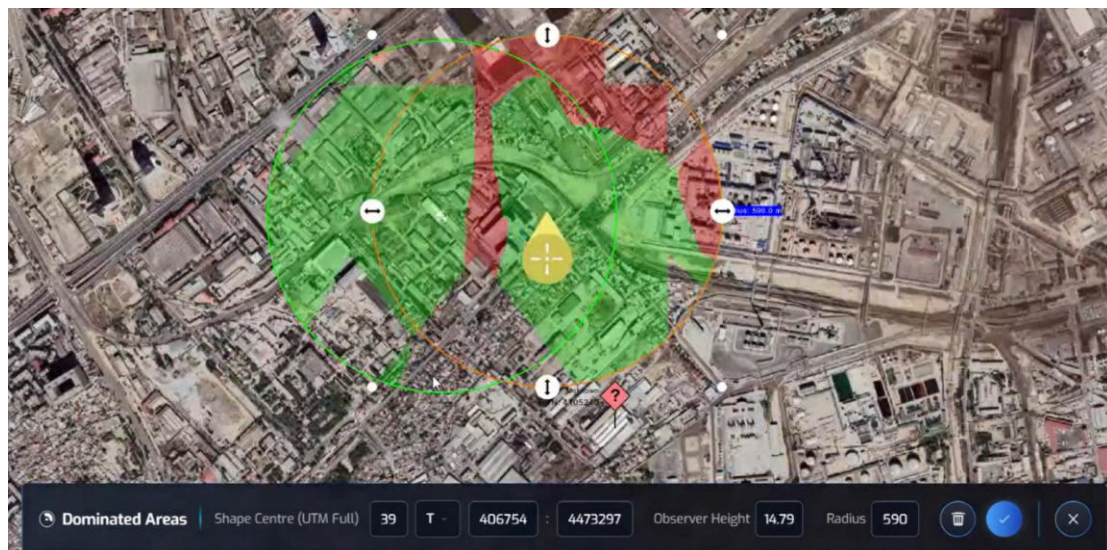
The **Dominated Areas** tool is activated.



- c. Tap on the map to set your area **center**.
- d. Move the cursor across the map and define the analysis area as required.

Result:

- 1) The system performs a terrain visibility analysis from the selected observation point.
- 2) The dominated areas are displayed on the tactical map using color-coded overlays according to the visibility analysis results.
- 3) Modify controls for changing the shape size appears.



NOTE

- The displayed colors indicate whether an area is visible (green) or masked (red) from the selected observation point according to available terrain elevation data.
- The accuracy of the analysis depends on the quality and resolution of the terrain elevation data loaded into the system.
- Any modifications to the observation point or analysis area automatically trigger an update of the Dominated Areas calculation and display.

e. To save the data, tap .

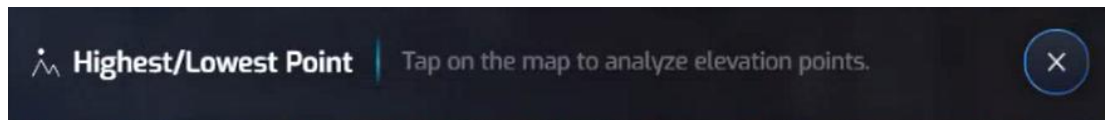
4.2.6.11 Viewing Highest/Lowest Point

The Highest/Lowest Point function enables the user to identify the highest and lowest terrain elevations within a selected area. This analysis assists in selecting observation posts, firing positions, communication sites, and route planning.

- a. Open the Terrain Analysis Tools (see section 4.2.6.5)
- b. Tap **Highest/Lowest Point**.

Result:

The Highest/Lowest Point tool is activated.



- c. Create a square on the tactical map covering the area to be analyzed.

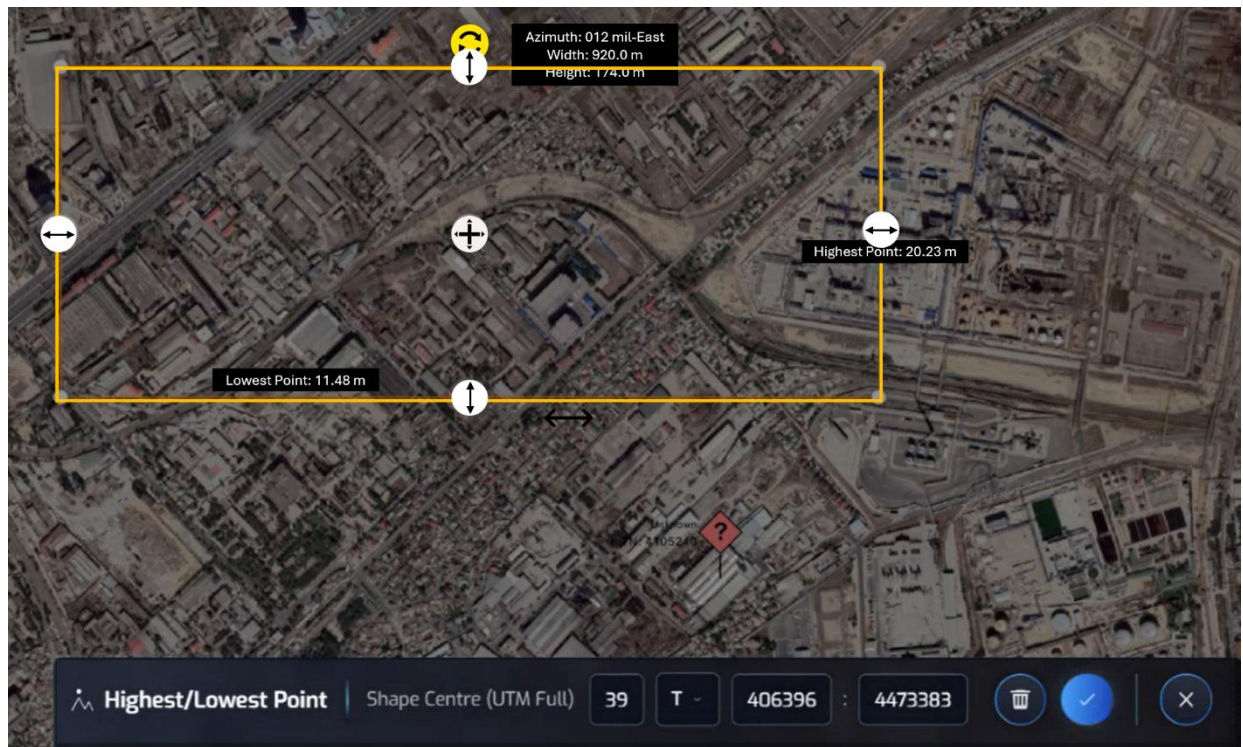
Result:

A square boundary is displayed on the map. The square is displayed using an **orange** outline and dynamically changes its size as the cursor is moved.

- d. Continue moving the cursor until the required analysis area is defined.
- e. Tap the map to finalize the analysis area.

Result:

- 1) The system analyzes the selected area and identifies the highest and lowest terrain points within the defined boundaries.
- 2) The highest and lowest points are displayed on the tactical map using dedicated markers and elevation data.
- 3) Modify controls () for changing the shape size appears.



NOTE

- The accuracy of the analysis depends on the quality and resolution of the terrain elevation data loaded into the system.
- Any modification to the analysis area automatically triggers a recalculation of the highest and lowest terrain points.

- f. To save the data, tap .

4.2.6.12 Local Graphics

The Local Graphics tool enables the user to create graphical annotations and tactical overlays directly on the tactical map. The tool can be used to draw and display various graphical objects, including lines, circles, rectangles, arrows or freehand drawings.

Local Graphics assists the user in mission planning, target marking, route designation, sector definition, area highlighting, operational briefing preparation, and situational awareness.

After a graphic object is created, the user can modify its visual and display properties, including color, thickness, style, position or dimensions.

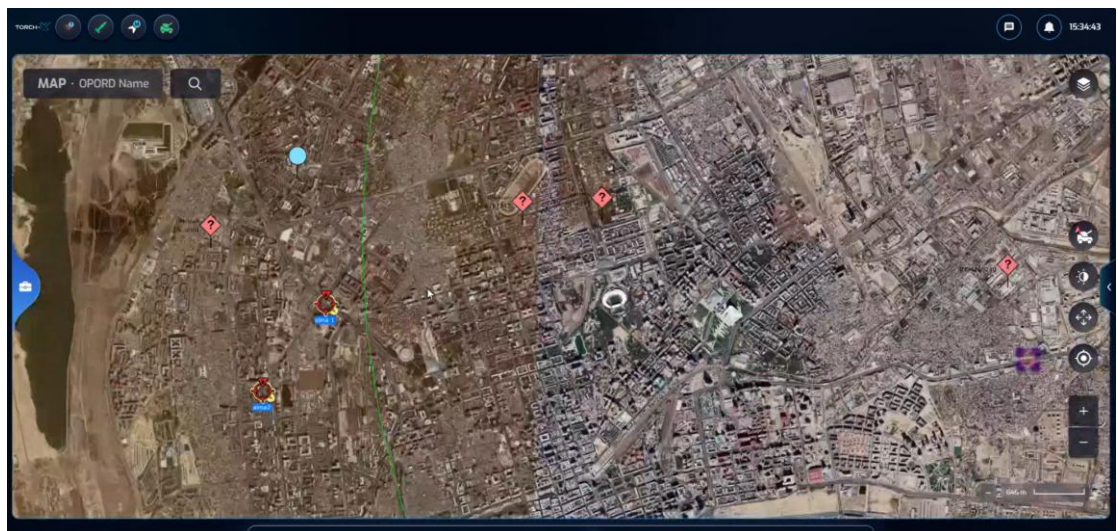
NOTE

Graphical objects created with the Local Graphics tool remain visible on the tactical map until they are deleted by the user or removed by system configuration.

- a. In the **Main Screens Navigation** toolbar, tap **Map**.

Result:

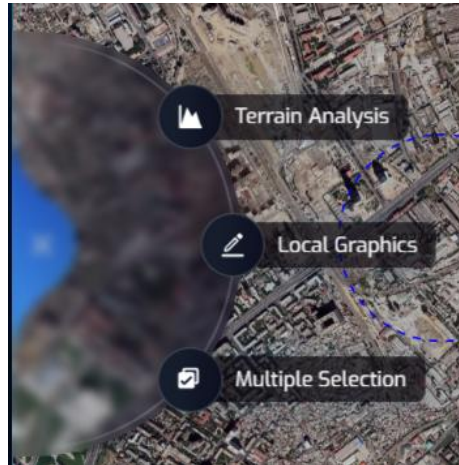
Map Screen appears:



- b. In the **Map** screen, tap the **Toolbox** icon ()

Result:

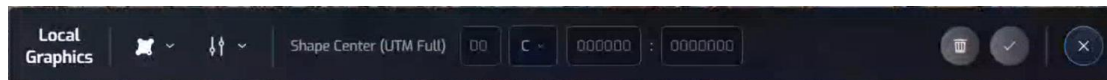
The Map Tools menu opens.



- c. Select **Local Graphics**.

Result:

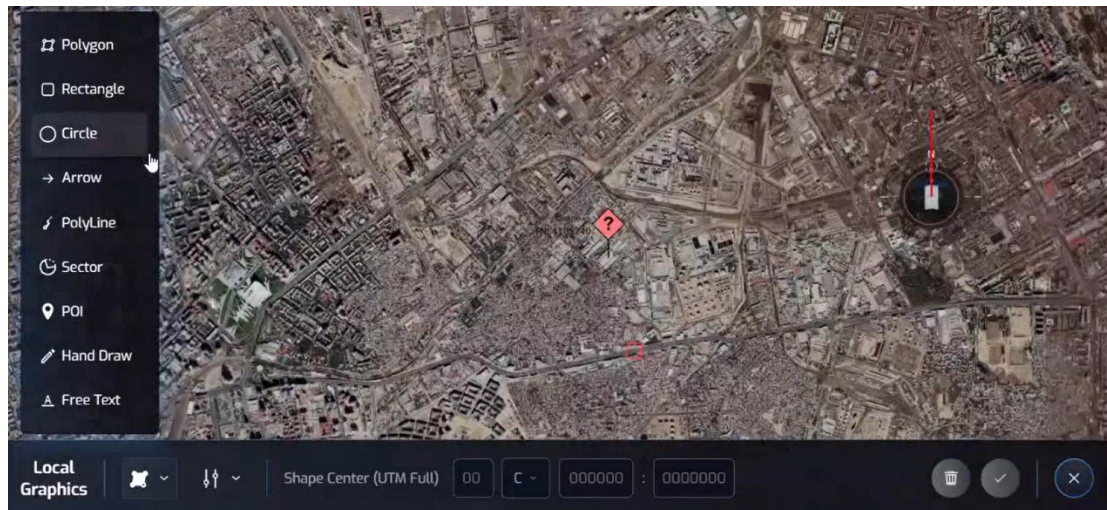
The available Local Graphics window is displayed.



- d. Tap the  icon.

Result:

A context menu opens displaying the available graphic objects and shapes that can be created on the tactical map.

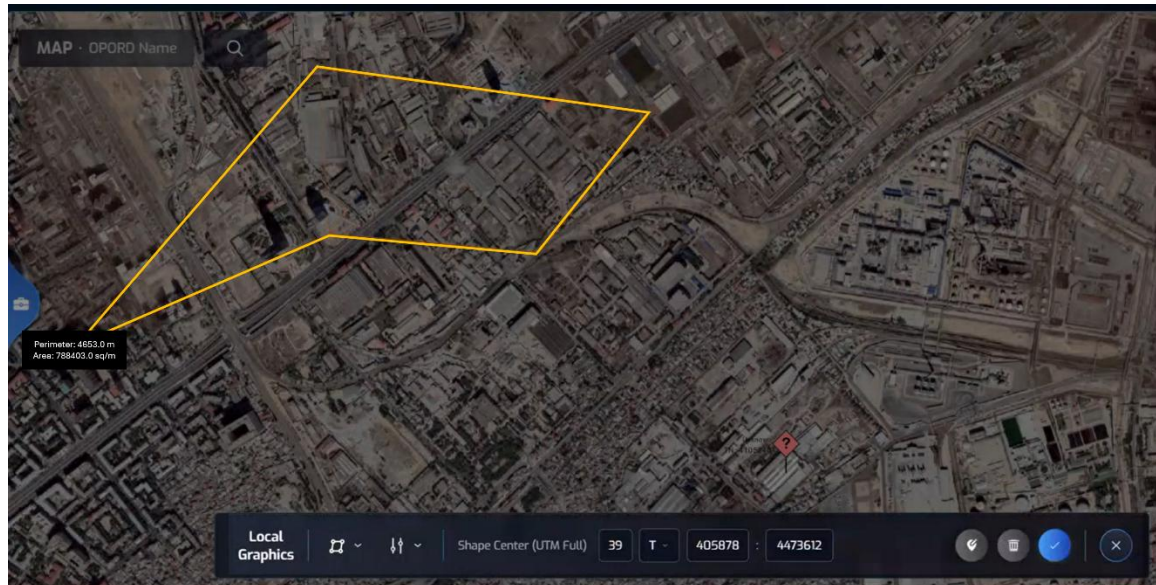


- e. Select the desired shape from the menu.

- f. Create the shape on the tactical map. For example, when creating a **polygon**, tap multiple locations on the map to define the required shape.

Result:

- 1) The shape is displayed on the tactical map.
- 2) During the creation process, the shape is displayed using an **orange** outline.
- 3) Shape information is displayed dynamically, including parameters such as **Perimeter** and **Area**.



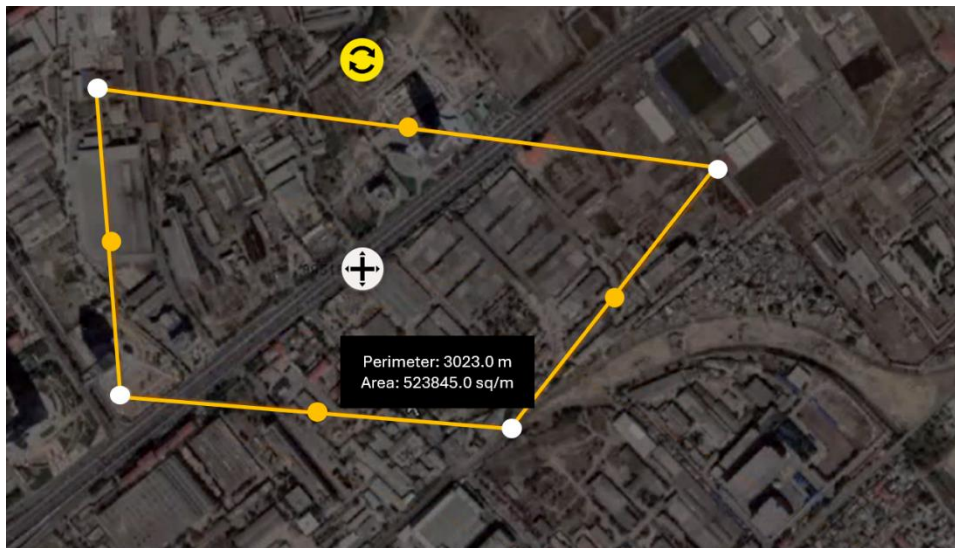
NOTE

The displayed measurements are updated automatically as additional points are added or existing points are repositioned.

- g. When shape is complete, double- tap an area in the map.

Result:

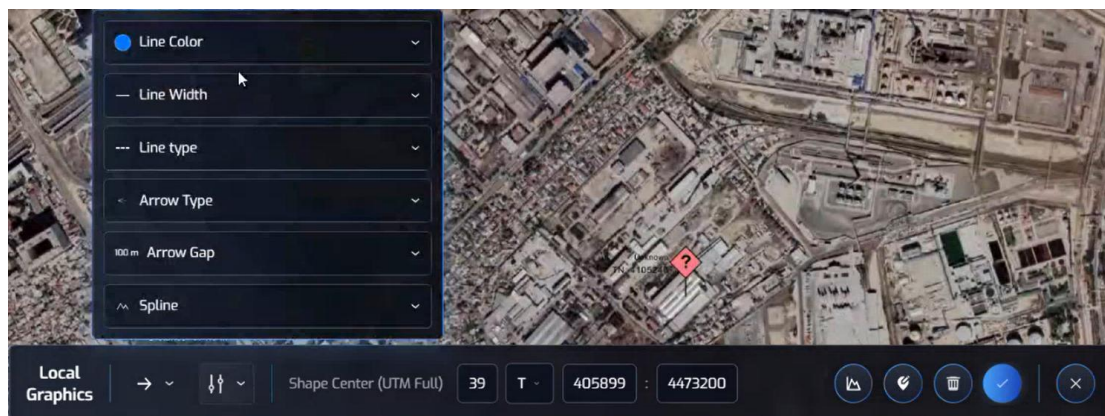
- 1) The shape is created on the map.
- 2) Shape modification aids appears (🔄) (⊕).



- h. To change the shape visual and display properties, tap the (⚙️) icon.

Result:

A context menu opens:



- i. Modify the display/visual properties as needed.
- j. When done, tap (✅).

Result:

The shape is displayed on the map.

NOTE

- The procedures for viewing a **Cross Section** (), **editing points** (), and **deleting** () a Line of Sight (LOS) are identical to those described for the Ruler tool.

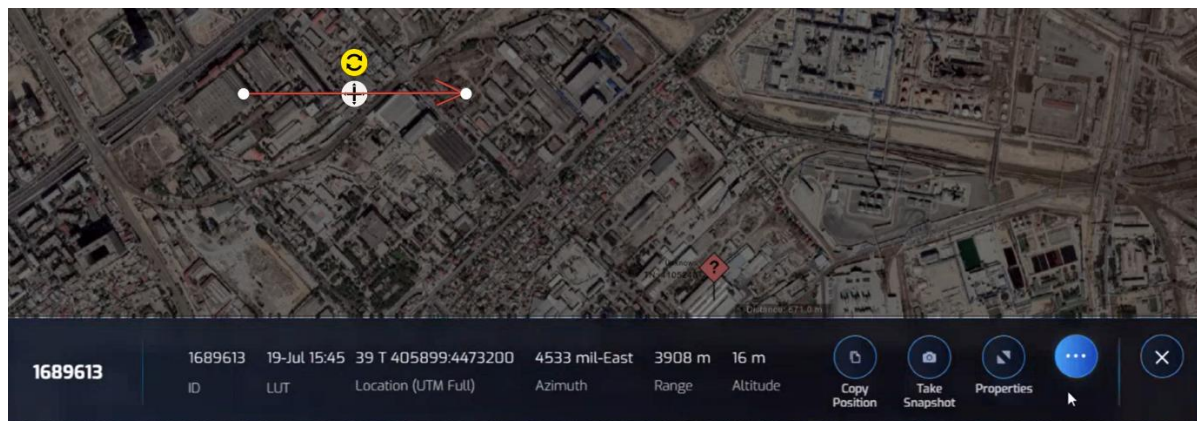
4.2.6.13 Actions on Local Graphic

Once a graphic object has been created, the user can view its properties, copy its location, and capture the current map view for documentation or operational use.

- After creating a graphic object on the tactical map, tap the graphic object.

Result:

The graphic object's details and properties are displayed.



- To view the graphic object's properties, tap **Properties**.

NOTE

The available properties depend on the selected graphic object type.

- c. To copy the graphic object's position, tap **Copy Position**.

NOTE

The location coordinates are copied to the system clipboard and can be pasted into other fields or applications.

- d. To capture the map display, tap **Take Snapshot**.

Result:

A snapshot of the current map view is created and displayed in the snapshot editor.

4.2.7 Chat Operations

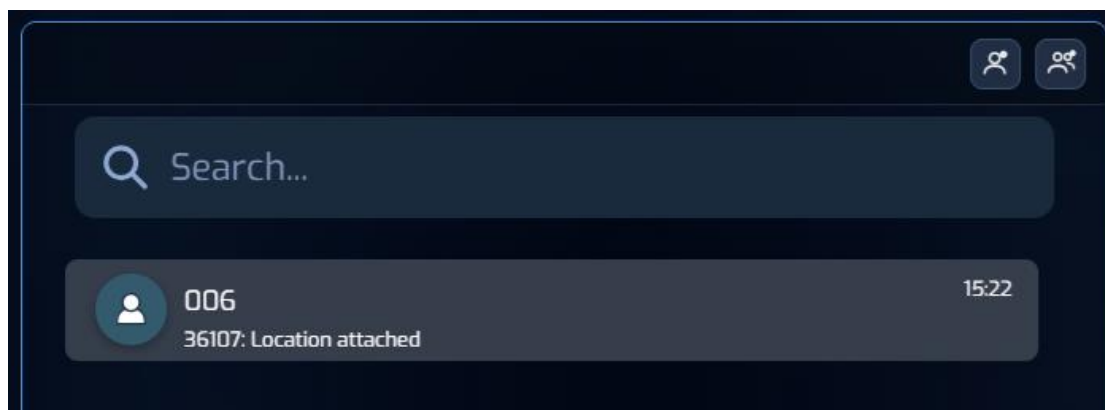
The chat window enables the DITA team to communicate with friendly forces via a convenient, interactive, and easy-to-operate interface. The user can start a conversation with a specific user or establish a user group focused on a defined subject, such as a combat team.

4.2.7.1 Opening the Chat Window

- a. Tap the **Chat** icon (.

Result:

The Chat window appears.

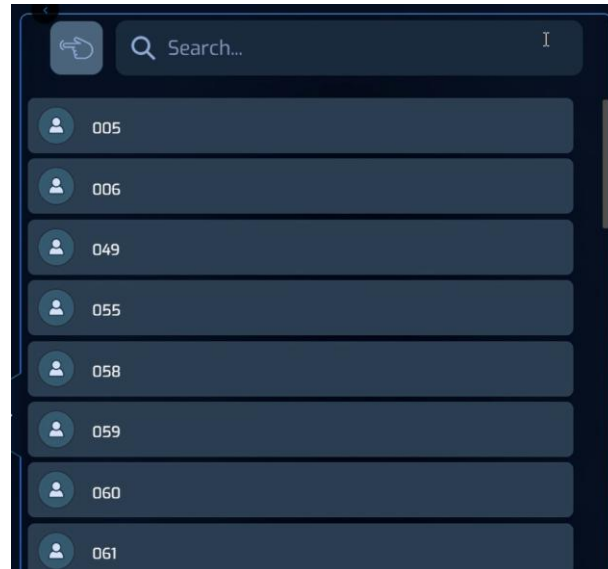


4.2.7.2 Initiating a Conversation with Network User

- a. In the **Chat** window, tap the  icon.

Result:

A window displaying all network users opens.



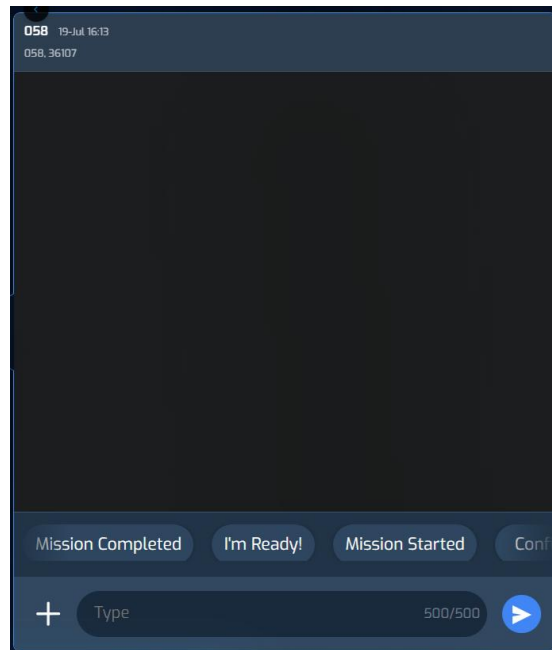
NOTE

- To find a specific user, type their details into the search bar. The list will automatically filter to show matching results.
- To select a user from the tactical map, tap the pick button () , and then select the user's platform directly from the map screen.

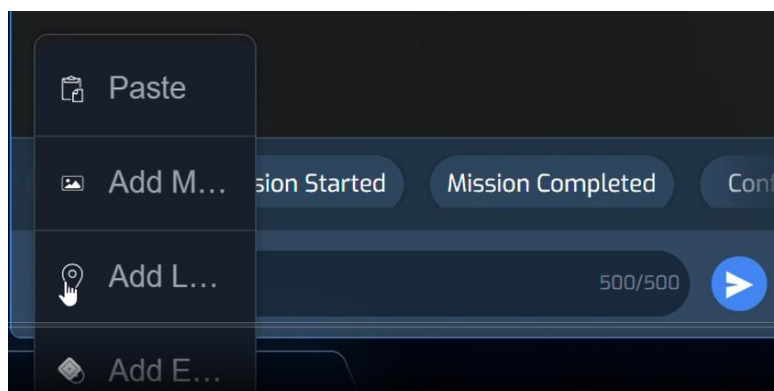
- b. Select the user with whom you want to initiate a conversation.

Result:

A private chat window with the selected user appears.



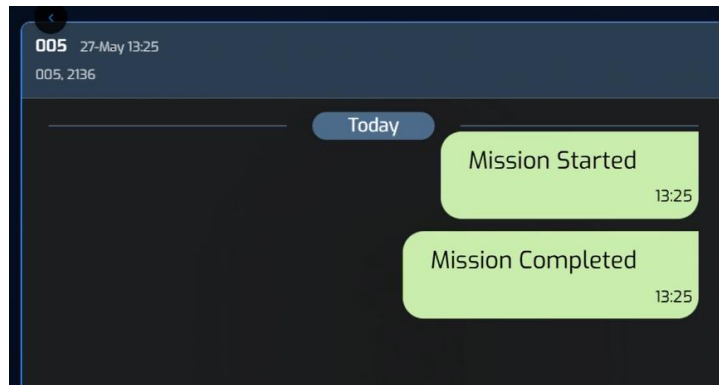
- c. Enter your message in the text input field, or select one of the defined Preset messages (e.g “Mission Started”).
- d. [Optional] To add an attachment, tap the "+" button and select one of the following options:
- **PASTE:** Inserts copied text data from the system clipboard into the input field.
 - **ADD MEDIA:** Enables the attachment of external imagery, sensor captures, or files to the message.
 - **ADD LOCATION:** Enables the user to select specific coordinates from the map.
 - **ADD ENTITY:** Enables the user to select a specific entity from the map.



- e. Tap the Send button (➤).

Result:

The message and any attached data are transmitted to the selected user.

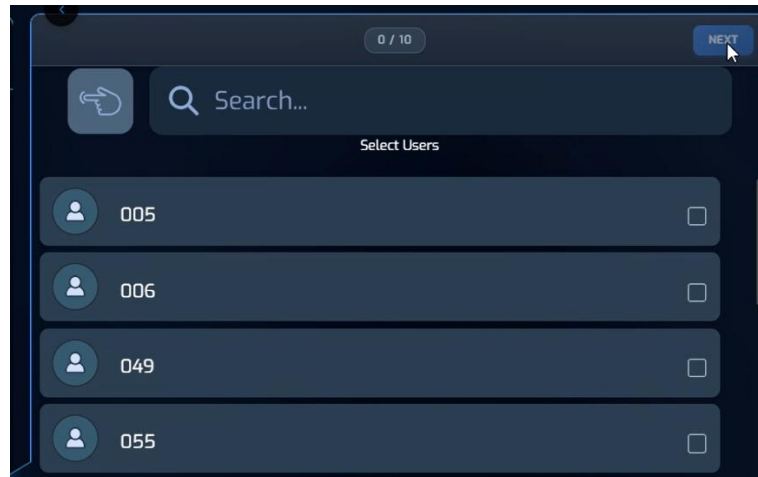


4.2.7.3 Creating a New Conversation Group

- a. In the **Chat** window, tap the New Group icon ().

Result:

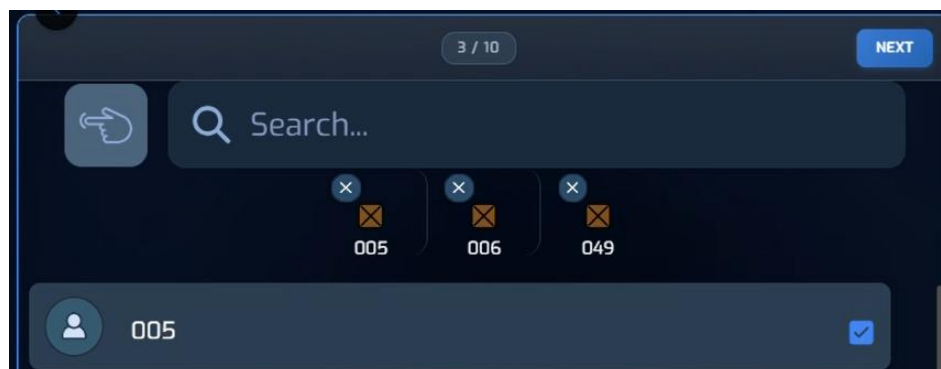
A window displaying all network users opens.



- b. Select up to ten (10) participants to add to the group conversation by tapping each user's record from the list.

Result:

The selected users, along with their unique IDs, are displayed and grouped above the main list.



- c. Tap **NEXT**.

Result:

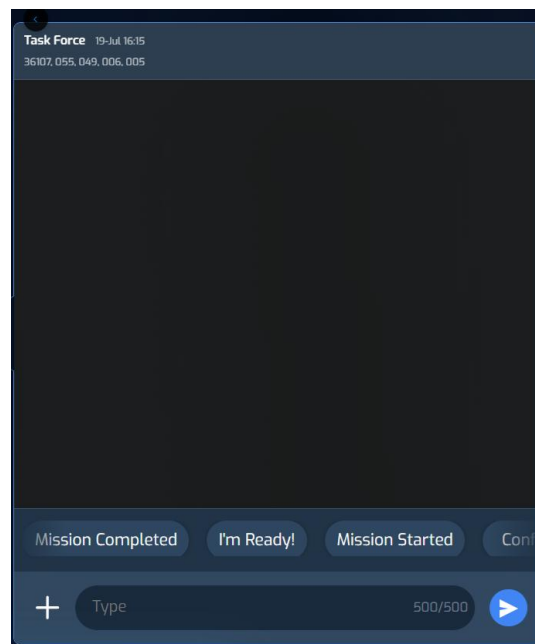
The New Group window appears.



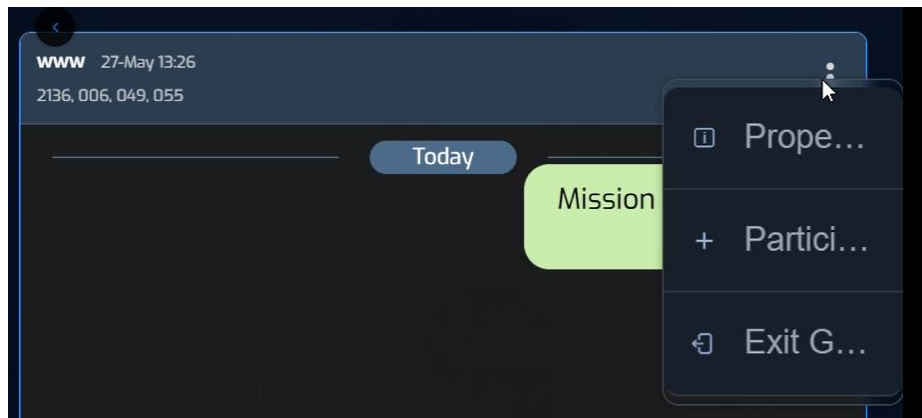
- d. In the Subject field, enter a name or topic for the group.
- e. Tap **CREATE**.

Result:

A new group chat window containing all selected participants appears.



- f. [Optional] In the active group chat window, tap the Action Icon (⋮) to access additional group management options:
- **Properties:** Opens the Add Participant window to view current participants, remove them from the group, or modify the group's subject/name.
 - **Participant:** Opens the chat window to select and add new participants to the group, or remove existing ones.
 - **Exit Group:** Disconnects the platform/unit from the active group.



4.2.8 TORCH-X-DITA Fire Mission Logic

This section describes the operational workflow of a fire mission within the C4I TORCH-X-DITA system, from mission creation through mission completion.

The **TORCH-X application** serves as the central command-and-control interface used to define the target, configure mission parameters, allocate ammunition, calculate firing data, and supervise mission execution.

Depending on the selected autonomy level, the system can support manual, standard, semi-automatic, or automatic mission execution while continuously monitoring platform readiness, safety constraints, weapon status, and firing progress.

The Fire Control System processes mission data and generates the required firing solution based on target information, ammunition selection, environmental inputs, and platform status.

Throughout the mission lifecycle, the operator can monitor mission status, apply fire corrections, modify ammunition allocation, select alternative firing solutions, recalculate mission data when required, and report mission progress.

Once all assigned rounds have been fired or the mission is terminated, the system updates the mission status and records the mission outcome for operational tracking and reporting purposes.

4.2.9 Meteorological (MET) Message Creation

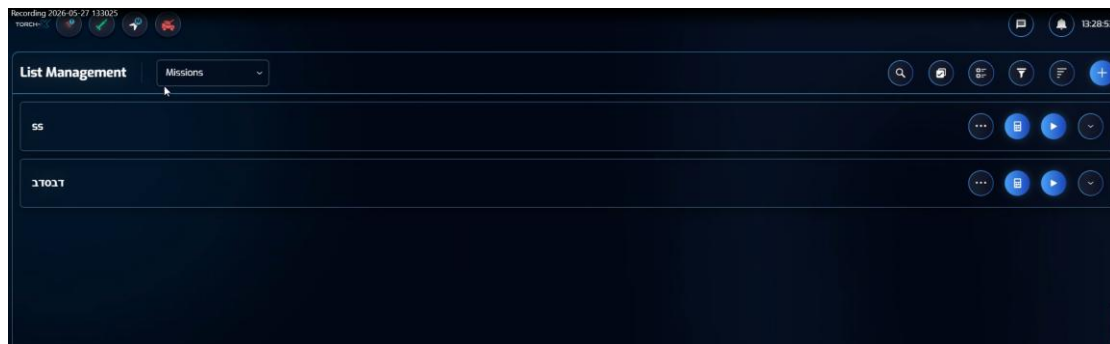
The process for creating **METEO11** and **METCM** meteorological messages is identical, except for certain fields that do not appear in **METCM** messages.

To create a new meteorological telegram, do the following:

- a. In the **Main Screens Navigation Bar**, tap **Data Center** button.

Result:

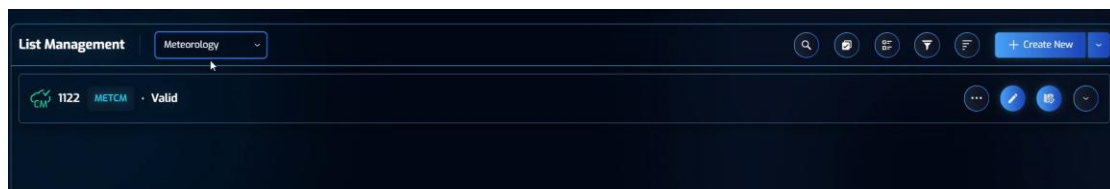
The **Data Center** screen appears.



- b. Tap the **List Management** menu and select the **Meteorology** option.

Result:

The Meteorology screen opens, displaying a list of meteorological messages that have been generated by or imported into the system.



- c. Tap the () in **Create New** button and select the **METEO11** option.

Result:

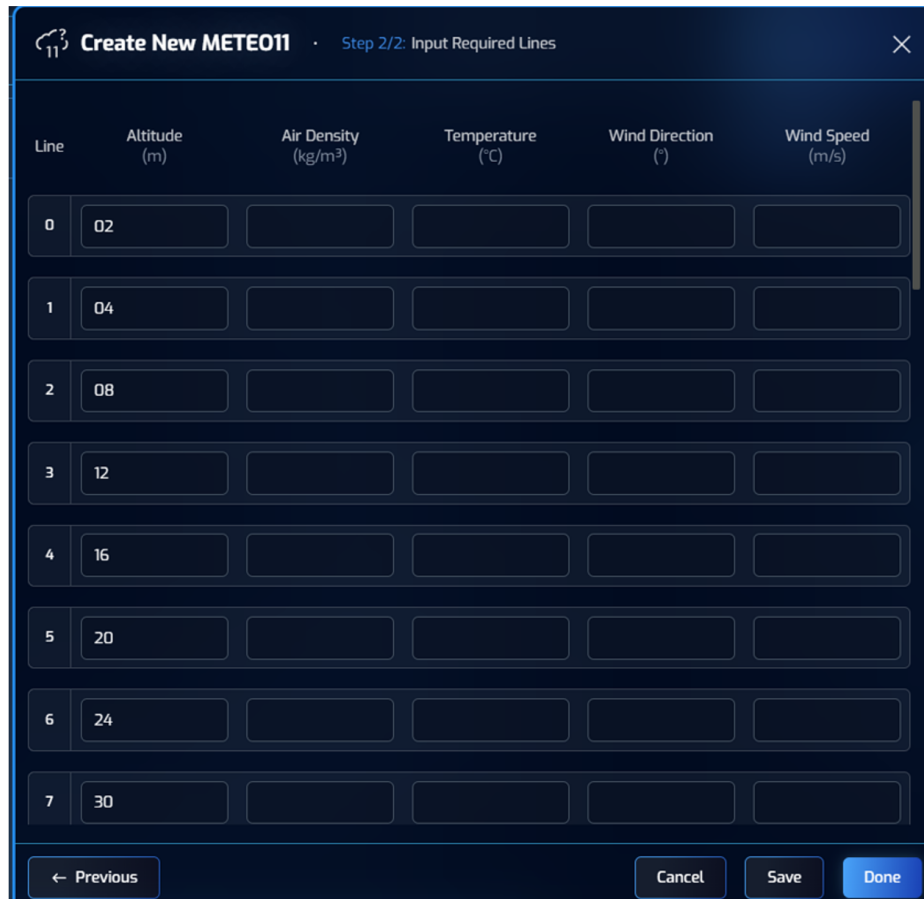
The **Create METEO11 Message** window opens.

- d. Select the message type (**Manual** or **Import**).
- e. In the **Telegram Name** field, enter a name for the Telegram.
- f. Define the station's location by performing one of the following actions in the **Location** field:
 - 1) Manual Entry: Manually enter the coordinates into the field.
 - 2) Map Pick: Tap the Pick icon (), and then select a specific point directly from the map screen.
- g. In the **Station Altitude** field, define the altitude of the meteorological station.
- h. In the **Start Time** and **Expiration Time** fields, define when the message creation time begins and when its validity expires.
- i. In the **Ground Pressure Deviation** field, enter the atmospheric pressure deviation value.
- j. In the **Ground Temperature Deviation** field, enter the temperature deviation value.

- k. Tap **Next**.

Result:

The system processes the parameters and proceeds to the next properties screen.



Line	Altitude (m)	Air Density (kg/m³)	Temperature (°C)	Wind Direction (°)	Wind Speed (m/s)
0	02				
1	04				
2	08				
3	12				
4	16				
5	20				
6	24				
7	30				

← Previous Cancel Save Done

- l. For each index number of the atmospheric layer/zone, enter the required data into the table. Pay strict attention to the specific units of measurement indicated at the top of each column.

- m. Tap **Done**.

Result:

The system processes the meteorological parameters, and the new message is displayed in the main Meteorology message list.

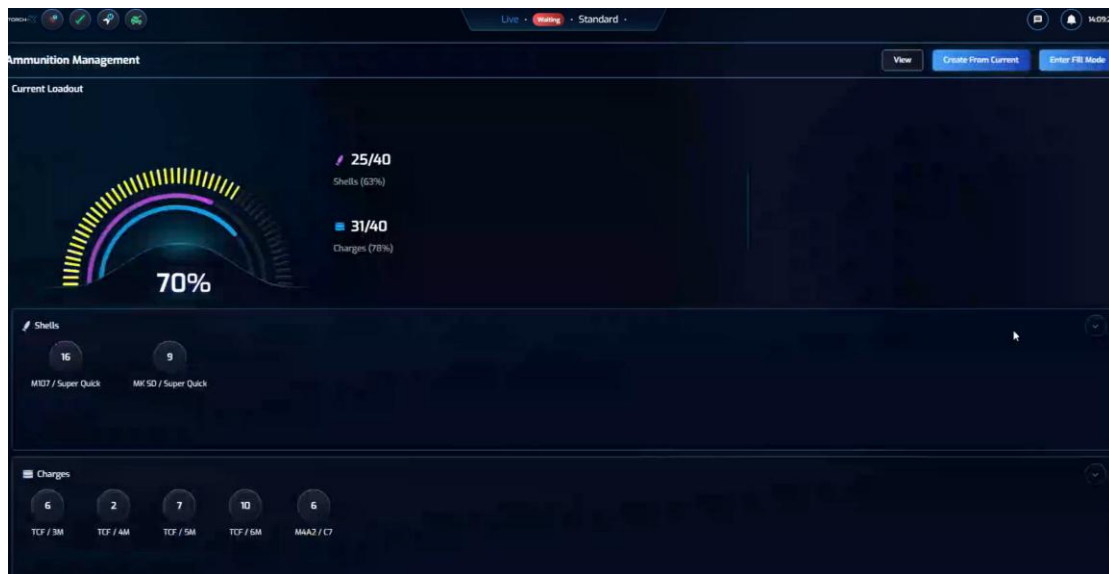
4.2.10 Ammunition Management

To view the current ammunition inventory and magazine allocation status, during an active mission, perform the following steps:

- In the **Main Screens Navigation Bar**, tap **Ammunition**.

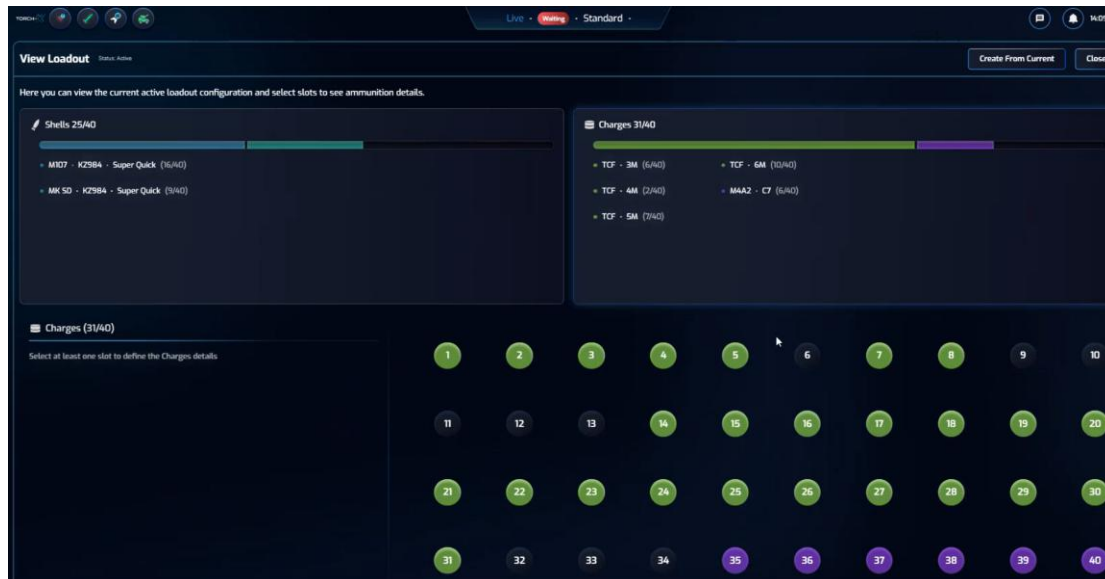
Result:

The **Ammunition Management** screen is displayed. The screen provides an overview of the current onboard ammunition inventory, including available shells and propellant charges.



b. Tap **View**.*Result:*

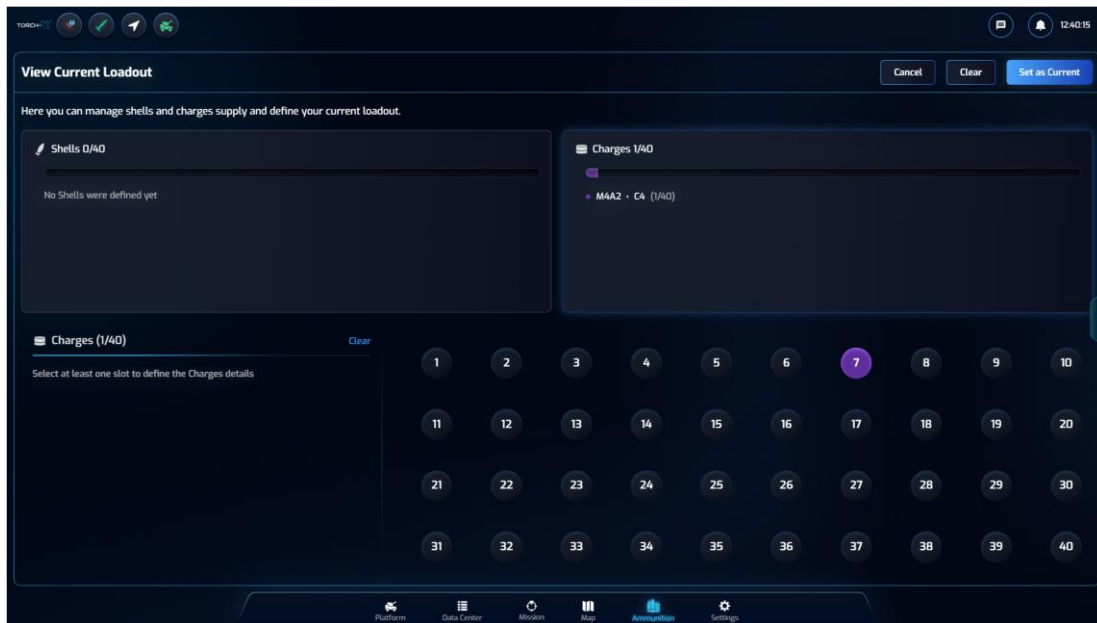
- 1) The **View Loadout** screen is displayed with a detailed view of the current ammunition loadout.
- 2) The ammunition magazine (Filling Boxes) layout is displayed, enabling the user to view which shells and propellant charges are assigned to each magazine slot.
- 3) The current ammunition inventory is displayed by ammunition type and quantity.

**NOTE**

The **View Loadout** screen is intended for monitoring and verification purposes and does not modify the current ammunition configuration.

c. [Optional] To change the current loadout, tap **Create from Current**.*Result:*

- 1) The **View Current Loadout** Screen is displayed.
- 2) The system enters fill mode, enabling ammunition loading and monitoring operations.



- d. Follow the steps described in section 4.2.5.3.

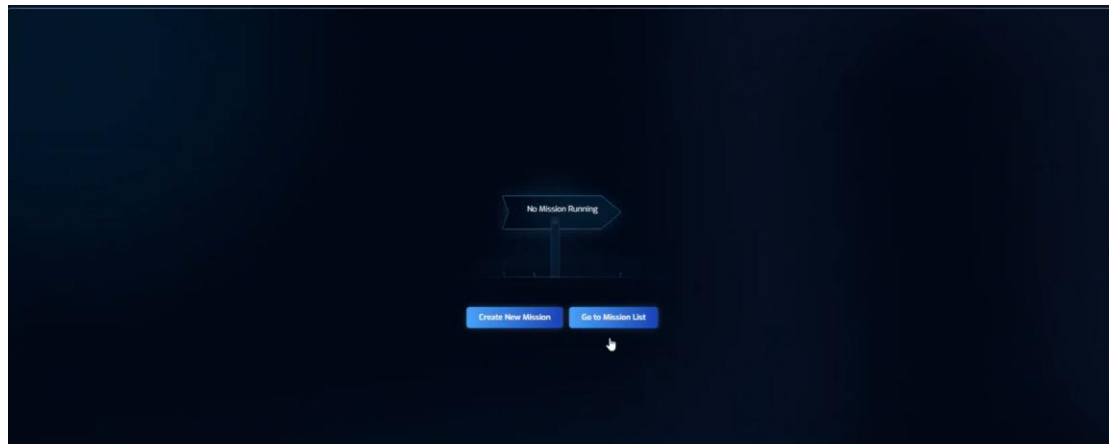
4.2.11 New Fire Mission Creation

A new mission can be created via the Map screen or the Mission screen.

- a. In the **Main Screens Navigation Bar**, tap **Mission** button.

Result:

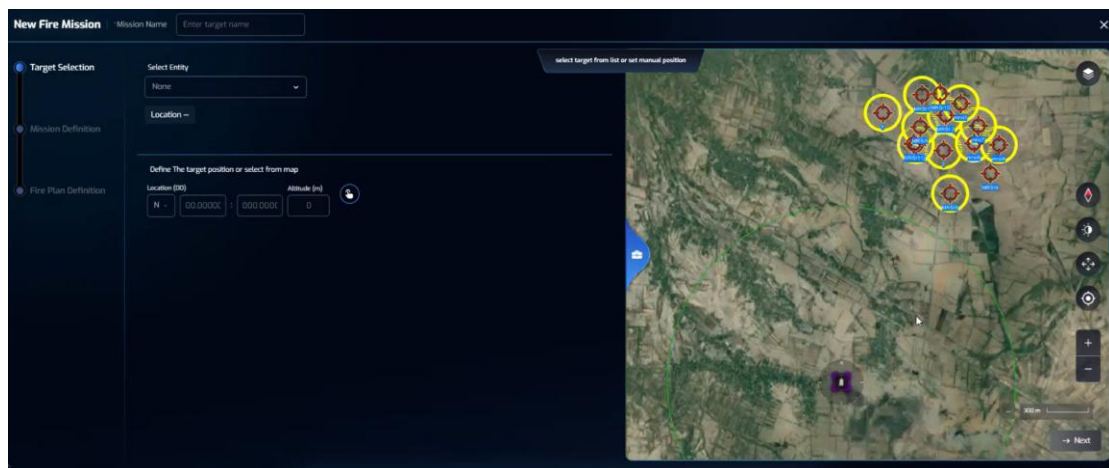
The **Mission** screen appears.



- a. Tap the **Create New Mission** button.

Result:

The **New Fire Mission** wizard opens at the **Target Selection** stage.



- b. In the **Mission Name** field, enter a name for the fire mission.
- c. In the **Select Entity** field, select the entity from the drop-down list.

Result:

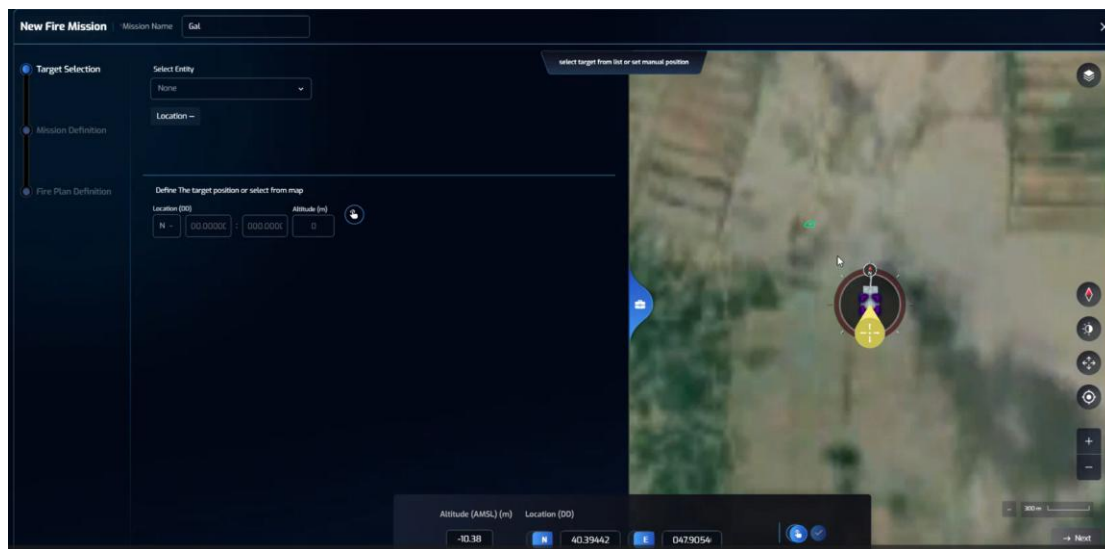
The **Location** field is updated with the entity position data.

- d. In the **Target Position** field, enter the target coordinates manually or

tap the pick icon ().

Result:

The target location annotation () and position window appears.



- e. Tap the position directly on the tactical map

Result:

The target's position and Altitude parameters appear in the position fields.

- f. Add graphics, objects, targets, quick missions and more using the map tools (see section 4.2.6)
- g. Tap **Next**.

Result:

The **Mission Definition** stage appears.

The screenshot shows the 'New Fire Mission' window with the 'Mission Definition' stage selected in the left sidebar. The main area contains the following fields:

- Purpose:** Radio buttons for 'Practice' (selected) and 'Live'.
- Trajectory Type:** Radio buttons for 'Low' (selected) and 'High'.
- Timing:** A date/time input field showing '06/22/26 13:27'.
- Mission type:** A dropdown menu with 'Fire For Effect' selected.

At the bottom, there are 'Previous' and 'Next' buttons. A small instruction box at the top right says 'Complete and select the fields below'.

- h. In the **Purpose** option, select the mission type: **Practice** or **Live**.

NOTE

Selecting Practice as the mission purpose affects the target **safety calculations**. Consequently, the target safety radius (displayed as the yellow circle around the target) is adjusted according to training-mode safety criteria rather than live-fire criteria.

- i. In the **Trajectory Type** field, select the trajectory: **High** or **Low**.
- j. In the **Timing** field, select or enter the required time.
- k. Tap **Next**.

Result:

The **Fire Plan Definition** stage appears.

The screenshot shows the 'New Fire Mission' window with the 'Fire Plan Definition' stage selected in the left sidebar. The main area is titled 'Fire Type' and contains the instruction 'Please Select a Fire Plan' with a large arrow pointing right. At the top right, there is a '+ Add fire plan' button. At the bottom, there are 'Previous', 'Play', and 'Save' buttons. A small instruction box at the top right says 'Select or add fire plan definitions'.

- I. Tap the **Add Fire Plan** button to add the required fire plans for the mission.

Result:

The **New Fire Plan** window opens.

- m. In the **Code Word** field, type a code word for the plan (for example: MRSI or MULTIPLE).
- n. In the **Rounds** field, select the number of rounds for the engagement using the + and - buttons.
- o. In the **Method of Burst** field, select Burst firing mode from the drop down-menu.
- p. Use the **Max Fire Rate** toggle button to select **YES** (for maximum fire rate) or **NO**.

NOTE

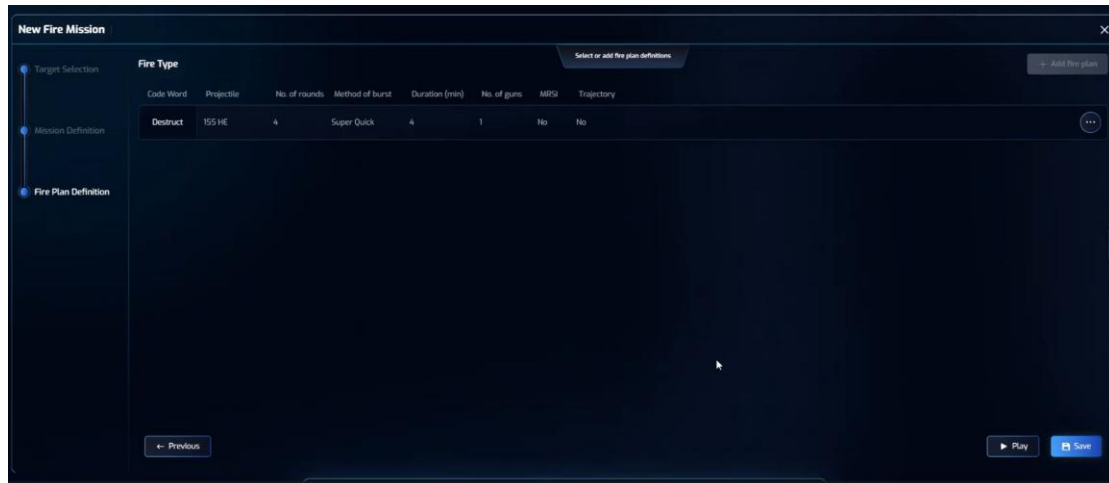
- Tapping **No** displays the Duration (min) field, enabling the user to set the fire rate in minutes.

- If you wish the firing plan to run at a sub-maximum rate, use the + / - buttons to select the interval (in minutes) between rounds.

- q. To define the mission as MRSI, select the **MRSI** checkbox.
- r. Tap **Add**.

Result:

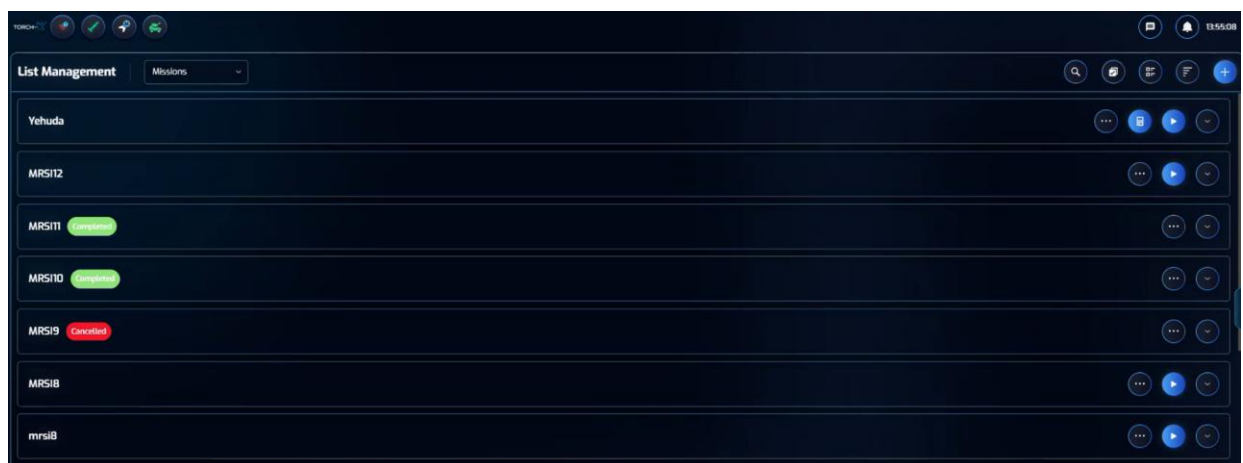
The fire plan is added to the fire plan definition list.



- s. Repeat the Steps above to add additional fire plans, if required.
- t. To save the mission, tap **Save** or tap **Play** to calculate, process and execute the mission immediately.

*Result: [If **Save** is tapped]*

The fire mission is saved and added to the mission list on the **Data Center** screen.

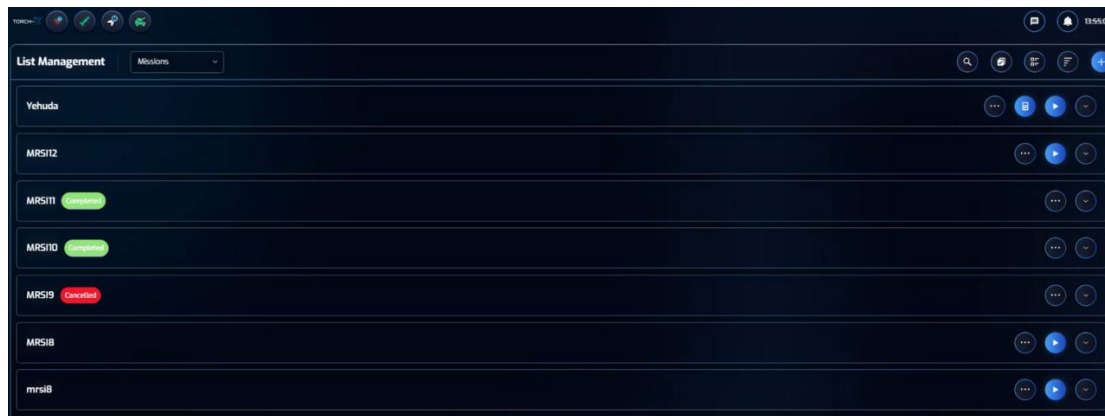


4.2.12 Execute Fire Mission (Standard Mode)

- a. In the navigation menu, tap **DATA CENTER**.

Result:

The **DATA CENTER** screen appears.

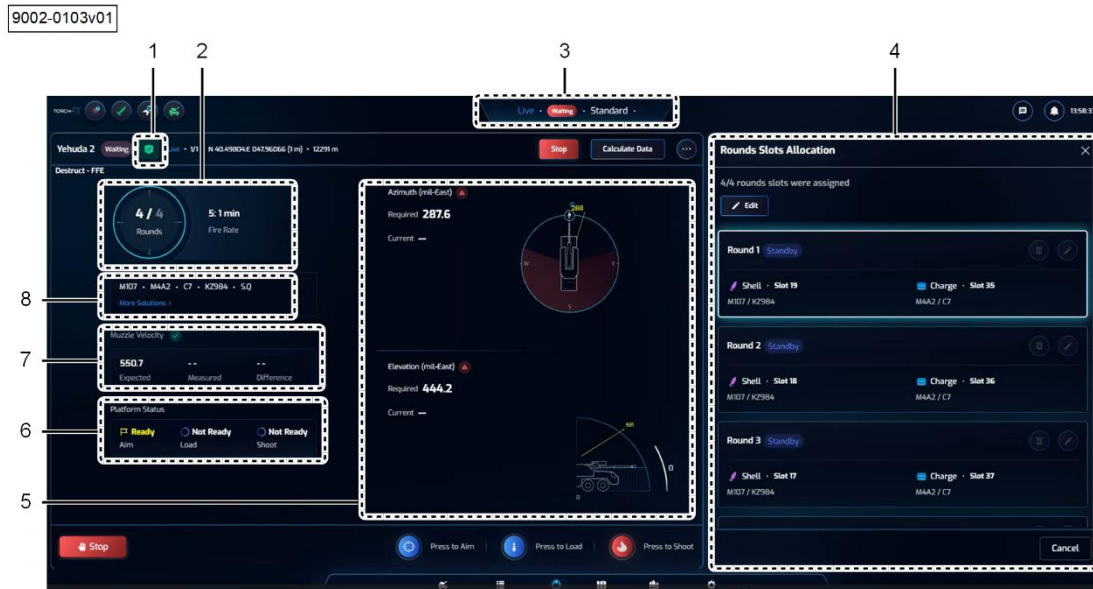


- b. Select the desired mission and tap the **PLAY** button ().

Result:

- 1) The mission is activated and calculated.
- 2) The **MISSION** screen is displayed with the following information:
 - a) The mission status (3), mission type (**Practice** or **Live**), and autonomy Level (e.g., **Standard**) appear at the top of the screen.
 - b) The allocated **Rounds** count for the mission (2).
 - c) The rounds and propellant charge types selected for the current firing mission (located on the right side of the screen in the **Rounds Slots Allocation** window) (4).
 - d) The mission safety status () (located in the mission menu) (1).
 - e) The **Required** and **Current** elevation and azimuth angles (5), including their corresponding infographics and the laying status:
 - (1) A visual indicator confirms that the system is **on-target** in azimuth ().
 - (2) A visual indicator confirms that the system is **on-target** in Elevation ().

- f) The optimal **Firing Solution** (8) chosen for the mission (round/ propellant / charge/ Fuze).
- g) The **Expected** and **Measured** Muzzle Velocity data (7).
- h) The **Platform Status** (Aim / Load/ Shoot) displays the system readiness (Ready/ Not Ready) (6).



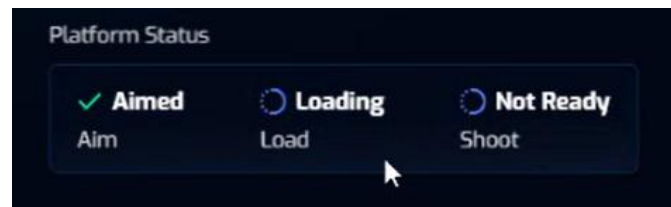
NOTE

If the **Rounds Slots Allocation** window is closed and you need to access it again, tap the action button (⋮) and select the **Ammunition** option.

- c. Tap the **Press to Aim** button.

Result:

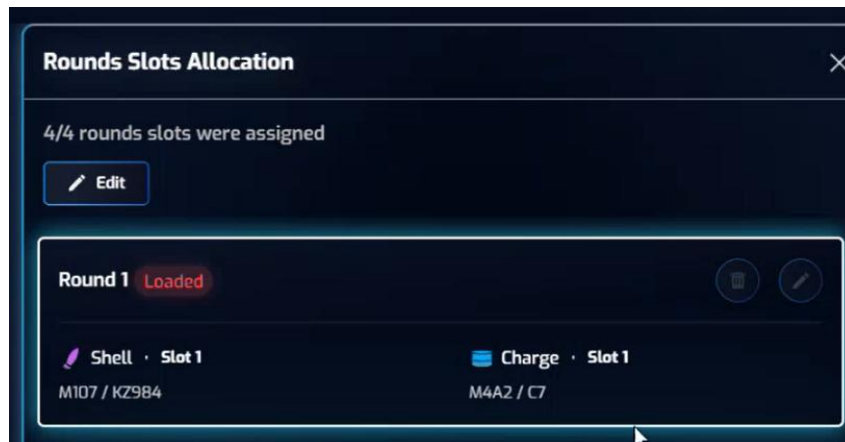
- 1) The system aligns and aims at the target. Upon completion, the **Aim** status under the Platform Status section changes to **Aimed**.



- d. Tap the **Press to Load** button.

Result:

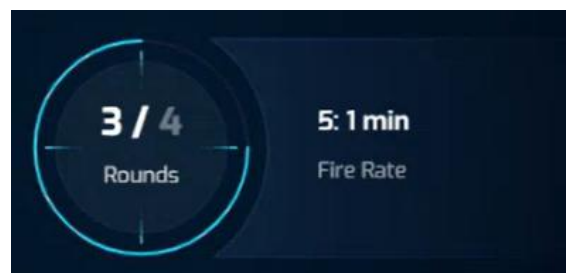
- 1) The system loads the shell and the propellant charge (based on the definitions in the Rounds Slots Allocation window). Upon completion, the **Load** status under the Platform Status section changes to **Loaded**.
- 2) The Round status in Rounds Slots Allocation window changes from **Standby** → **Loaded**.



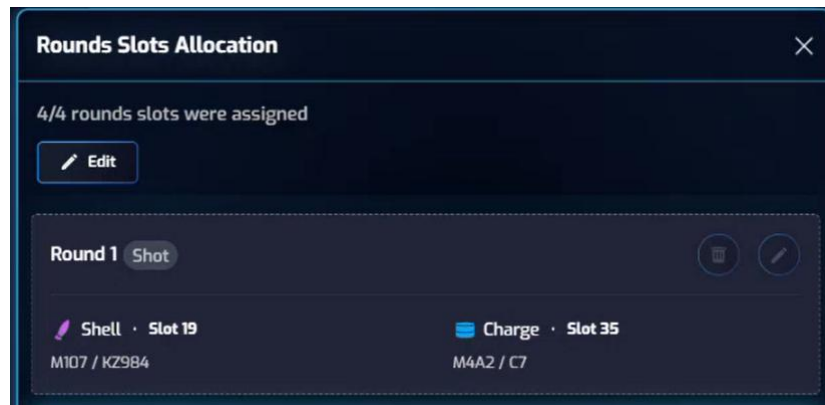
- e. Tap the **Press to Shoot** button.

Result:

- 1) The system fires the round.
- 2) the **Shoot** status under the Platform Status section changes momentarily to **Firing**, and then reverts to **Not Ready** until the next firing cycle.
- 3) The fired round is deducted from the remaining ammunition count allocated for the mission.



- 4) In the **Rounds Slots Allocation** window, the fired round is grayed out, and its status changes from **Standby** to **Shot**.



NOTE

If the system fails to detect that a round was fired and does not deduct the round, the user can manually override the status by tapping the **Actions** button (⋮) and selecting **Simulate Fire** option.

4.2.13 Execute Fire Mission (Automatic Mode)

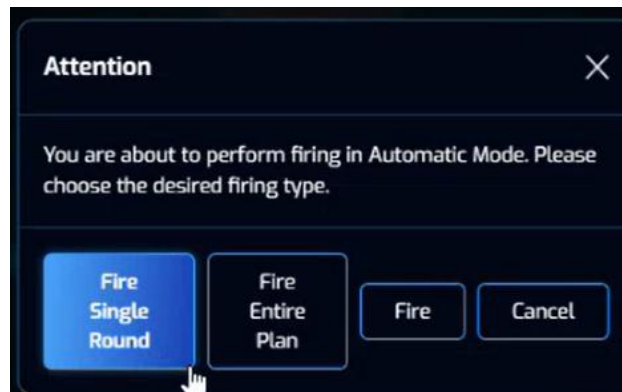
NOTE

- In **Automatic Mode**: The entire firing process (including the firing command and primer percussion) is executed by the DITA platform.
- In **Semi-Automatic Mode**: The system prepares itself for firing (Aiming/Loading), but the final firing command (**SHOOT**) is executed via the TORCH-X Application (C4I).

- In the active mission window, tap the **Actions** button (⋮), select **Mission Mode**, and then select **Automatic Mode**.

Result:

The **Attention** window appears.



- Select the desired option: **Fire Single Round**, **Fire Entire Plan**.
- Tap the **Fire** button.

Result:

- 1) The system executes the firing sequence according to the predefined firing plan.
- 2) The **Shoot** status under the Platform Status section changes momentarily to **Firing**, and then reverts to **Not Ready** until the next firing cycle.
- 3) The fired round is deducted from the remaining ammunition count allocated for the mission.



- 4) In the **Rounds Slots Allocation** window, the fired round is grayed out, and its status changes from **Standby** to **Shot**.

4.2.14 Execute Fire Mission (Semi-Automatic Mode)

NOTE

- In **Automatic Mode**: The entire firing process (including the firing command and primer percussion) is executed by the DITA platform.
- In **Semi-Automatic Mode**: The system prepares itself for firing (Aiming/Loading), but the final firing command (**SHOOT**) is executed via the TORCH-X Application (C4I).

- a. In the active mission window, tap the **Actions** button (⋮), select **Mission Mode**, and then select **Semi-Automatic Mode**.

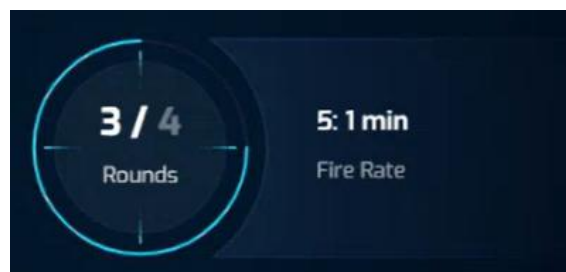
Result:

- 1) The system switches to **Semi-Automatic Mode**.
- 2) The system performs **Aiming** and ammunition **Loading** procedures, then waits for a firing command from the operator.

- b. Press the **Press to Shoot** button.

Result:

- 1) The system executes the firing sequence according to the predefined firing plan.
- 2) The **Shoot** status under the Platform Status section changes momentarily to **Firing**, and then reverts to **Not Ready** until the next firing cycle.
- 3) The fired round is deducted from the remaining ammunition count allocated for the mission.

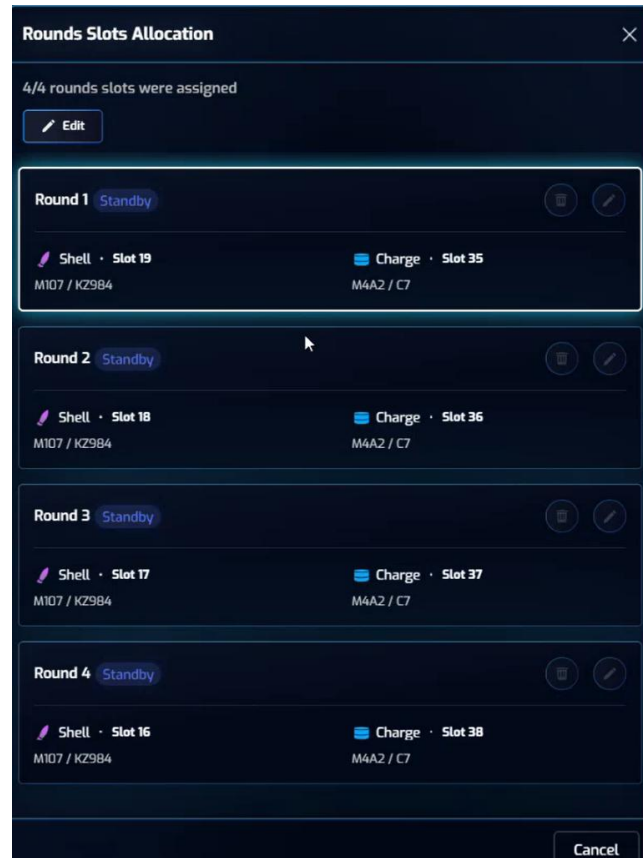


- 4) In the **Rounds Slots Allocation** window, the fired round is grayed out, and its status changes from **Standby** to **Shot**.

4.2.15 Active Mission Operations

4.2.15.1 Changing the Rounds Allocation

- a. If the **Rounds Slots Allocation** window is closed or hidden, open it by tapping the **Action** button (⋮) and selecting the **Ammunition** option.



- b. In the **Rounds Slots Allocation** window, tap **Edit**.

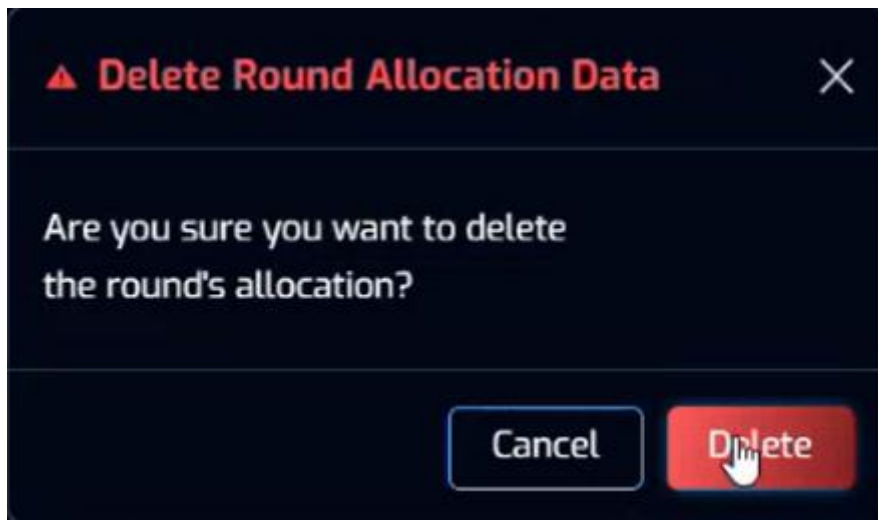
Result:

- 1) The window switches to Edit Mode.
- 2) An **Edit** icon (✎) and a **Delete** icon (🗑) are displayed next to each round in the allocation list.

- c. To **remove** a round from the allocation list, tap the Delete (🗑) icon.

Result:

Notification pop-up message appears:



- d. Tap **Delete**.

Result:

The selected round is removed from the mission's round allocation list.

- e. To modify an existing round, tap the **Edit** icon ().

Result:

The **Edit Round** window opens, enabling the user to modify the round's (shells tab) ammunition and charge configuration (charges tab).



NOTE

Changes made to the round allocation may affect the calculated firing solution and mission ballistic data.

- f. In the **Shells** tab, select the required shell by tapping the appropriate slot.

Result:

The selected shell is highlighted and assigned to the round being edited.

- g. Tap the **Charges** tab.

- h. Select the required charge by tapping the appropriate slot.

Result:

The selected charge is highlighted and assigned to the round being edited.

- i. Tap **Apply**.

Result:

The selected shell and charge configuration (if applicable) replaces the previously assigned round in the Rounds Slots Allocation window.

NOTE

Only ammunition items available in the current onboard loadout can be selected and assigned to a round.

CAUTION

Changing the shell type, fuze, propellant, or charge may affect the calculated firing solution and mission execution parameters. Reprocess (Calculate Data) the mission, if required, to ensure that the updated ammunition configuration is reflected in the firing data.

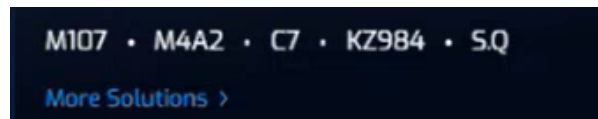
- j. Tap **Save**.

Result:

The Rounds Slots Allocation window updates and displays the current allocated rounds.

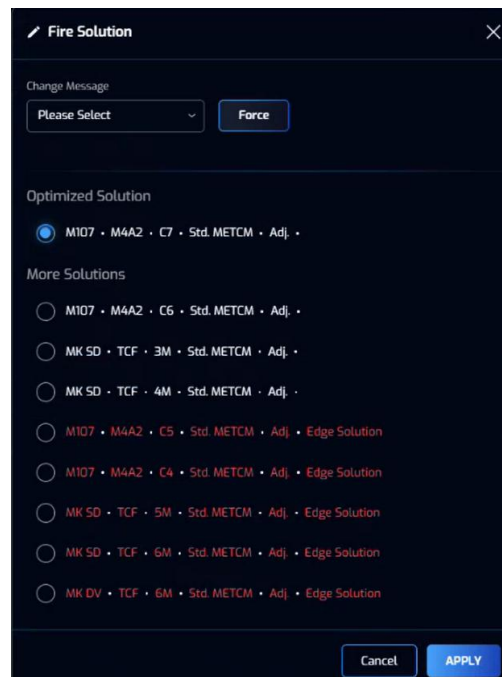
4.2.15.2 Selecting Alternative Fire Solution

- a. In the **Active Mission** window, tap the **More Solutions** field:



Result:

The **Fire Solution** window opens.



NOTE

Fire solutions displayed in red text indicate firing solutions that cannot be calculated accurately or do not meet operational, ballistic, or safety constraints.

- b. In the **Change Message** drop-down menu, select the required meteorological message.
- c. Tap **Force**.

- d. In the **More Solutions** section, select the required fire solution from the list of available solutions.
- e. Tap **Apply**.

Result:

- 1) The selected fire solution is applied to the active fire mission.
- 2) The mission data is recalculated and processed to match the selected fire solution.
- 3) The **Rounds Slots Allocation window** is updated and displays the revised round allocation for the mission based on the selected fire solution.

NOTE

If the mission data is modified while a **round is loaded** or a loading sequence is in progress, the system may require the mission to be reprocessed before the updated fire solution can be executed.

4.2.15.3 Processing the Mission (Restart)

If any modifications are made to the mission data (such as changing rounds slots allocation, modifying the firing solution, or any other updates), the user can reprocess the mission and firing data.

To process the mission data:

- a. Tap the **Actions** button () and select the **Process** option.

Result:

The mission data is recalculated, and the updated values are displayed.

4.2.15.4 Inserting New Fire Correction

- From the **Active Mission** screen, tap the **Actions** button (⋮) and select **Grid Correction** option.

Result:

The **Grid Correction Request** window opens.

- Enter the correction data and tap **Save**.

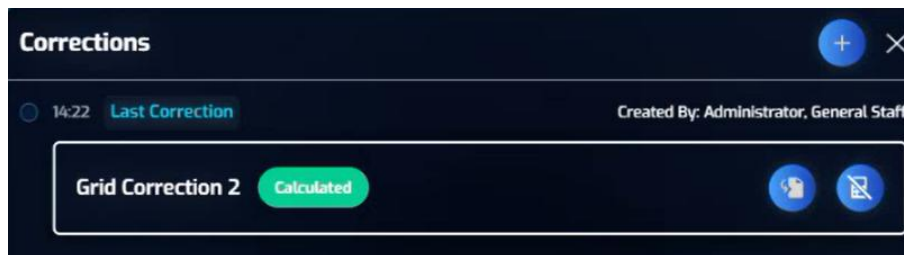
Result:

The correction appears in the Corrections window, along with the correction time, source (where it was received from), calculation status (whether it was calculated or not), and the user who entered the correction.

- c. Tap the **Calculate** button ().

Result:

- 1) The Fire correction is calculated and applied to the mission data for ballistic calculations and to the Azimuth and Elevation Calculated values.
- 2) The correction status changes to **Calculated**.



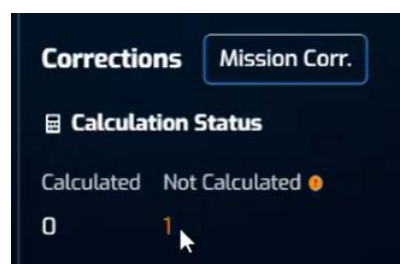
NOTE

- The presence of an entered correction in the **Corrections** window does not mean it is applied to the firing data. It will only take effect after tapping

Calculate ().

- The system alerts the user about fire corrections that were entered but not calculated (the warning appears in the Mission Safety window under the **Corrections** section).

Yehuda 2 Waiting  Live • V1 • N 40.49804E 047.96066 (1 m) • 12291 m



4.2.15.5 Calculate Fire Mission

- a. On the **Active** mission screen, tap the **Calculate Data** button.

Result:

The mission data is calculated, and updated values appear in the mission and firing data fields.

NOTE

The system will not calculate new mission data after tapping **Calculate Data** button if a round is currently Loaded or in the process of being loaded.

4.2.15.6 Concluding and Reporting a Mission

Once all rounds have been fired, report the completion of fire to the commander.

- a. Tap **Actions** button (⋮) and select **Report to Commander**.

Result:

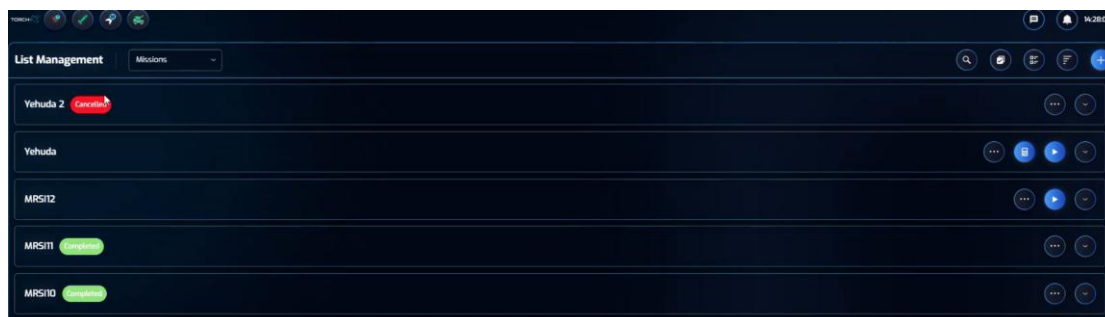
The report window opens.

- b. In the opened window, select **Rounds Complete**.

- c. Tap the **Actions** button (⋮) and select **End Mission**.

Result:

- 1) The mission is concluded.
- 2) On the **Data Center** screen, the mission status changes to **Completed**.

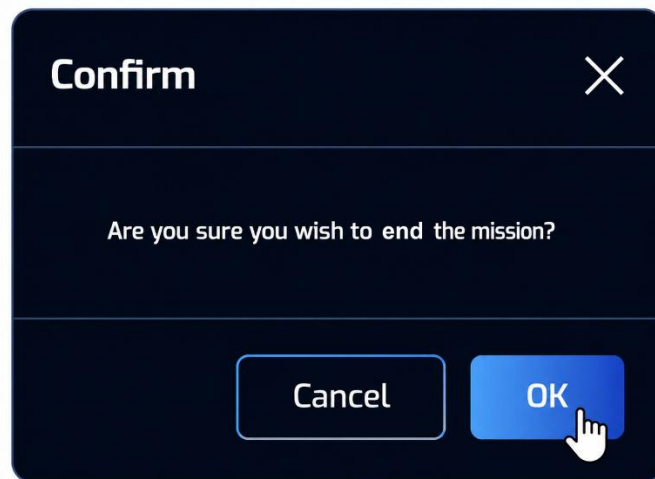


4.2.15.7 Cancelling a Mission

- a. Tap **Actions** button (⋮) and select **Cancel Mission**.

Result:

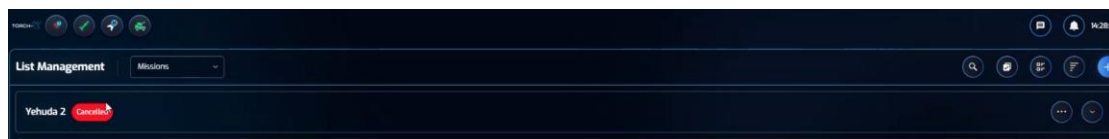
The **Confirm** window opens.



- b. In the opened window, tap **OK**.

Result:

- 1) The mission is cancelled.
- 2) The active mission window closes.
- 3) On the **Data Center** screen, the mission status changes to **Cancelled**.

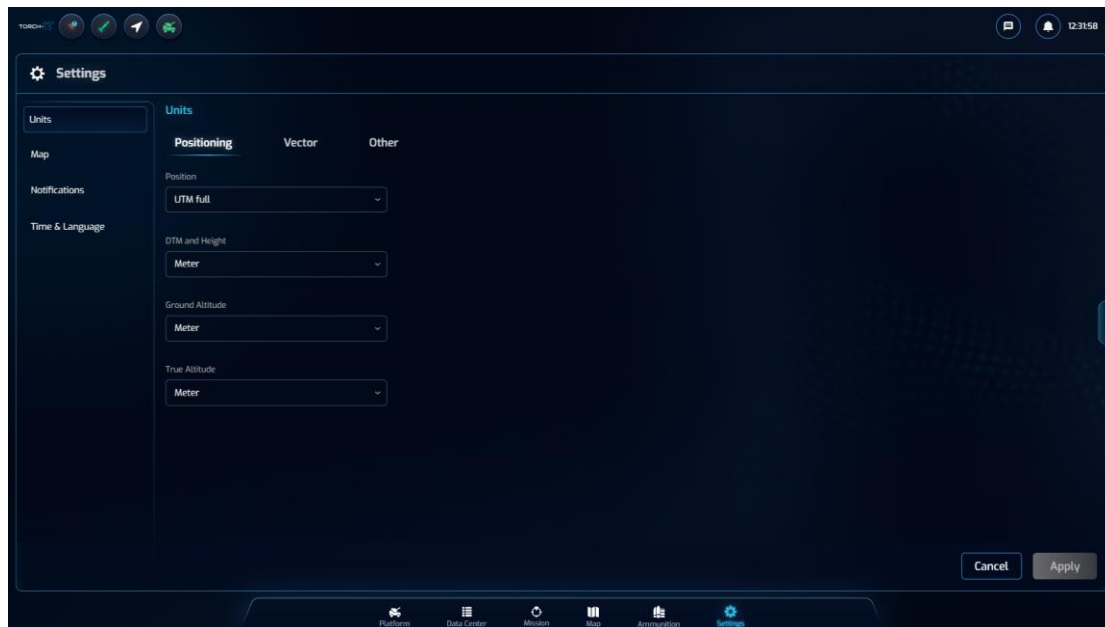


4.2.16 UI Language Toggle

- a. In the screen navigation bar, tap the **Settings** button.

Result:

The **Settings** screen appears.



- b. In the left menu, tap **Time & Language**.

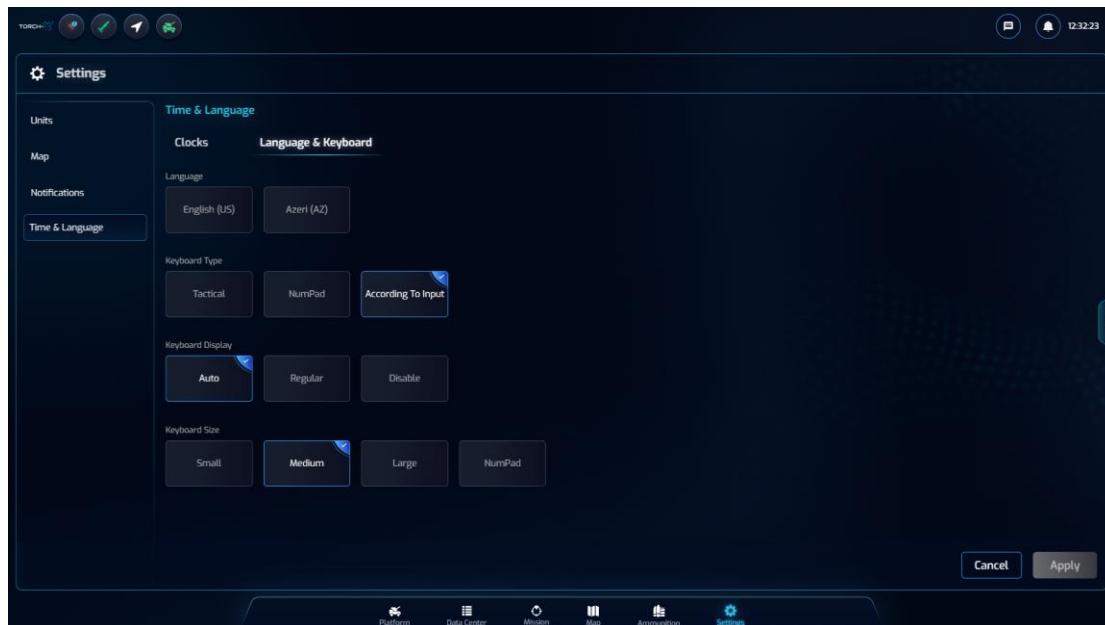
Result:

The **Time & Language** screen appears.

- c. Tap the **Language & Keyboard** tab.

Result:

The **Language & Keyboard** menu/section opens.



- d. In the **Language** section, tap **Azeri (AZ)**.

Result:

The application restarts and opens in the selected language.